

# The Pilot's Radio Handbook

Edition

24



Dietlind Lempp



**Dietlind Lempp**

# **The Pilot's Radio Handbook**

**24<sup>th</sup> edition**

**An easy to understand guide  
to radio procedures for pilots  
and  
an essential textbook  
for passing the exam and skills tests  
for the Restricted and General  
Radio Telephony Certificates  
(Aeronautical)**

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# Preface

Good radio procedures are essential. Not only do they help to streamline air traffic, they are essential to the safety of aircraft. It is an absolute necessity that pilots understand the air traffic system and are able to work smoothly within it.

Lack of knowledge leads to fear. Many aspiring pilots are afraid of the subject “radio” and dread having to sit the examinations for the Restricted or General Radio Telephony Certificates (ROC).

Once the examinations are over and real life begins, the fear is often still there. This book has been written to help pilots understand what “radio” is all about and to take the mystery out of the procedures associated with it. It should help both the novice and the more experienced pilot to understand the air traffic control system better, and therefore make flying safer and more enjoyable.

A while ago I received the following e-mail:

**“Thinking that I'll need to sit through hours of boring material, I was so surprised by the fact that the book was such an interesting read, fascinating actually. It really tied together all the info I took in on my theory course.**

**I read it 4 times over a few days, and took the test with great confidence.  
I passed with a score of 87%!!!!!!!  
All thanks to your book!**

**All the best, kind regards**

**Talie Raviv”**

I hope that many more pilots will use this book not only to study for the exams, but also to read it again every now and then, as a companion in their flight bag and as a reference for anything to do with the dreaded subject “radio”.

This 24<sup>th</sup> edition has been updated as usual to reflect the changing environment, mainly the chapter on ADS-B as well as Appendix 1.

All links to websites given in this book were correct and tested at the time of writing (November 2025). But, as with all internet-based information, this could change at any time.

Updates to this latest edition of the Radio Handbook are published when necessary. They can be downloaded free of charge from the website [www.radiobooks.co.za](http://www.radiobooks.co.za). There is also some other useful information to be found on the website, and readers are encouraged to visit from time to time.

Readers can e-mail me at [dlempp@gmail.com](mailto:dlempp@gmail.com) with any suggestions, discrepancies, irregularities, mistakes, oversights or any other comments whatsoever. The author appreciates readers' inputs.

Last, but not least, I wish to express my gratitude to everybody who has helped me make the book better over the years. The Beechcraft Debonair ZS-AIR on the cover belongs to Dr. Mike Brown, and Bruce Perkins kindly supplied the photograph of the aircraft. Several readers e-mailed me with suggestions of what to include in the book or where to expand on the explanations. It is much appreciated.

# *Dietlind Lempp*

Cape Town, November 2025

**N.B. Please note that all information in this book is subject to change, especially frequencies, airspaces, transition altitudes and the like. Therefore this book should not be used for flight planning purposes without additional reference to the valid and up-to-date official publications.**

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# Abbreviations

Aviation is fraught with abbreviations, which are often very daunting to understand for the novice. Hopefully the list below will help to clear up the confusion.

|         |                                                              |
|---------|--------------------------------------------------------------|
| A/c     | Aircraft                                                     |
| ABAS    | Air Based Augmentation System for GNSS Instrument Approaches |
| Abm     | Abeam                                                        |
| ACAS    | Airborne Collision Avoidance System                          |
| ACC     | Area Control Centre                                          |
| ADA     | Advisory Area (not used anymore)                             |
| ADF     | Automatic Direction Finding                                  |
| ADNL    | Additional (Jeppesen)                                        |
| ADR     | Advisory Route (not in use anymore)                          |
| ADS     | Automatic Dependent Surveillance                             |
| AFB     | Air Force Base                                               |
| AFI     | Africa Region (ICAO) for communication purposes              |
| AFIS    | Aerodrome Flight Information Service                         |
| AGL     | Above Ground Level                                           |
| AIC     | Aeronautical Information Circular                            |
| AIM     | Aeronautical Information Management                          |
| AIP     | Aeronautical Information Publication                         |
| AIP SUP | AIP Supplement                                               |
| AIRAC   | Aeronautical Information Regulation and Control              |
| AIS     | Aeronautical Information Service                             |
| ALERFA  | Alert Phase                                                  |
| ALT     | Altitude                                                     |
| AMSL    | Above Mean Sea Level                                         |
| APCH    | Approach                                                     |
| APP     | Approach Control                                             |
| ARP     | Aerodrome Reference Point                                    |
| ATA     | Aerodrome Traffic Area (now obsolete) - see ATZ Class G      |
| ATC     | Air Traffic Control                                          |
| ATFM    | Air Traffic Flow Management                                  |
| ATIS    | Automatic Terminal Information Service                       |
| ATM     | Air Traffic Management                                       |
| ATNS    | Air Traffic and Navigation Services Company                  |

|          |                                                                                          |
|----------|------------------------------------------------------------------------------------------|
| ATPL     | Airline Transport Pilot Licence                                                          |
| ATS      | Air Traffic Services                                                                     |
| ATSU     | Air Traffic Services Unit                                                                |
| ATU      | Air Traffic Unit                                                                         |
| ATZ      | Aerodrome Traffic Zone Class C (controlled) or G (uncontrolled)                          |
| AWY      | Airway                                                                                   |
| B-VLOS   | Beyond Visual Line of Sight (for drones)                                                 |
| CAA      | Civil Aviation Authority, also see SACAA                                                 |
| CAMU     | Central Airspace Management Unit                                                         |
| CAR      | Civil Aviation Regulations                                                               |
| CATS     | Civil Aviation Technical Standards                                                       |
| CAVOK    | Ceiling And Visibility OKay                                                              |
| CDFA     | Constant Descent Final Approach                                                          |
| CDI      | Course Deviation Indicator                                                               |
| COM      | Communications                                                                           |
| CPL      | Commercial Pilot Licence                                                                 |
| CTA      | Control Area                                                                             |
| CTL      | Control                                                                                  |
| CTOT     | Calculated take-off time                                                                 |
| CTR      | Control Zone                                                                             |
| DA       | Decision Altitude – for 3D Instrument Approaches                                         |
| DETRESFA | Distress Phase                                                                           |
| DF       | Direction Finding                                                                        |
| DH       | Decision Height – for 3D Instrument Approaches                                           |
| DME      | Distance Measuring Equipment                                                             |
| DST      | Daylight Saving Time                                                                     |
| EAT      | Expected Approach Time (leaving a beacon or holding waypoint for an instrument approach) |
| EFIS     | Electronic Flight Instrument System                                                      |
| ELBA     | Emergency Locator Beacon Aircraft                                                        |
| ELT      | Emergency Locator Transmitter                                                            |
| EOBT     | Estimated off-block time (aircraft starts taxiing to the runway)                         |
| ETA      | Estimated Time of Arrival (overhead the destination or a waypoint)                       |
| FAD      | Danger Area South Africa                                                                 |
| FAP      | Prohibited Area South Africa                                                             |
| FAR      | Restricted Area South Africa                                                             |
| FIC      | Flight Information Centre                                                                |
| FIR      | Flight Information Region                                                                |
| FIS      | Flight Information Service                                                               |
| FL       | Flight Level                                                                             |
| FMS      | Flight Management System                                                                 |
| FPL      | Flight Plan, Filed Flight Plan                                                           |
| GBAS     | Ground Based Augmentation System – for GNSS Approaches                                   |
| GCA      | Ground Controlled Approach                                                               |
| GFA      | General Flying Area                                                                      |
| GLS      | GBAS Landing System – a Precision Instrument Approach                                    |
| GMT      | Greenwich Mean Time (see also UTC)                                                       |

|           |                                                                      |
|-----------|----------------------------------------------------------------------|
| GND       | Ground                                                               |
| GNSS      | Global Navigation Satellite System                                   |
| GPS       | Global Positioning System (based on American satellites)             |
| HF        | High Frequency                                                       |
| hPa       | Hectopascal                                                          |
| IATA      | International Air Transport Association                              |
| ICAO      | International Civil Aviation Organisation                            |
| ICASA     | Independent Communications Authority of South Africa                 |
| IFBP      | IATA In-Flight Broadcast Procedure                                   |
| IFR       | Instrument Flight Rules                                              |
| ILS       | Instrument Landing System – a precision Instrument Approach          |
| IMC       | Instrument Meteorological Conditions                                 |
| INCERFA   | Uncertainty Phase                                                    |
| INFO      | Information                                                          |
| INO       | Indian Ocean Region (ICAO) for communication purposes                |
| INS       | Inertial Navigation System                                           |
| IRS       | Inertial Reference System (a “modern” INS)                           |
| IRU       | Inertial Reference Unit                                              |
| ISA       | International Standard Atmosphere                                    |
| kHz       | Kilohertz                                                            |
| kts       | Knots                                                                |
| LF        | Low frequency                                                        |
| LNAV      | Lateral Navigation – a Non-Precision Instrument Approach (2D)        |
| LNAV/VNAV | A non-precision Approach with vertical Guidance (3D)                 |
| LORAN     | a long range navigation system (obsolete)                            |
| LP        | Localizer Performance – GNSS approach without vertical guidance (2D) |
| LPV       | Localizer Performance with vertical guidance (3D)                    |
| LSA       | Localizer Sensitive Area                                             |
| LVO       | Low Visibility Operations                                            |
| LVP       | Low Visibility Procedures                                            |
| MAPt      | Missed Approach Point                                                |
| MAUW      | Maximum all-up weight                                                |
| mb        | Millibar (same as hPa)                                               |
| MDA       | Minimum Descent Altitude on a 2D Approach                            |
| MDH       | Minimum Descent Height on 2D Approach                                |
| MEA       | Minimum En-Route Altitude                                            |
| MET       | Meteorological or meteorology                                        |
| METAR     | Meteorological Aerodrome Report                                      |
| MF        | Medium Frequency                                                     |
| MFD       | Multi Function Display                                               |
| MHz       | Megahertz                                                            |
| MIL       | Military                                                             |
| MLS       | Microwave Landing System                                             |
| MORA      | Minimum Off-Route Altitude                                           |
| MSA       | Minimum Sector Altitude (conventional approach procedures)           |
| MSL       | Mean Sea Level                                                       |
| MTOW      | Maximum take-off weight                                              |

|         |                                                                      |
|---------|----------------------------------------------------------------------|
| NAV     | Navigation                                                           |
| NDB     | Non-Directional Beacon                                               |
| nm      | Nautical Miles                                                       |
| NOSIG   | No significant change (in the expected weather)                      |
| NOTAM   | Notice to Airmen                                                     |
| NPL     | National Pilot Licence                                               |
| OBS     | Omni Bearing Selector                                                |
| OCA     | Obstacle Clearance Altitude                                          |
| OCH     | Obstacle Clearance Height                                            |
| OCT     | Onward Clearance Time (leaving a beacon or significant point)        |
| PAR     | Precision Approach Radar                                             |
| PANS    | Procedures for Air Navigation (ICAO)                                 |
| PBN     | Performance Based Navigation                                         |
| PIB     | Pre-flight Information Bulletin                                      |
| PIREP   | Pilot Report                                                         |
| PPL     | Private Pilot's Licence                                              |
| PLB     | Personal Locator Beacon                                              |
| PTT     | Press to talk (switch)                                               |
| QBI     | IFR is compulsory (outdated)                                         |
| QDM     | Magnetic heading (zero wind)                                         |
| QDR     | Magnetic bearing                                                     |
| QFE     | Atmospheric pressure at aerodrome elevation                          |
| QNE     | Standard altimeter setting (USA)                                     |
| QNH     | Altimeter subscale setting to obtain elevation when on the ground    |
| QTF     | Position of aircraft according to bearings taken by a ground station |
| RA      | Resolution Advisory                                                  |
| RAIM    | Receiver Autonomous Integrity Monitoring (for GNSS approaches)       |
| RAS     | Radar Advisory Service                                               |
| RCF     | Radio Communication Failure                                          |
| RIS     | Radar Information Service                                            |
| RMK     | Remark                                                               |
| RNAV    | Area Navigation                                                      |
| RNP     | Required Navigation Performance                                      |
| ROC     | Radiotelephony Operator's Certificate                                |
| RPAS    | Remotely Piloted Aircraft Systems (Drones) – also called UAS         |
| RPL     | Remote Pilots Licence                                                |
| RT      | Radio Telephony                                                      |
| RTE     | Radio Telephony Examiner                                             |
| RTF     | Radiotelephony                                                       |
| R-VLOS  | Restricted Visual Line of Sight (for Drones)                         |
| RVR     | Runway Visual Range                                                  |
| RVSM    | Reduced Vertical Separation Minima                                   |
| RWY     | Runway                                                               |
| SACAA   | South African Civil Aviation Authority (often just called CAA)       |
| SACAR   | South African Civil Aviation Regulations (often just called CAR)     |
| SA-CATS | South African Civil Aviation Technical Standards (also just CATS)    |
| SAR     | Search and Rescue                                                    |

|        |                                                                |
|--------|----------------------------------------------------------------|
| SARPS  | Standards and Recommended Practices (as used by ICAO)          |
| SAST   | South African Standard Time                                    |
| SAT    | South Atlantic Region (ICAO) for communication purposes        |
| SBAS   | Space Based Augmentation System – for GNSS approaches          |
| SELCAL | Selective Calling System                                       |
| SID    | Standard Instrument Departure                                  |
| SMC    | Surface Movement Control                                       |
| SRA    | Special Rules Area                                             |
| SSB    | Single Sideband                                                |
| SSR    | Secondary Surveillance Radar                                   |
| STAR   | Standard Terminal Arrival Route                                |
| SVFR   | Special Visual Flight Rules                                    |
| TA     | Transition Altitude                                            |
| TA     | Traffic Advisory                                               |
| TAA    | Terminal Arrival Altitude (for RNAV approaches)                |
| TACAN  | UHF tactical air navigation aid (Military VOR and DME)         |
| TAS    | True Airspeed                                                  |
| TCAS   | Traffic Collision Avoidance System                             |
| TIBA   | Traffic Information Broadcast Aircraft                         |
| TIS    | Traffic Information System                                     |
| TL     | Transition Level                                               |
| TMA    | Terminal Control Area                                          |
| TWR    | Aerodrome Control Tower, Aerodrome Control                     |
| TWY    | Taxiway                                                        |
| UAS    | Unmanned Aircraft System (also called RPAS)                    |
| UHF    | Ultra High Frequency                                           |
| UNL    | Unlimited                                                      |
| UTC    | Universal Co-ordinated Time                                    |
| VDF    | Very High Frequency Direction Finding                          |
| VFR    | Visual Flight Rules                                            |
| VHF    | Very High Frequency                                            |
| VMC    | Visual Meteorological Conditions                               |
| VNAV   | Vertical Navigation                                            |
| VOR    | VHF omnidirectional radio range (Navigational Beacon)          |
| VORTAC | VOR and TACAN combination                                      |
| WAAS   | Wide Area Augmentation System – for GNSS approaches (USA only) |

# Introduction

## 0.1 Legal Stuff

### 0.1.1 The Transmitter

All aviation radios must be licenced in the state of registration of the aircraft. In South Africa this licencing is administered by the Independent Communications Authority of South Africa (ICASA). It is usually done by the aircraft owner or operator, who also pays the once-off application fee plus the annual licencing fees. A valid “Radio Station Licence” issued by ICASA must be carried in the aircraft.

DRONE pilots - except those operating only R-VLOS (restricted visual line-of-sight) and only privately (not for gain) - need a handheld aviation transceiver, which needs to be licenced by ICASA in the same manner as for radios built into manned aircraft.

### 0.1.2 The Radio Operator

CAR 61.03.1(c), CAR 61.04.1(c), CAR 61.04.1(c), CAR 61.05.1(c), CAR 61.11.2(d) etc.

The person speaking over the radio apparatus in an aircraft or on the ground is called the “operator of radio equipment”, and needs to be licenced by the state in which the aircraft is registered. This licencing function is part of the duties of the Independent Communications Authority of South Africa (ICASA), who delegates the task of examining for and issuing the Radio Telephony Certificates described in paragraph 0.2 below to the South African Civil Aviation Authority (SACAA). These certificates are not issued as separate licences, but are simply an endorsement in the pilot’s flying licence issued by the SACAA.

## 0.2 The Radiotelephony Operator's Certificates

AIC C009/2021 (category 30)

The syllabi for the Restricted and the General Radiotelephony Operator's Certificates are published as South African Civil Aviation Technical Standards, SA-CATS-FCL 61, Appendix 1.5 and 1.5a. Both syllabi are covered in this textbook.

- The syllabi can be found on the CAA website <https://www.caa.co.za>. Click on Legal Information at the bottom, then click Legislation, then under Legislation click here. Click Technical Standards (Personnel Licensing). Click SA-CATS 61 Pilot Licensing. Scroll all the way down to the appendices and click on Appendix 1.5: for the restricted, or Appendix 1.5a: for the general ROC syllabus.

The **Restricted Radiotelephony Operator's Certificate** is required before a person can obtain a PPL and fly under VFR (i.e. in good weather); a National Pilot Licence (NPL) to fly aircraft up to 600 kg all-up-weight, microlights, gyroplanes, hang-gliders, paragliders or touring motor gliders; a Glider Pilot Licence to fly gliders; a Free Balloon Pilot Licence to fly balloons; or a Remote Pilot Licence (RPL) to fly drones.

The **General Radiotelephony Operator's Certificate** becomes necessary if a pilot wants to obtain an Instrument Rating or become a Commercial Pilot or Airline Transport Pilot.

To obtain either Radiotelephony Operator's Certificate candidates must:

- ! pass the relevant SACAA written examination at one of the SACAA accredited test centres (usually affiliated with flight schools), and
- ! pass a skills test in radio procedures with a Radio Telephony Examiner (RTE). This test consists of:
  - completing a flight plan form for a Visual Flight Rules (VFR) flight for the Restricted ROC or an Instrument Flight Rules (IFR) flight for the General ROC, and
  - “flying” that flight in a radio simulator approved for the purpose. This includes (in addition to all the routine radio communications) deciphering the Morse code identification of at least one navigational beacon, an emergency, as well as a radio communication failure, and
- ! have the application form for the ROC endorsed by the examiner conducting the final flight test for the relevant pilot licence.
- Candidates studying for the SACAA-exams are strongly advised to read through the whole book at least once (candidates for the Restricted Certificate omit the grey shaded blocks), then carefully go through the respective syllabus item by item and re-read the relevant paragraphs. A tick-box is provided after each item for your convenience.

## 0.2.1 Syllabus for the Restricted ROC

SA-CATS-FCL 61 Appendix 1.5

The written examination for the Restricted “Radiotelephony Operator’s Certificate” (Restricted ROC) consists of 25 questions that must be answered in one hour with a pass mark of 75%.

It requires a good general knowledge of the rules and procedures governing visual flying. The candidate must know how to use the radio installation in small aircraft; he must be able to fly in different airspaces and give accurate position reports. He must know about Flight Plans, how to complete and file them. He must know the altimeter setting procedures, how Search and Rescue (SAR) works, and must be able to identify and interpret aerodrome traffic signals. He must also have a good working knowledge of the Aeronautical Information System as published by the SACAA.

| Reference in the SACAA-syllabus |                                                                       |                                         | Reference in this book | done |
|---------------------------------|-----------------------------------------------------------------------|-----------------------------------------|------------------------|------|
| <b>1</b>                        | <b>Classification of airspace and associated air traffic services</b> |                                         |                        |      |
| 1.1                             | Definitions and abbreviations                                         |                                         | Abbreviations          |      |
|                                 | 1.1.1                                                                 | Flight Rules                            | chapter 3              |      |
|                                 |                                                                       | Visual Flight Rules (VFR)               | paragraph 3.1          |      |
|                                 |                                                                       | Instrument Flight Rules (IFR)           | paragraph 3.2          |      |
|                                 |                                                                       | Special VFR (SVFR)                      | paragraph 3.3          |      |
| 1.2                             | ICAO classification of airspace                                       |                                         | paragraph 1.2          |      |
|                                 | 1.2.1                                                                 | Classes in use in South Africa          | paragraph 1.2          |      |
| 1.3                             | Airspace structure                                                    |                                         | chapter 1              |      |
|                                 | 1.3.1                                                                 | Flight information regions              | paragraph 1.1          |      |
|                                 | 1.3.2                                                                 | Controlled airspace                     | paragraph 1.2.1        |      |
|                                 | 1.3.3                                                                 | Aerodrome Traffic Zones                 | para 1.3.1.1           |      |
|                                 |                                                                       | Control zones                           | paragraph 1.3.1.2      |      |
|                                 |                                                                       | Terminal Control Areas                  | paragraph 1.3.2.1      |      |
|                                 |                                                                       | Control Areas                           | paragraph 1.3.2        |      |
|                                 |                                                                       | Airways                                 | paragraph 1.3.3        |      |
| 1.4                             | Uncontrolled airspace                                                 |                                         | paragraph 1.2.3        |      |
|                                 | 1.4.1                                                                 | Aerodrome traffic areas                 | paragraph 1.3.1.3      |      |
|                                 |                                                                       | Flight information routes               | Paragraph 1.3.3.3      |      |
|                                 |                                                                       | Advisory airspace                       | paragraph 1.2.3        |      |
|                                 |                                                                       | Danger, Restricted and Prohibited Areas | 1.3.4.2 – 1.3.4.4      |      |
|                                 |                                                                       | Special Rules Areas                     | paragraph 1.3.4.1      |      |
| 1.5                             | Recognition of airspace on aeronautical charts                        |                                         | chapter 1              |      |
| 1.6                             | Air Traffic Services                                                  |                                         | chapter 2              |      |
|                                 | 1.6.1                                                                 | Objectives of air traffic services      | paragraph 2.1          |      |
|                                 | 1.6.2                                                                 | Air traffic control service             | paragraph 2.2.1        |      |

|          |                                                               |                    |
|----------|---------------------------------------------------------------|--------------------|
|          | Advisory service                                              | paragraph 2.2.2    |
|          | Flight information service                                    | paragraph 2.2.2    |
|          | Alerting service                                              | p. 2.2.3, chap. 12 |
|          | Aerodrome flight information service                          | paragraph 2.2.2.1  |
| 1.7      | Signals for aerodrome traffic                                 | paragraph 7.10     |
| 1.8      | Visual ground signals                                         | para 7.10.1        |
| <b>2</b> | <b>Altimeter setting procedures</b>                           |                    |
| 2.1      | Definitions                                                   | paragraph 4.1      |
| 2.2      | Semi-circular rule                                            | para 5.1 and 5.3   |
| 2.3      | Practical application                                         | paragraph 4.4      |
| 2.4      | Transition altitude                                           | paragraph 4.2      |
| 2.5      | Transition level                                              | paragraph 4.3      |
| <b>3</b> | <b>Air traffic services flight plan</b>                       |                    |
| 3.1      | Applicability                                                 | paragraph 6.1      |
| 3.2      | Filing requirements                                           | paragraph 6.1      |
| 3.3      | Validity                                                      | paragraph 6.3      |
| 3.4      | Amendments to ETAs and changes in TAS                         | paragraph 6.4      |
| 3.5      | Methods of filing                                             | paragraph 6.2      |
| 3.6      | Completion of flight plan form                                | paragraph 6.6      |
| 3.7      | Closing of flight plan                                        | paragraph 6.5      |
| 3.8      | Search and rescue                                             | paragraph 12       |
|          | Methods of indicating on a flight plan                        | para 6.6 and 12.1  |
|          | Phases                                                        | paragraph 12.2     |
|          | Method of cancellation                                        | paragraph 6.5      |
| <b>4</b> | <b>Technical</b>                                              |                    |
| 4.1      | Basic principles of operation of VHF radios                   | paragraph 13.1     |
| 4.2      | VHF Frequency Band                                            | paragraph 13.2     |
| 4.3      | Factors affecting reception and transmission range            | paragraph 13.4.1   |
| 4.4      | Location and use of frequencies for various types of airspace | p. 8.9, 8.10, app. |
| 4.5      | Morse code                                                    | paragraph 7.4      |
| <b>5</b> | <b>Practical use of VHF radio installations</b>               |                    |
| 5.1      | Use of all switches and controls on the radio tuning panel    | para 8.1, 13.5.2   |
| 5.1.1    | Use of squelch control                                        | para 8.1, 13.5.2   |
| 5.1.2    | Use of volume control                                         | para 8.1, 13.5.2   |
| 5.1.3    | Changing frequency incl. 25 and 8.3 kHz spacing               | paragraph 13.5.2   |
| 5.2      | Use of audio control panel                                    | para 8.1, 13.5.6   |
| 5.3      | Methods of donning headsets                                   | para 8.1, 13.5.4   |
| 5.4      | Location and use of press to talk switches                    | paragraph 8.1      |
| 5.5      | Location and use of handheld microphone                       | paragraph 13.5.4   |
| 5.6      | Microphone technique                                          | para 8.1, 13.5.4   |
| 5.7      | Radio checks                                                  | paragraph 10.1     |
| 5.8      | Transmission technique                                        | paragraph 8.1      |

|          |                         |                                                              |                   |  |
|----------|-------------------------|--------------------------------------------------------------|-------------------|--|
|          | 5.9                     | Use of transponder, terminology and emergency codes          | paragraph 9.2     |  |
| <b>6</b> | <b>Terminology</b>      |                                                              |                   |  |
|          | 6.1                     | Phonetic alphabet, numerals and Q codes                      | p. 7.2, 7.3, 7.6  |  |
|          | 6.2                     | Call signs and methods of addressing various RTF stations    | paragraph 7.7     |  |
|          | 6.3                     | Standard words & phrases, test procedures, readability scale | para 8.8, 7.8     |  |
|          | 6.4                     | Listening out requirements and establishing contact          | para 8.1 and 8.2  |  |
|          | 6.5                     | Mandatory read back instructions                             | paragraph 8.4     |  |
|          | 6.6                     | Categories of messages and order of priority                 | paragraph 7.9     |  |
|          | 6.7                     | Time                                                         | paragraph 7.5     |  |
| <b>7</b> | <b>Radio procedures</b> |                                                              |                   |  |
|          | 7.1                     | Use of Radio on the ground                                   | para 8.2 and 8.10 |  |
|          | 7.1.1                   | Obtaining and complying with taxi instructions               | paragraph 8.10    |  |
|          | 7.1.2                   | What to expect from Air Traffic Control (ATC)                | Chapters 2 and 8  |  |
|          | 7.1.3                   | Reading back all "hold short of" instructions                | paragraph 8.8.3   |  |
|          | 7.1.4                   | The avoidance of runway incursions                           | paragraph 8.8.3   |  |
|          | 7.1.5                   | Meaning of "give way to other aircraft"                      | paragraph 8.3.3   |  |
|          | 7.2                     | Obtaining and complying with take-off instructions           | paragraph 8.10    |  |
|          | 7.2.1                   | Importance of understanding "line up behind"                 | paragraph 8.8.3   |  |
|          | 7.2.2                   | Importance of reading back the take-off clearance            | paragraph 8.4     |  |
|          | 7.2.3                   | Importance of reading back other instructions                | paragraph 8.4     |  |
|          | 7.2.4                   | Radio procedures at unmanned aerodromes                      | paragraph 8.10.2  |  |
|          | 7.3                     | Departure procedures                                         | paragraph 8.10    |  |
|          | 7.3.1                   | What to expect in respect of departure procedures            | paragraph 8.10    |  |
|          | 7.3.2                   | Calls to be made on leaving the aerodrome circuit            | paragraph 8.10    |  |
|          | 7.4                     | En-route procedures                                          | paragraph 8.9     |  |
|          | 7.4.1                   | What call should be made to which station and when           | paragraph 8.9     |  |
|          | 7.4.2                   | In-flight broadcast procedure in uncontrolled airspace       | paragraph 8.9.1.2 |  |
|          | 7.4.3                   | Making a position report                                     | paragraph 8.7     |  |
|          | 7.4.4                   | Obtaining relevant weather information, use of ATIS          | paragraph 8.3     |  |
|          | 7.4.5                   | Making appropriate weather reports (PIREPS)                  | paragraph 8.7     |  |
|          | 7.4.6                   | Controlled airspace vs Flight Information Service            | paragraph 8.7     |  |
|          | 7.4.7                   | Transponder use                                              | paragraph 9.2     |  |
|          | 7.4.8                   | Obtain bearing, heading and position from ATC/FIS            | paragraph 15.1    |  |
|          | 7.4.9                   | Altimeter setting procedures                                 | chapter 4         |  |
|          | 7.4.10                  | Relaying messages for other stations                         | paragraph 8.12    |  |
|          | 7.5                     | Arrival and traffic pattern procedures                       | paragraph 8.10    |  |
|          | 7.5.1                   | Knowing what to expect                                       | paragraph 8.10    |  |
|          | 7.5.2                   | Arrival clearance/instructions                               | paragraph 8.10.1  |  |
|          | 7.5.3                   | Calls and ATC instructions for joining the circuit           | paragraph 8.10.1  |  |
|          | 7.5.4                   | Calls to be made in the circuit                              | paragraph 8.10    |  |
|          | 7.5.5                   | Calls to be made on vacating the runway                      | paragraph 8.10    |  |

|          |                                        |                                                     |                  |
|----------|----------------------------------------|-----------------------------------------------------|------------------|
| <b>8</b> | <b>Communications failure</b>          |                                                     |                  |
| 8.1      | Initial actions to be taken            |                                                     | paragraph 10.1   |
|          | 8.1.1                                  | Check volume control                                | paragraph 10.1   |
|          | 8.1.2                                  | Check headset connections                           | paragraph 10.1   |
|          | 8.1.3                                  | Check microphone                                    | paragraph 10.1   |
|          | 8.1.4                                  | Make a test call                                    | paragraph 7.8    |
|          | 8.1.5                                  | Check electrical system                             | para 10.1, 13.6  |
| 8.2      | Use alternate frequencies              |                                                     | paragraph 10.1   |
|          | 8.2.1                                  | Try previous frequency                              | paragraph 10.1   |
|          | 8.2.2                                  | Try 121.5                                           | paragraph 10.1   |
|          | 8.2.3                                  | Try in-flight broadcast procedure frequency         | paragraph 10.1   |
| 8.3      | Communication failure procedures       |                                                     | paragraph 10.2   |
|          | 8.3.1                                  | Land at an uncontrolled airport                     | paragraph 10.2   |
|          | 8.3.2                                  | Follow published procedures for controlled airports | paragraph 10.2.2 |
| <b>9</b> | <b>Distress and urgency procedures</b> |                                                     |                  |
| 9.1      | MAYDAY calls                           |                                                     | paragraph 11.1   |
|          | 9.1.1                                  | Definition of MAYDAY                                | paragraph 11.1   |
|          | 9.1.2                                  | When to use                                         | paragraph 11.1   |
|          | 9.1.3                                  | Contents of a MAYDAY call                           | paragraph 11.1   |
|          | 9.1.4                                  | How to make                                         | paragraph 11.1   |
|          | 9.1.5                                  | Frequencies to use                                  | paragraph 11.1   |
|          | 9.1.6                                  | Requirements on hearing a MAYDAY call               | paragraph 11.1   |
|          | 9.1.7                                  | Cancellation of MAYDAY call                         | paragraph 11.1   |
| 9.2      | PAN calls                              |                                                     | paragraph 11.2   |
|          | 9.2.1                                  | Definition of PAN                                   | paragraph 11.2   |
|          | 9.2.2                                  | When to use                                         | paragraph 11.2   |
|          | 9.2.3                                  | Contents of a PAN call                              | paragraph 11.2   |
|          | 9.2.4                                  | How to make                                         | paragraph 11.2   |
|          | 9.2.5                                  | Frequencies to use                                  | paragraph 11.2   |
|          | 9.2.6                                  | Requirements on hearing a PAN call                  | paragraph 11.2   |
|          | 9.2.7                                  | Cancellation of PAN call                            | paragraph 11.2   |

## 0.2.2 Syllabus for the General ROC

SA-CATS-FCL 61 Appendix 1.5a

The written examination for the General “Radiotelephony Operator’s Certificate” (General ROC) consists of 50 questions that must be answered in one hour with a pass mark of 75%.

Candidates for the General ROC have to demonstrate a deeper knowledge of the subjects above as well as additional knowledge with regard to technical aspects of

radio theory and flights according to Instrument Flight Rules (IFR). The syllabus below is in addition to the syllabus for the Restricted ROC above.

| Reference in the SACAA-syllabus |                                                                                                                                     | Reference in this book | done |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------|------|
| <b>1</b>                        | <b>BASIC RADIO THEORY</b>                                                                                                           |                        |      |
|                                 | <b>a Electromagnetic waves</b>                                                                                                      |                        |      |
|                                 | Frequency, wave length, cycle, phase, amplitude                                                                                     | paragraph 13.1         |      |
|                                 | Frequency bands                                                                                                                     | paragraph 13.2         |      |
|                                 | Sidebands, double sideband, single sideband                                                                                         | para 13.1, 13.4.2      |      |
|                                 | Band width                                                                                                                          | paragraph 13.2         |      |
|                                 | Carrier wave, modulation, demodulation                                                                                              | paragraph 13.1         |      |
|                                 | Amplitude modulation                                                                                                                | paragraph 13.1         |      |
|                                 | Frequency modulation                                                                                                                | paragraph 13.1         |      |
|                                 | Pulse modulation                                                                                                                    | paragraph 13.1         |      |
|                                 | <b>b Antennas</b>                                                                                                                   |                        |      |
|                                 | Characteristics                                                                                                                     | paragraph 13.5.2       |      |
|                                 | Polarisation                                                                                                                        | paragraph 13.1         |      |
|                                 | Polar diagram                                                                                                                       | paragraph 13.1         |      |
|                                 | Types of antennas                                                                                                                   | paragraph 13.5.2       |      |
|                                 | <b>c Wave propagation</b>                                                                                                           |                        |      |
|                                 | Ground waves                                                                                                                        | paragraph 13.4.2       |      |
|                                 | Direct waves                                                                                                                        | paragraph 13.4.1       |      |
|                                 | Sky waves                                                                                                                           | paragraph 13.4.2       |      |
|                                 | Ionosphere, critical angle, skip distance                                                                                           | paragraph 13.4.2       |      |
|                                 | Dead space, refraction                                                                                                              | para 13.2, 13.4.2      |      |
|                                 | Fading                                                                                                                              | paragraph 13.2         |      |
|                                 | Factors affecting propagation (reflection, absorption, attenuation, coastline, mountain, static)                                    | para 13.2, 15.2        |      |
| <b>2</b>                        | <b>PRINCIPLES OF RADIO OPERATION</b>                                                                                                |                        |      |
|                                 | a Basic principles of operation of VHF radios                                                                                       | paragraph 13.5.3       |      |
|                                 | b VHF Frequency Band                                                                                                                | paragraph 13.2         |      |
|                                 | c Factors affecting reception and transmission range of VHF radios                                                                  | paragraph 13.4.1       |      |
|                                 | d Basic principles of operation of HF radios                                                                                        | paragraph 13.5.3       |      |
|                                 | e HF Frequency Band                                                                                                                 | paragraph 13.2         |      |
|                                 | f Factors affecting reception and transmission range of HF radios                                                                   | paragraph 13.4.2       |      |
| <b>3</b>                        | <b>AIRSPACE</b>                                                                                                                     |                        |      |
|                                 | <b>a Air traffic services</b>                                                                                                       |                        |      |
|                                 | Objectives of air traffic services                                                                                                  | paragraph 2.1          |      |
|                                 | Air traffic control services, advisory service in flight information service, alerting service aerodrome flight information service | paragraph 2.2          |      |
|                                 | b ICAO airspace classification and associated air traffic services                                                                  | para 1.2, chap 2       |      |
|                                 | <b>c Airspace structure</b>                                                                                                         |                        |      |
|                                 | Flight Information Regions (FIRs)                                                                                                   | paragraph 1.1          |      |
|                                 | Controlled airspace (Aerodrome traffic zones, Control zones, Terminal control areas, Control areas, Airways)                        | para 1.2.1, 1.3        |      |

|   |   |                                                                                                                                                             |                   |  |
|---|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--|
|   |   | Avoidance of unauthorized entry into controlled airspace                                                                                                    | paragraph 1.2.1   |  |
|   |   | Uncontrolled airspace (Aerodrome traffic areas, Flight information routes, Advisory airspace, Danger, Restricted and Prohibited areas, Special Rules Areas) | para 1.2.3, 1.3   |  |
|   | d | Recognition of airspace / classes of airspace on aeronautical charts                                                                                        | paragraph 1.3     |  |
| 4 |   | <b>SA CIVIL AVIATION REGULATIONS PART 91 AS FOLLOWS:<br/>(Including Technical Standards)</b>                                                                |                   |  |
|   | a | Recall: Mandatory radio communication in controlled airspace                                                                                                | paragraph 1.2.1   |  |
|   | b | Recall: Mandatory radio communication in advisory airspace                                                                                                  | paragraph 1.2.3   |  |
|   | c | Recall: Compliance with rules of air and air traffic control clearances and instructions                                                                    | paragraph 1.2.1   |  |
|   | d | Explain: Signals                                                                                                                                            | paragraph 7.10    |  |
| 5 |   | <b>RADIOTELEPHONY PHRASEOLOGY AND PROCEDURES</b>                                                                                                            |                   |  |
|   | a | Time - purpose and use of Universal Time Co-ordinated (UTC)                                                                                                 | paragraph 7.5     |  |
|   | b | Visually identify Morse code with the use of morse code tables                                                                                              | paragraph 7.4     |  |
|   | c | Standard speech abbreviations (words and phrases)                                                                                                           | paragraph 8.8     |  |
|   | d | ATU/ATS call signs                                                                                                                                          | paragraph 7.7.1   |  |
|   | e | Test procedures and readability scale                                                                                                                       | paragraph 7.8     |  |
|   | f | Listening out requirements and establishing of contact                                                                                                      | paragraph 8.3     |  |
|   | g | Acknowledgement of receipt                                                                                                                                  | paragraph 8.4     |  |
|   | h | Corrections and repetitions                                                                                                                                 | paragraph 8.1     |  |
|   | i | Radiotelephony phraseology common to all ATS units                                                                                                          | paragraph 8.8     |  |
|   | j | Categories of messages and order of priority                                                                                                                | paragraph 7.9     |  |
|   | k | Relaying messages for other stations                                                                                                                        | paragraph 8.12    |  |
|   | l | <b>Traffic Information Broadcast by Aircraft (TIBA) procedures<br/>(from RSA AIP)</b>                                                                       |                   |  |
|   |   | Explain the purpose and content of a TIBA radio call                                                                                                        | paragraph 8.9.2   |  |
|   |   | Explain the TIBA procedure                                                                                                                                  | paragraph 8.9.2   |  |
|   |   | Recall the frequencies to be used in a TIBA call (excluding dedicated frequencies listed for aerodromes without ATSUs)                                      | Appendix II       |  |
|   | m | Describe the IATA Inflight Broadcast Procedure (IFBP) in the AFI region                                                                                     | paragraph 8.9.3   |  |
|   | n | Explain briefly how to obtain relevant weather information (the use of ATIS)                                                                                | paragraph 8.2     |  |
|   | o | Explain briefly how to make appropriate weather reports (PIREPS)                                                                                            | paragraph 8.7     |  |
|   | p | <b>Explain the practical use of a transponder and associated radio procedures</b>                                                                           |                   |  |
|   |   | Radar terminology                                                                                                                                           | para 9.2, 9.3     |  |
|   |   | Knowledge and use                                                                                                                                           | paragraph 9.2     |  |
|   |   | Emergency codes                                                                                                                                             | paragraph 9.2     |  |
|   | q | Describe TCAS/ ACAS phraseology                                                                                                                             | paragraph 14.3.3  |  |
|   | r | Describe the meaning and use of the following "Q" codes (QDM, QDR, QFE, QNE, QNH, QTF)                                                                      | para 4.1 and 7.6  |  |
| 6 |   | <b>WAKE TURBULENCE</b>                                                                                                                                      |                   |  |
|   | a | Describe the different categories                                                                                                                           | paragraph 6.6.3.3 |  |
|   | b | Describe separation and phraseology                                                                                                                         | paragraph 6.6.3.3 |  |
|   | c | Application                                                                                                                                                 | paragraph 6.6.3.3 |  |
| 7 |   | <b>FILING OF FLIGHT PLANS &amp; SEARCH AND RESCUE (SAR)</b>                                                                                                 |                   |  |
|   | a | Applicability and filing requirements                                                                                                                       | paragraph 6.1     |  |

|          |          |                                                                                                                                  |                  |  |
|----------|----------|----------------------------------------------------------------------------------------------------------------------------------|------------------|--|
|          | b        | Procedures for the submission of a flight plan                                                                                   | paragraph 6.2    |  |
|          | c        | Completion of flight plans                                                                                                       | paragraph 6.6    |  |
|          | d        | The application of Search and rescue (SAR) elements                                                                              |                  |  |
|          |          | Describe the 3 SAR phases                                                                                                        | paragraph 12.2   |  |
|          |          | Methods of indicating SAR on a flight plan                                                                                       | paragraph 6.6.9  |  |
|          |          | Methods of cancelling SAR                                                                                                        | paragraph 6.5    |  |
| <b>8</b> |          | <b>EMERGENCY AND URGENCY PHRASEOLOGY AND PROCEDURES</b>                                                                          |                  |  |
|          | <b>a</b> | <b>Identification and interception of aircraft</b>                                                                               |                  |  |
|          |          | Explain the principles to be observed during an interception                                                                     | paragraph 7.10.4 |  |
|          |          | Explain the action by an intercepted aircraft                                                                                    | paragraph 7.10.4 |  |
|          |          | Radio communication during interception                                                                                          | paragraph 7.10.4 |  |
|          |          | Visual interception signals                                                                                                      | paragraph 7.10.4 |  |
|          | b        | Explain the 5 golden rules for aircraft subject to an emergency                                                                  | paragraph 12.3.1 |  |
|          | c        | Explain the use of VHF Direction-finding equipment (VDF)<br>(Procedure to obtain bearings, headings and position from ATC / FIS) | paragraph 15.1   |  |
|          | <b>d</b> | <b>Describe the procedures and signals to be used in the following cases:</b>                                                    |                  |  |
|          |          | An aircraft observing another aircraft in distress                                                                               | paragraph 11.1   |  |
|          |          | An aircraft intercepting a distress call                                                                                         | paragraph 11.1   |  |
|          | <b>e</b> | <b>Recall the frequencies to be guarded by search aircraft during SAR action:</b>                                                |                  |  |
|          |          | 500 kHz/2182 kHz / 8364 kHz/121.5 Mhz/243 Mhz                                                                                    | paragraph 12.6   |  |
|          | <b>f</b> | <b>Distress communications: MAYDAY calls</b>                                                                                     |                  |  |
|          |          | Explain the meaning and purpose of a MAYDAY call                                                                                 | paragraph 11.1   |  |
|          |          | Explain when a MAYDAY call is required                                                                                           | paragraph 11.1   |  |
|          |          | Describe the contents of a MAYDAY call                                                                                           | paragraph 11.1   |  |
|          |          | Recall dedicated frequencies                                                                                                     | paragraph 11.1   |  |
|          |          | Describe the cancellation of MAYDAY                                                                                              | paragraph 11.1   |  |
|          | <b>g</b> | <b>Urgency communications: PAN calls</b>                                                                                         |                  |  |
|          |          | Explain the meaning and purpose of a PAN call                                                                                    | paragraph 11.2   |  |
|          |          | Explain when a PAN call is required                                                                                              | paragraph 11.2   |  |
|          |          | Describe the contents of a PAN call                                                                                              | paragraph 11.2   |  |
|          |          | Recall dedicated frequencies                                                                                                     | paragraph 11.2   |  |
|          |          | Describe the cancellation of a PAN call                                                                                          | paragraph 11.2   |  |
|          | <b>h</b> | <b>Emergency Locator Transmitter (ELT)</b>                                                                                       |                  |  |
|          |          | Identify the frequencies allocated to ELT devices                                                                                | paragraph 12.4   |  |
|          |          | Explain the correct use of and rules for activation of an ELT                                                                    | paragraph 12.3.3 |  |
|          | <b>i</b> | <b>Survivor action after crash or forced landing:</b>                                                                            |                  |  |
|          |          | Identify the Ground / Air emergency signalling codes                                                                             | paragraph 7.10.5 |  |
|          |          | Describe the Ground / Air visual signal codes                                                                                    | paragraph 12.3.4 |  |
| <b>9</b> |          | <b>RADIO COMMUNICATIONS FAILURE (RCF)</b>                                                                                        |                  |  |
|          | a        | Explain the basic fault-finding actions following suspected RCF                                                                  | paragraph 10.1   |  |
|          | b        | Describe RCF procedures (VFR in VMC)                                                                                             | paragraph 10.2   |  |
|          | c        | Explain the procedure for transmitting blind                                                                                     | paragraph 10.1   |  |

## 0.3 About this book

### Chapters 1 to 13

cover the syllabus for both the Restricted and the General Radiotelephony Operator's Certificates (ROCs).

Some sentences or paragraphs within these chapters apply to Instrument Flight Rules (IFR) and are only needed for the General ROC. **Candidates for the Restricted Certificate do not need to learn these.** They are usually set in a grey box such as this one.

### Chapter 14

gives additional information for candidates for the General ROC. **Candidates for the Restricted Certificate do not need to learn this chapter.**

### Chapter 15

gives some information about Navigational Aids. **Paragraph 15.1 is part of the syllabus for both ROCs.** The remainder of the chapter is not strictly speaking part of the syllabus, and is included for information and deeper understanding.

### The appendices

are for flight planning and for looking up special procedures or frequencies. They are intended mainly for the pilot, be it recreational or professional, who keeps this "Radio Book" as a reference in his flight bag.

### Nice-to-know stuff

" is often given as a note like this

### Examples

Are usually printed in italics and serve to illustrate a point.

### References

to official publications are provided at the beginning of most paragraphs directly below the paragraph heading for anybody wanting to do more research. These references point to the source of the relevant information and were correct at the time of printing this book (November 2025).

## 0.4 Aeronautical Information

### 0.4.1 ICAO Documentation

The International Civil Aviation Organisation (ICAO) was established by the Chicago Convention on International Civil Aviation in 1944, and is a specialized agency of the United Nations with headquarters in Montreal, Canada. ICAO promotes understanding and security through co-operative aviation regulation. Contracting countries, also called member states, undertake to abide by the Standards and Recommended Practices (SARPS) published by ICAO as Annexes to the Chicago Convention. Member states are required to give notification of differences to ICAO standards, and are invited to notify differences from recommended practices in the Annexes. The differences declared by South Africa are listed in the South African AIP GEN 1.7.

The Annexes and Documents referenced in the compilation of this book are:

- Annex 2 - Rules of the Air
- Annex 5 - Units of Measurement
- Annex 10 - Aeronautical Telecommunications
- Annex 11 - Air Traffic Services
- Annex 12 - Search and Rescue
- Annex 15 - Aeronautical Information Services
- Doc 4444 - PANS-ATM – Air Traffic Management
- Doc 8126 – Aeronautical Information Services Manual
- Doc 8400 - PANS-ABC – Abbreviations and Codes

ICAO documentation can be purchased from the ICAO website (at a price), found on various websites on the internet (usually for free – but beware of out-dated documents!), or viewed at some ATC centres.

### 0.4.2 CARs – Civil Aviation Regulations of 2011

The South African Civil Aviation Regulations override all other South African aviation publications. They are divided into “Parts”; each Part dealing with a specific issue like Aviation Medicals, Air Carrier Operations in large aeroplanes, Operation of Gliders, Personnel Licensing etc. The “Aviation Legislation of South Africa” is available in six volumes containing CARs, CATS and other South African aviation legislation from LexisNexis in Durban - in hardcopy or as an internet subscription.

- “ CARs can also be found on the CAA website <https://www.caa.co.za>. Click on Legal Information at the bottom, then click Legislation, and under Legislation click here. Click Regulations. Click Civil Aviation Regulations, 2011, and then click the Part you want to read.

### 0.4.3 CATS – Civil Aviation Technical Standards

The CATS contain explanatory notes to the CARs. While the CARs might mention that all aircraft have to carry at least one fire extinguisher, the CATS will specify, under the same paragraph number, how many fire extinguishers must be carried by different size aircraft, and prescribe the properties of such fire extinguishers.

- “ The CATS can be found on the CAA website <https://www.caa.co.za>. Click on Legal Information at the bottom, then Legislation, and under Legislation click here. Scroll down and click on Technical Standards (General), (Flight Operations), or (Personnel Licensing) as required.

### 0.4.4 AIP – Aeronautical Information Publication

AIP GEN 0.1, AIP GEN 3.1

The AIP contains information of a lasting character essential to air navigation. The AIP is obtainable from the CAA in hardcopy and includes a quarterly amendment service. Alternatively, an AIP is kept at ATC centres and at flying schools and clubs, as well as on the CAA-website.

- **General** (GEN) contains general information such as some regulations, sunset and sunrise tables, units of measurement, abbreviations, location indicators etc.
- **Enroute** (ENR) contains navigational information such as altimeter setting procedures, approach procedures, and airspace information.
- **Aerodromes** (AGA) contains information about licenced airfields and heliports in South Africa.

The AIP is amended every three months, on the 15<sup>th</sup> of January, April, July and October.

- “ The AIP can be found on the CAA website <https://www.caa.co.za>. Click Industry on the menu on the top, click Aeronautical Information on the drop-down menu, click Aeronautical Information Publication on the menu below. Click the Download button on the right for the section you want.

Unfortunately, the download cannot be searched, and it is very cumbersome to scroll through the three parts of between 280 and 600 pages each. A good work-around is to download and save the three parts as pdf documents, which can be easily searched.

**Caution: a new version has to be downloaded in its entirety every three months in January, April, July and October.**

### 0.4.5 AIP SUP - AIP Supplement

AIP GEN 3.1

The AIP SUP contains information about temporary changes of three months or longer, and other operational information with long texts or graphics.

AIP SUPs are named Snnn/yy – where nnn is a sequential number within a year, and yy is the year, e.g. S064/25 is the 64<sup>th</sup> AIP SUP issued in the year 2025.

- “ AIP SUPs can be found on the CAA website <https://www.caa.co.za>. Click Industry on the menu on the top, click Aeronautical Information on the drop-down menu, click Index Of AIP SUP on the menu below. Tip: change to “Show All entries” to see all current entries.

## 0.4.6 NOTAM – Notice to Airmen

AIP GEN 3.1

NOTAMs contain information about short temporary changes (ICAO recommendation: less than 3 months) pertinent to navigation or safety of aircraft as well as information which will at a later stage be included in the AIP SUP or the AIP.

NOTAMs are classified as:

- Class A of interest to international flights and international airports
- Class B for regional flights to neighbouring countries
- Class C of interest to domestic flights only
- Class D about military airfields and military operations
- Class E about heliports, helipads and helistops
- Class F regarding administrative matters, hand amendments etc.

NOTAMs are named Znnnn/yy, with “Z” being the class A to F, “nnnn” being a sequential number within a class and year, and “yy” the year, e.g. A0473/25 is the 473<sup>rd</sup> class A NOTAM issued in the year 2025.

- “ NOTAMs can be found on the ATNS file2fly system. Go to [www.atns.co.za](http://www.atns.co.za), click on File2Fly Self Briefing Service. You need to register, free of charge. – A Pilot Information Briefing (PIB) including all relevant NOTAMs can also be obtained by phoning 0860 FLY NOW (0860 359 669). – Or go to the CAA website <https://www.caa.co.za>. On the menu on the top click Industry, on the drop-down menu click Aeronautical Information, then Notam Summaries. Scroll down and click on Notam Summary.

## 0.4.7 AIC – Aeronautical Information Circular

AIP GEN 3.1, AIC Series A 001/2014 dated 6 Feb 2014

AICs contain explanatory or advisory information regarding technical, administrative or legislative matters of civil flying, which does not qualify for inclusion into the AIP, AIP SUP or NOTAMs.

AICs are numbered in series A to F, with consecutive numbers per calendar year, e.g. A 007-2014. The old numbering was in categories with a sequential number, e.g. 12.4 – and there are still many valid AICs with the old numbering.

| Series (new) | Category (old)    | Topic                   |
|--------------|-------------------|-------------------------|
| • Series A   | Category 10 to 19 | General                 |
| • Series B   | Category 20 to 27 | Operation of Aircraft   |
| • Series C   | Category 30 to 31 | Personnel Licensing     |
| • Series D   | Category 40 to 45 | Air Navigation Services |
| • Series E   | Category 50 to 52 | Aerodromes              |
| • Series F   | Category 60 to 61 | Airworthiness           |

“ AICs can be found on the CAA website <https://www.caa.co.za>. Click Industry on the menu on the top, click Aeronautical Information on the drop-down menu, click Index Of \_\_\_\_\_ AIC's (sic). Tip: change to “show 100 entries” to see more on one page.

AICs are not much used anymore, and it seems that every department within the SACAA publishes important content as “Notices” under its own heading on the SACAA website (see next paragraph).

0.4.8 SACAA website

The website of the South African Civil Aviation Authority (SACAA) <https://www.caa.co.za> has a wealth of information. Some quick links are found as clickable blocks at the bottom of the home page. Most other useful information is accessible through the “Industry” tab on the menu at the top. Unfortunately, the site is not very user-friendly, but with a bit of ingenuity the required information can be found.

0.4.9 ATNS website

The Air Traffic Navigation Services (ATNS) website <https://atns.com> gives a wealth of information about their products and services. Of special interest is:

- “File2fly Self Briefing Service” for filing flight plans as mentioned in chapter 6 of this book.
- “RSA Airspace in 3D” where pilots can download a .kmz file to be used with Google Earth. After opening Google Earth different types of airspaces (TMAs, ACC/FIS sectors, airways etc.) can be selected or deselected - and clicking on a specific area will bring up the name of the airspace with lower and upper limits and the frequency to be used in that airspace – a very useful feature. Unfortunately, at the time of writing this edition (November 2025) the link seemed to be broken; hopefully it will be repaired soon.

# 1

# Airspaces

## 1.1 Flight Information Regions

CAR 172.02, AIP ENR 2.1

South Africa and adjacent oceans are divided into two Continental and one Oceanic Flight Information Regions (FIRs). The country identifier for South Africa is set by the International Civil Aviation Organisation (ICAO) to be 'FA', therefore the names of the FIRs are prefixed with FA. The three Flight Information Regions are:

- ! the Cape Town FIR (FACA)
- ! the Johannesburg FIR (FAJA)
- ! the Johannesburg Oceanic FIR (FAJO).

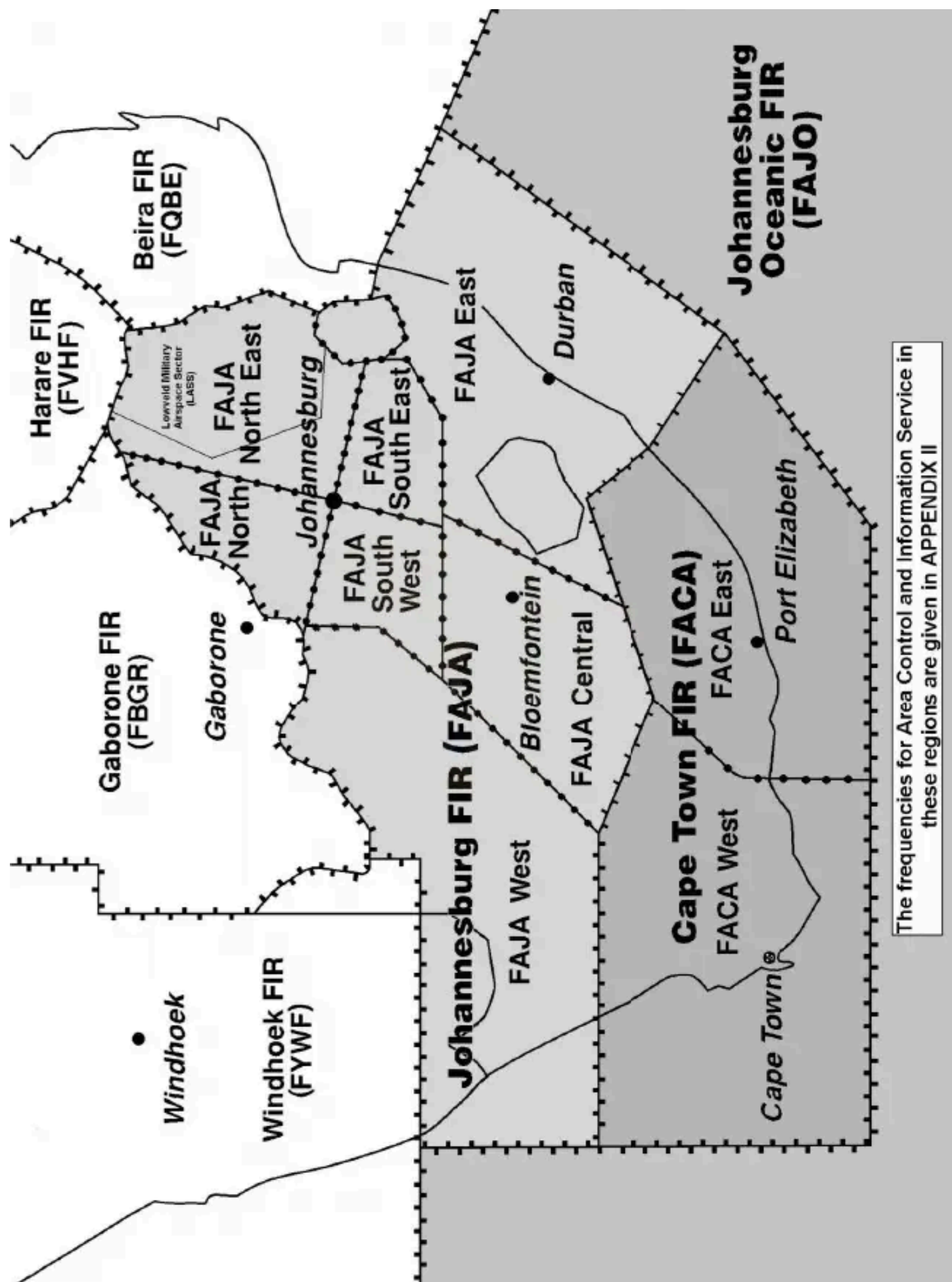
The two continental FIRs are further split into several sectors:

| Flight Information Region |                      | Sectors            | Approximate Geographical Area      |
|---------------------------|----------------------|--------------------|------------------------------------|
| FACA                      | Cape Town            | West               | mostly Western Cape                |
|                           |                      | East               | mostly Eastern Cape                |
| FAJA                      | Johannesburg         | North              | western Northern Province          |
|                           |                      | North-East         | eastern Northern Province          |
|                           |                      | South-West         | parts of North West                |
|                           |                      | South-East         | parts of Mpumalanga                |
|                           |                      | West               | Northern Cape                      |
|                           |                      | Central            | western Free State                 |
|                           |                      | East               | KwaZulu Natal / eastern Free State |
| FAJO                      | Johannesburg Oceanic | Surrounding oceans |                                    |

Adjoining FIRs to the north are:

| Country    | Flight Information Region | Flight Information Centre |
|------------|---------------------------|---------------------------|
| Namibia    | FYWF                      | Windhoek (FYWH)           |
| Botswana   | FBGR                      | Gaborone (FBSK)           |
| Zimbabwe   | FVHF                      | Harare (FVRG)             |
| Mozambique | FQBE                      | Beira (FQBR)              |

## FLIGHT INFORMATION REGIONS IN SOUTHERN AFRICA

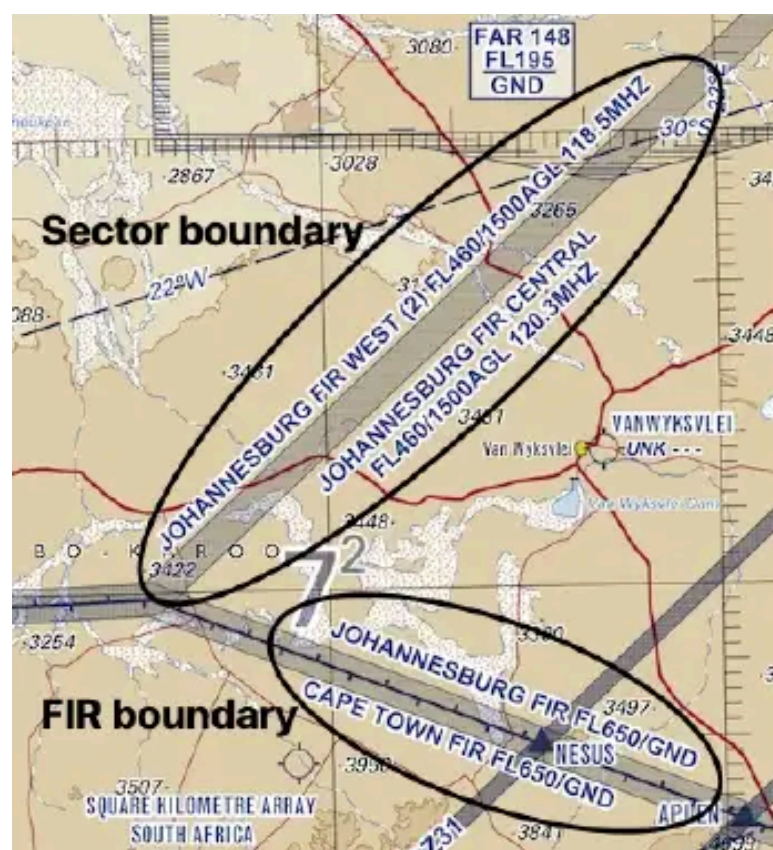


Flight Information Regions extend from the surface of the earth to Flight Level 650 (+/- 65,000 feet), but the individual sectors of the FIRs only extend up to FL460.

FIR boundaries are marked on aviation charts with a solid line with little stubs on either side

Sector boundaries are shown on aeronautical topographical maps as a wide grey shaded line.

Within the three South African FIRs are different **types** of smaller airspaces, which are discussed in paragraph 1.3 below.



These types of airspaces each belong to a certain **class** of airspace, which is fully explained in paragraph 1.2 below.

## 1.2 Classes of Airspace

AIP ENR 1.4 for airspace classification

ICAO has established seven classes of airspace: A, B, C, D, E, F, and G; and all airspaces within a Flight Information Region are declared by the state as belonging to one of these seven possible classes. Airspaces in South Africa are regulated and described in the South African Civil Aviation Regulations (SACAR) Part 172, and are listed in the AIP.

Each different **type** of airspace listed in paragraph 1.3 is assigned as belonging to one of the **classes** of airspace listed in the table below.

The **class** of airspace determines:

- ! **who may or may not fly in that particular airspace**  
this would be flights adhering to Visual Flight Rules (VFR) or Instrument Flight Rules (IFR) – see chapter 3
- ! **what level of service a pilot may expect from of Air Traffic Control**  
Control, Information or Alerting Service – see chapter 2
- ! **the responsibilities of the pilot flying in that airspace**  
two-way radio contact, adhere to instructions, give accurate estimates - see chapter 2

Classes of airspace - overview:

| Class | IFR traffic |                                                     | VFR traffic |                                                     |
|-------|-------------|-----------------------------------------------------|-------------|-----------------------------------------------------|
|       | Control?    | Service given                                       | Control?    | Service given                                       |
| A     | yes         | Separation of all traffic                           | Not allowed |                                                     |
| B     | yes         | Separation of all traffic                           | yes         | Separation of all traffic                           |
| C     | yes         | Separation of all traffic                           | yes         | Separation from IFR<br>Info about other VFR flights |
| D     | yes         | Separation from other IFR<br>Info about VFR flights | yes         | Information only                                    |
| E     | yes         | Separation from other IFR<br>Info about VFR flights | no          | Information only                                    |
| F     | no          | Advisory only                                       | no          | Information only                                    |
| G     | no          | Information only                                    | no          | Information only                                    |

1.2.1      **Controlled Airspace (Class A to D)**  
CAR 91.06.16, AIP ENR 2.1 (in FIRs), AIP ENR 3.1 (for airways)

Controlled Airspace is a three-dimensional block of air with lateral boundaries as well as lower and upper limits, in which the movement of aircraft is controlled by Air Traffic Control. Controlled airspace can be declared as belonging to one of the classes A, B, C, or D (no class E airspace in South Africa) and will be listed in the South African AIP as such.

- All aircraft flying in **controlled** airspace must:
- ! not enter this airspace without clearance from Air Traffic Control (ATC)
  - ! be equipped with a radio and talk to the appropriate Air Traffic Service Unit (ATSU) on the designated frequency
  - ! keep a listening watch on the designated frequency at all times
  - ! strictly adhere to Air Traffic Control instructions (except in emergencies), i.e. may not change routing or altitude without permission
  - ! navigate accurately
  - ! supply ATC with accurate position estimates.

A pilot wishing to fly into controlled airspace must contact ATC before entering and must maintain a continuous listening watch on the designated frequency. **Under no circumstances may a pilot enter controlled airspace without a clearance,** even when a Flight Plan has been filed and activated. – Only in exceptional cases (for example an aircraft with an unserviceable radio being ferried to a maintenance facility) may ATC grant permission for an aircraft to fly in controlled airspace without being able to maintain two-way communications.

Controlled airspace is depicted on both VFR and IFR en-route charts (and on most moving-map GPS units).

- ! **Class A** airspace is solely for IFR traffic, VFR flight is not permitted. All traffic is separated by ATC from all other traffic.

Class A airspace in South Africa comprises:

- in the two continental FIRs all airspace between FL200 and FL460,
- in the Oceanic FIR all airspace between FL245 and FL460,
- also most conventional air routes between FL145 and FL200.

- ! In **Class B** airspace IFR and VFR aircraft are permitted, and all traffic is separated from all other aircraft by Air Traffic Control. Currently there is no class B airspace in South Africa, but could be declared if and when the need arises.

- ! **Class C** airspace comprises most of the remainder of the controlled airspace in South Africa. In class C airspace both IFR and VFR flights are allowed and controlled, IFR traffic will be separated from all other traffic, and VFR traffic will be separated from IFR traffic. VFR traffic will not be separated from other VFR traffic, but Air Traffic Control will advise a VFR pilot about other conflicting VFR traffic. It is the VFR pilot's responsibility to keep a good look out and avoid other VFR traffic.

- ! **Class D** airspace at the moment only comprises the Grand Central ATZ, where both IFR and VFR traffic are allowed and controlled, where IFR traffic is separated from other IFR traffic only, and where both IFR and VFR traffic receive information about VFR traffic.

## 1.2.2 Class E Airspace

**Class E** airspace (not used in South Africa) : both IFR and VFR flights are permitted. IFR flights are controlled, and receive traffic information about VFR flights. VFR flights are uncontrolled and receive traffic information as far as practical.

## 1.2.3 Uncontrolled Airspace (Class F and G)

CAR 91.06.17

Classes F and G are **uncontrolled** airspace, in which:

- Pilots may talk to Flight Information Service (FIS)
- FIS will give the pilot information, but not positive control
- Pilots are responsible for their own traffic separation.

- ! **Class F** airspace is also known as Advisory Airspace. IFR and VFR flights are permitted, and both receive advisory and traffic information service. Aircraft must be equipped with a radio capable of two-way communications with the

responsible Air Traffic Service Unit. If out of range of an ATSU, the aircraft must broadcast its position as described in chapter 8. ICAO considers this a temporary measure only until advisory service can be replaced by air traffic control. There is no permanent Class F airspace in South Africa at present, but any section of airspace may be declared as class F, for example for an air show, as published in a NOTAM or in an AIP SUP.

**Class G** airspace is also known as Information Airspace and comprises the remainder of all airspace that has not been declared as belonging to any of the other classes. This includes all airspace above FL460.

Although aircraft flying in Class G airspace do not legally need to have a working radio on board, it is good practice to listen out and transmit regular TIBA (Traffic Information Broadcast Aircraft) messages (see paragraph 8.9.2).

## 1.3 Types of Airspace

AIP ENR 2

Individual airspaces can be established by the state where needed for the protection of air traffic, be it around airfields, over whole areas, or along air routes. These individual airspaces have a name, lateral as well as vertical boundaries, and belong to any one of the classes of airspace discussed in the previous chapter.

Airspaces are depicted on aviation maps with their name, upper and lower limit, class of airspace and sometimes the frequency.

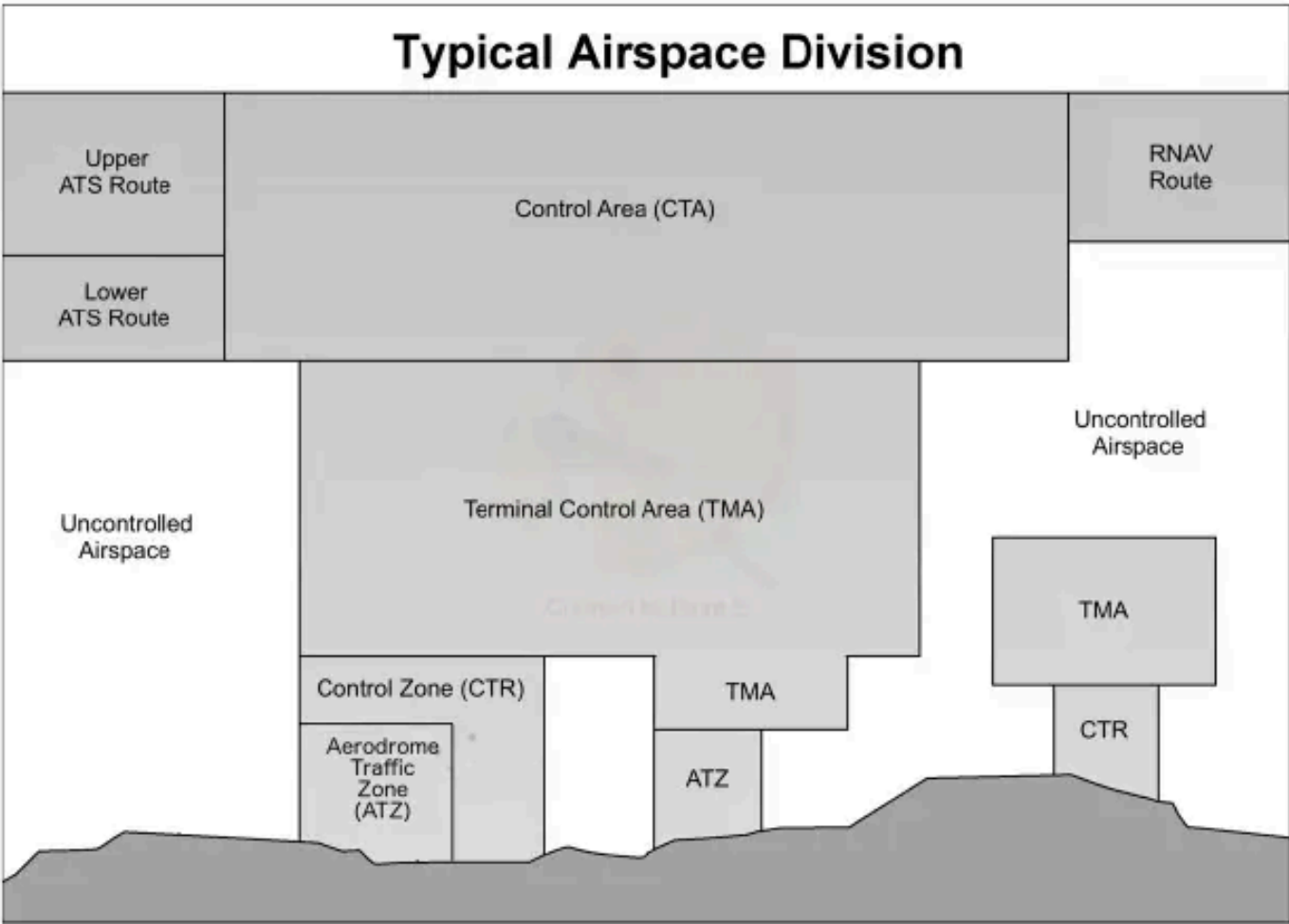
Example:

Kimberley TMA B  
Upper limit FL145  
Lower limit FL085  
Class C  
Frequency 119.4 MHz



Airspaces are also listed in the South African AIP, in section ENR 2 and AGA, with more information than depicted on charts, such as:

- ! the co-ordinates of the airspace boundaries,
- ! the Air Traffic Service Unit (ATSU) responsible for this airspace,
- ! the call sign of the ATSU,
- ! the language spoken in this airspace,
- ! the appropriate frequency, and
- ! the hours of operation of the relevant ATSU.



The different types of airspace are explained in the following paragraphs.

**1.3.1 Airspaces around Airfields**

CAR 172.02, AIP AD (for individual airfields)

Airspaces around airfields are established for the protection of traffic taking off, landing and flying in the circuit. These airspaces have a radius of at least 5 nm from the airfield, but are often shaped like a race-course pattern, elongated along the axis of the runway(s). They always extend from the ground to an upper limit, at least 700' above ground level (AGL), but usually to around 1000 to 1500 feet AGL.

**1.3.1.1 Aerodrome Traffic Zones (ATZ)**

CAR 172.02

Aerodrome Traffic Zones are usually established at smaller airfields, with or without an instrument approach procedure. They can be **Class C, D or G**.

Temporary ATZs of any class may be established when needed, for air shows, flying competitions or other events. These will be published in NOTAMs.

ATZs are depicted on aviation charts with dotted lines.

It is important to note that ATZs Class C and D are controlled airspace, radio is compulsory and the instructions of the Air Traffic Controller must be followed.

In an ATZ Class G the Air Traffic Service Unit only supplies Aerodrome Flight Information Service (AFIS); i.e. passes on information about the airfield but does not control aircraft. However, aircraft flying in an ATZ Class G, although this is uncontrolled airspace, must still be equipped with radio and talk to AFIS.

Example 1:

Virginia ATZ Class C

A semi-circular area with 3 nm radius around Virginia airfield in Durban  
Ground Level to 1000' Altitude (ALT).

Example 2 (without picture):

Grand Central ATZ Class D

An irregularly shaped area around Grand Central airfield in Johannesburg, between the OR Tambo CTR and the R55 road  
Ground Level to 7000' Altitude (ALT).

Example 3:

Port Alfred ATZ class G

A circle around the airfield with radius 5 nm  
Ground Level to 4500 feet  
Altitude (ALT)

(the clue that this is an ATZ class “**G**” lies in the word “**AFIS**” – because AFIS only exists in Class G airspace – see paragraph 2.2.2.2)



### 1.3.1.2 Control Zones (CTR)

CAR 172.02

Control Zones are established around larger airfields for the protection of IFR traffic. They could stand alone or may encompass one or more ATZs Class C or D. All Control Zones in South Africa at the moment are **Class C** airspace.

CTRs are depicted on aviation charts with a dashed line.

Example:

Bram Fisher CTR (Bloemfontein)

A rectangular pattern approx. 12 by 27 nm

Ground level to 6500' ALT

Frequency 120.8 MHz



### 1.3.1.3 Aerodrome Traffic Area (ATA)

Aerodrome Traffic Areas are still listed in the syllabi for the RT exam, but they do not exist anymore and have been replaced by ATZ Class G.

## 1.3.2 Control Areas

CAR 172.02, AIP ENR 2.1

Control Areas always extend from at least 700' above ground level (AGL) to a specified upper limit. They are always Class A or C, i.e. **controlled airspace**.

### 1.3.2.1 Terminal Control Area (TMA)

Terminal Control Areas may be established at busy centres with frequent IFR traffic to protect aircraft climbing out from or descending into a busy airfield, and are positioned above one or more CTRs and / or ATZs. **All TMAs in South Africa are class C airspace.** Since a TMA must be at least 700' above the ground or water, TMAs are often “staggered” like an upside-down wedding cake with different lower limits to accommodate terrain. In that case they are called “TMA A”, “TMA B”, “TMA C” etc. for ease of reference, but it's still really only one TMA with different lower limits.

TMAs are depicted on aeronautical charts with a solid blue shaded line.

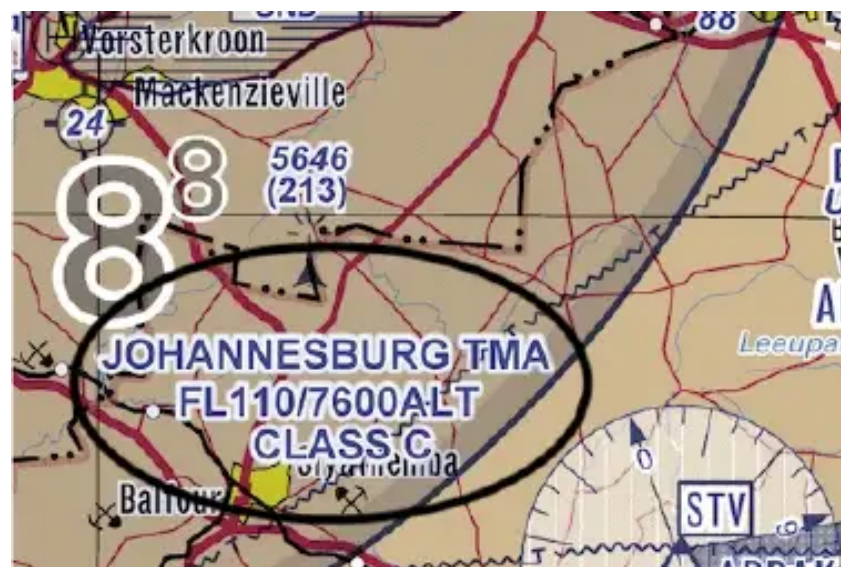
Example:

Johannesburg TMA

More or less a circle with a radius of 35 to 40 nm with a cut-out for the Waterkloof TMA and an extension towards Witbank

From 7600' ALT to FL110

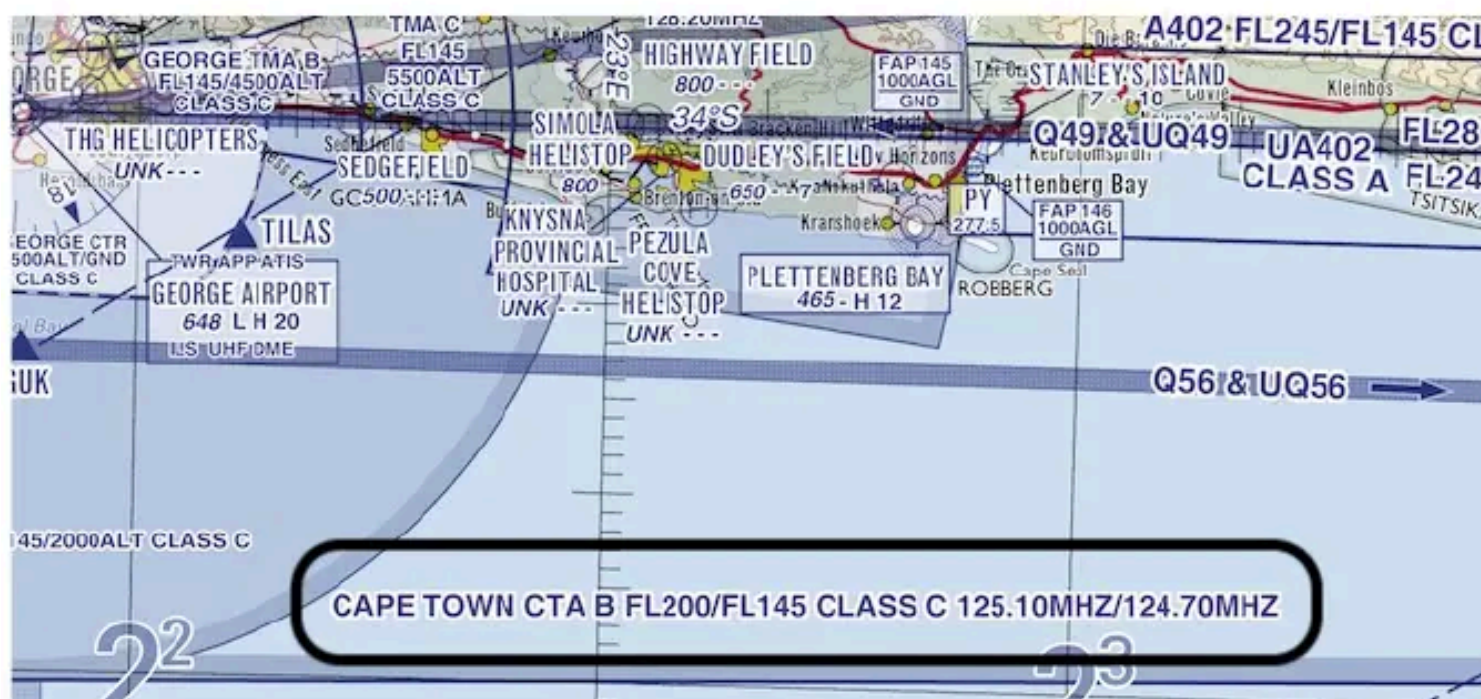
Class C



### 1.3.2.2 Control Area (CTA)

CAR 172.02, AIP ENR 2.1

Control Areas are usually established above one or more TMAs at busy centres, at the confluence of different air routes. They extend from a lower (at least 700' above the ground or water) to an upper limit. Existing lower limits are between FL105 and 145, and the upper limit is usually FL195 or FL200. At present all CTAs in South Africa are class C airspace.



Example: Cape Town Control Area (CTA) Area B  
 This sausage-shaped CTA sits above the TMAs of George, Port Elizabeth and East London, and is roughly 400 nm long and 90 nm wide  
 From FL145 to FL200  
 Frequency 125.1 MHz in the western and 124.7 MHz in the eastern sector

### 1.3.3 Routes

AIP ENR 3

Air Routes are a set path along which aircraft travel. Before Global Navigation Satellite System (GNSS), aircraft had to navigate from one ground-based navigation aid to the next. These beacons could only be built where an accessible site was available, and therefore the air routes often had “kinks” in them.

After GNSS was invented air routes could be defined anywhere, independent of ground-based physical beacons, by placing points at convenient locations, defined simply by their co-ordinates. These points have been assigned five-letter ICAO coded designators, such as RAGUL or APMIN.

With that air routes have become much more direct, saving time and fuel. But, although aircraft could now travel directly from departure to destination in a straight line it is often more convenient for traffic management (and safety) if aircraft are channelled along established routes.

These routes are discussed in the following paragraphs. They can be defined by navigational ground-based beacons or by GNSS points. They have a certain width (usually 10 nm), and a specified lower and upper limit. They can be bi-directional (aircraft can travel in both directions along the same route) or uni-directional (basically a one-way street). And they can be any class airspace as discussed in previous paragraphs.

### 1.3.3.1 Air Traffic Service Routes (Airways - AWY)

AIP ENR 3.1 and 3.2

Air Traffic Service Routes (the “conventional” Airways) are defined by ground-based navigational aids or GNSS-points, they have lower and upper limits, a certain width (usually 10 nm), are often (but not always) bi-directional, and are controlled airspace, either class C or class A.

Airways are named by a letter followed by a two- or three-digit number, with the letter being:

A, B, G, R when they are part of a regional network, and

H, J, V, W when they are not part of a network and just connect two or more cities.

Lower ATS routes in South Africa usually extend from FL145 to FL245. Some are class C airspace below FL200, and class A above that, while others are class A throughout.

Upper ATS Routes often sit above the lower routes. They take the same numbering, but are prefixed by U (Upper), i.e. UA (Upper Alpha), UB (Upper Bravo) etc., and extend usually from FL245 to FL280. They are always class A airspace.



Example: A402 (Alpha Four Zero Two) and UA402 (Upper Alpha Four Zero Two)  
 Cape Town, George, Port Elizabeth, East London, Durban to Mauritius  
 Varying limits along the length of the Airway, in this example:  
 A402 from FL145 to FL245, Class A and  
 UA402 from FL245 to FL280. Class A

- “ The strange lettering dates from when airways were named by colour: A (amber), B (blue), G (green), R (red), W (white) – when airway UW94 would have been pronounced “Upper White 94”. Today we call the same airway “Upper Whiskey 94”. Note that the “U” in Upper Airways is pronounced as Upper (not as Uniform), e.g. Upper Whiskey nine four.

VFR pilots may fly in airways, as long as they are Class C airspace (remember that VFR flight is not allowed in Class A airspace). Check AIP ENR 3.1 or on a VFR map for the class of airspace and the vertical limits of the airway. However, since Class C airspace is controlled, a flight plan has to be filed.

The exception is when a VFR pilot wants to only cross an airway Class C at more or less right angles, in which case a flight plan is not necessary (see paragraph 6.1). However, the airway Class C is still controlled airspace, and therefore the pilot needs to contact the relevant ATSU well in advance before crossing the airway on the appropriate area control frequency (see Appendix II under “En Route”). And if the pilot cannot reach the ATSU or permission is not granted, the pilot must descend below the lower limit of the airway before crossing.

### 1.3.3.2 RNAV Routes

AIP ENR 3.3, AIC 25.7

Area Navigation (RNAV) routes are the “modern” version of Airways, usually defined solely by GNSS-points. Aircraft wishing to fly along RNAV routes must be capable of accurate navigation over long distances, by means of an IFR-certified GPS or a Flight Management System (FMS), and the points defining the routes must be in the aircraft’s database. Flight crew as well as Air Traffic Controllers must be trained in RNAV procedures. Aircraft flying along RNAV routes in South Africa must be capable of adhering to Required Navigation Performance 4 (RNP4) – this means that the aircraft has to be within 4 nautical miles of the centreline of the route at least 95% of the time. RNAV routes are class A airspace, in which only IFR traffic is allowed.

- “ RNAV routes are also flown over the oceans in the Indian Ocean Strategic Partnership to Reduce Emissions (INSPIRE) and Atlantic Ocean Random RNAV Area (AORRA) with laterally offset tracks, User Preferred Routes (UPR) and other special procedures, which go beyond the scope of this book.

Area Navigation Routes usually extend from FL200 or FL245 to FL 460, mostly in a straight line without having to route from beacon to beacon. The more busy ones are directional (one-way). RNAV routes are prefixed with the letters:

L, M, N, P when they form part of a regional network, and  
Q, T, Y, Z when they are not part of a network.

Like airways, RNAV routes also have their upper counterparts UL, UN etc.

Example: Q10 (Quebec One Zero) from RAGUL near Johannesburg to Ceres near Cape Town, FL200 to FL 245, 10 nm wide, Class A  
UZ29 (Upper Zulu Two Nine) from Greytown near Durban to Standerton near Johannesburg, FL245 to FL460, Class A

### 1.3.3.3 Information Routes

AIP ENR 3.1

Flight Information Routes are the **uncontrolled** counterpart of Airways and can be established along lesser used tracks. They follow the same naming conventions as ATS Routes, but carry the suffix “F” to denote them as Information Routes Class G. There is only one Information Route established in South Africa at present.

Example: R987F (Romeo Nine Eight Seven Foxtrot)

From Nieuwoudtville VOR via Aggeneys VOR and significant point IBRUN to KEBAT at the Windhoek FIR boundary

Width 10 nm, bi-directional

Lower limit FL095, upper limit FL195

Class G

### 1.3.4 Other Regulated Airspaces

AIP ENR 2.2

Various other types of **uncontrolled** Class G airspaces can be established:

#### 1.3.4.1 Special Rules Areas (SRA)

CAR 172.02, AIP ENR 2.2

At large centres, especially where a number of flying schools are operating, the sheer volume of visual traffic can easily lead to frequency congestion. Training aeroplanes and helicopters, drones, small aircraft flying to and from the main or satellite airfields are usually restricted to the airspace below the TMA, causing further congestion.

Therefore, the airspace below most of the South African TMAs has been declared as Special Rules Airspace. Other areas, although not official SRAs, also have their own “special rules” – for the purpose of simplicity, they are all discussed in this chapter.

In these special areas one or more of the following could be established:

- ! dedicated frequencies for broadcast by aircraft
- ! special routings to and from main or secondary airports in the area
- ! special routings for aircraft bypassing the main airport in the area
- ! standard VFR departures and arrivals (see also paragraph 8.11)
- ! special routings to and from the General Flying Areas
- ! significant points used by VFR traffic for routing and reporting position
- ! special radio failure procedures
- ! speed limits
- ! altitude restrictions

Pilots flying aircraft VFR in these Special Rules Areas **must** familiarise themselves with the applicable special rules contained in the South African Aeronautical Information Publication (AIP), in section ENR 2.2 (and ENR 3.4 and 3.5).

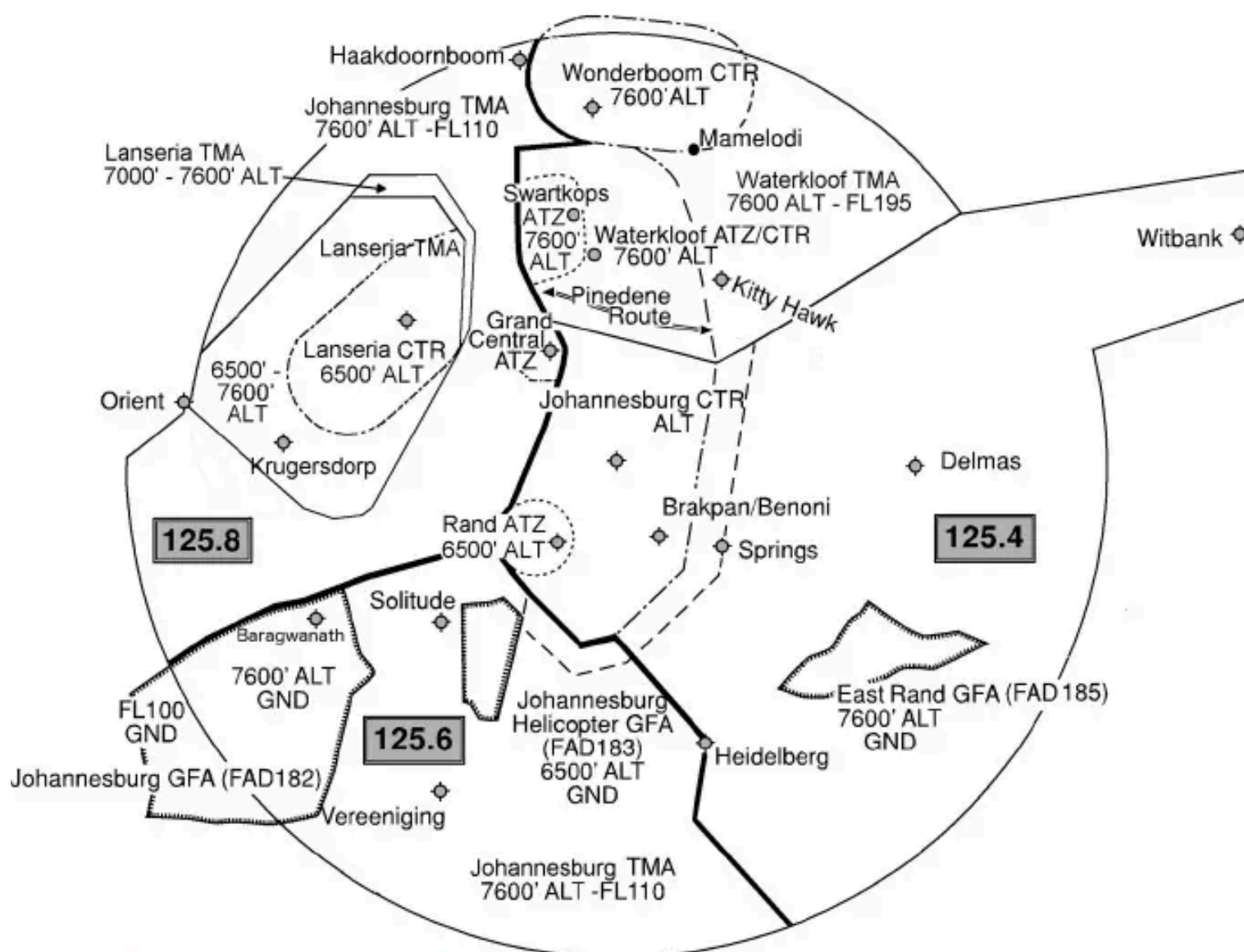
The currently (June 2024) existing Special Rules Areas and other areas or specific routings are:

- ! **Bloemfontein** area – including routes to and from New Tempe airfield  
AIP ENR 2.2 section 1.1 starting on page ENR 2.2-1
- ! **Cape Town Special Rules Area**  
AIP ENR 2.2 section 2.4 to 2.6 starting on page ENR 2.2-37
- ! **Durban Special Rules Area**  
AIP ENR 2.2 section 1.6 starting on page ENR 2.2-9
- ! **East London Special Rules Area**  
AIP ENR 2.2 section 2.1 and 2.2 starting on page ENR 2.2-24
- ! **George Special Rules Area**  
AIP ENR 2.2 section 2.7 and 2.8 starting on page ENR 2.2-44
- ! **Hoedspruit Special Rules Area**  
AIP ENR 2.2 section 1.10 starting on page ENR 2.2-22
- ! **Johannesburg Special Rules Area** – including Pretoria, Waterkloof, the Buffer zone and the Kyalami and Pinedene VFR routes  
AIP ENR 2.2 section 1.3 to 1.5 starting on page ENR 2.2-4  
AIP ENR 3.5 section 3.5.2.1 starting on page ENR 3.5-2 (Pinedene route)  
AIP ENR 3.4 section 3.4 starting on page ENR 3.4-1 (Helicopter routes)
- ! **Kimberley** area – including John Weston airfield  
AIP ENR 2.2 section 1.2 starting on page ENR 2.2-2
- ! **Kruger Special Rules Area** – including Nelspruit airfield  
AIP ENR 2.2 section 1.7 starting on page ENR 2.2-13
- ! **Malelane Special Rules Area**  
AIP ENR 2.2 section 1.8 starting on page ENR 2.2-20
- ! **Port Elizabeth Special Rules Area**  
AIP ENR 2.2 section 2.3 starting on page ENR 2.2-30
- ! **Sabi-Sands Special Rules Area**  
AIP ENR 2.2 section 1.9 starting on page ENR 2.2-21
- ! **Upington** aerodrome – mandatory VFR routes  
AIP ENR 2.2 section 1.11 on page ENR 2.2-24

For frequencies to be used in the above areas see also Appendix II of this book.

Pilots wishing to fly in any of the above areas and unfamiliar with that area are advised to:

- carefully read the relevant pages in the AIP
- print (or load on iPad) the relevant pages to have them available in flight
- program the relevant user points into the aircraft's GPS
- plan their flight according to the prescribed routings



Example: Johannesburg SRA comprising all **uncontrolled** airspace below the Johannesburg, Waterkloof and Lanseria TMAs, excluding General Flying Areas. The SRA is split into three sectors with three different TIBA frequencies.

#### 1.3.4.2 Prohibited Area (FAP)

CAR 91.06.19, CAR 172.02, AIP ENR 5.1

Prohibited Areas may **not be entered by manned aircraft under any circumstances**. Commercial drones may enter with permission. They are generally very small and could be national monuments, explosives factories etc. They have lateral limits and usually extend from ground to an upper limit. Prohibited areas are named FA (South Africa), followed by P (Prohibited), and a two or three digit number.

Example:

FAP68 (Prohibited Area 68) in Johannesburg - Modderfontein Explosives Factory

A circle with a radius of 1 nm

Lower limit Ground - Upper limit 1000' above ground.



### 1.3.4.3 Restricted Area (FAR)

CAR 91.06.20, CAR 172.02, AIP ENR 5.1

Restricted Areas may only be entered **with permission** from the appropriate authority, usually military, as given in AIP ENR 5. They have certain lateral as well as lower and upper limits.

Restricted areas are named FA (South Africa), followed by R (Restricted), and a two or three digit number.

Restricted Areas are class G airspace.

FARs are marked on aeronautical charts with a thin solid blue boundary with hash marks on the inside of the boundary.



Example: FAR23 (Restricted Area 23) - Bottelduim Military Shooting Range

A circle north of Upington with a radius of 14 nm

Lower limit Ground, upper limit FL195

### 1.3.4.4 Danger Area (FAD)

CAR 172.02, AIP ENR 5.1

Danger Areas can also be explosives factories or shooting ranges, but usually are military or civilian general flying training areas (GFAs). They have certain lateral as well as lower and upper limits. Danger areas are named FA (South Africa), followed by D (Danger), and a two or three digit number. Pilots may enter Danger Areas, but should be vigilant and keep a good lookout. Danger Areas are **class G** airspace.

FADs are marked on aeronautical charts with a thin solid blue boundary line with hash marks on the inside of the boundary.

DRONE pilots can expect manned aircraft flying very low in Flying Training areas, and should keep a good look-out, and a listening watch on the relevant frequency.



Example: FAD107S (Danger Area 107 South)

Bloemfontein Military Helicopter Southern Flying Area

Lower limit Ground, upper limit 6000' ALT

### 1.3.4.5 Advisory Area (ADA)

CAR 172.02

Advisory Areas are provided for, but currently do not exist in South Africa. They would be Class F (advisory) airspace, in which a radio is mandatory and pilots must listen out on the relevant frequency.

# 2

# Air Traffic Services

## 2.1 Objectives

CAR 172.01.2

The objectives of Air Traffic Services are:

- ! to prevent collisions between aircraft
- ! to prevent collisions between aircraft on the manoeuvring area (basically the taxiways, but **not** the apron) of an airport and obstructions on such area
- ! to expedite and maintain an orderly flow of air traffic
- ! to provide advice and information useful for the safe and efficient conduct of flight
- ! to notify appropriate organisations regarding aircraft in need of Search and Rescue (SAR – see chapter 12) and assist such organisations as required.

## 2.2 Types of Services

CAR 172.01.4, AIP GEN 3.3

Air Traffic Service in South Africa is usually provided by the Air Traffic and Navigation Services Company (ATNS) under the authority of the Minister of Transport. Military airfields fall under the jurisdiction of the Air Force. And at some smaller civilian airfields the owner or operator of the airfield may provide Aerodrome Flight Information Service (AFIS).

Civilian Air Traffic Service is provided by two Flight Information Centres (FICs):

- ! Johannesburg, controlling the two Johannesburg FIRs, and
- ! Cape Town, controlling the Cape Town FIR.

These two centres provide three types of Air Traffic Services:

- ✈ **Control Service** , comprising:
  - Area Control Service
  - Approach Control Service
  - Aerodrome Control Service

**Information Service** , comprising

- Flight Information Service (FIS), including Advisory Service
- Aerodrome Flight Information Service (AFIS)

✈ **Alerting Service** .

## 2.2.1 Control Service

CAR 91.06.15, CAR 91.06.15-18

Control Service is provided to IFR traffic in class A, B, C and D airspace and to VFR flights in class B, C and D airspace; further to all Special VFR flights and to all aerodrome traffic at controlled airfields (for an explanation of the terms IFR, VFR etc. see chapter 3). Simply stated, Control Service is provided in all controlled airspaces and controls air traffic, meaning it issues clearances and instructions (headings, speeds, altitudes), which must be adhered to. If a particular instruction cannot be complied with, Air Traffic Control must be notified immediately.

Example: ATC instructs a VFR pilot to climb to 6500 feet in controlled airspace.

Unknown to ATC, but visible to the pilot, the cloud base is 6000 feet.

The pilot must advise ATC: "Unable to comply due to cloud at 6000 feet" , after which ATC will issue another clearance to resolve any traffic conflict.

### 2.2.1.1 Area Control Service

Area Control Service controls VFR and IFR flights in CTAs, Air Service Routes and RNAV Routes. It will issue instructions such as Flight Levels or altitudes, headings or routes to be flown and may issue minimum or maximum speed limits. It will also pass on any other useful information, for example the weather at the destination or en route, information about hazardous weather conditions (e.g. icing) received from other pilots, any unserviceability of facilities at destination or en route, as well as Onward Clearance Times (OCTs) if en-route delays are expected, or Expected Approach Times (EATs) if there is a delay with the approach.

The radio call sign for Area Control Service is the name of the Flight Information Centre or Sector, followed by the words "Control" or "Area", which is often omitted.

Examples : Johannesburg Area Control, Johannesburg Central, Cape Town Control, Cape Town Area West

### 2.2.1.2 Approach Control Service

Approach Control Service controls all flights in a TMA. Clearances and instructions include Flight Levels or altitudes, headings, QNH (see paragraph 4.1), Transition

Level (see paragraph 4.3) and runway in use to inbound traffic, turns after take-off to departing traffic, plus any other useful information.

In Visual Meteorological Conditions (VMC, see chapter 3), i.e. in good weather, departing aircraft are given Approach Control Service from the time they are handed over from Aerodrome Control (Tower) until they get passed on to Area Control, and vice versa when arriving.

In Instrument Meteorological Conditions (IMC, see chapter 3), i.e. in bad weather, departing IFR and Special VFR flights are subject to Approach Control from the moment they enter the runway for take off until they are passed on to Area Control, and on arrival from the moment they are handed over from Area Control until they have landed and cleared the runway. In other words, in IMC Approach Control also controls traffic in the CTR. The pilot still speaks to "Tower", but the Tower controller has to co-ordinate all clearances with the Approach controller.

The radio call sign for Approach Control Service is composed of the name of the Flight Information Service Unit providing the service and the word APPROACH or RADAR. At smaller centres, where Approach and Aerodrome Control Service are combined, the call sign for Approach Control is the same as for Aerodrome Control, i.e. name of the airfield and the word TOWER.

At busy centres there might be several Approach controllers, each being responsible for a certain sector. There could also be a dedicated ILS radar controller (Director), who only vectors aircraft onto the final approach for the ILS. At present this only happens in Johannesburg and Cape Town at peak times.

Example: In the Cape Town TMA Approach Control is provided by "Cape Town Approach" and in the Overberg TMA by "Bredasdorp Approach".

### 2.2.1.3 Aerodrome Control Service

Aerodrome Control Service controls all traffic in CTRs and ATZs Class C and D in Visual Meteorological Conditions (VMC), i.e. aircraft on the taxiways (but not on the apron) and on the runways as well as aircraft flying in the circuit.

In Instrument Meteorological Conditions (IMC) Aerodrome Control Service is provided only to aircraft on taxiways. Traffic in the air and on the runway in IMC still speaks to the Aerodrome Controller (Tower), but he only relays instructions from the Approach Controller, and all movements in the CTR, ATZ and on the runway must be approved by Approach Control.

At larger airports Aerodrome Control may be divided into Aerodrome Control proper with the call sign TOWER, controlling aircraft on the runway in use and in the circuit area, and into Surface Movement Control (SMC) with the call sign GROUND,

controlling aircraft on the taxiways. The words TOWER or GROUND are preceded by the name of the airfield, e.g. "Shaka Tower" at King Shaka Airport in Durban or "Cape Town Ground" at Cape Town International Airport.

At very busy airports there might be several TOWER and GROUND frequencies, controlling different runways and taxiways. There may also be a CLEARANCE DELIVERY, giving after departure clearances to aircraft.

## 2.2.2 Information Service

### 2.2.2.1 Flight Information Service (FIS)

Flight Information Service is provided in all airspace to all traffic. Information service will provide information about other known traffic, as well as pass on useful information, for example about weather, the position of unmanned balloons, changes in the serviceability of navigational aids, runway condition etc.

In controlled airspace the Air Traffic Controller will provide information service in addition to the "normal" control service.

In uncontrolled airspace aircraft will report their position as they do in controlled airspace, and Flight Information Service will answer in a similar manner as Control Service does in controlled airspace. However, FIS does not issue clearances or instructions and does not provide traffic or terrain separation, but may warn pilots about known conflicting traffic.

The radio call-sign of Flight Information Service in uncontrolled airspace is the name of the Flight Information Centre or sector providing the service, followed by the word "Information", e.g. "Johannesburg East Information", usually abbreviated to "Johannesburg East", or sometimes just "Information".

### 2.2.2.2 Aerodrome Flight Information Service

Aerodrome Flight Information Service (AFIS) is given to all aircraft operating within an Aerodrome Traffic Zone Class G (i.e. uncontrolled airspace). Aerodrome Flight Information Service, as the name implies, will issue information only, not give clearances or instructions. The phraseology will be "the runway is clear" instead of "cleared to land". Pilots should use similar wording, e.g. say "ready to taxi" instead of "request taxi".

Aerodrome Flight Information Service may declare an airfield IMC, which means closed to VFR traffic due to weather.

AFIS will pass on to pilots any necessary information about the airfield, the weather, and other traffic. AFIS will inspect the airfield and mark temporary obstructions. It

might provide emergency flare-paths for night landings when necessary. AFIS will also relay messages between pilots and operators (e.g. a charter company) as well as between pilots and Flight Information Centres or Area Control Centres (e.g. IFR clearances). AFIS will maintain a NOTAM filing system for pilot briefing purposes.

The call-sign of an AFIS is the name of the airfield plus the word RADIO, e.g. "Port Alfred Radio". At present, AFIS is offered at airfields such as Port Alfred, Oudtshoorn, Beaufort West etc., usually by the resident flight training school.

### 2.2.2.3 Advisory Service

Air Traffic Advisory Service refers to the provision of flight information service to IFR aircraft in class F airspace.

### 2.2.3 Alerting Service

CAR 91.06.15-18

Alerting Service is provided in all airspaces by Air Traffic Service Units (ATSUs) to aircraft in need of Search and Rescue (SAR). See chapter 12.

## 2.3 Air Traffic Flow Management

AIP ENR 1.9.2

Air traffic worldwide is increasing steadily. Airspace, however, is finite, limiting the number of aircraft which can safely be accommodated on certain popular routes. Capacity is further limited by the number of available runways, parking bays and airport terminal facilities such as immigration, baggage handling etc. In addition, adverse weather such as thunderstorms might limit capacity.

Air Traffic Flow Management (ATFM) is a way of managing the airspace and the existing capacities. It is provided in South Africa by the Central Airspace Management Unit (CAMU), which is part of ATNS.

ATFM will plan the orderly and efficient flow of traffic by a three phase approach:

- ! Strategic Flow Management identifies demand well in advance, taking into account historical data, seasonal variations, holiday periods, special events etc.
- ! Pre-Tactical Flow Management will think about measures to aid the traffic flow a couple of days before a perceived capacity problem.
- ! Tactical Flow Management is applied on the day of operation, when all flight plans have been received and the actual weather is known.

### 2.3.1 The Airport Slot System

AIP ENR 1.9.1

To limit congestion a slot system has been established at the three major international airports in South Africa: OR Tambo in Johannesburg, Cape Town International Airport, and King Shaka in Durban. IFR flights departing from or arriving at one of these airports have to apply for a departure or arrival slot. VFR flights also need to apply for slots at OR Tambo and Cape Town International Airport.

The capacity of the three major airports will be established from time to time.

For scheduled airline traffic the operator (airline) will apply for slots twice a year for a summer and a winter schedule. Once-off flights are allocated the remaining slots on a first-come first-served basis. By filing a flight plan early rather than at the last minute the chance of obtaining the desired slot is much better.

The time stipulated by the pilot on the flight plan is an estimated off-block time (EOBT). CAMU will assign a Calculated Take-Off Time (CTOT) according to standard assumed taxi times of 10 minutes for Durban, 11 minutes for Cape Town and 15 minutes for Johannesburg (for heavy aircraft these agreed taxi times are slightly longer). The allocated CTOT for a specific flight can be found on the CAMU web site, communicated by e-mail or SMS when a flight plan has been filed via file2fly, or the pilot can phone 0860 FLY NOW (0860 359 669) for his CTOT.

It is up to the pilot to get his departure clearance, call for start-up, and ask for taxi instructions (remember the agreed taxi times at the various airports) to be ready for take-off as close as possible to his allocated CTOT. ATC has the discretion to get aircraft off between 5 minutes early and 10 minutes late; i.e. CTOT-5 to CTOT+10 minutes.

If it becomes apparent that a pilot will miss his CTOT for whatever reason he should change his flight plan's EOBT as soon as possible and obtain a new CTOT. This will not be a problem if departing Cape Town at 4 o'clock in the morning. But if the pilot is looking for a new CTOT during busy times at OR Tambo, the next available CTOT could be something like three hours later.

## 2.4 Air Traffic Incident Reporting

To improve the Air Traffic System and in the interest of safety pilots can file an Air Traffic Incident Report form and report any deficiencies or unserviceabilities of the Air Traffic System to ATNS.

# 3

# Flight Rules

A pilot can elect to fly under either one of the following two sets of **Flight Rules** :

- ! **VFR** **Visual Flight Rules**
- ✈ **IFR** **Instrument Flight Rules**

Flights under Instrument Flight Rules (IFR) have to adhere to more stringent requirements in respect of pilot licensing, aircraft equipment and fuel reserves.

The terms VFR and IFR must not be confused with the prevailing **Meteorological Conditions** (the weather):

- ! **VMC** **Visual Meteorological Conditions** ("good" weather)
- ✈ **IMC** **Instrument Meteorological Conditions** ("bad" weather)

VFR flight is only allowed in VMC, and not in class A airspace, and may not be operated at transonic or supersonic speeds unless specifically authorised to do so. IFR flight is possible in VMC or IMC in all airspaces.

## 3.1 Visual Flight Rules (VFR)

CAR 91.06.21, CAR 91.06.23, AIP ENR 1.2, AIC 40.4 (outdated)

An aircraft may be flown according to Visual Flight Rules (VFR) when:

- ! the pilot can maintain visual contact with the ground by day, or identifiable objects by night, **and**
- ! the aircraft is not above cloud covering more than " of the ground within a 5 nm radius of the aircraft, **and**
- ! the minima given in the table on the next page with regard to ceiling, visibility and distance from cloud can be adhered to.

If all three rules above cannot be adhered to, the aircraft may not be flown VFR.

An aircraft flying above a solid layer of cloud in glorious sunshine is flying in VMC, since visibility and distance from cloud are all good enough for VMC (Americans call this "VFR on top"). However, in South Africa, it may not be flown under VFR, because it is over more than " of cloud and the pilot cannot maintain visual contact with the ground.

| VMC minima                                                                | Ceiling*   | Forward<br>Visibility | Distance from cloud |                |
|---------------------------------------------------------------------------|------------|-----------------------|---------------------|----------------|
|                                                                           |            |                       | horizontally        | vertically     |
| FL100 to FL195                                                            |            |                       |                     |                |
| Aeroplanes and helicopter                                                 |            | 8 000 m               | 1 500 m             | 1 000 feet     |
| Between 1000' AGL / 3000' AMSL (whichever is higher) and FL100            |            |                       |                     |                |
| Aeroplanes and helicopter                                                 |            | 5 000 m               | 1 500 m             | 1 000 feet     |
| Below 1000' AGL / 3000' AMSL (whichever is higher)<br>in Class C airspace |            |                       |                     |                |
| Aeroplanes                                                                |            | 5 000 m               | 1 500 m             | 1 000 feet     |
| Helicopter                                                                |            | 2 500 m               | 1 500 m             | 1 000 feet     |
| in uncontrolled airspace                                                  |            |                       |                     |                |
| Aeroplanes                                                                |            | 5 000 m               | Clear of cloud      |                |
| Helicopter                                                                |            | 1 500 m               | Clear of cloud      |                |
| CTR or ATZ                                                                |            |                       |                     |                |
| Aeroplanes                                                                | 1 500 feet | 5 000 m               | 600 m               | 500 feet       |
| Helicopter                                                                | 1 500 feet | 5 000 m               | 300 m               | Clear of cloud |

\* **“Ceiling”** is defined as the height above ground or water of the base of the lowest layer of cloud below 20000' covering more than half the sky.

In a CTR or ATZ the Air Traffic Service Unit will determine whether the zone is VMC or IMC. In all other airspaces it is the pilot's responsibility to decide whether flight under VFR is possible or not, and therefore those minima should be memorised by every VFR pilot.

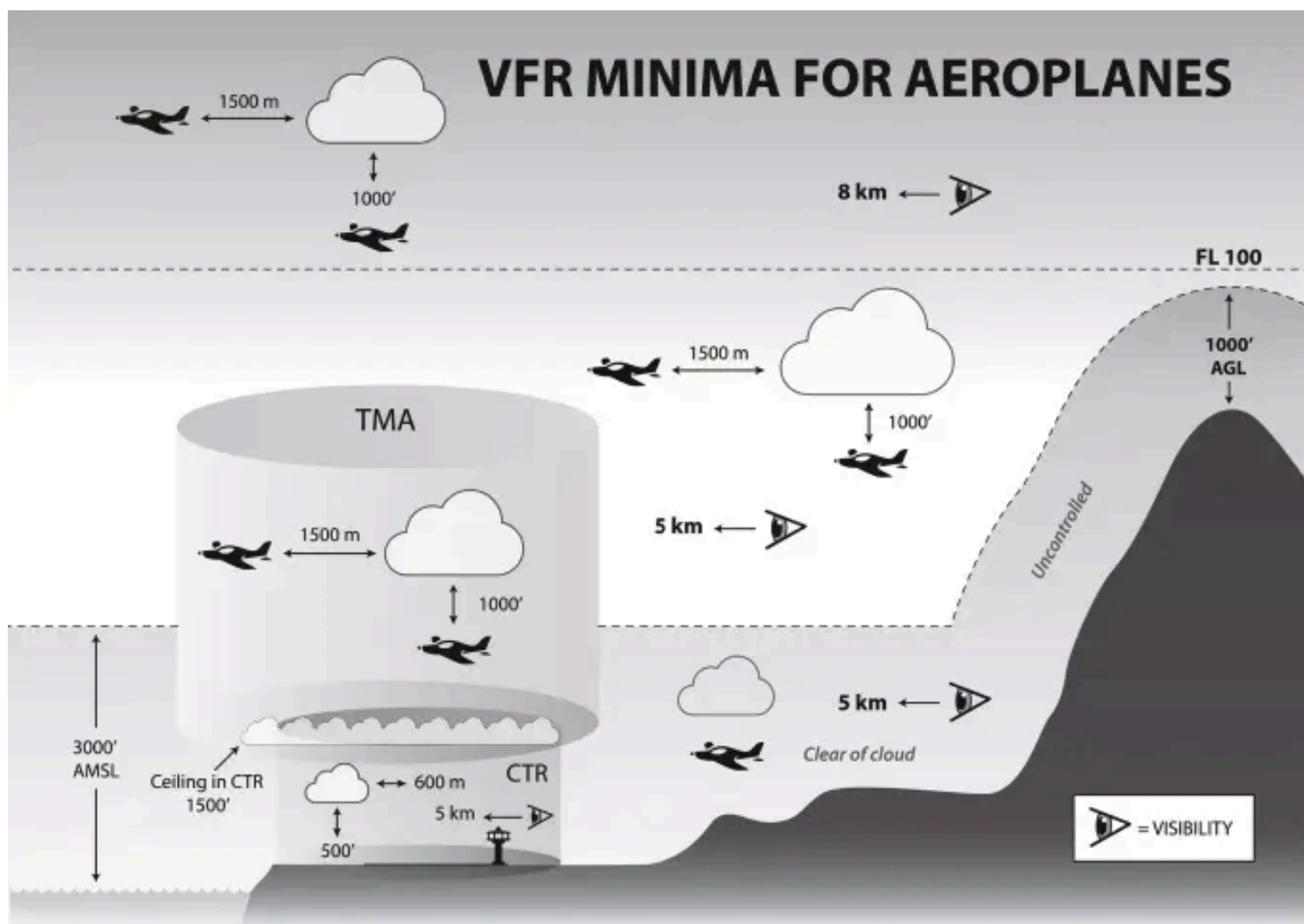
**Exception 1:** an aeroplane may take off and leave for a cross-country flight or arrive back from a cross-country flight and land in a ATZ Class C, D or G if the ceiling is at least 500 feet, the visibility at least 5 km, and the aircraft is in radio contact with the Tower or AFIS.

Example 1: The ceiling in the Lanseria **CTR** is 1000 feet. VFR is not possible (but see paragraph 3.3 for Special VFR).

Example 2: The ceiling in the Rand **ATZ** Class C is 1000 feet. It is legal to take off and leave the ATZ for a VFR cross-country flight, but it is not legal to do circuits at Rand under these weather conditions.

**Exception 2:** a helicopter may fly in uncontrolled airspace in a visibility of less than 1500 m if it flies slowly enough to observe other traffic and avoid obstacles.

**Exception 3:** a helicopter may, at an airfield or heliport, hover in ground-effect or hover-taxi if the visibility is at least 100 metres.



Only figures for aeroplanes are given – apologies to helicopter pilots.

## 3.2 Instrument Flight Rules (IFR)

CAR 91.06.24-27, AIP ENR 1.3, AIC 40.4 (outdated)

If the weather is worse than the minima given in the previous paragraph, Instrument Meteorological Conditions (IMC) prevail, and aircraft must be flown according to Instrument Flight Rules (IFR). Aircraft must also be flown under IFR in all Class A airspace, regardless of the weather. To do this the pilot must be instrument rated and the aircraft equipped for instrument flight. It also means larger fuel reserves must be carried than on a VFR flight and certain other rules have to be complied with.

### 3.3 Special VFR (SVFR)

CAR 91.06.22, AIP ENR 1.2.2, AIC 40.4 (outdated)

Air Traffic Control (ATC) may grant Special VFR to an aeroplane to fly in a CTR that has been declared IMC (i.e. when the cloud base is lower than 1500' or the visibility less than 5000 metres), under the following circumstances:

- ! by day only,
- ! the ceiling is at least 600',
- ! ground visibility is at least 1500 metres (at the aerodrome reference point),
- ! the aircraft stays clear of cloud, and
- ! the aircraft is in radio contact with the Control Tower and obeys instructions and does not delay an IFR flight.

Whether or not ATC will grant Special VFR to a flight depends on the traffic situation in the CTR. Special VFR may be used to leave for a cross-county flight if the weather outside the CTR is good enough for the aircraft to continue in VMC, or to enable an aircraft to return from a cross-country flight. It is also used to enable aircraft to cross a CTR or fly within a CTR for special purposes, for example traffic surveillance. Special VFR will not be granted to training flights to do circuits within the CTR.

When granting a Special VFR clearance the Air Traffic Controller will apply the more stringent separation standards associated with IFR flights, possibly resulting in delays for the SVFR flight. Note also that because the CTR is IMC, the approach controller (rather than the aerodrome controller) issues all clearances for flights in the CTR. This clearance is relayed to the aircraft by the aerodrome controller, who has to communicate with the approach controller via the intercom before being able to grant any clearance or issue any instruction.

Pilots must note that, when granted a Special VFR clearance, it is the pilot's responsibility to **stay clear of cloud and within sight of the ground**. If this is not possible, the controller must be notified immediately and the flight terminated. Normal minimum heights must still be observed, i.e. over built-up areas the aircraft must stay at least 1000 feet above the highest obstacle within a 2000' radius. A Special VFR clearance ends at the CTR boundary, thereafter the normal VFR minima as stated in paragraph 3.1 apply.

Special VFR can only be granted in a CTR, not in an ATZ – but refer to the exceptions in paragraph 3.1 above.

A helicopter may be granted Special VFR under the following circumstances:

- # by day only,
- # the ceiling is at least 300',
- # in a visibility of at least 800 metres offshore,
- # when it can remain clear of clouds,
- # if it is flown slowly enough to avoid other traffic or obstructions,
- # the helicopter can still observe all minimum height rules, and
- # when the special VFR helicopter flight does not unduly delay any IFR flight.

# 4

# Altimeter Setting Procedures

Atmospheric pressure decreases by approximately one hectopascal (hPa) for every 30 feet increase in altitude (up to about 10 000 feet). Therefore, we can use a barometer to measure altitude, simply by calibrating the instrument to indicate altitude in feet rather than atmospheric pressure in hectopascal. Since actual atmospheric pressure varies from day to day and place to place, the altimeter subscale must be set to a certain datum or pressure level. The datum selected will determine what exactly the altimeter is going to indicate.

When taking off, flying relatively low-level, and when landing it is sensible for a pilot to see the altitude above sea level on his altimeter, and the pilot should set the actual sea-level pressure of the day (QNH) in the subscale of the altimeter.

En route, high above any terrain, with aircraft coming from various places with different local QNH settings, it is imperative that everybody is on one standard setting. Pilots will then not necessarily know their exact altitude, but since everybody is on the same setting, relative separation between aircraft is ensured. This standard setting has been chosen to be 1013.2 hPa (see definitions below).

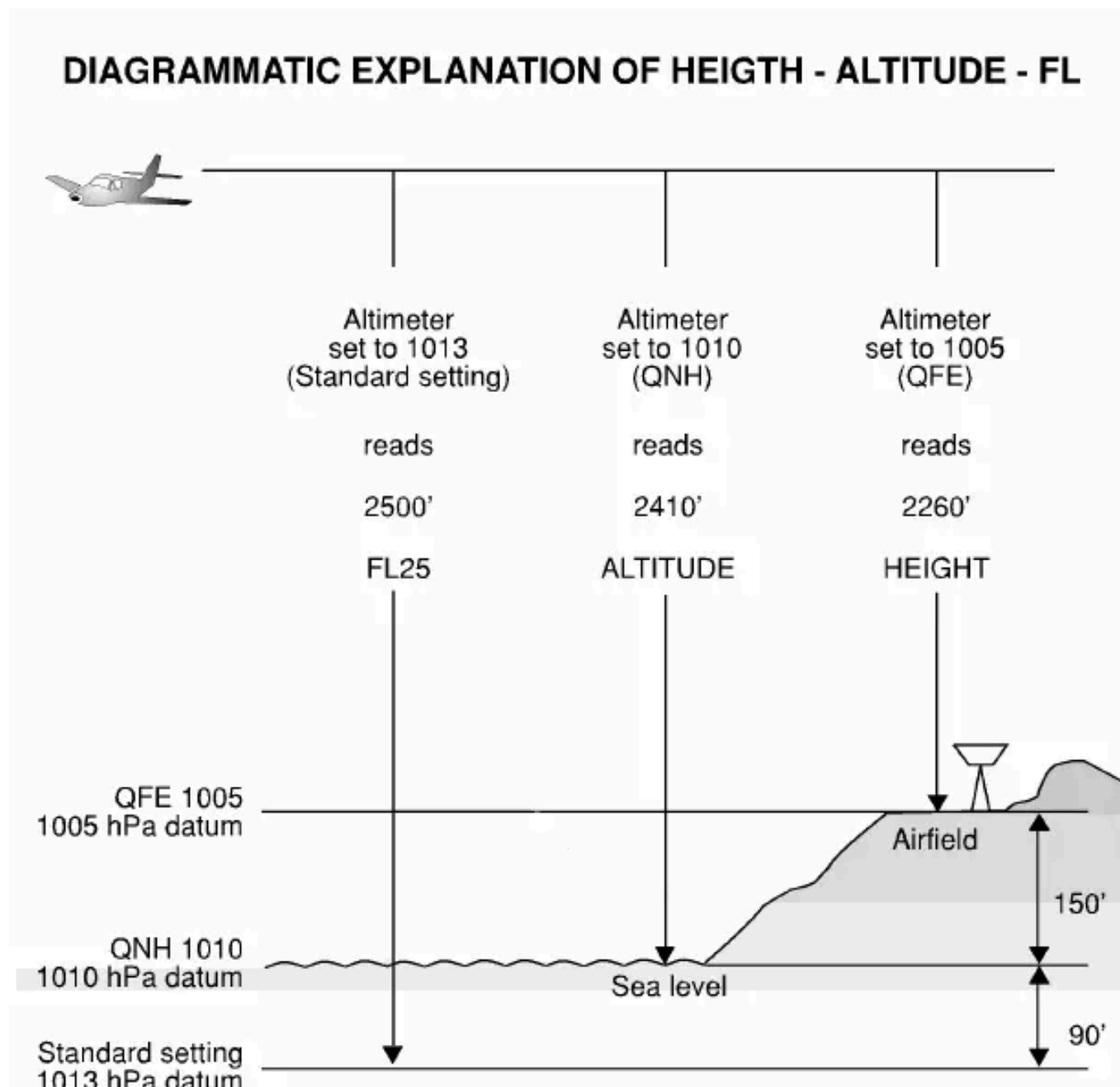
The following explains when and where the different altimeter settings are to be used.

" A hectopascal is the same as a millibar, and is a measurement for pressure.

## 4.1 Definitions

AIP GEN 2.2, AIP ENR 1.7, AIC 20.2

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>QNH</b>                          | Actual present atmospheric pressure at a certain place corrected for sea level. The datum is <b>sea level</b> ; therefore the altimeter will show <b>ALTITUDE</b> (= height above sea level). When on the ground at an airfield, the altimeter shows the ELEVATION of that airfield.                                                                                                                           |
| <b>QFE</b>                          | Actual atmospheric pressure at a particular place. The datum is <b>ground level</b> ; therefore the altimeter will show <b>HEIGHT</b> above ground. When on the ground at that place (the airfield), the altimeter shows zero.                                                                                                                                                                                 |
| <b>1013.2 hPa Standard Setting.</b> | An arbitrary datum (sea level pressure in the International Standard Atmosphere ISA), which can be below, at, or above sea level, depending on the prevailing atmospheric pressure. Vertical distance from this datum is called <b>Pressure Altitude</b> and is expressed as a <b>FLIGHT LEVEL</b> , whereby the last two zeroes are dropped, e.g. altimeter shows 7500 feet = FL75.                           |
| <b>Altitude</b>                     | Vertical distance between sea level and an object above the earth's surface, e.g. an aircraft in flight. Shown on the altimeter when the present QNH for that particular area is set in the altimeter subscale.                                                                                                                                                                                                |
| <b>Elevation</b>                    | Vertical distance between sea level and a point on the earth's surface.                                                                                                                                                                                                                                                                                                                                        |
| <b>Height</b>                       | Vertical distance between the ground and the aircraft. Height will be shown on the altimeter only when QFE is set on the subscale. The height shown is the height above the airfield to which the QFE applies. If the aircraft then flies over a mountain, the altimeter will still show height above the airfield, not height above the mountain.                                                             |
| <b>Flight Level</b>                 | Vertical distance between the 1013.2 hPa level and the aircraft, also called Pressure Altitude. A Flight Level will be shown on the altimeter only when the Standard Setting of 1013.2 hPa is set in the subscale. Flight Levels theoretically start at the 1013.2 hPa level as Flight Level zero and are separated by a pressure interval corresponding to 500 feet in the International Standard Atmosphere. |
| <b>Level</b>                        | Generic term for height, flight level and altitude. When asked to “report your level”, valid answers might be: “! maintaining flight level 75” or “! maintaining 2000 feet” or “! maintaining 500 feet AGL” .                                                                                                                                                                                                  |
| <b>QNE</b>                          | is not defined anymore by ICAO. - The <b>South African</b> AIP in Gen 2.2 defines QNE as the indicated height on landing with the altimeter subscale set to 1013.2 hPa. - In the <b>USA</b> (and by some SA instructors!) QNE is used as the standard altimeter setting of 1013.2 hPa.                                                                                                                         |



The aircraft in the diagram flies at a constant altitude. Depending on the setting of the altimeter subscale the pilot will read different indications off his altimeter.

## 4.2 Transition Altitude

AIP ENR 1.7.3 (outdated), AIC 20.2 (outdated), Instrument Approach Charts

Transition Altitude (TA) is the altitude at which the pilot of a climbing aircraft changes the subscale setting of his altimeter from local QNH to Standard Setting 1013.2 hPa.

The Transition Altitude is published in the AIP and on instrument charts for all major aerodromes in South Africa (see the list below), where it is set to be at least 1000' above all obstacles within a 25 nm radius around the airport. All other airfields within a radius of 25 nm of those listed will also use the same Transition Altitude.

**Transition Altitude for airports with published Instrument Approach Procedures and all other airfields within a 25 nm radius of those listed:**

|                                  |       |                                       |       |
|----------------------------------|-------|---------------------------------------|-------|
| Alexander Bay*                   | 6000' | Newcastle                             | 9500' |
| Bhisho*                          | 7000' | Phalaborwa*                           | 3500' |
| Bloemfontein (Bram Fischer)      | 8000' | Pietermaritzburg (Oribi)              | 8500' |
| Cape Town International          | 7500' | Pietersburg (civil)*                  | 9000' |
| Durban (King Shaka)              | 5500' | Pilanesberg International             | 7400' |
| East London (King Phalo)         | 7000' | Plettenberg Bay**                     | 6700' |
| George                           | 8000' | Polokwane International               | 9000' |
| Giyani*                          | 5500' | Pongola*                              | 6500' |
| Graaff Reinet**                  | 9000' | Port Elizabeth (Chief Dawid Stuurman) | 5500' |
| Greytown**                       | 7500' | Pretoria (Wonderboom)                 | 8000' |
| Johannesburg (OR Tambo)          | 8000' | Pullenshope/Hendrina*                 | 8000' |
| Johannesburg (Rand)*             | 8000' | Richards Bay                          | 3500' |
| Kimberley Airport                | 7000' | Secunda*                              | 8000' |
| Kleinsee*                        | 4500' | Sishen                                | 8000' |
| Kruger Mpumalanga Intern'l       | 9000' | Skukuza                               | 9000' |
| Ladysmith*                       | 9500' | St Francis Field**                    | 8500' |
| Lanseria International           | 8000' | Tswalu*                               | 8000' |
| Lichtenburg*                     | 8000' | Tutuka Power Station*                 | 8000' |
| Lydenburg*                       | 7500' | Tzaneen*                              | 8500' |
| Mafikeng (George Dick Montshioa) | 7000' | Ulundi*                               | 7100' |
| Makhado AFB**                    | 8000' | Upington International                | 5500' |
| Malelane                         | 7500' | Venetia Mine*                         | 7000' |
| Margate                          | 5000' | Vereeniging*                          | 8000' |
| Musina*                          | 7000' | Waterkloof AFB**                      | 8000' |
| Mthatha                          | 7500' | Welkom*                               | 7000' |
| Overberg AFB**                   | 5000' | Witbank**                             | 8000' |

\* these airfields do not have published Instrument Approach Procedures anymore, but still have Transition Altitudes listed in AIP ENR 1.7 or AIC 20.2.

\*\* these airfields are not listed in AIP ENR 1.7, but have published instrument approach procedures with a Transition Altitude.

For all airfields which are **not** listed in the table above and are **not** within 25 nm of those listed above: the Transition Altitude must be determined by the pilot himself as follows:

! In **VMC**: 2000' above ground level (AGL)

! In **IMC**: the lowest safe cruising altitude

## 4.3 Transition Level

AIP ENR 1.7.4, AIC 20.2

Transition Level (TL) is the Flight Level at which a descending aircraft must change the subscale setting of its altimeter from 1013.2 hPa to the local QNH.

The Transition Level must always be at least 1000 feet above the Transition Altitude. Therefore, the TL is not fixed, but changes (slightly), dependent on the local pressure on the day (QNH) – to guarantee that the Transition Layer is at least 1000'.

The Transition Level is:

- ! for the airports listed in paragraph 4.2 and for all airfields within 25 nm of those: as given by the controller or broadcast on the ATIS and at least 1000' above the listed Transition Altitude for that airfield.
- ! for all other airfields:
  - in **VMC**:  
the nearest flight level above 3000' AGL.

- in **IMC**:  
the nearest Flight Level 500 feet above the lowest safe altitude

## 4.4 Practical Application

AIP ENR 1.7.5 to 1.7.10, AIC 20.2

When starting up at an aerodrome the pilot will set the local QNH in the subscale of his altimeter. The local QNH can be obtained in various ways, depending on what is available at the airfield:

- ! from the ATIS (see paragraph 8.2) of the aerodrome or a nearby airport
- ! from the aerodrome controller
- ! by setting the altimeter to the (known) elevation of the airfield.

The pilot will then take-off and express his vertical position as an **altitude**. He will remain on QNH:

- ! when flying circuits and landings
- ! when on a local flight, e.g. a sightseeing flight or local training flight
- ! when remaining below 1500' AGL on a low-level cross-country flight.

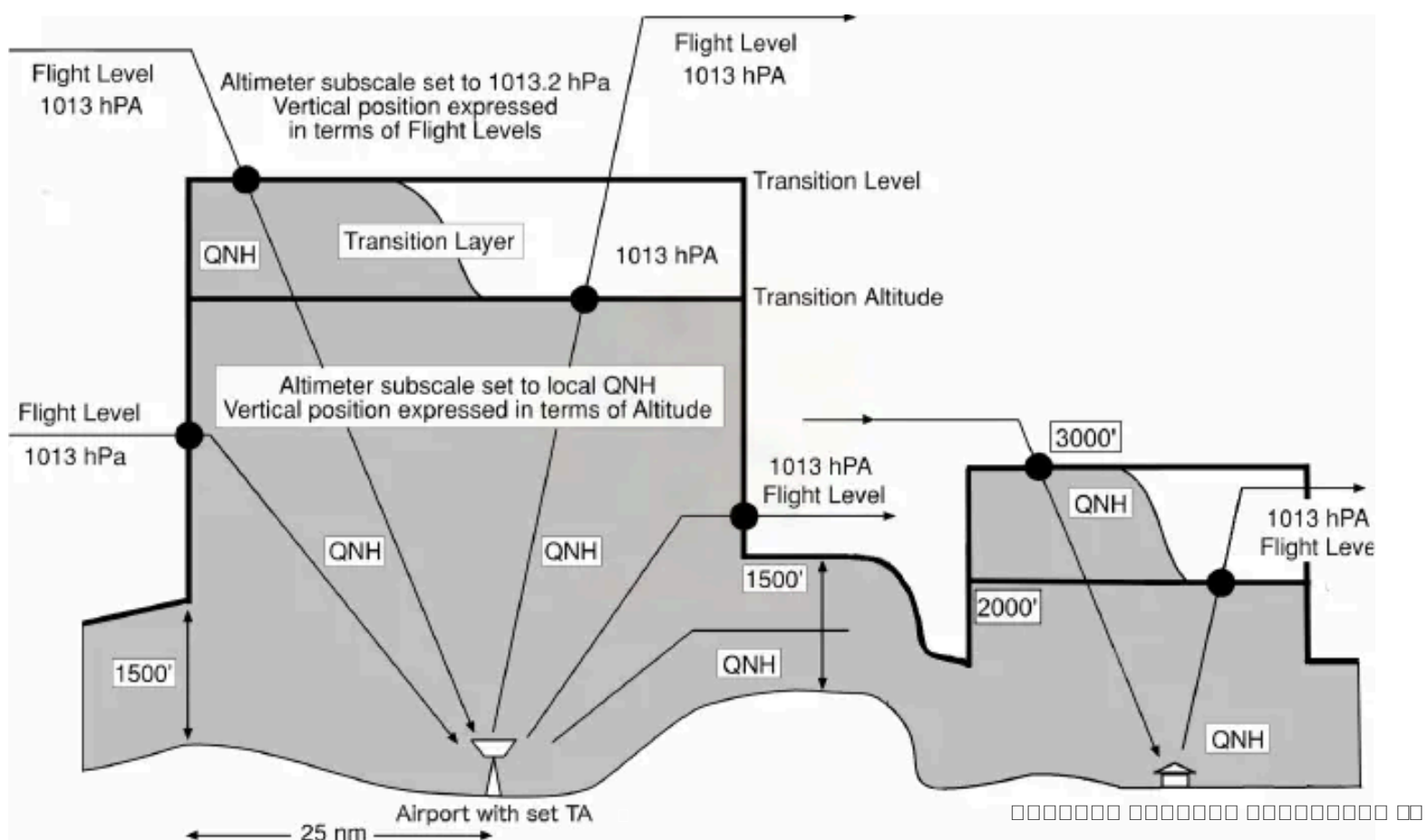
On a cross-country flight above 1500' AGL the pilot must change the subscale of his altimeter to standard setting 1013.2 hPa when **climbing** through the **Transition Altitude** as stated in paragraph 4.2 or when 25 nm from the departure airfield, whichever comes first. Thereafter, he will express his vertical position as a **Flight Level**.

When returning to land from a cross-country flight conducted above 1500' AGL the pilot must change the subscale of his altimeter to the local QNH when **descending** through the **Transition Level** as stated in paragraph 4.3 or when within 25 nm of the destination airfield. The local QNH can be obtained:

- ! from the ATIS (see paragraph 8.2) of the aerodrome or a nearby one
- ! from the aerodrome controller
- ! by setting the local area QNH as given in a meteorological briefing before the flight
- ! by asking another pilot in a nearby aircraft over the radio.

- In practice, most IFR pilots taking off from or landing at controlled airfields change the subscale to 1013.2 hPa as soon as they are cleared to climb to a Flight Level – and back to local QNH as soon as they are cleared to descend to an altitude.

Note that no aircraft should be flying **level** within the Transition Layer, i.e. between the Transition Altitude and the Transition Level.



# 5

# Semi-Circular Rule

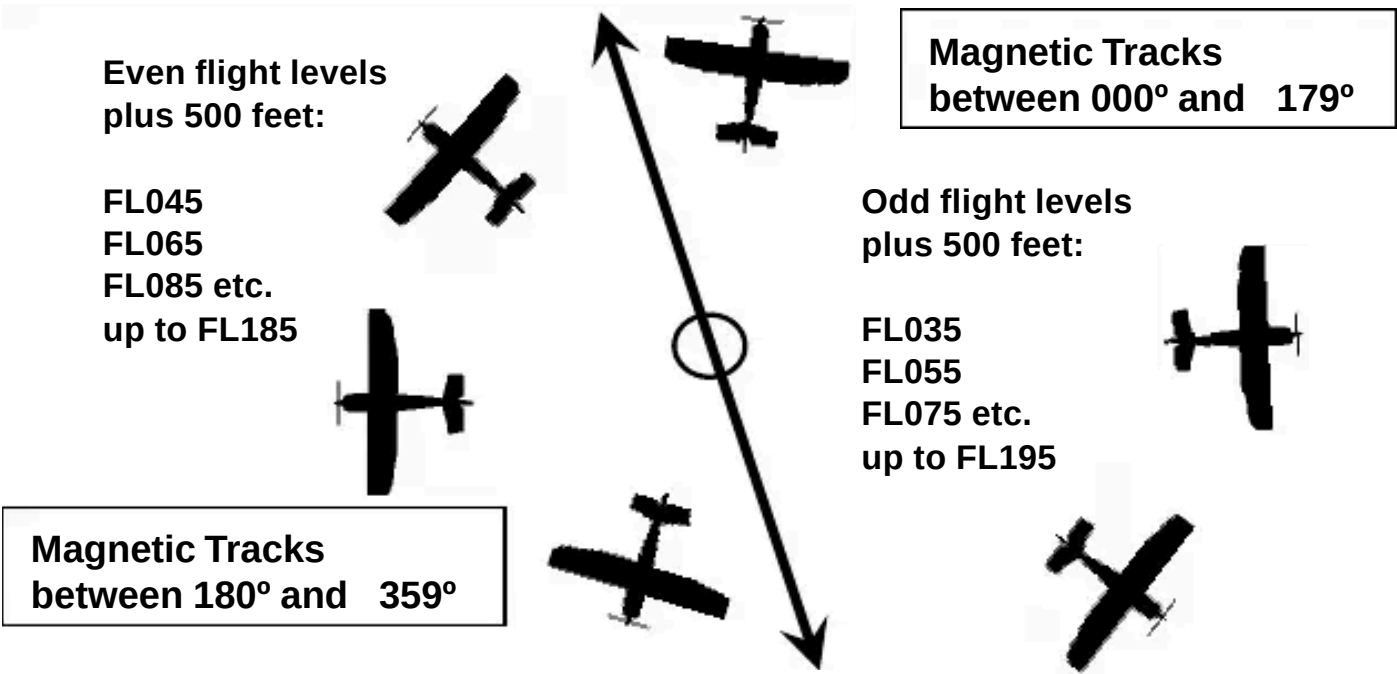
To ensure vertical separation between aircraft, all aircraft cruising (i.e. flying level in cruising flight) **at or above 1500' AGL** in uncontrolled airspace have to adhere to the semi-circular rule.

In controlled airspace ATC generally expects aircraft to also only request levels complying with the semi-circular rule, but may assign specific altitudes or non-standard flight levels.

## 5.1 VFR Flights

CAR 91.06.33, AIP ENR 1.7.11, AIC 20.2

All VFR flights cruising above 1500' AGL on **magnetic tracks** between 000° and 179°, i.e. flying in a roughly easterly direction, have to fly at odd flight levels plus 500'.



All VFR aircraft cruising above 1500' AGL on **magnetic tracks** between 180° and 359°, i.e. flying in a roughly westerly direction, have to fly at even flight levels plus 500'.

Remember that VFR flight is only allowed up to and including FL195.

- “ Whether the Flight Levels below FL100 are pronounced with or without the leading zero seems to be a matter of debate. However, ICAO in Doc 9432 gives as an example of correct phraseology “ ! cross A1 at Wicken FL70 ! ” – which seems to indicate that the leading zero in Flight Levels below FL100 is not necessary.

## 5.2 IFR Flights

CAR 91.06.33, AIP ENR 1.7.11, AIC 20.2

All IFR flights on magnetic tracks between 000° and 179°, i.e. flying in a roughly easterly direction, have to fly at odd flight levels.

And all IFR aircraft flying on magnetic tracks between 180° and 359°, i.e. flying in a roughly westerly direction, have to fly at even flight levels.

Altimeters are simply barometers and actually measure pressure as explained in chapter 4. In the lower atmosphere altimeters are quite accurate and a 1 hPa pressure drop corresponds quite well to a 30 foot gain in altitude. But air becomes much thinner at higher altitudes and there is less pressure difference between, say, 22000 and 23000 feet than there is between 2000 and 3000 feet.

What this means is simply that altimeters are less accurate at altitude, and therefore at higher altitudes we must either make altitude-keeping more stringent or we must increase the vertical separation between aircraft.

**Between flight levels 290 and 410** most airspace in the world has been declared RVSM (reduced vertical separation minima) airspace. For flight in RVSM airspace aircraft and pilots have to be “RVSM compliant” – see paragraph 14.3.1.

**Above FL410** vertical separation is increased to 2000' between aircraft on opposing tracks and 4000' between aircraft flying in roughly the same direction.

### 5.3 Table of Flight Levels

(IFR levels for the General Licence only)  
ICAO Annex 2 Appendix 3, AIP ENR 1.7.11, AIC 20.2

| Magnetic Track:                                                                                                                      | 180° to 359°  |        | 000° to 179°  |        |
|--------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|---------------|--------|
|                                                                                                                                      | ← Flying West |        | Flying East → |        |
| Flight Rules:                                                                                                                        | IFR           | VFR    | IFR           | VFR    |
| FLIGHT LEVEL 015 to FLIGHT LEVEL 195                                                                                                 |               |        |               |        |
| Various airspace classes<br>(controlled and uncontrolled)<br>VFR and IFR flights are allowed<br>Vertical separation is at least 500' | FL 020        | FL 025 | FL 030        | FL 015 |
|                                                                                                                                      | FL 040        | FL 045 | FL 050        | FL 035 |
|                                                                                                                                      | FL 060        | FL 065 | FL 070        | FL 055 |
|                                                                                                                                      | FL 080        | FL 085 | FL 090        | FL 075 |
|                                                                                                                                      | FL 100        | FL 105 | FL 110        | FL 095 |
|                                                                                                                                      | FL 120        | FL 125 | FL 130        | FL 115 |
|                                                                                                                                      | FL 140        | FL 145 | FL 150        | FL 135 |
|                                                                                                                                      | FL 160        | FL 165 | FL 170        | FL 155 |
|                                                                                                                                      | FL 180        | FL 185 | FL 190        | FL 175 |
| FLIGHT LEVEL 200 to FLIGHT LEVEL 280                                                                                                 |               |        |               |        |
| Class A airspace<br>IFR flights only<br>Vertical separation is at least 1000'                                                        | FL 200        |        | FL 210        |        |
|                                                                                                                                      | FL 220        |        | FL 230        |        |
|                                                                                                                                      | FL 240        |        | FL 250        |        |
|                                                                                                                                      | FL 260        |        | FL 270        |        |
|                                                                                                                                      | FL 280        |        |               |        |
| FLIGHT LEVEL 290 to FLIGHT LEVEL 410                                                                                                 |               |        |               |        |
| Class A RVSM airspace<br>IFR flights only<br>Vertical separation is at least 1000'<br>Aircraft & pilots must be RVSM compliant       | FL 300        |        | FL 290        |        |
|                                                                                                                                      | FL 320        |        | FL 310        |        |
|                                                                                                                                      | FL 340        |        | FL 330        |        |
|                                                                                                                                      | FL 360        |        | FL 350        |        |
|                                                                                                                                      | FL 380        |        | FL 370        |        |
|                                                                                                                                      | FL 400        |        | FL 390        |        |
| FLIGHT LEVEL 430 upwards                                                                                                             |               |        |               |        |
| Class A airspace - IFR flights only<br>Vertical separation is at least 2000'                                                         | FL 430        |        | FL 450        |        |
|                                                                                                                                      |               |        |               |        |
| Class G airspace (uncontrolled)<br>Vertical separation is at least 2000'                                                             | FL 470        |        | FL 490        |        |
|                                                                                                                                      | FL 510        |        | FL 530        |        |
|                                                                                                                                      | FL 550        |        | FL 570        |        |
|                                                                                                                                      | FL 590        |        |               |        |

# 6

# Flight Plans

A Flight Plan (form CA172-04) is a coded description of the details of a flight, which is provided to Air Traffic Service Units before a flight is commenced. It aids in traffic planning and provides information in case Search and Rescue becomes necessary.

## 6.1 When a Flight Plan is Mandatory

CAR 91.03.4, AIP ENR 1.10, AIC D003/2022, AIC D005/2017 (for drones)

A Flight Plan **may** be filed for any flight; but it **must** be filed for the following flights:

- ! All flights conducted wholly or partly in airspaces class A to F, except:
  - a local flight
  - a flight crossing an Air Traffic Service Route at right angles
  - a VFR flight taking off in a CTR or ATZ to fly to an unmanned airfield or vice versa, and not entering any other controlled or advisory airspace
- ! All flights between 2 airfields where an ATSU is operating
- ! All international flights
- ! All commercial flights: charter, scheduled flights and cargo flights
- ! All flights for which alerting action is required
- ! All flights (IFR and VFR) to/from Cape Town International (AIP AD FACT 2.23(3))

Example: A flight from Dawid Stuurman Airport, Port Elizabeth to Plettenberg Bay at 1000' AGL will be leaving the Dawid Stuurman CTR (upper limit 2000') and not enter any other controlled airspace. No Flight Plan necessary.

Another flight on the same route, but at FL45 will be entering the Port Elizabeth TMA (lower limit 2000' AGL. A Flight Plan **is** necessary.

A third flight from the same airport, this time to George, coastwise at 1000' will not be entering the Port Elizabeth TMA, but will, at the end of the flight, enter the George CTR. Therefore, a Flight Plan **is** necessary.

**DRONE PILOTS** must file a Flight Plan on the file2fly system for all flights conducted in controlled airspace. Furthermore, before filing the Flight Plan, they must obtain permission from the ATNS Central Airspace Management Unit (CAMU) and obtain a

CAMU reference number. Go to the ATNS website <http://www.atns.com> and click on the “Flexible Use of Airspace Icon” to apply. Drone pilots must also phone the relevant Air Traffic Services Unit prior to the flight and discuss the intended operation.

## 6.2 How to File a Flight Plan

CAR 91.03.4, AIP ENR 1.10, AIC D003/2022, AIC D005/2017 (for drones)

### 6.2.1 Before Departure

A Flight Plan may be filed up to 5 days (120 hours) before the flight (recommended), but must be filed **at least 30 minutes before departure** for a domestic flight or at least 60 minutes before departure for an international flight. It can be filed in any of the following ways:

- ! in person or by telephone with the nearest Air Traffic Service Unit (ATSU)
- ! by telephone with the Aeronautical Information Management (AIM) Service Centre in Johannesburg at 0860 359 669 (0860 FLY NOW)
- ! via the file2fly system on the ATNS website <https://file2fly.atns.co.za/aes/login.jsp> (this is the preferred method)

When filing a flight plan in person or via the telephone, the pilot should have the information ready in the correct order and format. It is probably best to fill in a flight plan form and then just read the information off.

After having filed the flight plan, the pilot must obtain a reference number for this flight plan. It sometimes happens that flight plans get lost in the system, and finding it will be made easier with the reference number. The reference number will be provided by the person taking down the details of the flight plan – or will be sent via e-mail or SMS when filing on the file2fly website.

However, the reference number does not automatically imply that the flight plan has been accepted. Acceptance is provided verbally if the flight plan has been filed telephonically or by the “accepted” or “processed” status indicated on file2fly.

A separate Flight Plan must be filed for different legs of a flight, e.g. when an intermediate fuel stop is required.

- “ File2fly does not always use the exact terminology that is prescribed in the relevant AIC – but it does send a detailed error message, so that pilots can then change to the specific format required by file2fly for certain items. – This book lists the terminology according to ICAO and AIC D005/2022.

## 6.2.2 Filing a Flight Plan in Flight

CAR 91.03.4, AIP ENR 1.10, AIC D003/2022

Filing a flight plan in flight is only permissible in exceptional circumstances. Possibly there were no communication facilities at the point of departure; or the pilot's plans change during the flight due to weather or other unforeseen circumstances. In this case the pilot must transmit the following details on first contact with an ATSU, but **at least 10 minutes** before entering controlled or advisory airspace:

- ! Aircraft registration
- ! Flight rules
- ! Type of aircraft
- ! Aerodrome of departure
- ! Cruising speed and flight level
- ! Route to be followed and estimates as applicable
- ! Aerodrome of destination and estimated time of arrival
- ! Alternate aerodrome (for IFR flights)
- ! Alerting action required
- ! Fuel endurance (if alerting action is requested)
- ! Total number of persons on board
- ! Name, address and telephone number of the owner or operator of the aircraft.

## 6.3 Activation of the Flight Plan

CAR 91.03.4, AIC D003/2022

A filed flight plan is only a notification that a pilot intends to go on a specific flight, and will simply be deleted if not **activated** within 60 minutes of the planned departure time. All a pilot has to do to activate the flight plan is to establish first contact with an Air Traffic Service Unit (ATSU) over the radio.

Therefore, if a flight is delayed for more than 60 minutes, the pilot must either amend the old flight plan before the hour is up or file a new one. Also, when taking off from a remote airfield and flying relatively low level it could be impossible to activate the flight plan within one hour after the stated departure time. In this instance the pilot should telephone an ATSU prior to take-off and advise them of the anticipated departure time, thus activating the Flight Plan. Alternatively, he could activate the Flight Plan with a relayed message (see paragraph 8.12) when airborne.

Filing a flight plan and activating it still does not allow the pilot to enter controlled airspace later during the flight. This clearance has to be given explicitly by the controller to the pilot over the radio.

## 6.4 Changes in Flight

CAR 91.03.4, AIP ENR 1.10 (outdated), AIC D003/2022 (outdated)

A pilot flying in controlled airspace may at any time request a change to the flight plan, e.g. to avoid adverse weather. This requested change must be approved by ATC before the pilot may implement it. If, due to an emergency, the aircraft deviates in any respect from the filed Flight Plan or subsequent clearance without prior permission, ATC must be notified as soon as possible.

If the aircraft is off track, heading must be adjusted to regain track as soon as possible.

If during flight the True Airspeed (TAS) changes by more than 10 kts or 0.02 Mach from the TAS nominated in the Flight Plan (e.g. due to a failed engine in a twin-engine aircraft) or if the estimated time at the next reporting point or destination is expected to change by more than 2 minutes, ATC must be notified via the radio.

- „ ICAO and SACAR 91.03.4 require pilots to notify ATC if the estimated time over the next reporting point changes by more than 2 minutes. AIP ENR 1.10, AIP GEN 1.7 (ICAO differences) and AIC D003/2022 still state 3 minutes. **2 minutes is correct.**

In uncontrolled airspace the pilot may deviate from his Flight Plan as he wishes, but should advise the relevant Information Service of any changes.

## 6.5 Closing of a Flight Plan

CAR 91.03.4, AIC D003/2022

A Flight Plan will be closed automatically when landing at a controlled airfield.

When Search and Rescue was requested and the destination is an uncontrolled airfield, the pilot must specifically close his flight plan with ATC before or after landing. This may be done over the radio by transmitting something like "field in sight, landing in 5 minutes, please close my flight plan" to ATC.

Alternatively, a flight plan may be closed by telephone after landing. State the call sign and the departure and destination airfields – not the flight plan reference number. The controller does not have that reference number in front of him.

Example: "Good morning, this is the pilot of ZS-ABC. Please cancel Search and Rescue for my flight from Pofadder to Pampoensfontein".

A flight plan can also be closed by asking another aircraft for a relay. See 8.12.

Failure to close a flight plan in which alerting action was requested could result in an unnecessary and costly search, for which the pilot may be held responsible.



In the explanations on the next 18 pages or so, the following conventions have been used:

! **The most common cases for private VFR flights are given first and in bold. These should be studied by candidates for the Restricted Radio Licence.**

! Less commonly used specifications are printed in normal font. Most of these are for interest only and not usually used by small aircraft.

! Specifications in shaded blocks (such as this one) are only applicable to the **General Radio Telephony Certificate** and need not be studied by candidates for the Restricted Radio Licence.

! Specific instructions for Remotely Piloted Aircraft Systems (RPAS) are prefixed by the word **DRONES** or **DRONE PILOTS**.

### 6.6.1 Item 7 – Aircraft Identification

Insert the aircraft radio call sign (refer also to paragraph 7.7.2) to be used in flight as up to 7 alphanumeric characters without hyphens or symbols:

#### **ZSABC Zulu Sierra Alpha Bravo Charlie**

The registration marks of the aircraft without the hyphen

**DRONE PILOTS** insert the registration of the drone, e.g. ZTABC

#### **BE55ABC Beechcraft Baron 55 Alpha Bravo Charlie**

The ICAO aircraft type designator followed by the last three characters of the registration. Specify the full registration in item 18 under remarks RMK/

**ZZZZ** for irregular call signs on internal flights only. Specify the name in item 18 under remarks (RMK/), e.g. RMK/Mozart or RMK/SAFAIR TRAINER

#### **SFR801 Safair Eight Zero One**

The 3-letter ICAO airline designator plus the flight number. Specify the registration in item 18 under remarks: RMK/

#### **SFRFGA Safair Foxtrot Golf Alpha**

The 3-letter ICAO airline designator plus the last three letters of the aircraft registration marks. Specify the full registration in item 18 under remarks: RMK/ - mostly used for training or test flights

Example: an aircraft with call-sign ZS-ABC

6.6.2 Item 8 – Flight Rules and Type of Flight

6.6.2.1 Flight Rules

Insert the Flight Rules as one letter:

**V** **VFR flights**  
and **DRONE** flights conducted at a VFR height

- I** IFR flights  
and **DRONE** flights Beyond Visual Line of Sight (B-VLOS), or those  
RPAS flights climbing above Transition Altitude

**Y** IFR first, followed by one or more changes in flight rules  
The point(s) at which the flight rules change are specified in Item 15

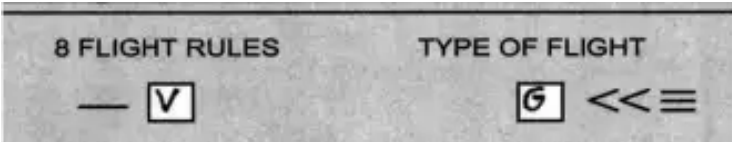
**Z** VFR first, followed by one or more changes in flight rules  
The point(s) at which the flight rules change are specified in Item 15

6.6.2.2 Type of Flight

Insert the type of flight as one letter

- G** **General Aviation**
- N** Non-scheduled air transport (charter flights: passengers or freight)
- S** Scheduled air transport (airlines: passengers or freight)
- M** Military flights
- X** Any other than the above, for example **DRONE** flights, fire fighting,  
medical flights etc. The status of the flight must be specified in item 18  
under STS/ (see the list of “other” flights under STS/ in item 18). If none of  
the options under STS/ fit, specify the nature of the flight under RMK/.

Example: an aircraft on a General Aviation flight adhering to Visual  
Flight Rules



6.6.3 Item 9 – Number, Type, Wake Turbulence Category

6.6.3.1 Number of Aircraft

**Leave blank if the Flight Plan is for one aircraft only.**  
Insert the number of aircraft if more than one.

### 6.6.3.2 Type of Aircraft

Insert the appropriate ICAO aircraft type designator

or insert

ZZZZ if no designator has been assigned (typical for drones), or if there is more than one type in the formation. Specify the type(s) in item 18 under TYP/, e.g.: if the formation comprises 2 Cessna 172 and 3 Cessna 182 specify: TYP/ 2 C172 3 C182. **DRONE PILOTS** insert ZZZZ and specify in item 18: TYP/multi-rotor, or TYP/aeroplane or TYP/helicopter.

### 6.6.3.3 Wake Turbulence Category

Insert

**L** Light, for aircraft with a maximum all-up weight of up to 7,000 kg  
**M** Medium, for aircraft above 7,000 kg but less than 136,000 kg  
**H** Heavy, for aircraft of 136,000 kg or more

Example: 1 aircraft, Piper Arrow, Wake Turbulence category Light.

| 9 NUMBER               | TYPE OF AIRCRAFT                   | WAKE TURBULENCE CAT.             |
|------------------------|------------------------------------|----------------------------------|
| — <input type="text"/> | <input type="text" value="P128R"/> | / <input type="text" value="L"/> |

See paragraph 8.10.1.1 for a further explanation of wake turbulence.

### 6.6.4 Item 10a - Communication & Navigation Equipment

Insert up to 64 alphanumeric characters (in any order) to indicate the Communication and Navigation **capabilities** of the aircraft and crew.

**Capability** in this context means:

- the aircraft must be suitably equipped and the equipment serviceable,
- the crew must be trained and qualified to use the equipment, and
- the equipment must be CAA approved, if necessary, e.g. for RVSM.

This means that an individual aircraft could have different indicators, depending on who the pilot is. If the pilot is a VFR good weather pilot, the equipment might be listed as “VO” (VHF, VOR), while an instrument rated pilot, who is trained and licenced to fly the Instrument Landing System (ILS) might specify “VOL” or “S” (VHF, VOR, ILS).

For clarity the equipment below is split into three lists (voice communication, data link communication and navigation equipment), but

that is arbitrary – the order of the entries is not prescribed. Please remember that candidates for the Restricted Radio Licence should only encounter the items **outside** the grey shaded blocks.

6.6.4.1 General

- N**      **No serviceable COM or NAV capabilities** – if used, this must be the only letter in Item 10a, no other characters are allowed
- S**      **Standard equipment:** this includes **VHF (V), VOR (O) and ILS (L)** or as prescribed by ATC for the route to be flown
- Z**      **other equipment** – this must be specified in item 18 under COM/ or NAV/ or DAT/. Common uses for Z are:
- a non-certified GPS. Then specify in item 18: NAV/GPS
  - a handheld radio for **DRONE PILOTS** . Then specify in item 18: COM/handheld VHF

6.6.4.2 Voice Communication Equipment

|    |                                                                                     |
|----|-------------------------------------------------------------------------------------|
| H  | HF ( <b>H</b> igh <b>F</b> requency) radio voice communications                     |
| M1 | ATC RTF SATCOM (INMARSAT) – voice communications via the INMARSAT satellite network |
| M2 | ATC RTF SATCOM (MTSAT) – voice via the MTSAT satellite network                      |
| M3 | ATC RTF SATCOM (Iridium) – voice via the Iridium satellite network                  |
| U  | UHF voice communications                                                            |

- V**      **VHF voice communications – the normal VHF radio**
- Y**      VHF radio with 8.33 kHz spacing, i.e. a radio that allows the selection of frequencies with three digits behind the decimal point, such as 118.008, 118.017, 118.025 etc.

6.6.4.3 Data Link Communication Equipment

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E1 | FMC WPR ACARS – <b>F</b> light <b>M</b> anagement <b>C</b> omputer - <b>W</b> ay <b>P</b> oint <b>R</b> eporting - <b>A</b> ircraft <b>C</b> ommunications <b>A</b> ddressing <b>R</b> eporting <b>S</b> ystem. Some operators can receive position reports from their aircraft via satellite as part of their flight monitoring regime. These position reports can be forwarded to ATC and used to replace HF voice reports. ACARS is a digital datalink system for transmission of short, relatively simple messages between aircraft and ground stations via radio or satellite |
| E2 | D-FIS ACARS - <b>D</b> atalink - <b>F</b> light <b>I</b> nformation <b>S</b> ervices - <b>A</b> ircraft <b>C</b> ommunications <b>A</b> ddressing <b>R</b> eporting <b>S</b> ystem<br>Flight information services (weather reports, operational data) can be transmitted via ACARS, a digital datalink system for transmission of short text between aircraft and ground stations via radio or satellite                                                                                                                                                                           |

|    |                                                                                                                                                                                                                                                     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E3 | PDC ACARS - <b>P</b> re- <b>D</b> eparture <b>C</b> learance - <b>A</b> ircraft <b>C</b> ommunications <b>A</b> ddressing <b>R</b> eporting <b>S</b> ystem – The pre-departure clearance can be printed or displayed in the cockpit                 |
| J1 | CPDLC ATN VDL Mode 2 - <b>C</b> ontroller <b>P</b> ilot <b>D</b> ata <b>L</b> ink <b>C</b> ommunications - <b>A</b> eronautical <b>T</b> elecommunication <b>N</b> etwork - <b>V</b> HF <b>D</b> igital Mode 2 – Data-link between ATC and aircraft |
| J2 | CPDLC FANS 1/A HFDL - CPDLC <b>F</b> uture <b>A</b> ir <b>N</b> avigation <b>S</b> ervices 1/A <b>H</b> igh <b>F</b> requency <b>D</b> ata <b>L</b> ink – Datalink via HF, mostly used in oceanic airspace and in polar regions                     |
| J3 | CPDLC FANS 1/A VDL Mode A – as above, but via VHF                                                                                                                                                                                                   |
| J4 | CPDLC FANS 1/A VDL Mode 2 – as above, European version                                                                                                                                                                                              |
| J5 | CPDLC FANS 1/A SATCOM (INMARSAT) – as above, communications are via satellite using the INMARSAT satellite network                                                                                                                                  |
| J6 | CPDLC FANS 1/A SATCOM (MTSAT) – as above, using the Japanese MTSAT satellite network                                                                                                                                                                |
| J7 | CPDLC FANS 1/A SATCOM (Iridium) – as above using Iridium satellites                                                                                                                                                                                 |

#### 6.6.4.4 Navigation Equipment

|   |                                                                                                                                                                                                                                                                                                                         |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A | GBAS Landing system - <b>G</b> round <b>B</b> ased <b>A</b> ugmentation <b>S</b> ystem<br>GNSS augmented by a differential ground station, which improves GNSS accuracy for approach, landing, departure, and surface operations within the area of coverage of the ground station                                      |
| B | LPV = APV with SBAS - <b>L</b> ocalizer <b>P</b> erformance with <b>V</b> ertical guidance = <b>A</b> Pproach with <b>V</b> ertical guidance with <b>S</b> pace <b>B</b> ased <b>A</b> ugmentation <b>S</b> ystem. A non-precision approach flown as an ILS look-alike with vertical guidance and possibly lower minima |
| C | LORAN C – <b>L</b> ong <b>R</b> ange <b>N</b> avigation C – LF navigation - obsolete                                                                                                                                                                                                                                    |

|          |                                                                                                                                               |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D</b> | <b>DME - Distance Measuring Equipment</b> – see paragraph 15.5                                                                                |
| <b>F</b> | <b>ADF - Automatic Direction Finding</b> – see paragraph 15.2                                                                                 |
| <b>G</b> | <b>GNSS - Global Navigation Satellite System</b> – see paragraph 15.6<br>If external augmentation is used, specify this in item 18 under NAV/ |

|          |                                         |
|----------|-----------------------------------------|
| <b>I</b> | <b>INS (Inertial Navigation System)</b> |
| <b>K</b> | <b>MLS (Microwave Landing System)</b>   |
| <b>L</b> | <b>ILS (Instrument Landing System)</b>  |

|          |                                                  |
|----------|--------------------------------------------------|
| <b>O</b> | <b>VOR (VHF Omni Range)</b> – see paragraph 15.3 |
|----------|--------------------------------------------------|

|   |                                                                                                                                  |
|---|----------------------------------------------------------------------------------------------------------------------------------|
| R | RNP (Required Navigation Performance) – if used, the performance level must be specified in item 18 following the indicator PBN/ |
| T | Tacan (Navigational aid)                                                                                                         |
| W | RVSM (Reduced Vertical Separation Minima) approved – regardless of whether an RSVM flight level is requested or not              |
| X | MNPS (Minimum Navigation Performance Specification) approved for flights over the North Atlantic Ocean                           |

6.6.5 Item 10b – Surveillance Equipment

Insert a stroke (/) and then any of the following to indicate the SSR (Secondary Surveillance Radar = Transponder) equipment and the ADS (Automatic Dependent Surveillance) capability, up to 20 characters.

For clarity the list below is split into two (Transponder equipment and ADS), but that is arbitrary – the order of the entries is not prescribed.

6.6.5.1 Transponder Equipment

The following transponder entries are inclusive. For example, if an aircraft has an “S” Transponder, it automatically also has “I” and “P” as well, but the latter two are not specified.

- N No Transponder at all
- A Mode A, transmitting just an identifying code (4 digits = 4096 codes)
- C Mode A and C (altitude encoding), transmitting an identifying code as well as the aircraft altitude
- X Mode S, transmitting no extra information
- I Mode S, transmitting the aircraft identification only
- P Mode S, transmitting the pressure altitude only
- S Mode S, transmitting both the aircraft identification and the altitude

|   |                                                                                                                              |
|---|------------------------------------------------------------------------------------------------------------------------------|
| E | Mode S, including aircraft identification, pressure altitude and extended squitter (ADS-B) capability                        |
| H | Mode S, including aircraft identification, pressure altitude and enhanced surveillance capability                            |
| L | Mode S, including aircraft identification, pressure altitude, extended squitter (ADS-B) and enhanced surveillance capability |

6.6.5.2 ADS-B and ADS-C Capabilities

ADS-B (Automatic Dependent Surveillance – Broadcast) is basically a “super” transponder linked to GPS. ADS-B “out” periodically broadcasts

information such as aircraft identification, position, altitude or speed, which is then displayed on the controller's screen. ADS-B "in" means that the aircraft can receive data about other traffic in the vicinity.

ADS-C (C for Contract) uses the same systems on board an aircraft to one or more specific ATSUs for surveillance and/or route conformance monitoring.

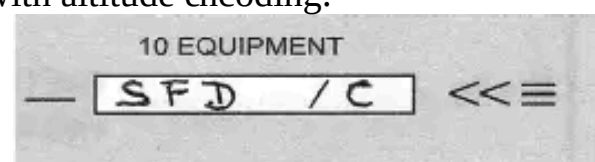
See paragraph 9.3 for more information about ADS-B.

Any indication of ADS-B or ADS-C cannot be present without an indication of a transponder.

|       |                                                                                                                                                                                                                                      |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B1    | ADS-B ( <b>A</b> utomatic <b>D</b> ependent <b>S</b> urveillance <b>B</b> roadcast) with dedicated 1090 MHz ADS-B "out" capability only – the transponder broadcasts aircraft identification, current position, altitude, speed etc. |
| B2    | ADS-B with dedicated 1090 MHz ADS-B "out" and "in" capability – as above, but the aircraft also receives and displays the position of other aircraft in the vicinity.                                                                |
| U1    | ADS-B "out" capability using UAT ( <b>U</b> niversal <b>A</b> ccess <b>T</b> ransceiver) – essentially the same as B1, but intended for use of general aviation aircraft below 18000' (available in USA)                             |
| U2    | ADS-B "out" and "in" capability using UAT – essentially the same as B2, but intended for use of general aviation aircraft below 18000' (USA)                                                                                         |
| V1    | ADS-B "out" capability using VDL Mode 4 – the same as B1 and U1 above, but using the VDL Mode 4 system (now superseded by B1 or U1)                                                                                                  |
| V2    | ADS-B "out" and "in" capability using VDL Mode 4 – the same as B2 and U2 above, but using the VDL Mode 4 system (superseded by B1 or U1)                                                                                             |
| D1    | ADS-C with FANS 1/A ( <b>F</b> uture <b>A</b> ir <b>N</b> avigation <b>S</b> ystem) capabilities                                                                                                                                     |
| G1    | ADS-C with ATN ( <b>A</b> eronautical <b>T</b> elecommunication <b>N</b> etwork) capability.                                                                                                                                         |
| Note: | the indicators above can be combined, i.e. an aircraft can have "B2" and "U2", and would indicate both. But if an aircraft has "B2" it automatically includes "B1", and "B1" does not need to be specified.                          |

**DRONE PILOTS** flying in uncontrolled airspace typically specify Z/N. Z for other COM equipment (and in item 18 "COM/VHF handheld radio") and N for no transponder. Drones flying in controlled airspace normally need at least a Mode C transponder.

Example 1: an aircraft with VHF radio, VOR, ILS (= Standard equipment), ADF, DME and a transponder with altitude encoding.



6.6.6 Item 13 - Departure Aerodrome and Time

6.6.6.1 Departure Aerodrome

Insert the ICAO four letter location indicator of the departure aerodrome (see Appendix I for ICAO location indicators)

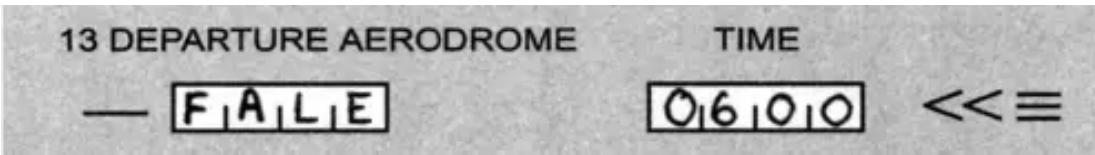
or insert

**ZZZZ** if the departure airfield or place has no location indicator - and specify the name and co-ordinates of the departure airfield/place in item 18 under DEP/. This would typically be the case for **DRONE** flights, for example DEP/Brakpan Mall 2503S 02659E (no seconds required).

6.6.6.2 Time of Departure

Insert the estimated off-block time in UTC (off-block time is the time the aircraft starts taxiing to the holding point). Helicopters or **DRONES** insert the take-off time.

Example: The aircraft will depart Durban at 0600 UTC, which is 8 am South African Time.



6.6.7 Item 15 – Cruise Speed, Level and Route

6.6.7.1 Cruising Speed

Insert the planned cruise True Airspeed (TAS) for the first or the whole cruise portion of the flight as:

- N** followed by four figures, expressing the speed in knots ,  
e.g. N0115 = 115 knots
- M** followed by three figures, expressing the speed as a Mach number, e.g.  
M082 = Mach 0.82

6.6.7.2 Level

Insert the planned cruising level for the first part or the whole flight in one of the following options:

- F** followed by three figures denoting the Flight Level,  
e.g. F085 = Flight Level 85.
- A** followed by three figures denoting the altitude in hundreds of feet,  
e.g. A015 = 1500 feet Altitude.
- VFR** for VFR flights in uncontrolled airspace. The aircraft must still adhere to the semi-circular rule at or above 1500' AGL in uncontrolled airspace.

**DRONE** flights in uncontrolled airspace might not be cruising at a constant altitude. State either maximum altitude or simply enter "VFR" here.

- ii AIC D005/2017 states that **DRONE PILOTS**, if they specified VFR above, must enter their level/height/altitude in Item 18 as, for example "VFR/400FT AGL" – but VFR/ is not a permitted indicator for item 18 – see paragraph 6.6.9 - so rather use RMK/VFR400FTAGL".

### 6.6.7.3 Route

Route planning is done on VFR charts for a VFR flight and on Enroute Charts for IFR flights. Both types of charts depict obstacles (terrain, mountains) and air routes. After the pilot has selected a route and his altitude or flight level the planned route should be accurately described, using the conventions stated below, separated by a space:

**If the aircraft wants to follow a designated ATS or RNAV Route insert the designator of the route,** e.g. **G467**. Also state the point at which you want to join the route (unless the departure airfield is on the route) and at which point you want to exit the route (unless the destination airfield is on the route). If an aircraft wants to fly along a route, but below the vertical limits of the route, it should be treated as point-to-point (see below).

**If the aircraft wants to fly direct from point to point, insert these points separated by "DCT" (direct).** "DCT" need **not** be written if:

- the two points are both on an ATS route, or
- the two points are given as co-ordinates or as bearing and distance.

When using DCT outside of designated ATS routes points should **not be further apart than 200 nm or 30 minutes flying time** and should be towns within 10 nm of track. Over oceans and deserts pilots should define these points by latitude and longitude.

For trans-ocean flights on east-westerly routes between 70°N and 70°S these points should be formed by the intersections of half or whole degrees of latitude with meridians spaced at intervals of 10 ° of longitude.

Flights on east-westerly routes outside of 70°N and 70°S specify these points as the intersections of parallels of latitude with meridians spaced at intervals of 20 ° of longitude not more than one hour's flight time apart.

For flights operating predominantly in a north-south direction these intermediate points must be the intersections of whole degrees of longitude with specified parallels of latitude spaced at 5 degrees.

**DRONE PILOTS** could enter the co-ordinates of the take-off location and enter the radius of the operational area in item 18 under RMK/. Or they could specify an operational “box” by giving the co-ordinates of the four corners in degrees and minutes only (11 digits), no seconds necessary – see paragraph 6.6.7.4 below.

6.6.7.4 Significant points

All of the following can be used as significant points to describe a route:

- ! The code of a **navigational aid** , e.g. KYV (Kimberley VOR),
- ! **The ICAO coded designator of a significant point,** e.g. ETOSA,
- ! Co-ordinates in degrees as 7 characters: degrees latitude (2 figures), **followed** by N or S, and degrees longitude (3 figures), **followed** by E or W, e.g. 26S028E,
- ! Co-ordinates in degrees and minutes as 11 characters: Latitude in degrees and minutes (4 figures), **followed** by N or S, and longitude in degrees and minutes (5 figures), **followed** by E or W, e.g. 2610S02723E,
- !! Describe a point which has a common name, say a little town, by the name of the town and its co-ordinates. That will help ATC understand where you want to go.
- ! Bearing and distance from a navigational aid or significant point, followed by the magnetic bearing from it as 3 figures, followed by the distance as 3 figures in nautical miles. Example: GRV270040 means a point 40 nm from VOR GRV (George) on the 270 degree radial.

Example 1: Along a route: from Durban (FALE) to East London (FAEL) within airway A402. Both departure and destination airports are on the route, so they need not be mentioned.

|                   |         |         |
|-------------------|---------|---------|
| 15 CRUISING SPEED | LEVEL   | ROUTE   |
| N 0 1 2 0         | F 1 6 0 | A 4 0 2 |

Example 2: Point to point: from Upington (FAUP) to Sishen VOR (SSV) to Kuruman (FAKU) to Reivilo (FARI). Upington and Sishen VOR are connected by an ATS route (but you are flying below the route) - no “DCT” is necessary before “SSV”. The

destination is not mentioned in the route, because it will be stated in item 16.

| 15 CRUISING SPEED | LEVEL   | ROUTE            |
|-------------------|---------|------------------|
| N 0 1 2 0         | F 0 7 5 | SSV DCT FAKU DCT |

Example 3: Mixed point-to-point and ATS route: from Beaufort West (FABW) direct to Victoria West VOR (VWV), then along airway A405 to Kimberley (FAKM). DCT is needed before VWV, because Beaufort West and VWV are not connected by a route. Again, the destination will only be mentioned in item 16.

| 15 CRUISING SPEED | LEVEL   | ROUTE        |
|-------------------|---------|--------------|
| N 0 1 2 0         | F 1 5 0 | DCT VWV A405 |

#### 6.6.7.5 Changes in Speed or Flight Level

A change of either speed or Flight Level has to be noted with the significant point at which these occur. If either one changes, both speed and level have to be repeated exactly as above, preceded by the significant point at which the change occurs, followed by an oblique stroke (/).

Example: an aircraft flying from Lanseria (FALA) to Bethlehem (FABM), TAS 180 knots, initially at an altitude of 7000' below the Johannesburg TMA via Heidelberg VOR (HGV), then climbing to FL105 and routing direct to the destination.

| 15 CRUISING SPEED | LEVEL   | ROUTE                     |
|-------------------|---------|---------------------------|
| N 0 1 8 0         | A 0 7 0 | DCT HGV / N0180 F 105 DCT |

#### 6.6.7.6 Change of Flight Rules

A change of Flight Rules ("Y" or "Z" in item 8) must be clearly marked with the point at which this change occurs, followed by a stroke (/), followed by the new speed and Flight Level (if necessary), followed by a space, followed by the word "VFR" if changing to VFR, or "IFR" if changing to IFR.

Example: an aircraft flying from Durban to Hluhluwe, via Richards Bay, TAS 150 kts, IFR at FL130 below Airway B529 to Richards Bay (RBV), and then VFR at FL35 direct to Hluhluwe should be described as:

| ROUTE                          |
|--------------------------------|
| ITMIL RBV / N0150 F035 VFR DCT |

6.6.7.7 Cruise Climb

Long-haul flights are often too heavy initially to climb immediately to their optimum altitude. They will climb higher when sufficient fuel is burned and the aircraft has become lighter – this is called a cruise climb. A cruise climb should be indicated as follows:

C / {point at which the cruise climb is to start} / { speed to be maintained during the climb} {starting level of the cruise climb} { ending level of cruise climb}

Example: cruise climb from a point 30 S 24 E at a speed of 0.82 Mach, from FL290 to FL350.

15 CRUISING SPEED

LEVEL

ROUTE

→

... C/30S 024E / M082F290F350 ...

6.6.8 Item 16 – Destination, Elapsed Time, Alternates

6.6.8.1 Destination Aerodrome

Insert the ICAO four letter location indicator of the destination aerodrome (see Appendix I for ICAO location indicators)

or insert

ZZZZ if no location indicator has been assigned - and specify the name of the destination airfield in Item 18 under DEST/. This would typically be the case for **DRONE** flights.

6.6.8.2 Total Estimated Elapsed Time

Insert the estimated time for the whole flight, from take-off to overhead the destination for VFR flights, or to overhead the navigational beacon serving the destination for IFR flights, in hours and minutes (4 digits).

6.6.8.3 Destination Alternate Aerodromes

Insert the ICAO location indicator(s) of the destination alternate aerodrome(s), i.e. airfield(s) you wish to proceed to if a landing at the destination is not possible,

or insert

ZZZZ if no location indicator has been assigned - and specify the name of the alternate airfield(s) in Item 18 under ALTN/.

Example: a flight to Kimberley of a total duration of 2 hours and 10 minutes, with Bloemfontein as an alternate aerodrome.

|                          |                      |                |
|--------------------------|----------------------|----------------|
| 16 DESTINATION AERODROME | TOTAL EET<br>HR. MIN | ALTN AERODROME |
| — FAKM                   | 0210                 | → FABL         |
| 18 OTHER INFORMATION     |                      |                |

## 6.6.9 Item 18 – Other Information

Insert one or more of the following indicators, **in the order given below**. Insert each indicator only once – if there are two items behind any one indicator, separate the two items with a space.

**0** if there is no other information.

STS/ Reason for special handling by ATS; necessary if X was specified under Type of Flight (item 8). If more than one reason, state “STS/” and string all the special reasons together, separated by a space:

|         |                                                                                                     |
|---------|-----------------------------------------------------------------------------------------------------|
| ALTRV   | flights with an altitude reservation                                                                |
| ATFMX   | <b>flights with approved ATC exemptions</b> – including <b>DRONES</b> flying in controlled airspace |
| FFR     | fire fighting flights                                                                               |
| FLTCK   | flights checking or calibrating navigational aids                                                   |
| HAZMAT  | flights carrying hazardous materials                                                                |
| HEAD    | flights with a head of state on board                                                               |
| HOSP    | for a medical flight declared by medical authorities                                                |
| HUM     | for a flight operating on a humanitarian mission                                                    |
| MARSA   | for flights where the military assumes responsibility for separation                                |
| MEDEVAC | for life critical medical emergency evacuation                                                      |
| NONRVSM | for non-RVSM capable flights operating in RVSM airspace                                             |
| SAR     | for flights engaged in Search and Rescue missions                                                   |
| STATE   | for flights engaged in military, customs or police services                                         |

Example 1: for a routine medical flight in a non RVSM equipped aircraft wanting to operate in RVSM airspace:  
STS/HOSP NONRVSM

Example 2: for a **DRONE** flight in controlled airspace:  
STS/ATFMX RPAS CAMU ref. no. xxx

PBN/ Performance Based Navigation capabilities: necessary if R was specified in item 10a. Up to 8 capabilities can be strung together, without spaces, behind “PBN/”, i.e. up to 16 characters. Use common sense: if an aircraft is capable of B1, this would include B2, B3, B4 etc. and these do not need to be specified.

- A1 RNAV 10 (RNP 10) – the aircraft can remain within 10 nm of an assigned track at least 95% of the time
- B1 RNAV 5 with all permitted sensors
- B2 RNAV 5 with GNSS sensors
- B3 RNAV 5 with DME/DME sensors
- B4 RNAV 5 with VOR/DME sensors
- B5 RNAV 5 with INS or IRS sensors
- B6 RNAV 5 with LORAN C sensors
- C1 RNAV 2 with all permitted sensors
- C2 RNAV 2 with GNSS sensors
- C3 RNAV 2 with DME/DME sensors
- C4 RNAV 2 with DME/DME/IRU (Inertial Reference Unit) sensors
- D1 RNAV 1 with all permitted sensors
- D2 RNAV 1 with GNSS sensors
- D3 RNAV 1 with DME/DME sensors
- D4 RNAV 1 with DME/DME/IRU (Inertial Reference Unit) sensors
- L1 RNP 4 (Required Navigation Performance)
- O1 Basic RNP 1 with all permitted sensors
- O2 Basic RNP 1 with GNSS sensors
- O3 Basic RNP 1 with DME/DME sensors
- O4 Basic RNP 1 with DME/DME/IRU sensors
- S1 RNP APCH – LNAV - for non-precision instrument approaches without vertical guidance
- S2 RNP APCH – LNAV/VNAV - for non-precision instrument approaches with vertical guidance
- T1 RNP AR (Authorisation Required) APCH with RF (Radius to Fix)
- T2 RNP AR (Authorisation Required) APCH without RF

Example: The typical Garmin G430-installation is capable of LNAV only and would be: PBN/B2C2D2O2S1 – see also chapter 15.6

- NAVI/ Navigation equipment, if Z or G was specified in item 10a
- RNAVX RNAV exempted
  - RNAVINOP RNAV inoperable
  - SBAS or other GNSS augmentation
  - other **any other as required. DRONE pilots insert NAV/GPS**

**COM/ Communications equipment, if Z was specified in item 10a**  
 EXM833 exempt from carrying a VHF radio with 8.33 MHz spacing  
 other **any other as required, e.g. for DRONES:**  
**COM/handheld radio**

**DAT/ Data capabilities, if Z was specified in item 10a**  
 CPDLCX exempt from carrying CPDLC equipment  
 other any other as required

**SUR/ Surveillance capabilities not specified in item 10b**

**DEP/ Departure aerodrome as the name and either the co-ordinates as degrees and minutes followed by N/S and degrees and minutes followed by E/W or bearing and distance from a navaid or significant point as stated in paragraph 6.6.7.4 (if ZZZZ was inserted in item 13)**  
 Example: Glen Lyon 2848S02223E

**DEST/ Destination aerodrome in the same way as given above under DEP/ (only necessary if ZZZZ was inserted in item 16).**

**DOF/ Date of flight as yymmdd (only necessary if the flight plan is filed more than 24 hours in advance).**

**REG/ Registration markings of the aircraft (only necessary if different from the radio call sign in item 7).**

**EET/ Estimated elapsed time: accumulated EETs (from departure time) to FIR boundaries. The boundary can be noted as the FIR designator for the FIR to be entered or as the navaid or significant point at the boundary (only if one or more FIR boundaries are crossed).**  
 Example: EET/FAJA0140 FQBE0310 means the boundary to the Johannesburg FIR will be crossed 1 hour 40 minutes after departure, and the Beira FIR will be reached 3 hours and 10 minutes after departure.

**SEL/ Selcal code, if the aircraft is equipped with SELCAL (see paragraph 14.3.3)**

**TYP/ Type(s) of aircraft - only necessary if ZZZZ has been specified in Item 9. For formation flights, precede the type by the number of aircraft of that type**  
 For **DRONES** , insert the type of drone, e.g. TYP/multirotor

**CODE/ The electronic aircraft ID as 6 hexadecimal characters, which has been originally defined for use with SSR Mode S transponders to each individual aircraft**

DLE/ En-route delay or holding as the significant point followed by the delay/holding time as hhmm

OPR/ **ATNS account number or name and contact details of the operator - necessary for the billing of Air Traffic Control charges**  
**For DRONE flights, insert OPR/RPAS flight and the telephone number of the pilot**

ORGN/ The originator's AFTN address

PER/ Aircraft performance category based on the stall speed as one of the letters A,B,C,D,E,H (H is for helicopters) - only necessary if the type and its performance are not known to ATC

ALTN/ **Destination alternate aerodrome if ZZZZ was inserted in item 16**

RALT/ The en-route alternate aerodrome, if necessary

TALT/ The take-off alternate aerodrome, if necessary

RIF/ Route details to revised destination, followed by its ICAO designation.

RMK/ **Search and Rescue requirements** , for example RMK/SARFABL1400.  
Details of **DRONE** flights, for example RMK/Survey pattern with radius 500m around the departure point  
**Any other plain language remark** , for example RMK/STUDENT SOLO NAVEX

RFP/ Q followed by a digit to indicate the sequence of the replacement flight plan being submitted (if necessary)

Example: A flight plan for the 28<sup>th</sup> of October 2022, crossing into the Johannesburg FIR 50 minutes after take-off, aircraft owner A. Flyer, Search and Rescue requested if Johannesburg ATC is not notified by 18:00 hours UTC.

|                  |                |                             |
|------------------|----------------|-----------------------------|
| DOF/ 221028      | EET/ FAJA 0050 | OPR/ A.FLYER (011) 123-4567 |
| RMK/SAR FAOR1800 |                |                             |

## 6.6.10 Item 19 – Endurance, POB etc.

### 6.6.10.1 Endurance

- E/** Insert the total fuel endurance in hours and minutes as a four figure group.  
 This is the time the aircraft can stay airborne with the amount of fuel actually on board at the beginning of this particular flight. **DRONE PILOTS** specify the battery endurance

### 6.6.10.2 Persons On Board

- P/** Insert the total number of people on board, passengers and crew.  
**TBN** Insert TBN (to be notified) if the total number of people is not yet known at the time of filing the flight plan.

### 6.6.10.3 Emergency Radio

- U - UHF** cross out if UHF frequency 243.0 MHz is not available  
**V - VHF** cross out if a VHF radio with frequency 121.5 MHz is not available  
**E - Elba** cross out if an Emergency Locator Transmitter is not available.  
 An Emergency Locator Beacon Aircraft / Emergency Locator Transmitter (ELT) is a radio fitted to the aircraft, which automatically sends out a signal on the international emergency frequencies 121.5 MHz and/or 406.0 MHz when it is subjected to a sudden deceleration (in other words when the aircraft has crashed).

### 6.6.10.4 Survival Equipment

- S** cross out S, P, D, M and J if no survival equipment is carried.  
**P** cross out P if polar survival equipment is not carried.  
**D** cross out D if desert survival equipment is not carried.  
**M** cross out M if maritime survival equipment is not carried.  
**J** cross out J if jungle survival equipment is not carried.

### 6.6.10.5 Jackets

- J** cross out J, L, F, U and V if no life jackets are carried.  
**L** cross out L if life jackets are not equipped with lights.  
**F** cross out F if life jackets are not equipped with fluorescein.  
**U** cross out U if life jackets are not equipped with UHF beacon (243.0 MHz)  
**V** cross out V if life jackets are not equipped with VHF beacon (121.5 MHz).

6.6.10.6 Dinghies

- D cross out indicators D and C if no dinghies are carried.
- Number insert number of dinghies or leave blank if no dinghies are carried.
- Capacity total capacity of all dinghies combined or blank if no dinghies are carried.
- C cross out if dinghies have no covers or if no dinghies are carried.
- Colour insert colour of covers or leave blank if no dinghies are carried or if dinghies have no covers.

6.6.10.7 Aircraft Colour And Markings

- A Insert colour of aircraft and any significant markings,  
e.g. white with blue and red stripes.

6.6.10.8 Remarks (Other Survival Equipment)

- N cross out if there are no remarks; otherwise indicate any other survival equipment carried, e.g. FAK (First Aid Kit), Strips (Signal Strips), Water.

6.6.10.9 Pilot-In-Command

- C Insert initial and surname of pilot-in-command.

6.6.10.10 Filed By

Insert the name of the person or agency filing the flight plan.

Example: An aircraft with 4 hrs endurance, 3 people on board. The aircraft is white with blue and yellow stripes. The pilot is Joe Bloggs. Handheld radio, First Aid Kit, strips and 5 litres water.

|                                                                             |                      |                  |  |                                         |                                     |                                     |
|-----------------------------------------------------------------------------|----------------------|------------------|--|-----------------------------------------|-------------------------------------|-------------------------------------|
| 19                                                                          | ENDURANCE<br>HR. MIN | PERSONS ON BOARD |  | EMERGENCY RADIO                         |                                     |                                     |
| E/                                                                          | 0400                 | → P/ 13          |  | UHF                                     | VHF                                 | ELBA                                |
| SURVIVAL EQUIPMENT                                                          |                      |                  |  |                                         |                                     |                                     |
| POLAR                                                                       |                      |                  |  | DESERT                                  | MARITIME                            | JUNGLE                              |
| → <input checked="" type="checkbox"/> / <input checked="" type="checkbox"/> |                      |                  |  | <input checked="" type="checkbox"/>     | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |
| DINGHIES<br>NUMBER                                                          |                      |                  |  | JACKETS /<br>LIGHT                      |                                     |                                     |
| → <input checked="" type="checkbox"/> /                                     |                      |                  |  | → <input checked="" type="checkbox"/> / |                                     |                                     |
| CAPACITY                                                                    |                      |                  |  | FLUORES                                 |                                     |                                     |
| →                                                                           |                      |                  |  | UHF                                     |                                     |                                     |
| COVER                                                                       |                      |                  |  | VHF                                     |                                     |                                     |
| → <input checked="" type="checkbox"/>                                       |                      |                  |  | →                                       |                                     |                                     |
| COLOUR                                                                      |                      |                  |  |                                         |                                     |                                     |
| →                                                                           |                      |                  |  |                                         |                                     |                                     |
| AIRCRAFT COLOUR AND MARKINGS                                                |                      |                  |  |                                         |                                     |                                     |
| A/ WHITE WITH BLUE AND YELLOW STRIPES                                       |                      |                  |  |                                         |                                     |                                     |
| REMARKS                                                                     |                      |                  |  |                                         |                                     |                                     |
| → N/ FAK / STRIPS / 5L WATER                                                |                      |                  |  |                                         |                                     |                                     |
| PILOT - IN - COMMAND                                                        |                      |                  |  |                                         |                                     |                                     |
| C/ J. BLOGGS                                                                |                      |                  |  |                                         |                                     |                                     |

## 6.7 Sample Flight Plans

The following pages show three sample flight plans.

**Sample Flight Plan 1:** A flight from Cape Town to Port Elizabeth, VFR in a Cessna Cutlass ZS-KSS, equipped with VHF, VOR, ADF, ILS, a non-IFR approved GPS and a 4-digit transponder with altitude encoding. Leaving Cape Town at 8 am local time, cruising speed 130 knots, Flight Level 95, routing via Greyton (GETEN) and George VOR (GRV). Total time en route 2:35. The endurance is 4:30 hours, three people on board, no special equipment for survival is carried, except the legally required First Aid Kit and strips. Search and Rescue is only required at departure, destination, and in controlled airspace. The aircraft is white and blue, the pilot is Joe Soap.

**Sample Flight Plan 2:** A private flight from Grand Central to Tzaneen, VFR in a RV7 ZU-CDF, equipped with VHF, a GPS and a 4-digit transponder with altitude encoding. Leaving Grand Central at 12 noon local time, cruising speed 120 knots, initially at 6000' via the Pinedene VFR route, then at Flight Level 095. Total time en route 1:30. The endurance is 5:00 hours, two people on board, no special equipment for survival is carried, except the legally required First Aid Kit and strips, as well as some water (H<sub>2</sub>O). The pilot wants Search and Rescue to start if he has not arrived at his destination one hour after his intended arrival time. The aircraft is white with red and yellow stripes, the pilot is Andrew Ball.

**Sample Flight Plan 3:** A non-scheduled flight by Icarus Charter from Durban's King Shaka Airport to Lanseria, IFR in a Cessna Mustang (C510) ZS-KLM, equipped with VHF with 8.33 kHz spacing, VOR, ADF, ILS, DME, an IFR approved GPS, a mode S transponder, ADS-B with "out" capability only and an ELT (Emergency Locator Transmitter). The aircraft is approved for RVSM (W), it is also capable of RNP LNAV/VNAV approaches. The aircraft and the pilot are able and certified for LNAV/VNAV approaches. Leaving Durban at 8 am local time on 21 September 2022, cruising speed 320 knots, Flight Level 320, routing via Greytown VOR (GYV), along the RNAV route Z29 to Standerton VOR (STV) and then to Lanseria. Total time en route is 1:15. Alternate airfield is OR Tambo Airport. Endurance 4:50 hours, 1 pilot and 3 passengers on board, First Aid Kit, signal strips and water carried. The pilot wants to be subject to Search and Rescue, if needed, throughout the flight. The aircraft is white and tan, the pilot is John Brown.





**SOUTH AFRICAN**

**CIVIL AVIATION**

**AUTHORITY**

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Form Number: CA 172-04

## AFTN FLIGHT PLAN

Tel: 0860 FLY NOW (0860 359 669) +27 11 928 6518 Fax: +27 (0)11 928 6514 SITA: JNBXTYF

|                                                                       |                            |                                              |                    |
|-----------------------------------------------------------------------|----------------------------|----------------------------------------------|--------------------|
| Priority                                                              | Addressee(s)               |                                              |                    |
| « ≡ F F ▶                                                             |                            |                                              |                    |
| Filing time                                                           | Originator                 |                                              |                    |
| Specific identification of addressee(s) and/or originator             |                            |                                              |                    |
| 3. Message Type                                                       | 7. Aircraft identification | 8. Flight rules                              | Type of flight     |
| « ≡ (FPL                                                              | Z U Q D F                  | - V                                          | G « ≡              |
| 9. Number                                                             | Type of Aircraft           | Wake turbulence cat                          | 10. Equipment      |
| -                                                                     | R V 7                      | L                                            | - V Z / C          |
| 13. Departure aerodrome                                               | Time                       |                                              |                    |
| - F A G C                                                             | 1 0 0 0 « ≡                |                                              |                    |
| 15. Cruising speed                                                    | Level                      | Route                                        |                    |
| - N 0 1 2 0                                                           | A 0 6 0 ▶                  | D C T 2553S02806E PINEDENE ROUTE 2555S02819E |                    |
| D C T 2555S02825E N0120F095 D C T                                     |                            |                                              |                    |
| 16. Destination aerodrome                                             | Total est                  | Alt. Aerodrome                               | 2nd alt. aerodrome |
| - F A T Z                                                             | Hr. Min. 0 1 3 0           | F A P P ▶                                    |                    |
| 18. Other information                                                 |                            |                                              |                    |
| - N A V / G P S D O F / 220906 O P R / A N D R E W B A L L 0111234567 |                            |                                              |                    |
| R M K / S A R F A O R E T A 1 H R                                     |                            |                                              |                    |
| Supplementary information (not to be transmitted in FPL messages)     |                            |                                              |                    |
| 19. Endurance                                                         | Persons on board           | Emergency Radio                              |                    |
| Hr. Min. E / 0 5 0 0                                                  | ▶ P / 2                    | UHF VHF ELT                                  |                    |
| Survival equipment                                                    |                            | Jackets                                      |                    |
| Polar Desert Maritime Jungle                                          |                            | Light Fluores UHF VHF                        |                    |
| ▶ S /                                                                 |                            | ▶ J                                          |                    |
| Dinghies                                                              |                            |                                              |                    |
| Number Capacity Cover Colour                                          |                            |                                              |                    |
| ▶ X /                                                                 |                            |                                              |                    |
| Aircraft Colour and Markings                                          |                            |                                              |                    |
| A / W H I T E R E D Y E L L O W                                       |                            |                                              |                    |
| Remarks                                                               |                            |                                              |                    |
| ▶ N / F A K S T R I P S W A T E R                                     |                            |                                              |                    |
| Pilot in command                                                      |                            | PILOT                                        |                    |
| C / A B A L L                                                         |                            |                                              |                    |
| FILED BY:                                                             |                            |                                              |                    |
| SIGNATURE AND CAPACITY                                                |                            | NAME IN BLOCK LETTERS                        | DATE               |
| PILOT                                                                 |                            | ANDREW BALL                                  | 3 SEP 2022         |

|                                                                                                                               |  |                                                      |  |
|-------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------|--|
| Priority                                                                                                                      |  | Addressee(s)                                         |  |
| 1. Filing time<br>000000                                                                                                      |  | 2. Originator<br>0000000000                          |  |
| 3. Message Type<br>(FPL)                                                                                                      |  |                                                      |  |
| 4. Number<br>00                                                                                                               |  |                                                      |  |
| 5. Departure aerodrome<br>FALE                                                                                                |  | 6. Time<br>0600                                      |  |
| 7. Aircraft identification<br>ZSKLM                                                                                           |  | 8. Flight rules<br>I                                 |  |
| 9. Type of Aircraft<br>C510                                                                                                   |  | 10. Equipment<br>YDFGLORW/SB1                        |  |
| 11. Cruising speed<br>N0320                                                                                                   |  | 12. Level<br>F320                                    |  |
| 13. Route<br>GYV Z29 STV DCT                                                                                                  |  |                                                      |  |
| 14. Destination aerodrome<br>FALE                                                                                             |  |                                                      |  |
| 15. Total est<br>Hr. Min.<br>0115                                                                                             |  | 16. Altn. Aerodrome<br>FAOR                          |  |
| 17. 2nd altn. aerodrome                                                                                                       |  |                                                      |  |
| 18. Other information<br>PBN/B2C2D2O2S2 DOF/220921 OPR/IKARUS CHARTER POBOX 123<br>RANDFONTEIN 0123 TEL 0834567890 RMK/SARNML |  |                                                      |  |
| 19. Supplementary information (not to be transmitted in FPL messages)                                                         |  |                                                      |  |
| 20. Endurance<br>Hr. Min.<br>E / 0450                                                                                         |  | 21. Persons on board<br>P / 4                        |  |
| 22. Survival equipment<br>Polar [X] Desert [X] Maritime [X] Jungle [X]                                                        |  | 23. Emergency Radio<br>UHF [X] VHF [V] ELT [E]       |  |
| 24. Dinghies<br>Number [X] Capacity [ ] Cover [X] Colour [ ]                                                                  |  | 25. Jackets<br>Light [L] Fluores [F] UHF [X] VHF [V] |  |
| 26. Aircraft Colour and Markings<br>A / WHITE AND TAN                                                                         |  |                                                      |  |
| 27. Remarks<br>N / FAK STRIPS H2O                                                                                             |  |                                                      |  |
| 28. Pilot in command<br>C / J BROWN                                                                                           |  |                                                      |  |
| FILED BY:                                                                                                                     |  |                                                      |  |
| <br>DISPATCHER                             |  | M MERTENS                                            |  |
| SIGNATURE AND CAPACITY                                                                                                        |  | NAME IN BLOCK LETTERS                                |  |
|                                                                                                                               |  | 18 SEP 2022                                          |  |
|                                                                                                                               |  | DATE                                                 |  |

# 7

# Terms, Codes and Signals

## 7.1 Units of Measurement

AIP GEN 2.1

South Africa uses the following units of measurement in aviation:

|   |                                                   |                          |         |
|---|---------------------------------------------------|--------------------------|---------|
| ! | Long distances, as in navigation                  | nautical miles           | nm      |
| ! | Short distances, as in runway length              | metres                   | m       |
| ! | Altitudes, elevations and heights                 | feet                     | ' or ft |
| ! | Horizontal speed of aircraft and wind             | knots                    | kts     |
| ! | Vertical speed                                    | feet per minute          | fpm     |
| ! | Surface wind direction (for take-off and landing) | degrees magnetic         | °M      |
| ! | All other wind directions                         | degrees true             | °T      |
| ! | Visibility, including runway visual range         | metres                   | m       |
| ! | Atmospheric pressure, altimeter setting           | hectopascal              | hPa     |
| ! | Temperature                                       | degrees Celsius          | °C      |
| ! | Weight                                            | kilograms or metric tons | kg or t |

## 7.2 Phonetic Alphabet

AIC D005/2018 , ICAO Doc 9432

When transmitting over the aircraft radio, all groups of letters, aircraft call signs, location indicators etc. have to be spelt out using the phonetic alphabet. Only letter groups which by everyday use have become unmistakable, e.g. ILS, ETA, CTR etc. may be spoken in the normal way.

Pilots must learn the phonetic alphabet by heart. The **bold** syllable is emphasised.

|              |   |                   |             |   |                           |
|--------------|---|-------------------|-------------|---|---------------------------|
| A - Alpha    | - | <b>al</b> fah     | I - India   | - | in dee ah                 |
| B - Bravo    | - | <b>bra</b> voh    | J - Juliet  | - | <b>jew</b> lee <b>ett</b> |
| C - Charlie  | - | <b>char</b> lee   | K - Kilo    | - | key loh                   |
| D - Delta    | - | <b>dell</b> tah   | L - Lima    | - | <b>lee</b> mah            |
| E - Echo     | - | <b>eck</b> oh     | R - Romeo   | - | row me oh                 |
| F - Foxtrot  | - | <b>foks</b> trot  | S - Sierra  | - | see air rah               |
| G - Golf     | - | <b>golf</b>       | T - Tango   | - | <b>tang</b> go            |
| H - Hotel    | - | hoh <b>tell</b>   | U - Uniform | - | <b>you</b> nee form       |
| M - Mike     | - | <b>mike</b>       | V - Victor  | - | <b>vik</b> tah            |
| N - November | - | no <b>vem</b> ber | W - Whiskey | - | <b>wiss</b> key           |
| O - Oscar    | - | <b>oss</b> cah    | X - X-ray   | - | <b>ecks</b> ray           |
| P - Papa     | - | pah <b>pah</b>    | Y - Yankee  | - | <b>yank</b> key           |
| Q - Quebec   | - | keh <b>beck</b>   | Z - Zulu    | - | <b>zoo</b> loo            |

## 7.3 Numerals

AIC D005/2018 , ICAO Doc 9432

All numbers are transmitted by pronouncing **each digit separately** , with some exceptions as stated below.

|           |   |               |                 |   |                    |
|-----------|---|---------------|-----------------|---|--------------------|
| 0 - zero  | - | <b>ze</b> ro  | 7 - seven       | - | <b>sev</b> en      |
| 1 - one   | - | <b>wun</b>    | 8 - eight       | - | <b>ait</b>         |
| 2 - two   | - | <b>too</b>    | 9 - nine        | - | <b>nin</b> er      |
| 3 - three | - | <b>tree</b>   | 100 - hundred   | - | <b>hun</b> dred    |
| 4 - four  | - | <b>fow</b> er | 1000 - thousand | - | <b>tou</b> sand    |
| 5 - five  | - | <b>fife</b>   | point - decimal | - | <b>deh</b> see mil |
| 6 - six   | - | <b>six</b>    |                 |   |                    |

|           |       |   |                             |
|-----------|-------|---|-----------------------------|
| Examples: | 118.9 | - | wun wun ait dehseemil niner |
|           | 06    | - | runway zero six             |
|           | 200 ° | - | heading too zero zero       |
|           | 1012  | - | QNH wun zero wun too        |

### 7.3.1 Hundreds and Thousands

**Transponder codes** and **altimeter setting** containing **only** whole thousands are transmitted by using the word “thousand”.

|           |      |   |                           |
|-----------|------|---|---------------------------|
| Examples: | 2000 | - | squawk too tousand        |
|           | 1000 | - | QNH wun tousand           |
| but:      | 2500 | - | squawk too fife zero zero |
|           | 900  | - | QFE niner zero zero       |

**Altitude** and **Flight Levels**, **cloud height**, **visibility** and **Runway Visual Range** (RVR) containing **only** whole hundreds **or** whole thousands are transmitted by using the words “hundred” and/or “thousand”.

Examples: 2 500' - altitude too tousand fife hundred feet  
 FL200 - Flight Level too hundred  
 15 000' - cloud height wun fife tousand feet  
 but: 850 m - visibility ait fife zero metres  
 FL120 - Flight Level wun too zero

### 7.3.2 Frequency pronunciation

ICAO Doc 9432

In countries which still use 25 kHz frequency spacing (South Africa, for example), VHF radio frequencies are spoken with as many decimals as specified. For example, 118.1 is pronounced “wun wun ait dahseemil wun” and 118.65 is spoken “wun wun ait dehseemil six fife”.

However, some countries use 8.33 kHz spacing (most of Europe) to allow for more frequencies within the allocated VHF-frequency band. In those countries VHF radio frequencies are pronounced with either only **one** decimal (if the second and third decimals are zeroes) or with **three** decimals, but never with only two decimals. For example: 118.1 is spoken as “wun wun ait deh-see-mil wun” but 118.65 is spoken as “wun wun ait deh-see-mil six fife zero” (note that Zimbabwe also enforces this rule).

### 7.3.3 The Clock Code

When providing information regarding **relative bearing using the 12-hour-clock**, the double-digit numbers are pronounced as TEN, ELEVEN, and TWELVE.

Example: Traffic is a Cherokee, three miles at eleven o'clock .

## 7.4 Morse Code

Morse code was originally used for Radio Telegraphy. Today it has been replaced by voice communications or data transmissions. But navigational aids such as NDB, VOR and ILS still send out their identifying letters in Morse code. Therefore, every pilot must be able to identify Morse code in order to decide whether or not his navigational radios are tuned to and receiving the correct station. Pilots are advised to carry a copy of the code below in their flight bag or on their clipboard.

|         |         |          |         |         |         |
|---------|---------|----------|---------|---------|---------|
| Alpha   | ! -     | Juliet   | ! - - - | Sierra  | ! ! !   |
| Bravo   | - ! ! ! | Kilo     | - ! -   | Tango   | -       |
| Charlie | - ! - ! | Lima     | ! - ! ! | Uniform | ! ! -   |
| Delta   | - ! !   | Mike     | - -     | Victor  | ! ! ! - |
| Echo    | !       | November | - !     | Whiskey | ! - -   |
| Foxtrot | ! ! - ! | Oscar    | - - -   | X-ray   | - ! ! - |
| Golf    | - - !   | Papa     | ! - - ! | Yankee  | - ! - - |
| Hotel   | ! ! ! ! | Quebec   | - - ! - | Zulu    | - - ! ! |
| India   | ! !     | Romeo    | ! - !   |         |         |

## 7.5 Time

AIP GEN 2.1, ICAO Doc 9432

To standardise time internationally and to avoid confusion between flights originating in different time zones, aviation uses only UNIVERSAL TIME CO-ORDINATED (UTC = Greenwich Mean Time GMT) and the 24-hour clock. Midnight is designated to be 0000 at the start of the day, and 2400 at the end of the day.

The earth turns 360° in 24 hours and therefore 15° per hour. It has been divided into 24 one-hour time zones, each comprising 15° of longitude, but adjusted for local geographical boundaries, and in some countries for Daylight Saving Time (DST).

Time zone Z (Zulu) is the same as UTC and covers Great Britain. To the east follows zone A (Alpha) or UTC+1, comprising most of Europe. Then it's zone B (Bravo) as UTC+2, comprising all of South Africa, then C (Charlie) as UTC+3 etc. There is no time zone J (Juliet), so finally time zone M (Mike) as UTC+12 covers some of the Pacific Ocean Islands. Starting again at zone Z = UTC = GMT and going west, zone N (November) is UTC-1, zone O (Oscar) is UTC-2 and so on until time zone Y (Yankee), comprising American Samoa, is equal to UTC-11.

Bravo time or South African Standard Time (SAST) is UTC+2 and always two hours later than Zulu time or UTC.

Examples:

|               |   |            |   |                        |
|---------------|---|------------|---|------------------------|
| 10.30 am SAST | = | 1030 SAST  | = | 0830 UTC               |
| 9.20 pm Bravo | = | 2120 Bravo | = | 1920 UTC               |
| noon SAST     | = | 1200 SAST  | = | 1000 UTC               |
| 1.25 am SAST  | = | 0125 SAST  | = | 2325 UTC (minus 1 day) |

Times are always spoken as individual digits, e.g. 1920 UTC is spoken "one niner two zero" over the radio and 2000 UTC is "two zero zero zero". If there is no confusion as to which hour is meant, only the minutes need to be spoken, i.e. 0957 becomes "five seven" as in "estimate PJV at five seven".

## 7.6 Q-Codes

ICAO Doc 8400, AIP GEN 2.2

Q-codes were invented in 1912 as short-cuts to communications in the days of Radio Telegraphy in Morse code. They are three-letter groups, all starting with "Q", representing often used phrases. Today, with good voice or data communication, Q-codes have become largely redundant and were abandoned by ICAO in 1999. A few Q-codes have survived, however, and are now listed as "abbreviations".

These are actually used, and should be known by pilots:

|            |                                                                                                                                                                                                                                                                                               |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>QDM</b> | Magnetic heading to a station in no wind                                                                                                                                                                                                                                                      |
| <b>QDR</b> | Magnetic bearing from a station                                                                                                                                                                                                                                                               |
| <b>QFE</b> | Atmospheric pressure at aerodrome elevation<br>With QFE set in the subscale of the altimeter, it will read zero when on the ground, or HEIGHT in flight. See also chapter 4.                                                                                                                  |
| <b>QNE</b> | Indicated height on landing with 1013.2 hPa set (SA only, non-ICAO)                                                                                                                                                                                                                           |
| <b>QNH</b> | Altimeter sub-scale setting to obtain elevation when on the ground<br>This is the present atmospheric pressure at a place corrected for sea level<br>With QNH set in the subscale of the altimeter it reads airfield elevation when on the ground, or ALTITUDE in flight. See also chapter 4. |
| <b>QTE</b> | True bearing from a station                                                                                                                                                                                                                                                                   |
| <b>QTF</b> | Position of an aircraft according to bearings taken by a ground station                                                                                                                                                                                                                       |

These are still defined by ICAO, but not normally used by pilots:

|            |                                                |
|------------|------------------------------------------------|
| <b>QDL</b> | A series of bearings                           |
| <b>QFU</b> | Magnetic orientation of runway                 |
| <b>QGE</b> | Distance to a station                          |
| <b>QJH</b> | Run a test tape or sentence (for use by ATCs)  |
| <b>QSP</b> | To relay free of charge (for use by ATCs)      |
| <b>QTA</b> | Cancel telegram number   !   (for use by ATCs) |
| <b>QUJ</b> | True track to a station                        |

„ Long abandoned by ICAO and South Africa, but sometimes still used by older pilots are:  
 QSY - change frequency to   !   and  
 QBI – IFR is in force at an airfield   !   (due to weather or for procedural reasons)

## 7.7 Call Signs

AIP ENR 1.8 (7), AIC D005/2018 , ICAO Annex 10 Vol II, ICAO Doc 9432

### 7.7.1 Ground Stations

Aeronautical ground stations can have the following call signs:

**DELIVERY**       reads the after departure clearance to departing IFR traffic. And to VFR pilots at Cape Town International and OR Tambo only.

**GROUND** provides surface movement control to aircraft on taxiways, but not on the apron. They take their name from the aerodrome, e.g. LANSERIA GROUND.

**TOWER** provides aerodrome control service to aircraft in a CTR or ATZ and on the runway. They take their name from the aerodrome, e.g. RAND TOWER, or DAVID STUURMAN TOWER.

**APPROACH** provides approach control service to aircraft in a TMA. They take their name from the town, e.g. CAPE TOWN APPROACH.

**RADAR** Radar control general.

**DEPARTURE** Radar control for departing aircraft.

**ARRIVAL / DIRECTOR** accepts arriving aircraft from approach control and vectors them onto the ILS, before handing them over to the aerodrome controller.

**CONTROL / AREA** provides area control service to aircraft in a CTA, airways or other parts of the FIR. They take their name from the Control centre, i.e. in the Johannesburg FIR the call-sign of area control is JOHANNESBURG AREA or simply JOHANNESBURG.

**APRON** allocates parking bays. They take their name from the aerodrome, e.g. KING PHALO APRON in East London.

**RADIO** provides aerodrome flight information service in ATZs Class G. They take their name from the airfield, e.g. PORT ALFRED RADIO.

**INFORMATION** provides flight information service in uncontrolled airspace. They take their name from the flight information centre or FIR sector, e.g. JOHANNESBURG INFORMATION or CAPE TOWN INFORMATION.

Services may be combined, either during quiet periods, or at airfields which are not busy enough to warrant having all positions filled all the time. For example, BLOEMFONTEIN TOWER gives aircraft their after departure clearances and provides surface movement control and aerodrome control. And CAPE TOWN AREA provides both control and information service at night.

## 7.7.2 Aircraft Call Signs

Aircraft stations may use one of the call signs as described in paragraph 6.6.1 of this book in the chapter about Flight Plans.

The most commonly used call signs are:

! **The registration marking of the aircraft.**

For example ZS-BCV, pronounced Zulu Sierra Bravo Charlie Victor. According to ICAO and AIC D005/2018 this may be abbreviated to the first character of the nationality marking plus at least the last two characters, e.g. ZS-BCV becomes ZCV, and N6593W becomes N93W. In practice, however, in South Africa and some other countries, call signs are abbreviated to the last three characters, omitting the country registration letters, e.g. ZS-BCV becomes BCV. - With ZS-, ZT- and ZU-registered aircraft in South Africa **it is important for pilots to use the full call sign on first contact with any one controller** . Only once the controller uses an abbreviated call sign may the pilot also use this abbreviated call sign until being transferred to a different station.

! **The name of the operating agency plus the flight number** .

Examples are SAFAIR 234, SPEEDBIRD 456 (British Airways), MERCY 3, or SKYAD 1. No abbreviations possible.

! **A special call sign agreed to by ATC or a code for military aircraft.**

Examples are RAVEN, CEMAIR TRAINER, or LIGHTNING FORMATION.

If an air traffic controller realises at any one time that confusion could exist between aircraft with similar sounding call signs, he could temporarily assign different call-signs to the two confusing aircraft.

### 7.7.3 Call Sign Appendices

AIP ENR 1.8 (7)

To alert ATC to special handling requirements, aircraft should use the following prefixes or suffixes when establishing first contact with the ATC. The prefix or suffix will not be repeated by ATC and does not have to be mentioned by the pilot on subsequent calls to the same controller.

Suffix **Heavy** for heavy aircraft on first contact with Clearance Delivery, Ground, Tower and Approach services. Example: “Tower, Speedbird 123 Heavy, final 20”.

Suffix **Super** for super heavy aircraft (Airbus 380-800) on first contact with Clearance Delivery, Ground, Tower and Approach services. Example: “Ground, Singapore 123 Super”.

Suffix **Hospital** for all medical flights, which need priority, but can accept delays of up to 10 minutes. The Flight Plan should state STS/HOSP in item 18 . Example: “Approach, Medair 123 Hospital, passing 2000 feet”.

Suffix **Medevac** for all medical emergency flights, which need absolute priority. The Flight Plan should state STS/MEDEVAC in item 18

Prefix **Student** for all **ab-initio** student pilot training flights **with** an instructor on board. Not to be used for advanced training, conversions etc.

Prefix **Solo Student** for all ab-initio solo student pilot training flights **without** an instructor on board. Example: "Ground, solo student ZS-ABC request taxi".

Prefix **Helicopter** for all helicopters, especially those whose call-sign does not start with H or R.

Prefix **Glider** for all gliders, especially those whose call-sign does not start with G. Example: "Information, Glider ZS-MNO, good morning".

Prefix **Motor Glider, Balloon** or **Airship** for motor gliders, balloons or airships.

Suffix **Mayday** for an aircraft in distress when calling a subsequent station (after the original Mayday call). Example: "Tower, Airline 123 Mayday, final 01".

Suffix **Pan** for an aircraft in an urgency situation when calling a subsequent station.

## 7.8 Test Procedures

AIC D005/2018 , ICAO Doc 9432

When there is any doubt as to the proper operation of radio equipment, the following test procedures must be used. The aircraft wanting to test its equipment shall say:

Identification of station called  
Aircraft call sign  
"Radio check"  
The frequency.

The reply to this test procedure is:

Aircraft call sign  
Identification of station calling  
"Reading you" plus the number according to the readability scale below.

The following readability scale shall be used:

- 1 unreadable
- 2 readable now and then
- 3 readable, but with difficulty
- 4 readable
- 5 perfectly readable

Example: Aircraft: Cape Town Ground, Zulu Sierra Mike Golf Golf, Radio check  
on one two one decimal niner  
Reply: Mike Golf Golf, Cape Town Ground, reading you five.

## 7.9 Categories / Priority of Messages

AIC D005/2018

If Air Traffic Control is busy, messages will be dealt with in the following order of priority, i.e. a message with a higher priority will be dealt with first:

1. **Distress** messages (suffix MAYDAY), see paragraph 11.1
2. **Urgency** messages (suffix PAN-PAN), see paragraph 11.1
3. Communications relating to **Direction Finding** (DF), see paragraph 15.1
4. **Flight Safety messages**, which are the “normal” radio messages between controllers and aircraft, i.e. position reports, clearances, instructions etc.
5. **Meteorological** messages
6. **Flight Regularity messages**, which are communications between an aircraft operating agency (e.g. a charter company) to one of their aircraft or vice versa about changes in flight schedules, spare parts required etc.

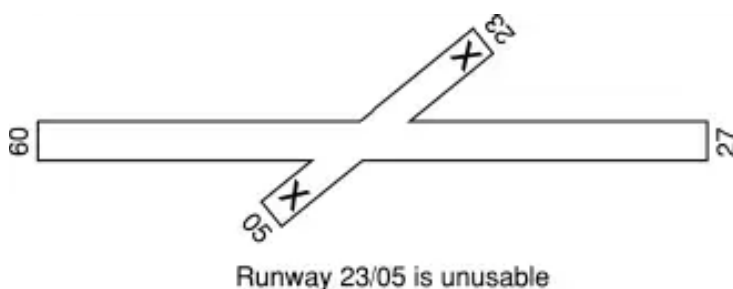
## 7.10 Signals to Air Traffic

SACATS 91.06.13, ICAO Annex 2 Rules of the Air (Appendix 1)

The following signals may be used for communication between ground and aircraft in flight.

### 7.10.1 Visual Ground Signals

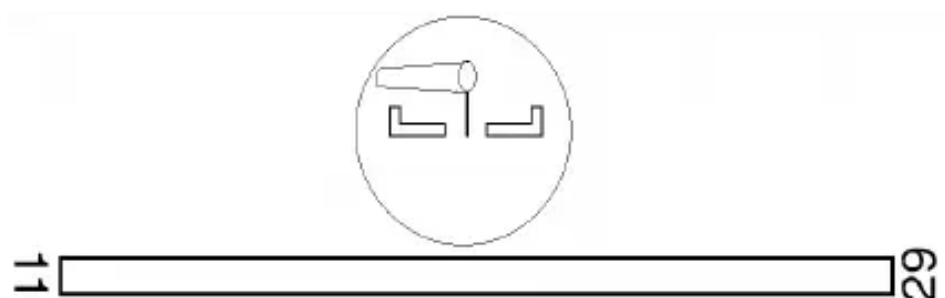
SACATS 91.06.13



**White or yellow crosses on runways or taxiways** denote unsafe areas, for example waterlogged runways or areas undermined by moles. The crosses will be displayed at both ends of the unserviceable area; and aircraft may not land, take-off or taxi in the area between

the crosses. By night, the unserviceable areas are marked with red lights. In addition, any runway or taxiway lights between the delimiting red lights will be extinguished.

**L-shaped markings** in the windsock circle (signals area) denote the circuit direction. In the picture all circuits are to be flown to the north, i.e. right hand circuits on runway 29 and left hand circuits on runway 11.





Left: this sign displayed in the windsock circle (signals area) means: landing is prohibited

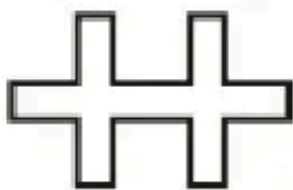


Right: observe special precaution when landing, probably due to the bad state of the airfield



A **white dumbbell** means: aircraft may use only runways and taxiways (not the grass, for example).

A **white dumbbell with black stripes** means:  
Aircraft must take-off and land on runways only, but can taxi anywhere (also on the grass, for example)



A **double white cross** means:  
Glider operations are in progress.

A **large black "A"** means:  
Agricultural operations (crop spraying) are in progress

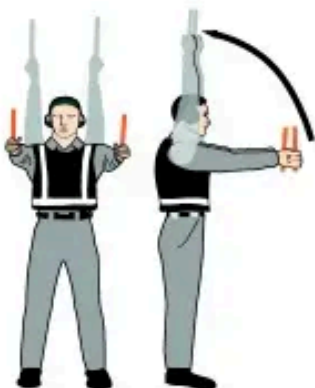


A **black "C" on a yellow background** indicates where a pilot has to report to after landing, usually to pay landing fees.

## 7.10.2 Marshalling Signals

SACATS 91.06.13

At larger airports an arriving aircraft will often be parked by Apron Services through signals from a marshaller. Pilots must understand and follow these signals. The most important ones for small aircraft are given below. The full list is in SACATS 91.06.13.



Arms above the head in a vertical position means:  
**Park in this bay**

Arms extended, bent at the elbows and moved up and down from chest height to head means:  
**Move straight ahead**





One arm extended sideways, the other arm bent at the elbow and moved up and down as above means:

**Turn towards the extended arm**

In this picture the pilot should turn to his left

Arms fully extended sideways and slowly moved to above the head until they cross means:

**Stop**



Open hand raised just above shoulder height, closing hand into a fist means:

**Set brakes**

Closed fist raised just above shoulder height, opening hand means:

**Release brakes**



Arms or wands fully extended above head, moving hands inward with a jabbing motion until hands touch means:

**Chocks are inserted**

Arms or wands fully extended above head, moving hands outward in a jabbing motion means:

**Chocks are removed**



Right arm up, conducting a circular motion, raised left arm points to the engine to be started means:

**Start engine**

Left arm up, right arm drawn from left to right shoulder across the throat in a slicing motion means:

**Cut engine(s)**



## 7.10.3 Lights

### 7.10.3.1 Use of aircraft lights

AIP ENR 1.8(6), CAR 91.06.10

To reduce incidences of runway incursions and increase situational awareness of an aircraft's intended movement, the following aircraft lights should be used by the pilot:

- Prior to starting the engine(s): Rotating Beacon ON
- Prior to push-back (or taxi): Navigation lights ON
- Commencement of taxi: Taxi lights ON
- Entering runway: Strobe lights ON
- When cleared for take-off: Landing light ON
- When crossing a runway: All of the above ON

The lights should be switched off again after landing in a similar manner, i.e.:

- Leaving the runway: Landing light and strobes OFF
- Once parked: Taxi lights and navigation lights OFF
- After engine shut-down: Rotating beacon OFF

### 7.10.3.2 Light Signals to Aerodrome Traffic

CATS 91.06.13 (4)

In case of radio failure the following light signals may be used to direct aircraft on or around an airfield, in flight or on the ground. These light signals are still valid and part of the Regulations, and must be learned. However, there is probably not a single civilian airport in South Africa still equipped with flare guns, and pilots must rather be acquainted with the radio failure procedures in chapter 10.

| Light signal given by aerodrome control to: | Aircraft in flight                               | Aircraft on the ground            |
|---------------------------------------------|--------------------------------------------------|-----------------------------------|
| <b>Steady green</b>                         | Cleared to land                                  | Cleared for take-off              |
| <b>Steady red</b>                           | Give way to other aircraft and continue circling | Stop                              |
| <b>Series of green flashes</b>              | Return for landing                               | Cleared to taxi                   |
| <b>Series of red flashes</b>                | Aerodrome unsafe, do not land                    | Taxi clear of landing area in use |
| <b>Series of white flashes</b>              | Land at this airfield, proceed to the apron      | Return to the starting area       |
| <b>Red pyrotechnic</b>                      | Do not land for the time being                   |                                   |

Aircraft in the air acknowledge the above light signals by rocking their wings during the day or by flashing the landing or navigation lights twice at night.

Aircraft on the ground acknowledge the signals by moving the ailerons or rudder during the day and by flashing the landing or navigation lights twice at night.

**7.10.3.3 Visual Warning Lights**  
CATS 91.06.13 (3)

A series of projectiles, fired from the ground at 10 second intervals, each showing, on bursting, red and green lights or stars means that an aircraft is about to enter a prohibited, restricted or danger area. The aircraft should leave the area in question.

**7.10.4 Interception Signals**  
AIP ENR 1.12, ICAO Annex 2 (Appendix 1)

It may sometimes be necessary to investigate the identity of an aircraft (e.g. one which may have strayed into foreign airspace). If this happens, the intercepted aircraft should squawk 7700 and attempt to establish radio contact on the international distress frequency 121.5 MHz or any other suitable frequency. The following phraseology should be used:

| <u>By intercepting aircraft:</u>    | <u>By intercepted aircraft:</u>      |
|-------------------------------------|--------------------------------------|
| CALL SIGN – what is your call sign? | CALL SIGN – my call sign is      ! . |
| FOLLOW – follow me                  | WILCO – understood, will comply      |
| DESCEND – descend for landing       | CANNOT – unable to comply            |
| YOU LAND – land at this aerodrome   | REPEAT – repeat your instruction     |
| PROCEED – you may proceed           | AM LOST – position unknown           |
|                                     | MAYDAY – I am in distress            |
|                                     | HIJACK – I have been hijacked        |
|                                     | LAND – I request to land at      !   |
|                                     | DESCEND – I require descent          |

If verbal communication is not possible for any reason the visual signals set out below will be used. If the intercepted aircraft is in distress it should use the distress signals mentioned in paragraph 7.10.5. Although interception is very rare, pilots should carry a copy of the signals below to be prepared for any eventuality (the signals are also part of the R/T certificate syllabus).

**Signals by intercepting aircraft and responses by intercepted aircraft**

**First Series:**

**Signal by interceptor:** Rocking wings while in front and to the left of the intercepted aircraft (and flashing navigation and landing lights at night), followed by a slow level turn onto a specific course.

**Meaning:** Follow me away from a prohibited area and to a landing terrain.

**Response by intercepted aircraft:** Rock the wings (show steady landing lights at night) and follow the interceptor.

**Second Series:**

**Signal by interceptor:** An abrupt breakaway manoeuvre from the intercepted aircraft consisting of a climbing turn of 90 ° or more without crossing the line of flight of the intercepted aircraft.

**Meaning:** You may proceed.

**Response by intercepted aircraft:** Rock the wings, show steady landing lights and continue your flight.

**Third Series:**

**Signal by interceptor:** Circling the landing area, lowering landing gear (at night also showing steady landing lights) and overflying the runway in use.

**Meaning:** Land at this aerodrome.

**Response by intercepted aircraft:** Lower the landing gear (at night also showing steady landing lights), follow the intercepting aircraft and if, after overflying the runway in use or helicopter landing area, landing is considered safe, proceed to land.

**Signals by intercepted aircraft and responses by interceptor**

**Fourth Series:**

**Signal by intercepted aircraft:** Rocking the wings and/or raising the landing gear (at night also flashing the landing lights) while passing over the landing area between 1000' and 2000' AGL and continuing circling.

**Meaning:** aerodrome you have designated is inadequate.

**Response by interceptor:** if it is desired that the intercepted aircraft follow the intercepting aircraft to an alternate aerodrome, the intercepting aircraft raises its landing gear and uses the Series 1 signals. If it is decided to release the intercepted aircraft, the intercepting aircraft uses the Series 2 signals.

**Fifth Series:**

**Signal by intercepted aircraft:** regular switching on and off of all available lights but in such a manner as to be distinct from flashing lights.

**Meaning:** cannot comply.

**Response by interceptor:** Use Second Series signals.

**Sixth Series:**

**Signal by intercepted aircraft:** irregular flashing of all available lights.

**Meaning:** In distress.

**Response by interceptor:** Use Second Series signals.

## 7.10.5 Distress and Urgency Signals

SACATS 91.06.13, AIP ENR 1.12

### Distress

The following signals may be used when an aircraft is in grave and imminent danger and in need of immediate assistance:

- ! SOS in Morse code ( · · · / — — — / · · · )
- ! the word MAYDAY (see also paragraph 11.1)
- ! rockets or shells throwing red lights, fired one at a time at short intervals
- ! a parachute flare showing a red light

### Urgency

The following signals are used when an aircraft is in difficulties and needs to land, but is not in need of immediate assistance:

- ! XXX in Morse code ( — · · — / — · · — / — · · — )
- ! the word PAN-PAN (see also paragraph 11.2)
- ! repeatedly switching the landing lights on and off
- ! repeated switching the navigation lights on and off in a manner that is distinct from normal flashing navigation lights

## 7.11 Airport Markings and Signs

ICAO Annex 14

### 7.11.1 Taxiway Signs

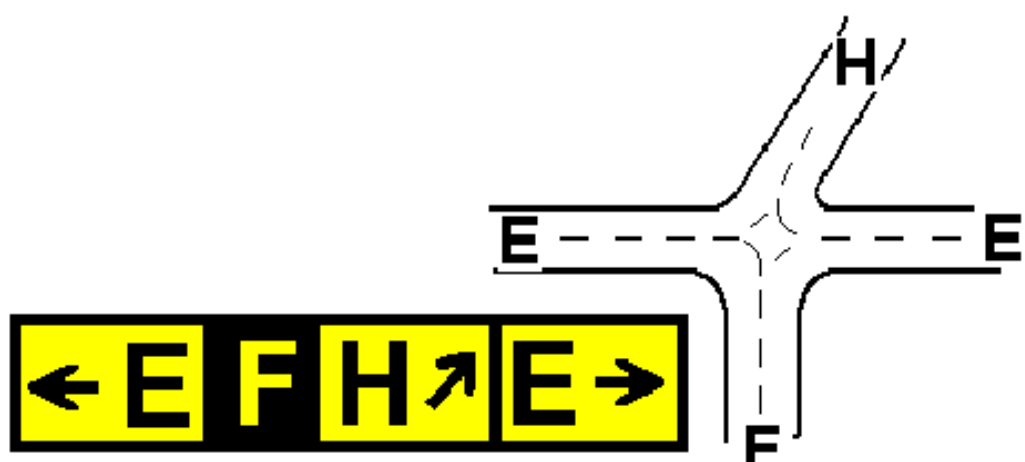
Taxiways at large airports are named with letters of the alphabet: A, B, C etc. All taxiway signage is **black and yellow**. Signs are positioned on the left side of a taxiway, approximately 40 metres before a taxiway intersection.

**Yellow letters on a black background** show on which taxiway an aircraft is.

**Black letters on a yellow background**, with an arrow next to them, show which taxiway is coming up ahead and in which direction that taxiway is going.

Example:

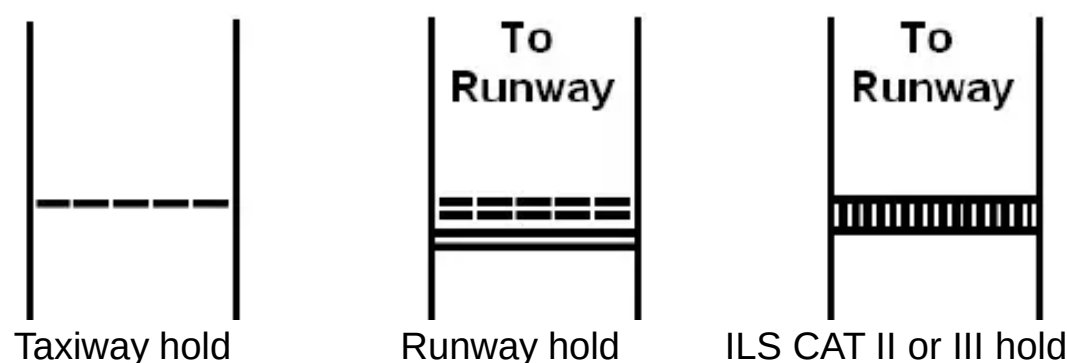
Aircraft is on Foxtrot.  
Echo is ahead left-right.  
Hotel is ahead,  
obliquely to the right.



### 7.11.2 Holding Point Markings

The intersections of taxiways with other taxiways or aprons, and at other points where aircraft might need to hold are marked with dashed holding lines painted in white across the taxiway.

Where a taxiway leads onto a runway are runway holding point markings, consisting of a double dashed line and a double solid line, with the double solid line towards the taxiway. As in road traffic, this means that the line may be freely crossed from the dashed (runway) side to the taxiway. Coming from the solid (taxiway) side, though, a pilot needs explicit clearance to cross that line and enter the runway, as in either “line up runway ! ” or “cleared take-off runway ! ” or “cross runway ! ”.



Runways with an ILS Category II or III have an ILS hold, marked with two solid lines with crossbars in between, similar to a ladder lying across the taxiway. The ILS hold will usually be further away from the runway than the simple runway hold. In very bad weather, which necessitates ILS Category II or III operations, holding aircraft have to be kept further away from the runway to avoid interference with the ILS signal. Approaching such a runway, an aircraft will be clearly instructed where it may taxi to with the words “taxi to the Cat II hold”, or “taxi to the runway hold”.

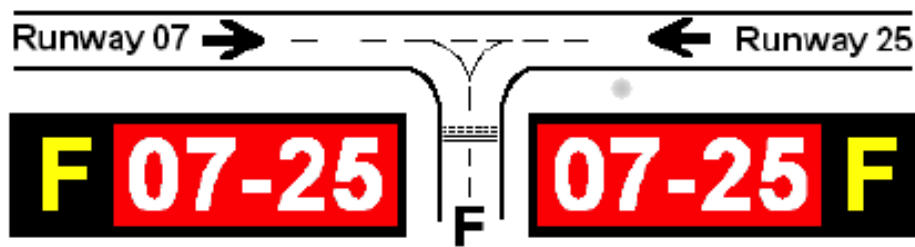
Stop bars are a line of red lights across the taxiway. When lit, no aircraft may cross, even with permission from the tower controller. If ATC wants the aircraft to cross, they must switch the stop bar lights off.

An aircraft vacating the runway after landing is only legally clear of the runway (and only then can another aircraft be cleared to land or take off) once the aircraft has taxied across the runway holding point lines.

### 7.11.3 Runway Names and Signage

Runways are designated according to the magnetic runway direction to the nearest 10 degrees, and with the trailing zero dropped. A runway aligned with a magnetic track of 034° would be designated as runway 03 and a runway with a magnetic track of 258° would be named runway 26.

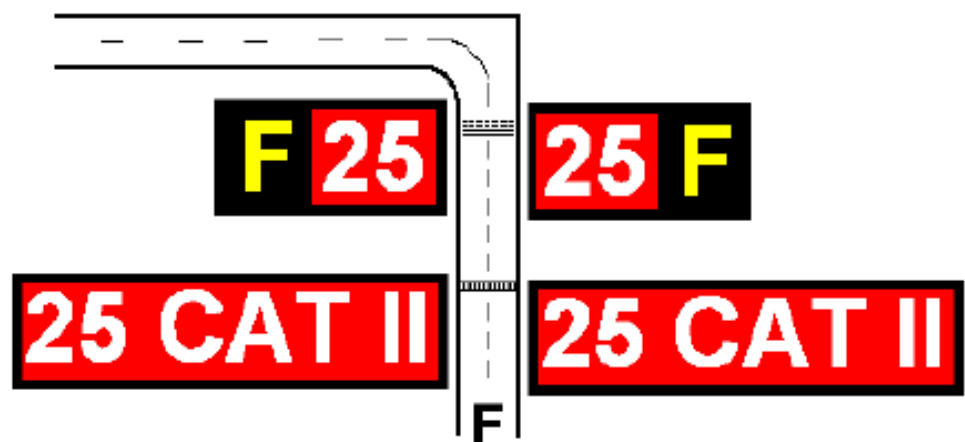
Runway signs are **white letters on a red background**, placed on both sides of the taxiway, clearly visible to the pilot when facing the runway.



Runway without an ILS or with only a Cat I ILS

The signs at the runway holding point, in line with the runway holding point markings, will show the taxiway on which the aircraft is in **yellow letters on a black background**, as well as the runway ahead in **white numbers on a red background**. The taxiway letter will be on the outside, furthest away from the taxiway. The runway numbers are arranged like the runways ahead. In the examples runway 07 runs from left to right, and runway 25 runs from right to left.

Note the two successive holding points in this example. The one first encountered would be called the “ILS Hold”, and the one closer to the runway is the “Runway Hold”.



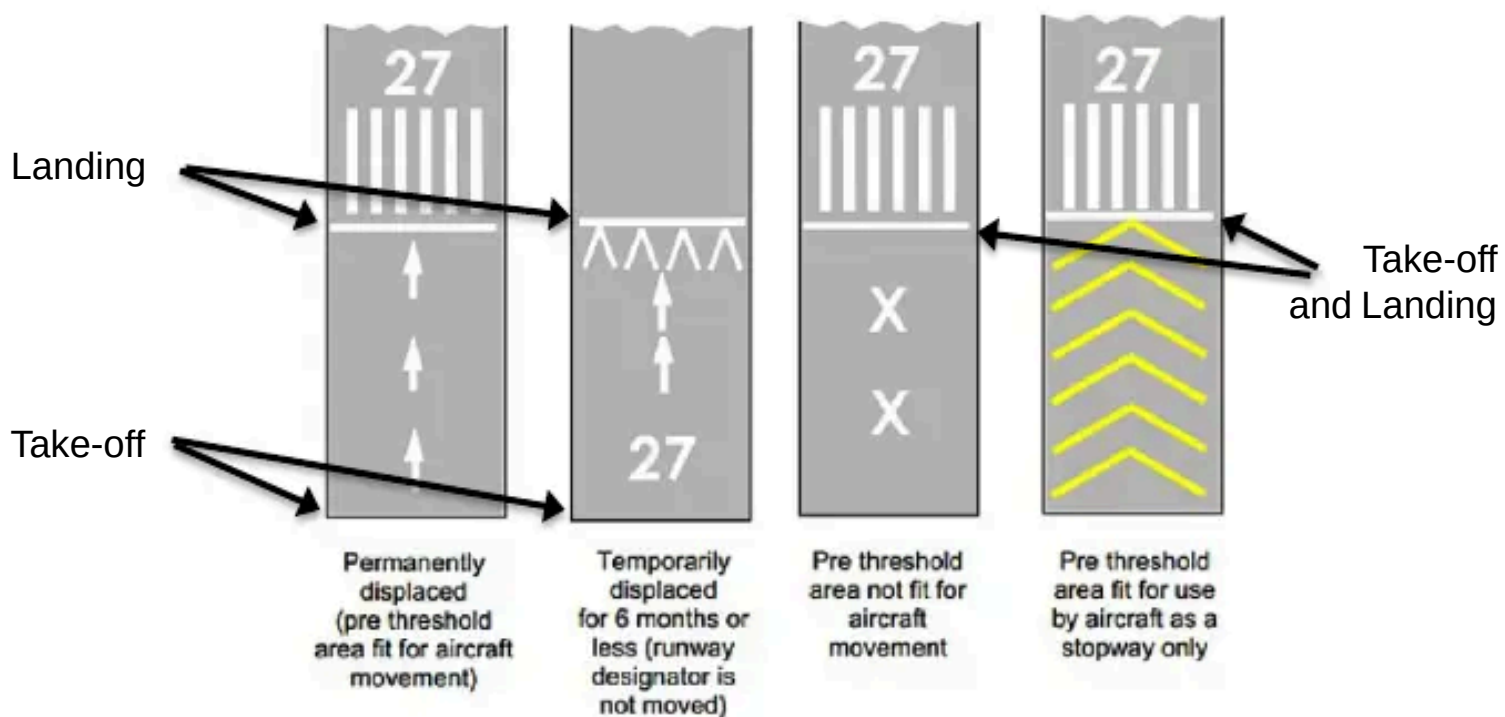
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## 7.11.4 Displaced Thresholds

Sometimes the official threshold of the runway is not at the actual beginning of the tar. This could be due to obstacles in the approach path, which would place the end of the (usually 3°) approach path beyond the actual beginning of the runway. In this case the threshold will be “displaced”, i.e. aircraft must not land before the determined and marked threshold. However, aircraft can still take-off from the beginning of the tar or hard surface – before the “threshold” for landing aircraft.

Another reason for displacing a threshold could be a physical problem with the tar or hard surface. It could have become unstable due to subsiding ground or any other physical reason. Depending on the reason that first bit of the runway may not be used at all - or it might only be taken into the calculations as a stopway for an emergency for aircraft taking off in the opposite direction.

See the illustration on the next page.



## 7.12 Speed Restrictions

AIP ENR 1.5

Speed restrictions apply in certain airspaces, stated as maximum permitted indicated airspeeds. If the safe indicated airspeed for a particular aircraft is more than the maximum permissible speed, the aircraft may be flown at the minimum safe speed.

- ! Within a CTR or ATZ Class C: Aircraft with reciprocating engines: 160 kts
- Turbine powered aircraft: 200 kts
- ! In the Johannesburg and Cape Town Special Rules Areas: 180 kts
- ! In uncontrolled airspace below FL100: 250 kts
- ! TMA speeds: when not on a STAR and within the following distances from Bloemfontein, Cape Town, Durban, East London, Johannesburg (OR Tambo) or Port Elizabeth:
  - within 50 nm of the respective VOR: 250 kts
  - within 15 nm of the respective VOR: 210 kts

Note also the following additional speed restrictions:

- ! maximum speeds in holding patterns paragraph 14.4.2
- ! maximum speeds on instrument approaches paragraph 14.4.4.6
- ! speed restrictions on a STAR paragraph 14.4.1

# 8

# Practical Communications

## 8.1 General Considerations

AIC D005/2018, ICAO Annex 10 Vol II chapter 5, ICAO Doc 9432

When speaking on the aircraft radio, consider the following general points:

- ! Make sure you have selected the correct frequency, turned the volume control up sufficiently and set the squelch (if not automatic).
- ! Make sure the radio is switched "on" and the audio panel switches are correctly set to headphones or speaker. Also make sure that the transmit switch on the audio panel (if fitted) is set to the correct radio.
- ! Prepare your message before you start speaking. For longer messages, such as AIREPs, you might want to write the message out before starting to speak.
- ! Anticipate the reply. If you expect detailed instructions or numbers, such as QNH, runway in use, another frequency etc., have pen and paper ready and write it down as you receive it. It will make the subsequent read-back easier.
- ! After changing frequency, listen out for a short while before pressing the push-to-talk switch, so that you do not interrupt other stations.
- ! Press the push-to-talk switch firmly so that it makes proper contact.
- ! Press the push-to-talk switch **before** starting to speak and release it only **after** you have finished speaking.
- ! Keep the microphone close to your mouth.
- ! Speak in a normal conversational tone, not too fast, not too slowly.
- ! Pronounce words clearly and distinctly.

- ! Use standard phraseology as much as possible. Try to put yourself into the controller's place; it will make it much easier to work out what information he needs from you. Don't be verbose.
- ! Make sure you have understood clearly what the controller requires. Don't be afraid to say "say again" if you have not understood the message perfectly. If you did get some, but not all of it, specify what it is you did not understand, e.g. "say again the QNH" rather than making the controller repeat the whole long message.
- ! After having made a mistake, don't give lengthy apologies, simply say "correction" and transmit the correct version.
- ! When asked something which you can't answer immediately, e.g. an estimate for the next reporting point, don't hum and haw; ask the controller to "stand by" , work out your answer, then call him/her back.
- ! If you don't receive an immediate reply, wait at least 10 seconds before calling again; the controller may be busy on the telephone, the intercom or another frequency.
- ! Think ahead and give the controller as much warning as possible of your intentions. When doing training circuits, let the controller know on downwind (rather than on final) that the next landing will be a full stop.
- ! When practicing instrument approaches, let him know when initially requesting clearance for the approach that you plan to do a missed approach followed by radar vectors onto the ILS or that you want to land off that approach or whatever the case may be. It helps the controller in traffic planning and makes his/her life easier.

Example: when requesting taxi and you are planning to do an intersection take-off or use the secondary runway state your request in the first transmission rather than halfway down the taxiway as an afterthought. "Stuurman Ground, Zulu Sierra Kilo Bravo Kilo requests taxi to runway 17, if available, for a VFR flight to Port Alfred".

## 8.2 ATIS Broadcast

AIP GEN 3.4(3)

Most larger airports offer an Automatic Terminal Information Service (ATIS) on a dedicated communication frequency. The ATIS broadcasts the current weather at the aerodrome, airport and ATC information, and anything else that pilots should know before take-off or landing - spoken either by a human or a synthetic computer voice.

Example of an ATIS broadcast

|   |                                                                          |                                                      |
|---|--------------------------------------------------------------------------|------------------------------------------------------|
| ! | The name of the ATC centre                                               | Cape Town                                            |
| ! | An identifying sequential letter in the phonetic alphabet, Alpha to Zulu | Information Hotel                                    |
| ! | Time in UTC (Universal time)                                             | 0456                                                 |
| ! | Departure/arrival procedures                                             | Expect SIDs and STARs in use                         |
| ! | Status of Control Zone                                                   | Zone VMC                                             |
| ! | Runway in use                                                            | Runway 19 for arrivals and departures                |
| ! | Transition Level                                                         | Transition Level 85                                  |
| ! | Wind direction and speed (magnetic)                                      | Wind 250 degrees, 4 knots                            |
| ! | Visibility                                                               | Visibility 10 kilometres or more                     |
| ! | Cloud cover (if any)                                                     | Broken cloud at 3800 feet                            |
| ! | Temperature                                                              | Temperature 20                                       |
| ! | Dew point                                                                | Dew point 15                                         |
| ! | QNH                                                                      | QNH 1016                                             |
| ! | Any expected weather change                                              | NOSIG                                                |
| ! | Special instructions                                                     | Confirm Cape Town Information Hotel on first contact |

- Some ATIS broadcasts are also available by telephone, for example:  
Cape Town International Airport on 021 939 1179  
OR Tambo International Airport on 011 928 6461

The ATIS will usually be updated and assigned the next letter of the alphabet every half hour. Before the first contact with Air Traffic Control at a centre where an ATIS is available, the pilot should listen to this broadcast and inform ATC accordingly. This applies both to departing and arriving aircraft.

Example: “King Shaka Ground, Zulu Sierra Alpha Bravo Charlie, received information Mike”.

## 8.3 Establishing Contact

AIC D005/2018

When transmitting **for the first time during any one flight** , establish contact by calling the station addressed, giving your full call sign, e.g.: "Cape Town Ground, Zulu Sierra Alpha Bravo Charlie". If the controller replies using an abbreviated form of

your call sign, i.e. "Alpha Bravo Charlie, Cape Town Ground", you may continue using the abbreviated form. See also Paragraph 7.7.2.

Remember to use prefixes and suffixes as explained in paragraph 7.7.3. For example, an ab-initio student with an instructor would say: "Lanseria Tower, Student Zulu Sierra Foxtrot November Delta".

If the ATIS (see paragraph 8.2) requests that any other information be given on first contact, the pilot should do so. If the ATIS says: "Pilots are to state altitude on first contact with Approach", the pilot should do exactly that.

Do not mention the **flight plan reference number** – the controller does not have the reference number in front of him. That number is only useful for finding your flight plan if it got lost in the system.

If an ATSU is establishing contact with you ("ABC, Kimberley Tower"), answer with the name of the station calling you, followed by your own call sign ("Kimberley Tower, ABC").

**After a lengthy pause in communications, or when calling a new station**, to which you have been transferred, establish contact by calling the station addressed, giving your full call sign, e.g.: "Bloemfontein Approach, Zulu Sierra Alpha Bravo Charlie". The reply will be "Alpha Bravo Charlie, Bloemfontein Approach". - Note that it is not necessary to state a frequency when establishing contact with a station to which you have been transferred. The previous controller, who transferred you, will have alerted the controller of the new frequency, so he will be aware of you and waiting for your call. The frequency should only be stated when there may be confusion as to which one you are calling on. This could happen if you don't get contact on the primary frequency and then try a secondary frequency. The controller of that second one would not be expecting your call, and since he might be monitoring more than one frequency it would help if you stated the frequency.

**If only a short time has elapsed between your last and the next call**, e.g. between downwind and final approach, or if it is assured that the station called will receive your call, it is not necessary to establish contact at all. In that case, you may go straight ahead with your message, e.g. "Alpha Bravo Charlie final one niner".

## 8.4 Acknowledgement of Receipt

AIC D005/2018

All **clearances** (take-off, after departure, en-route, approach, landing etc.), all **instructions** (runway in use, altitudes or flight levels, speeds etc.), and the **QNH** must be read back, that means they must be repeated exactly as received.

Pure information, e.g. the wind direction and speed, must not be read back. If that information is given on its own, without other instructions requiring read-back, it should be acknowledged by just stating the aircraft call-sign.

Traffic information should be acknowledged by saying either "looking out" if the pilot has not yet spotted the other traffic, or by "traffic in sight" if he has.

Example:

**at the holding point:**

Aircraft: Stuurman Tower, Alpha Bravo Charlie ready.

Tower: Alpha Bravo Charlie, wind 090/15, cleared take-off, report left downwind, traffic is a Cessna 152 joining left downwind.

Aircraft: Cleared for take-off, report left downwind, looking out, Alpha Bravo Charlie. (the wind is information which does not require repeating, the take-off clearance and the next reporting point are instructions and are read back, and the Cessna is acknowledged by 'looking out').

**on downwind:**

Aircraft: Alpha Bravo Charlie left downwind 08.

Tower: Alpha Bravo Charlie, number one to final.

Aircraft: Number one to final, Alpha Bravo Charlie .  
(the next reporting point is an instruction and must be read back).

**on final:**

Aircraft: Alpha Bravo Charlie final, touch and go.

Tower: Alpha Bravo Charlie, cleared touch and go.

Aircraft: Cleared touch and go, Alpha Bravo Charlie.  
(landing or touch-and-go clearance requires read-back).

Tower: Alpha Bravo Charlie, wind 095/20.

Aircraft: Alpha Bravo Charlie.  
(pure information is acknowledged by saying the aircraft call sign).

Broadcasts to "all stations" by an ATSU, e.g. a new QNH, do not have to be acknowledged unless the broadcasting station specifically requests this.

## 8.5 Transfer of Control

When a controller wants an aircraft to contact another Air Traffic Services Unit (ATSU) he will issue an instruction giving the name of the new station and its frequency and the time or point at which he wants the transfer to take place. This can either be immediately or at some specific point later in the flight.

Aircraft may not change frequency without permission from the present controller.

**Example 1 (immediate transfer):**

APP: Alpha Bravo Charlie, contact King Shaka Tower on 118.45.

means the aircraft must change frequency and establish communications immediately after the read back.

A/c: reading back :  
King Shaka Tower on 118.45, Alpha Bravo Charlie.  
The pilot will now change frequency to 118.45, listen out, and say:  
A/c: King Shaka Tower, Zulu Sierra Alpha Bravo Charlie.

**Example 2 (delayed transfer):**

TWR: Alpha Bravo Charlie cleared take-off, left turn out, contact Durban Approach on 125.75 passing 2000 foot.  
means the aircraft must change frequency only when climbing through 2000'.  
A/c: reading back:  
Cleared take-off, left out, contact Approach 125.75 passing 2000 feet.  
Now the aircraft will take off, turn left, and when it is passing 2000', will change frequency to 125.750, listen out, then say:  
A/c: Durban Approach, Zulu Sierra Alpha Bravo Charlie, passing 2000'.

## 8.6 Reporting Points

A reporting point may be:

- ! a geographical point, e.g. "report HBV" or "report leaving the CTR",
- ! a Flight Level or altitude, e.g. "report passing Level 150",
- ! a condition, e.g. "report when you have the Airbus on left base in sight",
- ! a time, e.g. "report at 20" means 20 minutes past the present hour.

Compulsory reporting points are marked on charts with solid triangles ( # ). They can be navigational aids, e.g. NDBs or VORs. They can also be points where an air-route crosses another airway or an airspace boundary, in which case they have usually been allocated a five-letter identifier, e.g. ETBUX or GERAX.

In the airfield environment, the compulsory reporting points are:

- ! Downwind (abeam the end of the runway in use)
- ! Final

! The outer marker or equivalent DME distance on an ILS approach.

On-request reporting points are marked with a hollow triangle ( \$ ). ATC will advise whether or not they wish the pilot to report at these points.

ATC may also ask a pilot to report at any other place, Flight Level, altitude or time.

# 8.7 The Position Report

ICAO Doc 4444 Appendix 1, AIP ENR 1.3

Position reports are compulsory in controlled airspace, in advisory airspace and for flights for which alerting action is requested. In practice, though, pilots also report their position in uncontrolled airspace, either to the Flight Information Service, or on the TIBA frequencies as a broadcast.

The format of the position report (AIREP) detailed below must be used for all reports to an ATSU, sometimes in abbreviated form. It is also used for reports to an ATSU when flying in uncontrolled airspace, and, in slightly modified form, for traffic **broadcasts** in uncontrolled airspace below 1500' AGL, in General Flying Areas, Special Rules Areas or when out of range of an ATSU.

It is important to know whether you are flying in **controlled** airspace, where your flight will be controlled, instructions from ATC have to be followed and you might be separated from other traffic, or whether you are flying in **uncontrolled** airspace, where you will receive traffic information only and will not be separated from other traffic. In both cases the actual communication between the aircraft and the ATSU will be almost identical.

Section 1 (Position report) of the AIREP is compulsory at each significant point. In addition, section 2 (Operating data) should be used when necessary, e.g. when no Flight Plan has been filed and when first contacting an ATSU. Section 3 (PIREP, Pilot weather report) is optional.

|            | Format                                                                                                  | Example                                                                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Section 1: | Station called<br>Identification<br>Position<br>Time<br>Flight Level/Altitude<br>Next position and time | Cape Town Approach<br>Juliet Golf Papa<br>Worcester<br>one zero two five<br>Flight Level eight five<br>estimate Wolseley at four one            |
| Section 2: | Destination and ETA<br>Endurance                                                                        | Estimate Cape Town at four niner<br>Endurance zero two three zero                                                                               |
| Section 3: | Air Temperature<br>Mean wind<br>Turbulence<br>Aircraft icing<br>Any other                               | Temperature zero three<br>Wind three zero zero at two three knots<br>Moderate turbulence<br>No icing<br>Thunderstorms approx. 30 nm north of GE |

The “position” in the AIREP above can be given as a town, a navigational aid, a significant point with a 5-letter indicator, as co-ordinates or as described in paragraph 6.6.7.4.

When broadcasting one's position in uncontrolled airspace, the AIREP must be given in full. In addition, It is helpful for other traffic to add the route as well as the aircraft make and model, and to end with one's call sign: "Traffic in the Robertson area, Cessna ZS-ABC on a flight from Worcester to Swellendam, over Robertson at 1235, Flight Level 75, estimate Bonnievale at 50. ABC."

In a non-radar environment the AIREP should always be given as completely as possible, even during climb and descent: "Upington, KSS passing 2 thousand feet, estimate Sishen at 40" - or: "Kimberley, ABC, top of descent, estimate Kimberley at 35".

When the aircraft is operating under a flight plan, flies in a radar environment and can be tracked on the controller's radar screen, position reports become much easier. In cruise, on first contact with a new station the pilot should just state his call sign and flight level: "Johannesburg West, Hummingbird 611, Flight Level 190". Subsequent position reports must give the call sign, the point and the time over that point: "Johannesburg East, NII, ITPOL, 2248".

## 8.8 Phraseology

AIC D005/2018

### 8.8.1 Standard Words

Whenever possible the standard words and phrases given below should be used. This makes communications easier, prevents misunderstandings and keeps talk over the radio short and precise. Pleasantries like "good morning" should be kept to a minimum. Banter, non-standard phrases or exchanges in a language other than English are neither professional nor safe.

For special terminology associated with RVSM, ACAS and other IFR procedures see chapter 14 .

|             |                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| Acknowledge | Let me know that you have received and understood this message. Read this message back more or less word for word           |
| Affirm      | Yes                                                                                                                         |
| Approved    | Permission is granted                                                                                                       |
| Break       | Indicates separation between different portions of a message                                                                |
| Break Break | Indicates separation between messages to different aircraft                                                                 |
| Cancel      | Invalidates the previously transmitted clearance                                                                            |
| CAVOK       | Ceiling and Visibility okay (pronounced Cav-oh-kay)                                                                         |
| Check       | Examine a system or procedure (normally no answer required)                                                                 |
| Cleared     | You are authorised to proceed with...                                                                                       |
| Confirm     | Have I correctly received... or Did you correctly receive...<br>(Note: the answer should usually be "affirm" or "negative") |

|                 |                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Contact         | Change frequency and establish contact with...                                                                                       |
| Correct         | That is correct                                                                                                                      |
| Correction      | I made an error. The correct version is...                                                                                           |
| Disregard       | Consider the last transmission as not sent                                                                                           |
| How do you read | How readable (understandable) is my transmission?                                                                                    |
| I say again     | I repeat for clarity or emphasis                                                                                                     |
| Maintain        | Continue as specified                                                                                                                |
| Monitor         | Listen out on frequency...                                                                                                           |
| Negative        | No or that is not correct or permission not granted                                                                                  |
| Out             | This exchange is finished, no response is required (used in HF communications only)                                                  |
| Over            | I am finished but expect a response from you (used in HF communications only)                                                        |
| Read back       | Read back all or a specified part of the message                                                                                     |
| Re-cleared      | The last clearance is superseded by this new clearance                                                                               |
| Report          | Pass me the following information...                                                                                                 |
| Request         | I ask permission or I would like to know                                                                                             |
| Roger           | I have received all of your last transmission (not to be used if the message requires reading back)                                  |
| Say again       | Repeat all or a specified part of your last transmission                                                                             |
| Speak slower    | Speak slower, I have difficulty understanding                                                                                        |
| Stand by        | Wait, I will call you back                                                                                                           |
| Unable          | I cannot comply with your request, instruction or clearance                                                                          |
| Wilco           | I understand your message and <b>will</b> comply with it (not to be used if the message requires reading back)                       |
| Words twice     | Since communications are difficult I will send every word or group of words twice or please send every word or group of words twice. |

### 8.8.2 Phraseology Used by Pilots

The following standard phrases should be used whenever possible. Only the phrases used by pilots when initiating a call are given; all other communications used by air traffic controllers should be read back exactly as received. Aircraft and ground station call signs have been omitted from these examples, but must, of course, be included in the actual message.

- ! Words in (round parentheses) indicate that specific information must be inserted to complete the phrase.
- ! Words in [square parentheses] indicate optional additional information.

to request a time check:

**REQUEST TIME CHECK**

to request aerodrome data (wind, temperature etc.), if there is no ATIS available:

**REQUEST DEPARTURE INFORMATION**

when unable to comply with an instruction:

**UNABLE TO COMPLY**

when the pilot wishes to change to another frequency:

**REQUEST CHANGE TO (frequency)**

to request taxi to a point on the airfield:

**[aircraft location] REQUEST TAXI TO (destination on aerodrome)**

Example: George Tower, Alpha Bravo Charlie at the hangars, request taxi to the fuel pumps.

to taxi for departure:

**[aircraft type] [wake turbulence category if heavy] [aircraft location] REQUEST TAXI [intentions]**

Example: Cape Town Ground, Tomahawk Zulu Sierra Kilo Kilo Charlie at SAFAIR, request taxi for training circuits.

or **[aircraft type] [wake turbulence category if heavy] [aircraft location] (flight rules) TO (aerodrome of destination) REQUEST TAXI [intentions]**

Example: Virginia Tower, Cessna Cardinal Zulu Sierra Mike Oscar Golf, on the apron, VFR to Hibberdene, request taxi.

to request crossing a runway:

**REQUEST CROSS RUNWAY (number)**

when ready for departure:

**READY**

Note the terminology: just “Ready” or “Ready for departure”, never “ready for take-off”

to request descent:

**REQUEST DESCENT [AT (time)]**

to request a visual descent:

**REQUEST VMC DESCENT**

to request a visual or straight-in approach:

**REQUEST VISUAL APPROACH**

**REQUEST STRAIGHT-IN APPROACH**

first transmission to TOWER for landing:

**[aircraft type] (position) (level) FOR LANDING**

Example: Cape Town Tower, Zulu Sierra Bravo Charlie Delta, Zeekoevlei at 1500 feet for landing.

reporting position in the circuit:

**(position in the circuit, e.g. DOWNWIND / FINAL)**

Example: Alpha Bravo Charlie left downwind 19.

when vacating the runway:

**RUNWAY VACATED**

when any change to the existing clearance would result in landing with less than the planned final reserve fuel (see also chapter 11):

**MINIMUM FUEL**

### 8.8.3 Phraseology Used by Controllers

AIC D005/2018

The following are sample phrases used by air traffic controllers. The list is by no means exhaustive, but this is an indication of what a pilot can expect. Pilots should read back all instructions, clearances and the QNH exactly as received.

A clearance issued by a controller always has a clearance limit, and does not absolve pilots from complying with the rules. The clearance limit could be stated, as in: “ABC, cleared Kimberley to Bloemfontein”. Or the limit could be implied, for example the following take-off clearance ends at the CTR boundary, where the pilot leaves controlled airspace: “ABC, cleared take-off runway 16, left turn out, not above 1000’, report leaving the CTR”.

The clearance limit on an instrument approach is either the Missed Approach Point (MAPt) if the aircraft continues to land, or, if the aircraft does not have the runway in sight and executes a missed approach the end of the missed approach procedure.

#### **WHEN READY**

take action when you are ready for it

Example: Descend to 5000 feet when ready.

#### **APPROVED**

you may do what you requested

Example: 4000 feet approved above the General Flying Area

#### **HOLD POSITION**

stay where you are

#### **HOLD SHORT OF (position)**

stop before reaching a certain point

Example: Hold short of the main apron

**BACKTRACK RUNWAY (number)**

enter and proceed along the runway against the take-off and landing direction

**FOLLOW (description of aircraft)**

Example: Follow the 737 to the holding point runway 27.

**GIVE WAY TO (description of aircraft)**

Let the other aircraft proceed first

**REPORT READY [FOR DEPARTURE]**

**ARE YOU READY FOR IMMEDIATE DEPARTURE?**

Note: if you answer "affirm", get going. ATC asks this to do you a favour and get you off, don't mess up by answering "affirm" and then taking your time. But don't let yourself be rushed: if you haven't finished your checks yet, answer "negative", do the checks in peace and wait. Do not be unsafe just to please the controller and/or save a few seconds.

**BEHIND (aircraft) LINE UP BEHIND**

Note: the two "behind" are there to make it very clear that the Falcon lands before the other aircraft taxis onto the runway

Example: Behind the landing Falcon line up behind.

**LINE UP AND WAIT**

taxi onto the runway and stop there

**CLEARED FOR TAKE-OFF**

**WIND (number) DEGREES (number) KNOTS**

Example: Wind 210 degrees 15 knots.

**CLIMB [or DESCEND] TO (level) TO REACH (level) BY (point) AT (number) FEET PER MINUTE [MINIMUM or MAXIMUM]**

Example: Climb to FL120, to reach 6000 feet by 10 DME TGV.

**CROSS (reporting point) AT (altitude) [OR ABOVE / BELOW]**

Example: Cross OBV at 3000 feet or above.

**REPORT LEAVING [or REACHING or PASSING] (level)**

Example: Report passing FL85.

**CONTINUE CLIMB [or DESCENT] TO (level)**

Example: Continue descent to FL55.

**MAINTAIN (level)**

Example: Maintain 1500 feet.

**MAINTAIN OWN SEPARATION**

ATC will not separate you from other traffic

**MAINTAIN VMC**

stay in Visual Meteorological Conditions

**REPORT AT (point)**

Example: Report at SLV.

**REPORT PASSING (point)**

Example: Report passing 2000 feet.

**REPORT RUNWAY IN SIGHT**

**TRAFFIC IS (information)**

Example: Traffic is a Cessna 210 joining left base.

**NO REPORTED TRAFFIC**

the controller does not know of any aircraft on a possible collision course with you (that does not necessarily mean that there is no other traffic).

**CONTACT (unit call sign) (frequency) (condition)**

Example: Contact Cape Town Approach 119.7 passing 2000 feet.

**MONITOR (unit call sign) (frequency)**

change frequency and listen out, do not call them

Example: Monitor Langebaan Approach 122.5.

**REMAIN THIS FREQUENCY**

continue listening on this frequency.

**SQUAWK (SSR code)**

dial the given code into your transponder and switch it on.

**COMMENCE APPROACH AT (time)**

**CLEARED STRAIGHT-IN APPROACH [RUNWAY (number)]**

**ENTER CONTROLLED AIRSPACE AT (level and/or time or point)**

Example: Enter controlled airspace at FL65.

**TURN RIGHT [or LEFT] HEADING (three digits)**

Example: Turn right heading 310.

**JOIN (RIGHT or LEFT) (position in circuit) (runway)**

Example: Join right downwind runway 12.

**EXTEND DOWNWIND**

**REPORT BEFORE TURNING BASE or REPORT READY FOR BASE TURN**

**NUMBER (number) TO (position) [FOLLOW (aircraft)]**

Example: Number two to final, follow the Cessna on left base.

**MAKE SHORT APPROACH**

**SHOW LANDING LIGHT**

switch landing light on for better identification.

**CONTINUE APPROACH**

after you have reported final, but cannot be given landing clearance, because somebody else is still on the runway, or a helicopter is still crossing.

**ORBIT RIGHT (or LEFT) FROM PRESENT POSITION**

execute a 360 degree turn in the given direction.

**GO AROUND**

apply power and go around – do not land.

**CAUTION WAKE TURBULENCE**

**CLEARED TO LAND [RUNWAY (number)]**

**CLEARED TOUCH AND GO**

**VACATE RUNWAY AT (taxiway)**

Example: Vacate right at Charlie.

**EXPEDITE**

hurry up (safely!).

**ALL STATIONS**

Used by controllers to broadcast a message. Readback is not necessary unless specifically requested.

Example: All Stations, QNH now 1023.

**STATION CALLING (ATSU station), SAY AGAIN YOUR CALLSIGN**

Used by ATC when they did not understand the callsign of the last caller

Example: Station calling Johannesburg Information, say again your callsign.

## 8.9 En-Route Reports

AIP ENR 1.3

### 8.9.1 When in Contact With an ATSU

An aircraft should be in contact with an ATSU

- ! in controlled airspace,
- ! in uncontrolled airspace above 1500' AGL – with some exceptions, see paragraph 8.9.2,
- ! in most military areas, e.g. Lowveld Military Airspace Sector (LASS).

#### 8.9.1.1 In Controlled Airspace

In controlled airspace the pilot must be in radio contact with an Air Traffic Controller and has to report at all compulsory reporting points. He must also report at non-compulsory reporting points or any other points when requested by ATC (see paragraph 8.6). - The pilot's reports are made in the form of the AIREP as detailed in paragraph 8.7. The aircraft will receive Control Service and has to follow instructions.

Example: a pilot on a VFR flight plan from Sutherland to Cape Town at FL105 is talking to Cape Town Information, who will transfer him to Cape Town Approach in the vicinity of Ceres, before he reaches the TMA boundary. He will then call Cape Town Approach on frequency 119.7:

A/c: Cape Town Approach, ZS-MIT

C.T.: MIT, Cape Town

A/c: MIT at Ceres 1045, FL105, estimate Cape Town 1110

C.T.: MIT, cleared to enter the Cape Town TMA at FL105, report for descent.

A/c: Cleared to enter the TMA, FL105, report for descent, MIT.

#### 8.9.1.2 In Uncontrolled Airspace

In uncontrolled airspace above 1500 feet AGL the pilot should communicate with the appropriate ATSU on the "Information" frequency (see Appendix II). The communication will take essentially the same format as in controlled airspace (i.e. the AIREP), and ATSU will pass on any relevant information about conflicting traffic, if known, as well as any other pertinent information, but will not control the flight.

Example: an aircraft flying from Upington to Kimberley at FL95 has just left the Upington ATZ. It is now in uncontrolled airspace in the Johannesburg FIR, West Sector. The pilot will talk to Information on frequency 118.55:

A/c: Information, ZS-EOA

Info: EOA, Information

A/c: EOA left the Upington ATZ en route to Kimberley, passing FL60, climbing to FL95, estimate the Kimberley TMA at 1120, and Kimberley at 40

Info: EOA, no reported traffic at FL95, report abeam Douglas.

## 8.9.2 When Not in Contact With an ATSU (TIBA)

When an aircraft is not in contact with an ATSU, pilots must broadcast their position and intentions on specific frequencies to ensure separation and avoid mid-air collisions. These broadcasts are called **Traffic Information Broadcasts by Aircraft (TIBA)**. Pilots should transmit their position, altitude and intentions in a format and manner that other pilots can understand and interpret correctly.

An aircraft will not be in contact with an ATSU:

**by design and intentionally:**

- ! in uncontrolled airspace below 1500' AGL,
- ! in Special Rules Areas (see paragraph 1.3.4.1),
- ! in Flying Training Areas,
- ! in the circuit of uncontrolled airfields.

**due to operational constraints:**

- ! when the "Information" frequency is not manned, for example most military areas outside of normal office hours,
- ! when out of range of an ATSU due to the aircraft's distance from the station (see paragraph 13.4 for range calculations) or because there is high ground between the aircraft and the station.

The TIBA message should be given in a similar format to the normal AIREP, in a way that other aircraft in the vicinity can understand and interpret. TIBA is used by pilots on different missions (cross-country, sight-seeing, training, surveys, crop spraying), and pilots must use common sense in describing their position as accurately and as meaningful as possible.

A broadcast should include:

- ! The general area – to alert other pilots in the same area that they should be listening closely to the following message. Say something like "Traffic in the Bethlehem area" rather than the often heard, but useless "Traffic on 124.8".
- ! The type of the aircraft – so that other pilots know what they should be looking out for and can get an idea of the speed of the broadcasting aircraft.
- ! The call sign – so that other pilots know who you are.
- ! The route of the aircraft (if on a cross-country flight) – again so that other pilots can form a mental picture of where the broadcasting aircraft is. Use a route description that other pilots can understand. "Upington to Kimberley" is meaningful, but few people would benefit from a route description such as "Pampoensfontein to Uncle Bennie's farm".
- ! The present position. Again, use a point that people are likely to know.
- ! The present altitude or flight level.
- ! The next point the aircraft is flying to (if applicable).
- ! The estimate at the next point (if applicable).
- ! The call sign again, in case the other traffic didn't catch it earlier.

**Broadcasts should not be acknowledged** unless there is a potential collision risk. “Alpha Bravo Charlie, this is X-Ray Yankee Zulu, we copy and will be looking out for you” is verbose and totally meaningless. Instead, rather give an AIREP yourself.

It is also not terribly useful to end every broadcast with the words “any other traffic please acknowledge” – it is meaningless, clutters up the radio waves, and takes time away from other pilots to make more useful transmissions. The onus is on the receiving pilot to perceive a potential collision risk and decide whether to acknowledge the broadcast or not.

Example 1: \_ an aircraft flying from Tswalu near Upington to Kimberley is still climbing. The pilot is not sure whether he is able to talk to ATC yet, he tries to establish communications with Johannesburg on 118.55, and, if not successful, will broadcast. This transmission could lead to interference with other aircraft on the same frequency: our Tswalu pilot in a relatively low flying aircraft out of range might broadcast at the same time as the ATSU talking to a higher flying or closer aircraft; the latter might receive both transmissions and is therefore not able to understand either. Pilots broadcasting on a manned frequency should listen out very carefully, try not to interrupt anybody, and keep their reports as short and to the point as possible.

A/c: Johannesburg, ZS-EOA – no reply - wait

A/c: Traffic in the Tswalu area: Piper Arrow ZS-EOA airborne Tswalu 1045, passing 6000' for FL95 en route to Kimberley, estimate the Kimberley TMA at 1120 and Kimberley at 40, ZS-EOA.

Example 2: an aircraft is on a pleasure flight along the west coast near Langebaan in a military restricted area. The pilot will try to establish communications with Langebaanweg on frequency 122.5:

A/c: Langebaanweg, ZS-EOA – no reply

A/c: Traffic in the Langebaan area: Piper Arrow ZS-EOA en route from Yzerfontein to Elandsbay coastwise, passing abeam Saldhana 1010, FL35, estimate abeam St Helena Bay at 1025.

Example 3: an aircraft flying from George to Port Elizabeth along the coast at 1000' AGL is in uncontrolled airspace at Plettenberg Bay. The pilot will broadcast on frequency 124.8:

A/c: Traffic in the Plettenberg Bay Area, Piper Arrow Lima Alpha Golf on a flight from George to Port Elizabeth coastwise. Plettenberg Bay 1045, 1000' AGL, estimate Stormsrivier 1105.

Example 4 : training flight:

A/c: Traffic in the Johannesburg GFA, Beech Sundowner India Tango Foxtrot five miles west of Everton, operating between ground and 1000 feet AGL on QNH 1028, practising precautionary landings.

### 8.9.3 IATA Inflight Broadcast Procedures (IFBP)

AIC D003/2014

Over many parts of Africa air traffic controlling is often not as efficient as it could be, and many Air Traffic Control Centres are not (yet) equipped with radar. Furthermore, VHF communications are sometimes not possible due to range problems, and HF communications are often plagued by static and interference. Also, many controllers and local pilots in this region are speaking French, Portuguese or Spanish with each other – which means that the pilots of an overflying aircraft often cannot understand those communications and cannot form a mental picture of other traffic around them.

The International Air Transport Association (IATA), the international industry trade group of airlines, has therefore advised its member airlines to adopt the IATA In-Flight Broadcasting Procedure (IFBP). This means pilots should listen out and make broadcasts in English on the VHF frequency 126.9 MHz (memorize the frequency!) **in addition** to the normal reporting procedures. Broadcasts should be made when necessary, 10 minutes before entering an FIR or crossing an air route, before changing Flight Level, and generally not less than 20 minutes apart.

These broadcasts take a very similar format to the normal AIREP as discussed in paragraph 8.7, with the addition of the direction of flight.

Example : All stations, this is Speedbird 123 in the Brazzaville FIR, flight level 400, northbound on UM998, position OXIMU at 0125, estimating position NAMOR at 0147. Speedbird 123, Airbus 330, flight level 400, northbound in the Brazzaville FIR.

## 8.10 Circuit Reports

### 8.10.1 Circuit Reports at Controlled Airfields

**The standard circuit is flown to the left** , so, when you want to do training circuits the controller just tells you: “Cleared for take-off, report downwind”, you must fly a left hand circuit and report on left downwind. If he wants you to fly a non-standard right-hand circuit, he will instruct you: “Cleared for take-off, report right downwind”. The circuit altitude is 1000’ AGL – unless otherwise noted in the AIP for that particular aerodrome.

The runway extends laterally up to the runway hold markings (see paragraph 7.11.2), and after landing you are not clear of the runway until you have crossed the runway hold markings, usually 50 m from the runway centre-line. No other aircraft should be given landing clearance until the preceding landing aircraft has crossed the runway

hold markings. During the day, when the runway is long enough for an aircraft to land while the preceding one is still on the runway, a “land after” clearance may be given, provided the second aircraft accepts this clearance.

At larger airfields there may be stop bars, i.e. a line of red lights across a taxiway, usually where this taxiway intersects with a runway. Under no circumstances may illuminated red stop bars be crossed, even if authorised to do so. Should this happen, request the controller to extinguish the stop bars before crossing them.

Depending on the airfield, one or more frequencies may be used for communication on and around the airfield. Smaller airfields will only have a Tower frequency, on which taxi and circuit instructions are given and clearances are transmitted. Busy airports may have a Tower (TWR) frequency for aircraft on the runway and in the air, a Surface Movement Control (SMC) frequency for controlling taxiing traffic, a Clearance Delivery frequency for transmitting clearances to departing traffic, and an Apron Office frequency for allocating parking bays to arriving aircraft.

Both VFR and IFR pilots must be familiar with the airport layout and taxiway names. Airport charts for the larger airfields in South Africa can be viewed on the CAA website under Industry > Aeronautical Information > Aeronautical Charts > Aerodrome Layout (usually called AD\_01) under the individual airport.

Air Traffic Controllers at controlled airfields will not only issue instructions, but will also pass on pertinent information about runway surface conditions (e.g. “runway damp” ), bird activity etc.

At a controlled airfield reports have to be made as follows:

#### **When departing:**

##### **VFR flights:**

- ! for engine start only when starting on the apron, not in the hangar area,
- ! for obtaining a departure clearance (Cape Town and OR Tambo only)
- ! for taxi,
- ! for departure (take-off),
- ! leaving the CTR or ATZ,
- ! any other as instructed by ATC

##### **IFR flights**

- ! for the departure clearance,
- ! for start,
- ! for taxi,
- ! for departure (take-off).

## When arriving:

### VFR flights

- ! field in sight or just outside the CTR/ATZ,
- ! downwind (abeam the upwind threshold or opposite the control tower) or where instructed. If the controller instructs you to join left or right base or even final, you do exactly that: descend to circuit altitude (if necessary), then fly direct to the instructed joining position,
- ! base (if requested by the controller),
- ! final – about 4 nm from the threshold (“long final” is 8 nm, “short final” is 2 nm),
- ! any other as instructed by ATC

**IFR flights** contact Tower when handed over from Approach

#### Example 1 (departing a smaller airfield with Tower only):

A/c: Virginia, Zulu Sierra Juliet Golf Papa  
 TWR: Juliet Golf Papa, Virginia, go ahead  
 A/c: Juliet Golf Papa, at the northern hangars, VFR to East London, three on board, request taxi  
 TWR: Juliet Golf Papa, taxi to the holding point runway 23, QNH 1021  
 A/c: QNH 1021, taxi to the holding point 23, Juliet Golf Papa  
 And a bit later:  
 A/c: Virginia, Juliet Golf Papa ready  
 TWR: Juliet Golf Papa, wind 280 at 10 knots, cleared take-off runway 23, report passing the Harbour entrance  
 A/c: Cleared take-off, report passing the Harbour entrance ! .

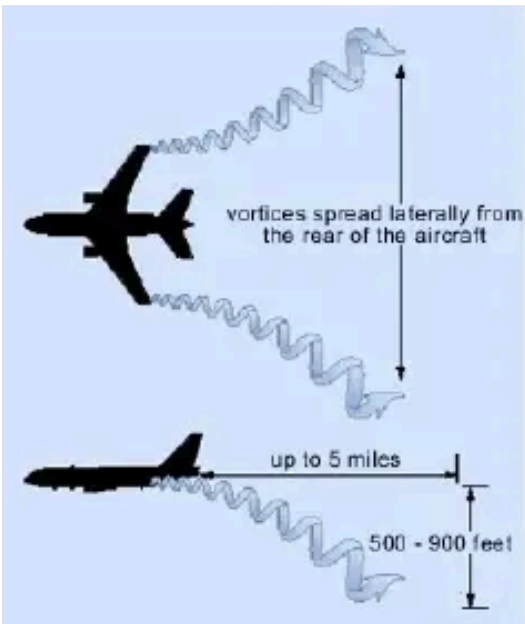
#### Example 2 (arriving at a larger airfield with TWR and SMC):

A/c: Cape Town Tower, Zulu Sierra Alpha Bravo Charlie  
 TWR: Alpha Bravo Charlie, Cape Town Tower  
 A/c: ABC north of the Bottelary, 1500', for landing  
 TWR: Alpha Bravo Charlie, join and report right downwind runway 01  
 A/c: Join and report right downwind 01, Alpha Bravo Charlie  
 And a bit later:  
 A/c: Cape Town Tower, Alpha Bravo Charlie right downwind 01  
 TWR: Alpha Bravo Charlie, report a Boeing 737 on a two mile final in sight  
 A/c: Traffic in sight, Alpha Bravo Charlie  
 TWR: ABC is number two to final 01, follow the Boeing 737 on a two mile final  
 A/c: Number two to final to follow a Boeing 737, Alpha Bravo Charlie  
 When on final:  
 A/c: Alpha Bravo Charlie final 01  
 TWR: Alpha Bravo Charlie cleared to land, wind 340/10  
 A/c: Cleared to land, Alpha Bravo Charlie  
 After the aircraft has landed:  
 TWR: ABC, left on Charlie, contact Ground 121.9 runway vacated

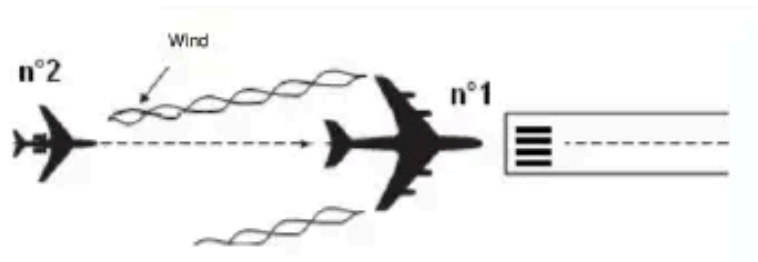
A/c: Left on Charlie, Ground 121.9 vacated, Alpha Bravo Charlie  
On 121.9 MHz:  
A/c: Cape Town Ground, Alpha Bravo Charlie, vacated on Charlie  
GND: Alpha Bravo Charlie, taxi Hotel to the hangars, expedite crossing Bravo  
A/c: Hotel to the hangars, expedite crossing Bravo, Alpha Bravo Charlie.

8.10.1.1 Wake Turbulence

Wake turbulence is generated in the form of wing tip vortices by all aircraft, due to the difference in pressure below and above the wing. At the wing tips the higher pressure from below the wing spills around to the top of the wing, creating a spiralling airflow called wing tip vortices.



Wing tip vortices are most pronounced in heavy aircraft and when the wings are at a high angle of attack, i.e. when the heavy aircraft is flying slowly during take-off and landing. They spread outwards and downwards, and also move with the wind or crosswind.



Wing tip vortices are potentially dangerous, especially if a lighter aircraft is caught in the vortices of a heavy aircraft. Therefore, Air Traffic Control will usually ensure that aircraft taking off and landing are separated by between 2 and 4 minutes, depending on the wake turbulence category of the aircraft and various other factors, such a parallel runways etc. The following table gives some examples for light aircraft:

|                                                                                                       |           |
|-------------------------------------------------------------------------------------------------------|-----------|
| Light aircraft landing behind a super heavy aircraft                                                  | 4 minutes |
| Light aircraft taking off after a super heavy aircraft or landing behind any medium to heavy aircraft | 3 minutes |
| Light aircraft taking off after a medium or heavy aircraft                                            | 2 minutes |

However, pilots of light aircraft must also be vigilant and take the following precautions. Pilots can also always ask ATC for greater separation.

**Precautions when taking off after a heavier aircraft:**

Vortices start at rotation and drop below the path of the aircraft. A pilot of a light aircraft taking off after a heavier one should note the point of rotation of the heavier aircraft, get airborne before that rotation point and turn upwind as soon as it is safe to do so.

**Precautions when landing behind a heavier aircraft:**

Vortices stop when the aircraft stops generating lift, i.e. more or less at the touch-down point. A pilot of a light aircraft landing behind a heavier one should stay above the heavier aircraft's flight path and touch down further down the runway from the heavier aircraft's touch-down point.

**8.10.2 Circuit Reports at Uncontrolled Airfields**

CAR 91.06.12

When landing or taking off at an uncontrolled airfield, pilots must broadcast their position and intentions on the applicable frequency.

Some airfields have been allocated specific frequencies, listed in Appendix I. Airfields situated within General Flying Areas use the frequency of that GFA, and airfields in Special Rules Areas usually use the Special Rules frequency. At all other airfields use **124.8 MHz** (see also Appendix II).

In VMC the following procedures should be adopted:

When **approaching an uncontrolled airfield to land** , the pilot should tune in and listen out on the appropriate frequency during the descent, say at about 20 nm from the airfield, preferably on a second radio. This will give him an idea about traffic as well as about the runway in use. When about 5 nm from the field he should broadcast his position, level or altitude and intentions. This is followed by overflying the airfield at 2000' AGL, doing left-hand turns only, observing the windsock and other traffic and making a radio call. After descending on the non-circuit (usually right) side of the runway and joining on crosswind, the pilot should usually transmit his position on downwind, base and final. Again, the standard circuit is to the left and the standard circuit altitude is 1000' AGL, unless otherwise stated in the AIP or the local airfield rules.

When **taking off** , the pilot tunes in and listens out before broadcasting his position on the airfield and his intentions. This is followed with transmissions before taxiing onto the runway for take-off and when leaving the circuit.

Example: Klerksdorp Traffic, Warrior Zulu Sierra Juliet Golf Papa, 5 miles west of the field at 6500 feet to join overhead  
 Klerksdorp Traffic, Juliet Golf Papa overhead at 6500 feet, to join left downwind runway 18  
 Klerksdorp Traffic, Juliet Golf Papa left downwind runway 18  
 Klerksdorp Traffic, Juliet Golf Papa left base runway 18  
 Klerksdorp Traffic, Juliet Golf Papa final runway 18  
 Klerksdorp Traffic, Juliet Golf Papa vacating runway 18.

and: Klerksdorp Traffic, Piper Warrior Zulu Sierra Juliet Golf Papa in the parking area, taxiing to the holding point runway 18

Klerksdorp Traffic, Juliet Golf Papa entering runway 18 for take-off for a flight to Lanseria

Klerksdorp Traffic, Juliet Golf Papa leaving the circuit to the north-east on a flight to Lanseria.

Landing at an unmanned airfield with a published instrument approach procedure, in IMC, pilots broadcast their intentions on the appropriate airfield frequency overhead the facility, before commencing the approach, when the inbound turn is completed, and commencing the missed approach, if applicable.

## 8.11 VFR Departures and Arrivals

Aircraft departing from or arriving at some of the larger airports in South Africa may be given special routings by ATC. These routings are published together with the Special Rules Areas (see paragraph 1.3.4.1 for a list of Special Rules Areas and where to find information about them).

Pilots need to familiarise themselves with these routings during the flight planning stage.

Example: The GREENBUSHES 08 VFR Departure from Chief Dawid Stuurman Airport (Port Elizabeth) to the west reads: "After departure runway 08 maintain runway track to 1000' ALT, then turn left track 350 and climb to 1500'. At 1500' ALT proceed direct to Greenbushes mast, at Kragga Kamma Road set course (left turn) for Maitland River Mouth, routing south of the N2 road. Report passing abeam Seaview". The Tower controller would instruct a pilot: ABC, after departure runway 08 follow the Greenbushes 08 VFR departure, wind 100 at 10 knots, cleared for take-off runway 08. – He will expect the pilot to fly exactly as described in the VFR Departure above, along the given routing and at the given altitudes, and will expect a call when the pilot passes Seaview.

## 8.12 Relay

Sometimes an aircraft is out of range of an ATSU, but needs to communicate with ATC. This can be done with a relay via another, higher flying aircraft. When in need of a relay, listen out on the frequency on which you should be talking to the controller. Note the call sign of anybody else who you hear transmitting on this frequency, and when there is radio silence, call this other aircraft and ask for a relay. The other aircraft will ask the controller for permission before accepting your relay.

Example : Shortly before landing at a small airfield where you know is no cell phone reception you would like to close your Flight Plan and cancel Search and Rescue. You hear Safair 345 talking to the Johannesburg North controller. After that conversation is finished, you say:

ABC: Safair 345, Zulu Sierra Alpha Bravo Charlie, request a relay

FA345: Johannesburg, Safair 345, request a relay for ABC

Area: Safair 345, go ahead with the relay

FA345: Alpha Bravo Charlie, go ahead

ABC: ABC, destination in sight, request to cancel Search and Rescue

FA345: Johannesburg, ABC has the destination in sight and requests that Search and Rescue be cancelled

Area: Thank you, Safair 345, please advise ABC Search and Rescue has been cancelled at 1735

FA345: ABC, your Search and Rescue has been cancelled at 1735

ABC: Thank you for the relay, enjoy your flight, ABC.

The same procedure applies if you are being asked to relay for another aircraft.

## 8.13 Helicopter Communications

AIP ENR 1.8 (7)

Communications for helicopters are basically the same as for fixed wing aircraft. The main difference being that helicopters usually do not take off or land on a runway, but in an uncontrolled general aviation area. In that area a helicopter will call the Tower Controller before take-off, and will be told the wind and any other pertinent information as well as something like “Hotel Echo Lima, lift off, remain west of the main runway, report airborne”.

When the helicopter returns for landing, the pilot will report to Tower before entering the CTR or ATZ. He will be given inbound clearance with any other pertinent information, such as the QNH or information about other fixed or rotary wing traffic. If the helicopter is going to land on a runway or taxiway, he will be given landing clearance like any other aircraft. If the helicopter lands in an uncontrolled area, though, he will only be passed information, and communications are very similar to Aerodrome Flight Information Service: “Hotel Echo Lima, land at your own discretion, the wind is zero five zero at 10 knots. Report safe on the ground”.

Most (but not all) older helicopter registrations in South Africa start with ZS-H.. (Helicopter) or ZS-R.. (Rotorcraft), e.g. ZS- **H**BF or ZS- **R**MT. New helicopters are registered as ZT-R... Those that do not conform to that should always prefix their call sign with “Helicopter” to alert ATC and/or other traffic, for example: “Helicopter ZS-ABC! ”.

## 8.14 Sample VFR Flights

The following are examples of two complete VFR flights. The first one is conducted mainly in controlled, and the second one totally in uncontrolled airspace. It cannot be emphasised enough that although procedures in controlled and uncontrolled airspace sound very similar, the implications are totally different and pilots must be aware of what type of airspace they are flying in.

In the following samples, groups of letters and figures have not been spelled out. It is understood, though, that these are spoken in individual letters according to the phonetic alphabet and that numbers are spoken separately, i.e. JGP is Juliet Golf Papa , 118.1 is one one eight decimal one etc.

For a typical IFR flight see paragraph 14.5.3.

### 8.14.1 A VFR Flight From Cape Town to George

This flight takes off and lands at a controlled airfield. Initially, the flight is in controlled airspace (Cape Town CTR and TMA), it then proceeds into uncontrolled airspace until it re-enters controlled airspace (the George TMA and CTR). A flight plan is mandatory for this flight, since it is conducted at least partly in controlled airspace. (VFR aircraft in Cape Town are usually kept below the TMA, so the flight below is not a “real-life” example. However, it demonstrates the principle).

#### In the parking area on 122.1:

A/c: Clearance Delivery, ZS-KKU  
 Delivery: KKU, Cape Town Clearance Delivery, go ahead  
 A/c: KKU, VFR to George as per flight plan, request clearance  
 Delivery: KKU is cleared Cape Town to George. After departure runway 19 follow the Bottelary 19 departure, contact Approach 119.7 when clear of the CTR. Your squawk is 4701  
 A/c: Cleared Cape Town to George, after departure 19 to follow the Bottelary 19 departure, contact Approach 119.7 when clear of the CTR, squawk 4701, KKU.  
 Delivery: Read-back correct, contact Ground 121.9 for start-up and taxi.  
 A/c: Contact Ground 121.9 for taxi, KKU

#### After start on 121.9:

A/c: Ground, ZS-KKU at the Safair hangar, received Information Golf, 2 on board, request taxi for E  
 GND: KKU, Golf is current, taxi via Hotel, left Bravo, right Echo, to the holding point runway 19 Echo  
 A/c: Taxi Hotel, Bravo, Alpha, Echo to the holding point runway 19 Echo, KKU.

**The aircraft starts taxiing, then sometime:**

GND: KKU, contact Tower 118.1 when ready for departure

A/c: Tower 118.1 when ready, KKU

**At the holding position** , and when ready to depart, the pilot changes frequency, listens out and calls Tower. Tower has been notified by Ground and is expecting the call – the aircraft does not need to establish contact, but can go straight ahead with the message.

A/c: Cape Town Tower, ZS-KKU ready

TWR: KKU cleared take-off, wind 210/15

A/c: Cleared take-off, KKU.

The aircraft takes off, follows the given Departure, and **when leaving the CTR** , says:

A/c: Tower, KKU leaving the CTR

TWR: Contact Approach on 119.7

A/c: Approach 119.7, KKU.

The pilot changes frequency to 119.7, listens out and calls:

A/c: Cape Town Approach, ZS-KKU

APP: KKU, radar identified on a squawk of 4701, climb to FL95, report reaching

A/c: Climb FL95, report reaching, KKU.

**When the aircraft has levelled off at FL95:**

A/c: Cape Town Approach, KKU, maintaining FL95

APP: KKU, report Caledon

A/c: Report Caledon, KKU.

**At Caledon** the aircraft is leaving controlled airspace:

A/c: Cape Town Approach, KKU abeam Caledon at 32, FL95, estimate George at 1036

APP: KKU, contact Cape Town Information on 131.125, good day

A/c: Information on 131.125, KKU.

**Immediately afterwards on 131.125:**

A/c: Cape Town Information, ZS-KKU, good morning

INFO: KKU, Cape Town, good morning. No reported traffic FL95, report at Swellendam

A/c: Report at Swellendam, KKU.

**At Swellendam:**

A/c: Cape Town Information, KKU Swellendam 0953, FL95, estimate Riversdale at 1011

INFO: No reported traffic FL95, contact George Approach on 128.2 ten minutes before entering controlled airspace

A/c: George Approach 128.2 ten minutes before entering their airspace, KKU.

**Just before Riversdale** the pilot changes frequency to 128.2 and listens out. When nobody else is talking, he says:

A/c: George Approach, ZS-KKU

APP: ZS-KKU, George Approach, go ahead

A/c: ZS-KKU, Riversdale 1012, FL95, estimate George at 37

APP: ZS-KKU, runway 11 in use at George, QNH 1023, route direct to final 11, report for descent

A/c: Report for descent, ZS-KKU

**When the pilot wants to descend:**

A/c: George Approach, ZS-KKU request descent

APP: ZS-KKU descend as required, contact George Tower 118.9 passing 4500', QNH 1020.

A/c: Descend as require, contact George Tower passing 4500', QNH 1020, ZS-KKU.

**When passing 4500' on the descent:**

A/c: George Tower, ZS-KKU passing 4500'

TWR: ZS-KKU, report final 11

A/c: Report final runway 11, ZS-KKU

**On final:**

A/c: Tower, ZS-KKU final 11

TWR: The wind is 100 at 15 knots, ZS-KKU cleared to land

A/c: Cleared to land, ZS-KKU.

**After the aircraft has landed and is still on the runway:**

TWR: ZS-KKU, next left, right Alpha to the apron, report marshaller in sight

A/c: Next left, right Alpha to the apron, report marshaller in sight, ZS-KKU.

**When the aircraft enters the apron** and can see the marshaller:

A/c: ZS-KKU has the marshaller in sight

TWR: Follow the marshaller, good day

A/c: Follow the marshaller, ZS-KKU.

## 8.14.2 A VFR Flight Lichtenburg to Bethlehem

This flight is conducted at FL75 entirely in uncontrolled airspace. Neither of the two airfields is situated within a General Flying Area or has been allocated a specific frequency, therefore the pilot uses 124.8 at both airfields for take-off and landing. After climbing through 1500' AGL, the pilot contacts the Flight Information Service for the Johannesburg South Sector on 119.5. Twenty miles before Bethlehem he crosses into the East Sector of the FIR and contacts the Flight Information Service for the Johannesburg East Sector on 129.1.

**In the parking area** , before starting to taxi, the pilot will listen out on 124.8 and say:

A/c: Lichtenburg, Warrior ZS-JGP taxiing to the holding point runway 18.

**At the holding point** when ready for take-off on 124.8:

A/c: Lichtenburg Traffic, JGP entering runway 18 for take-off. After take-off left turn out, climbing to flight level 075 for a flight to Bethlehem.

The aircraft takes off, turns out left, **and when leaving the circuit** , the pilot broadcasts:

A/c: Lichtenburg Traffic, JGP leaving the circuit, climbing to flight level 75 for a flight to Bethlehem.

When **passing 1500' AGL** the pilot changes to the FIS frequency for the Johannesburg FIR South Sector, 119.5, and says:

A/c: Johannesburg Information, ZS-JGP

FIS: JGP, Johannesburg Information, go ahead

A/c: Good morning. JGP is airborne Lichtenburg at 0955 for a flight to Bethlehem, climbing to FL75, estimate the sector boundary at 1110, Bethlehem at 1120

FIS: JGP, no reported traffic for your flight, report abeam Klerksdorp

A/c: Report abeam Klerksdorp, JGP.

**Abeam Klerksdorp** still on 119.5:

A/c: Johannesburg Information, JGP

FIS: JGP, Jo'burg Information, go ahead

A/c: JGP abeam Klerksdorp, 1020, FL75, estimate Bethlehem at 1120

FIS: JGP, traffic is a Cessna 210, Welkom to Lanseria FL95, he's estimating abeam Klerksdorp at 1025. Report at the sector boundary

A/c: Looking out. To report at the sector boundary, JGP.

**At the boundary** still on 119.5:

A/c: Johannesburg Information, JGP is at the boundary, 1110, FL75, estimate Bethlehem at 20

FIS: Thank you, contact Johannesburg Information on 129.1, good day

A/c: Information 129.1, good day.

The pilot tries to contact Jo'burg East on 129.1, but is out of range and gets no reply:

A/c: Johannesburg Information, ZS-JGP

After having listened out for a reply for a short while, still on 129.1:

A/c: Traffic in the Bethlehem area, Warrior JGP is approaching Bethlehem from the north-west, FL75, changing frequency to 124.8.

The pilot now announces his arrival to any possible traffic at Bethlehem on 124.8:

A/c: Bethlehem Traffic, Warrior ZS-JGP 5 nm northwest of Bethlehem at 7500 feet, joining overhead Bethlehem at 7500 feet in two minutes.

**When overhead** , after having seen the windsock and established the runway to use:

A/c: Bethlehem Traffic, JGP overhead Bethlehem at 7500 feet, to join left downwind runway 29.

**On downwind, base and final** depending on traffic on 124.8:

A/c: Bethlehem Traffic, JGP left downwind / left base / final runway 29.

**After landing** , when vacating the runway:

A/c: Bethlehem, JGP vacating runway 29, taxiing to the hangars.

### 8.14.3 Drone (RPAS or UAS) flights

Drone pilots, i.e. pilots of Remotely Piloted Aircraft Systems (RPAS) or Unmanned Aircraft Systems (UAS) – unfortunately, both terms are used in the current regulations - should describe their position and intentions as accurately as possible.

When flying in **controlled airspace** , drone pilots will have filed a flight plan, Air Traffic Control (ATC) will have been informed, will be in contact with the drone pilot, and will notify manned aircraft in the vicinity of the drone activities.

When flying in **uncontrolled airspace** , drone pilots will need to broadcast their position on the relevant TIBA frequency. A broadcast should be made every 5 to 10 minutes, and when the drone pilot sees or hears a manned aircraft in the vicinity. The call sign of the drone should be prefixed with the word “RPAS”. But since many pilots of manned aircraft are not familiar with the term, it is recommended to also include the word “Drone”.

Sample broadcasts in uncontrolled airspace:

Traffic in the Pretoria GF no1, this is RPAS ZT-TYZ, a drone operating 2 nm west of Freeway airstrip between ground and 400ft AGL for the next 20 minutes, TYZ.

Traffic at Sunset Beach, RPAS ZT-TUV, drone operations for a film shoot just west of the children’s play area, up to 200 feet above ground, ZT-TUV.

Traffic in the Petrusburg area, RPAS ZT-TAB, a drone, doing a survey flight in the area between the N8, the R48 and Liebenbergspan at 400 feet above ground, drone TAB.

Morningstar traffic, RPAS ZT-TMT, drone operations just north of the airfield over the polo field. We have the aircraft in the circuit in sight, and will remain at 50’ AGL until the aircraft has landed.

# 9

## Radar, SSR, ADS-B

### 9.1 Primary Radar

AIP ENR 1.6

Primary RADAR ( **RA**dio **D**etection **And** **R**anging) uses the principle of pulse technique to determine the bearing and distance of an object from a station. The transmitter sends out a narrow beam of radio pulses, sweeping horizontally through 360°. The pulses are reflected by objects, e.g. aircraft, the earth, mountains, or by precipitation (rain, hail). The reflected beam is received again, and the distance to the object is computed from the time the signal takes to return to the receiver..

Primary radar is a self-contained system, requiring no equipment other than the transmitter/receiver in the aircraft and no action by the pilot. It is susceptible to radar clutter from rain, cloud or terrain.

Primary RADAR is employed in various aviation applications:

- ! weather radar for detection of precipitation, especially in thunderstorms,
- ! distance measuring equipment (DME) to indicate to a pilot the aircraft's distance from a ground station,
- ! radio altimeter to show the pilot the height of the aircraft above ground,
- ! precision approach radar (PAR) as a ground controlled approach aid,
- ! basic primary radar used for detection of aircraft by the air traffic controller, although now largely replaced by SSR or ADS-B in civilian use.

### 9.2 Secondary Surveillance Radar (SSR)

AIP ENR 1.6

Secondary radar relies on a transmitter (interrogator) sending out a group of pulses which is received by a distant suitable receiver. These signals are recognised at the receiver, causing its transmitter (transponder) to reply on a different carrier frequency.

Secondary Radar is used in:

- ! secondary surveillance radar (SSR) for improved air traffic control,
- ! airborne collision avoidance in the Traffic Collision Avoidance System (TCAS).

Secondary surveillance radar has largely replaced primary radar as a means of identifying aircraft for air traffic control purposes. The advantages over primary radar are that aircraft can be positively identified with the aircraft's SSR code or call-sign displayed on the controller's radar screen. Furthermore, with all but the very basic transponders, ATC can also see the pressure altitude of the aircraft (i.e. the vertical difference between the 1013 hPa level and the aircraft), together with the aircraft's groundspeed and present rate of climb or descent. Thus it reduces verbal communication between pilot and controller and helps greatly in traffic separation and traffic flow control.

Mode S transponders answer only when interrogated, so that an air traffic controller can selectively enquire data from specific aircraft, thus reducing clutter on his screen and helping him to resolve traffic conflicts even further. Mode S transponders are also used in Traffic Collision Avoidance Systems (TCAS), where the aircraft's transponder interrogates the transponders of near-by aircraft and displays that information in the cockpit. Airborne Collision Avoidance Systems (ACAS II) take this a step further and also offer a visual and aural traffic resolution to the pilot, such as "climb, climb now" or similar.

The ground interrogator sends its group of pulses on a frequency of 1030 MHz and receives on 1090 MHz, while the aircraft's transponder sends on 1090 MHz and receives on 1030 MHz. Thus neither the ground nor the airborne equipment will receive its own reflected signals, thereby eliminating any radar clutter. The aircraft's equipment recognises the mode of interrogation by the time spacing between pairs of interrogating pulses, and it will reply with the code selected by the pilot on the transponder set by the inclusion or omission of any of up to 12 pulses.

ATC will assign a specific transponder code (theoretically 0000 to 7777) to an individual aircraft, which the pilot must select on the set. After start-up the pilot turns the transponder to STANDBY to warm up. The transponder must be selected to ALT when entering the runway on departure or when instructed by ATC to do so with the words "Squawk (code)".

When changing codes the pilot must first select STANDBY, then change the code, then switch the transponder back to ALT. This prevents spurious codes being transmitted while the pilot dials through different numbers.

When the controller's radar screen is quite cluttered with traffic, he may ask a pilot to "squawk ident", after which the pilot should press the IDENT button on his transponder set. This will result in the corresponding 'blip' on the controller's radar screen to fill in, thus enabling him to better identify a particular aircraft on his screen. After landing or when asked to "Squawk standby", the pilot must switch the transponder back to the STANDBY setting.

Three transponder codes have been assigned internationally for specific purposes. These can be memorised by the mnemonic "ICE" from lowest to highest:

|      |   |   |                                                    |
|------|---|---|----------------------------------------------------|
| 7500 | I | = | Interference, e.g. hijack or other act of violence |
| 7600 | C | = | <b>C</b> ommunications failure, radio failure      |
| 7700 | E | = | <b>E</b> mergency, aircraft emergency              |

The 7 civilian SSR-equipped Air Traffic Control centres in South Africa (Bloemfontein, Cape Town, Durban, East London, George, Johannesburg and Port Elizabeth) are integrated with military radar stations, so that there is almost complete radar coverage of all South African airspace above approximately 10000'.

Aeroplanes which have not been assigned a transponder code should squawk 2000 in all airspaces, except in the circuit of an uncontrolled airfield below 3000' AGL. The default squawk code for helicopters is 2600.

## 9.3 ADS-B

SACAR 91.05.05, SACATS 91.05.05, AIC D010/2024

Automatic Dependent Surveillance-Broadcast (ADS-B) is a surveillance technology in which an aircraft determines its own position via satellite navigation (GNSS) and periodically broadcasts it. This can be received by ATC and by other aircraft and thus replaces Secondary Surveillance Radar (SSR). ADS-B is "automatic" in that it requires no pilot or external input. It is "dependent" in that it depends on data from the aircraft's navigation system.

ADS-B operates on 1090 MHz like SSR, or at 978 MHz. It requires the aircraft to be equipped with a certified GNSS receiver as well as a datalink (the ADS-B unit). The ground infrastructure is significantly cheaper to install and operate than secondary radar installations.

ADS-B "out" means that an aircraft periodically broadcasts information about itself, such as identification, current position, altitude and velocity, through the onboard transmitter. This can be received by Air Traffic Control as well as by other aircraft.

ADS-B "in" is the added capability for aircraft to receive information from other aircraft (position, altitude etc.) as well as from ground-based stations (real-time weather, NOTAMs etc.).

ADS-B coverage is significantly better than SSR, since signals travel via satellite, and thus are not restricted by terrain. The disadvantages, as with all new technology, is that not all aircraft are equipped with ADS-B yet. ADS-B "out" is already compulsory in South African class A airspace, and it is envisaged to become compulsory in class C airspace by 2027.

## 9.4 Radar Service and Control

AIP ENR 1.6, AIC D 004/2024

Secondary Surveillance Radar (and/or ADS-B) may be used to provide aircraft with a variety of services, such as to

- ! separate traffic from each other,
- ! radar vector aircraft where necessary,
- ! monitor the aircraft's flight path and response to ATC instructions,
- ! warn a pilot about significant deviations from normal flight path,
- ! assist aircraft in emergencies,
- ! give warnings of other aircraft on possible conflicting flight paths.

When a pilot enters a radar environment, i.e. after taking off and climbing into a TMA or when laterally entering controlled airspace, the pilot should establish contact and state his altitude, e.g. "Approach, ABC passing 2000 feet". The controller will advise the pilot that he has identified the aircraft and verified its altitude with words such as "ABC, radar identified, 2 miles south of CTV". The pilot should check that he is in fact at the stated position. With that it is established that the aircraft has been positively identified and the aircraft can be provided with **Radar Service**. The aircraft can now be radar monitored, it will receive **Radar Advisory Service** (RAS) in class F airspace and **Radar Information Service** (RIS) in class G airspace. It will receive information about other traffic seen on radar, can be helped in an emergency, and can be placed under radar control when necessary. An aircraft provided with Radar Service will **not be separated from terrain**.

In controlled airspace, after an IFR aircraft has been radar identified, the controller can place it under **Radar Control** with the words "ABC radar identified, under radar control". From then on the aircraft **will be separated from terrain**, from other traffic and can be radar vectored. When radar control is no longer provided the controller will notify the aircraft with the words "ABC, radar control terminates". Radar Control only applies in controlled airspace; when an aircraft leaves controlled airspace, any radar control will automatically be terminated without the pilot being notified.

Radar Terrain Clearance charts are published in the Aerodrome section of the AIP and by commercial entities such as Jeppesen – they show the area around the airport with sectors marked with minimum altitudes. Radar service can only be provided above the altitudes shown and only when the aircraft has been positively identified.

# 10

# Radio Failure

## 10.1 Fault Finding

AIC D005/2018

When calling an ATSU and getting no reply, the pilot should at first presume that the "fault" lies with himself or his radio installation and should check the following:

- ! correct frequency selected,
- ! aircraft within range of the VHF station (see paragraph 13.4.1),
- ! aircraft master switch is ON,
- ! radio master switch is ON (if fitted),
- ! radio set is ON,
- ! audio panel microphone switch set to the correct radio (a common mistake),
- ! audio panel headset switch set to the correct radio (another common mistake),
- ! squelch set correctly, if fitted,
- ! volume control turned up sufficiently (also a common mistake!),
- ! headset plugged in properly,
- ! push-to-talk switch not stuck (press the push-to-talk switch quickly a few times and listen to the tell-tale "transmit-click"),
- ! battery not flat, alternator charging (check ammeter),
- ! relevant circuit breakers in,
- ! check that the radio is transmitting (many radios show a "T" or similar).

If, after a second call to the ATSU there is still no reply, the pilot should try the following:

- ! tapping the microphone lightly,
- ! another radio set (if fitted) or a handheld radio,
- ! another headset (if available),
- ! checking connections and fittings as far as possible,
- ! another frequency (previous frequency, TIBA frequency etc.).

The above should be used with common sense. For example, if there is total silence in a normally busy environment, the fault is most likely on the receiving side, e.g. volume not turned up, audio panel set incorrectly etc. If the pilot can hear other aircraft or ATC, the problem seems to be with transmitting, e.g. faulty microphone.

If the above fault finding procedures do not solve the problem, and if the aircraft is definitely within range of the ground station, the pilot must assume that he is experiencing a radio communication failure (RCF). He should now select 7600 (Radio failure) on his transponder.

If the pilot can hear transmissions from other stations the aircraft most likely is experiencing a **transmitter failure**. In that case ATC will hear the pilot and might request the aircraft to execute a specific manoeuvre, e.g. an orbit to the left, to switch the landing lights on, or to squawk ident to find out whether the pilot can hear the controller's transmissions. If that is successful, the controller will issue instructions, which the aircraft should follow.

If the radio is completely silent, it could be **either a receiver or a total radio failure**. Because the pilot is unable to distinguish between these two while in the air he should assume that the transmitter is working and transmit blind. Those transmissions are prefixed with "transmitting blind", e.g. "Rand, JGP transmitting blind, final 36" and should be made twice.

" **Transmitting blind** means a transmission from one station to another station in circumstances where two-way communication cannot be established but where it may be possible that the called station is able to receive the transmission. This is a radio failure situation. It must not be confused with **broadcasting**, which means a transmission of information relating to navigation that is not addressed to a specific station. These are, for example, the normal TIBA messages.

## 10.2 Radio Failure – VFR Flights

AIP ENR 1.5.2, SACATS 91.06.16, ICAO Annex 2 paragraph 3.6.5.2

### 10.2.1 In Uncontrolled Airspace

When the radio of a VFR flight fails in uncontrolled airspace, the pilot must remain in uncontrolled airspace. He may continue his flight to an uncontrolled airfield, squawk 7600, and switch the landing lights on. Approaching the airfield for landing he must adhere to normal joining procedures for uncontrolled airfields, make the appropriate radio calls, prefixing all messages with "transmitting blind" and keep a good lookout. If the aircraft was in contact with an ATSU before the radio failure or if a flight plan was filed, the pilot must contact the nearest ATSU by telephone after landing.

### 10.2.2 In Controlled Airspace

A VFR pilot flying in controlled airspace and experiencing a radio failure should squawk 7600 on the transponder and leave controlled airspace by the shortest possible route.

He should make a "transmitting blind" report to the ATSU he was in contact with before the failure, stating that the radio failed and what his intentions are. After that he must proceed to land at an uncontrolled airfield, observing all the rules for joining

the circuit, transmitting blind and keeping a careful lookout for other traffic. After landing, he must telephone the appropriate ATSU as soon as possible.

If proceeding to an uncontrolled airfield is not possible (at night, for example) the pilot should continue to a controlled airfield, switch the landing light on, join overhead the airfield 1000' above circuit altitude, and carefully observe other traffic. When safe, he should join the circuit, preferably on a left downwind (or as specified for that airfield), and space himself so that other aircraft are not endangered or inconvenienced. He should also make "transmit blind" reports, stating his position and intentions. Again, after landing, he must telephone ATC.

## 10.3 Radio Failure - IFR Flights

AIP ENR 1.5, SACATS 91.06.16, ICAO Annex 2 paragraph 3.6.5.2

### 10.3.1 When Following a SID or STAR

Aircraft cleared for a Standard Instrument Departure (SID) or a Standard Terminal Arrival Route (STAR) must comply with the specific Communications Failure Procedure as published with the SID or STAR on the relevant instrument chart.

The same applies to aircraft **approaching an airport where STARs are in use**, but which have not yet been cleared for a STAR (the pilots must select the most likely STAR for the most likely runway). The following procedures do NOT apply to such aircraft (for SIDs and STARs see paragraphs 14.2.1 and 14.4.1).

Note that STARs also provide for communications failure procedures to be followed in the event of a missed approach.

### 10.3.2 When cleared for the approach

Aircraft already cleared for an Instrument Approach Procedure and experiencing radio failure in IMC or VMC must comply with the Communication Failure Procedure stipulated on the approach plate. The same would apply to aircraft who have been told to "expect" a certain Instrument Approach Procedure.

If the relevant approach plate does not state a specific Communication Failure Procedure: aircraft already cleared for an Instrument Approach Procedure at FACT, FALF and FAOR and experiencing radio failure in IMC or VMC must continue the approach and land (AIP ENR 1.5.3, 5, and 6).

At other airports the aircraft should treat the communication failure on the approach like any other comms failure and proceed according to paragraphs 10.3.3 or 10.3.4. below as applicable.

### 10.3.3 In VMC

An aircraft on an IFR flight plan, not on a SID or STAR, not (yet) cleared for an instrument approach at FACT, FALE or FAOR **and** in VMC at the time of the radio failure shall continue in VMC and land at the nearest suitable aerodrome, following the same procedures as described under VFR Flights in paragraph 10.2.

“The nearest suitable aerodrome” for a light aircraft will usually be an uncontrolled airfield. For a large aircraft it might well be the original destination, e.g. a controlled airport. In that case and also if the aircraft is in VMC at the time of the radio failure, but cannot remain in VMC, the aircraft should follow the “in IMC” procedures below.

### 10.3.4 In IMC

The following applies to aircraft on an IFR flight plan in IMC, who are not following a SID or STAR, are not approaching an airfield where STARs with published communication failure procedures are in use, and are not yet cleared for an instrument approach procedure at FACT, FALE or FAOR:

1. Proceed according to the current flight plan.

If ATC has restricted the flight to a **lower** level than specified in the flight plan:

- a) if the flight takes place in a **non-radar environment**, climb to and/or maintain this last cleared level (or minimum safe altitude, if that is higher) and speed for **twenty minutes** following the aircraft's failure to report its position over a compulsory reporting point.
- b) if the flight takes place in a **radar environment**, climb to and/or maintain this last cleared level (or minimum safe altitude, if that is higher) and speed for a period of **seven minutes**, from the latest of:
  - i) the time the last assigned level (or minimum altitude) was reached,
  - ii) the time the transponder was set to 7600, or
  - iii) the aircraft was not able to report at a compulsory reporting point.

After that, the climb should be continued to flight plan level.

2. Proceed **at the Flight Plan level** to the navigational facility (usually a VOR) serving the aerodrome of destination and hold **at cruise** level over this facility until the last Expected Approach Time (EAT). If no EAT has been received hold until the Estimated Time of Arrival (ETA) according to the current flight plan. Then start the descent.
3. Complete the Instrument Approach Procedure for the destination aerodrome.
4. Land within 30 minutes after commencing the descent.

Example 1: The pilot originally gave the controller an ETA of 1005, no specific EAT (approach time) was mentioned.  
The pilot should fly to the destination VOR at flight plan level, hold in level flight until 1005, commence the descent and approach, and land before 1035.

Example 2: The ETA was 1310.  
The controller gave the pilot an EAT (approach time) of 1318.  
The pilot should fly to the VOR and hold there until 1318, commence the descent and approach, and land before 1348.

While carrying out the procedures outlined above, it makes sense to try to contact ATC via other means: satellite phone, mobile phone, ADS-B etc., if available - and if it is safe to do so.

# 11

# Emergency Communications

## 11.1 Distress

AIP GEN 3.6, AIP ENR 1.8 (7), AIC 40.2, AIC D005/2018, ICAO Doc 9976

Distress is defined as a situation in which an aircraft is in serious or imminent danger and **in need of immediate assistance**. A distress situation is, for example, an engine fire in flight or an engine failure in a single engine aircraft.

An aircraft in distress should squawk 7700 (unless it has been assigned a discreet transponder code by ATC) and make a radio call containing as much as possible of the following on the last frequency in use, on any other useful frequency, or on the VHF international emergency frequency 121.5 MHz (which is monitored by ATC, by airliners, the military and rescue organisations):

- ! the word MAYDAY spoken three times: “mayday, mayday, mayday”,
- ! the station addressed,
- ! the identification of the aircraft,
- ! the nature of the distress,
- ! the intentions of the pilot,
- ! present position, altitude and heading,
- ! any other useful information.

“ the word “mayday” has nothing to do with the 1<sup>st</sup> of May; it stems from the French “m’aidez”, which means “help me”

Example: Mayday, mayday, mayday, Cape Town Information, Juliet Golf Papa, engine fire, intend landing on a field north of the N2 approximately 5 nm west of Caledon, overhead that field at FL75, heading west.

A pilot observing another aircraft in distress shall keep this other aircraft in sight, note the position of the distressed aircraft, contact an ATSU ground station, and then carry out instructions from that ground station.

**Distress communications have absolute priority over all other traffic messages**, and all other communications shall cease until either the distress is cancelled or the distress traffic has been transferred to a different frequency. Normally, then, a pilot hearing a distress message should keep quiet and let other, more competent people deal with the distress. However, if a pilot hears a distress message that is not acknowledged by any other station, he shall first acknowledge to the distressed aeroplane, then try to relay to the nearest ATSU or any other station which may be in a position to help.

If the aircraft declaring the emergency is transferred to another frequency it is helpful to add the suffix “Mayday” to the call-sign at least on first contact with the new station – to make absolutely sure that the status of the distressed aircraft is known, e.g. “ZS-ABC Mayday ! ”.

A controller busy with an emergency on a particular frequency may impose radio silence, which means he wants all other, non-distress traffic, to remain silent to keep the frequency open for the distressed aircraft.

Should the aircraft that declared an emergency be able to resolve the problem and not be in need of assistance anymore, the pilot should transmit: “Cancel Mayday” and give the reason for the cancellation. Example: ABC, cancel Mayday, engine fire successfully extinguished”.

When the situation is resolved, or when the distress communications have been transferred to another frequency, the relevant ground station will cancel the distress communications with the words “Distress traffic ended” .

#### **Note on fuel emergencies:**

“Minimum fuel” must be declared as soon as it becomes clear to the pilot that any\_\_\_\_ change to the existing clearance, i.e. any further delay, would result in the aircraft landing with less than the planned final reserve fuel.

“Mayday mayday mayday fuel” must be declared as soon as it becomes clear to the pilot that following the existing clearance would result in the aircraft landing with less than the planned final reserve fuel.

Declaration of “minimum fuel” or “Mayday mayday mayday fuel” will alert controllers to a critical situation while there is still time to remedy the situation.

## 11.2 Urgency

AIP GEN 3.6, AIP ENR 1.8 (7), AIC 40.2, AIC D005/2018, ICAO Doc 9976

Urgency is defined as a situation concerning the safety of an aircraft, ship or other vehicle or of some person on board or within sight, but which **does not require immediate assistance**. An urgency situation might be a low oil pressure indication, or a passenger experiencing a medical emergency, or similar.

An aircraft sending out an urgency message should state as much as possible of the following on the last air/ground frequency in use, on any other useful frequency or on the VHF international emergency frequency 121.5 MHz:

- ! the word PAN-PAN spoken three times: “pan-pan, pan-pan, pan-pan”,
- ! the station addressed,
- ! the identification of the aircraft,
- ! the nature of the urgency,
- ! the intention of the pilot,
- ! present position, altitude and heading,
- ! any other useful information

Example: Pan-pan, pan-pan, pan-pan, Lanseria Tower, Foxtrot Uniform November, a passenger seems to have a heart attack. We are just south of Hartebeespoort Dam, 7000 feet, heading south. Intend returning to Lanseria for landing, request the shortest approach possible and request an ambulance on the apron upon landing.

Urgency communications have priority over all other traffic messages except distress, and all other communications shall take care not to interfere with the transmissions of urgency traffic. Normally, then, a pilot hearing an urgency message should keep quiet and let other, more competent people deal with the emergency. However, if a pilot hears an urgency message that is not acknowledged by any other station, he shall first acknowledge to the aeroplane in trouble, then try to relay to the nearest ATSU or any other station that may be in a position to help.

If the aircraft declaring the urgency is transferred to another frequency it is helpful to add the suffix “Pan” to the call-sign at least on first contact with the new station – to make absolutely sure that the status of the aircraft is known, e.g. “ZS-ABC Pan !”.

# 12

# Search & Rescue

Search and Rescue service in South Africa is provided by the South African Search and Rescue Organisation (SASAR) on behalf of the Department of Transport. The Aeronautical Rescue Coordination Centre (ARCC) in Johannesburg, when alerted by an ATSU or other “alerting post” (Police, Harbour Masters etc.), is responsible for searching for aircraft believed to be missing and aircraft that have been involved in a crash or forced landing, and to coordinate the rescue of survivors of aircraft accidents and forced landings.

## 12.1 Flights Subject to Search and Rescue

AIP GEN 3.6, AIC 40.2, AIC D003/2022 (App. B)

Search and Rescue (SAR) is **automatic** for:

- ! flights taking off from a **manned** aerodrome (where an Air Traffic Service Unit is in operation, i.e. a tower controller is on duty), when the departing aircraft fails to report within five minutes after being expected to do so, and no contact can be made,
- ! flights in **controlled airspace** which are 5 minutes overdue at reporting points (except flights crossing an airway at right angles, for which no Flight Plan has been filed), and
- ! flights to a **manned** aerodrome, which fail to arrive.

Additionally, Search and Rescue **can** be requested by the pilot for flights to/from an unmanned aerodrome and in uncontrolled airspace. This request must be stated in item 18 of the Flight Plan under RMK/SAR.... Remember though, that a Flight Plan is only a prior notification of an intended flight. Only once the flight plan has been activated with an actual time of departure will SAR be activated, should the need arise (see also paragraph 6.3).

Search and Rescue requirements must be stated in the Flight Plan in one of the following formats:

**RMK/SARNML** (SAR normal) can only be requested when the destination is a **manned airfield**. Search and Rescue will start when the aircraft:

- ! is 5 minutes overdue at reporting points in **controlled** airspace only, or
- ! does not arrive at the destination.

To specify RMK/SARNML is really only useful if the aircraft is in controlled airspace for the whole flight. If the aircraft leaves, for example, Lanseria, then flies in uncontrolled airspace until it finally enters the Bloemfontein TMA or CTR, nobody would come and look for it after leaving the Lanseria CTR or TMA until its scheduled arrival time in Bloemfontein.

**RMK/SARNMLENROUTE** (SAR normal en-route) can only be stated when the destination is a **manned airfield** and at least part of the flight is conducted in **uncontrolled airspace**. It must also be ensured that it will be possible to maintain 2-way radio contact with ATC throughout the flight. SAR will start when the aircraft:

- ! is 5 minutes overdue at reporting points in **all** airspace, or
- ! does not arrive at the destination.

RMK/SARNMLENROUTE would possibly be the safest way to request SAR for the above example flight from Lanseria to Bloemfontein. After leaving the Johannesburg Special Rules Area the pilot can speak to Johannesburg South Information and Johannesburg Central Information all the way, and is subject to SAR when 5 minutes overdue at every reporting point along the way.

**RMK/SARFAXxhmm** (SAR place time - where FAXx is the ATSU, and hhmm is the time at which Search and Rescue should start if it has not been cancelled by the pilot with the nominated ATSU) can only be specified when the destination is **unmanned**. SAR will start when the aircraft:

- ! is 5 minutes overdue at reporting points in **controlled** airspace, or
- ! at the time specified if SAR has not been cancelled by the pilot.

A pilot flies from Kleinsee to Vredendal, at 1500' AGL along the coast. He will most probably not be able to be in contact with any ATSU throughout the flight. He specifies RMK/SARFACT1230, so that somebody will at least search for him at 12:30 UTC, if he has not phoned Cape Town ATSU to close his Flight Plan. Since he probably will not be able to activate his Flight Plan in the air, he must phone ATC before take-off to activate his Flight Plan.

**RMK/SARFAXxhmmENROUTE** (where FAXx is the ATSU, and hhmm is the time at which Search and Rescue should start if it has not been cancelled by the pilot with the nominated ATSU) can only be specified when the destination is an **unmanned airfield** and at least part of the flight is conducted in **uncontrolled airspace**. In this case the pilot must ensure that it will be possible to make position reports throughout the flight. SAR will start when the aircraft:

- ! is 5 minutes overdue at reporting points in **all** airspace, or
- ! at the time specified if SAR has not been cancelled by the pilot.

An aircraft taking off from and landing again at Ulundi for conducting a survey in the area at 6000' could specify RMK/SARFALE1230ENROUTE, and make regular "operations normal" position reports every, say, 20 minutes. SAR would

be commenced when the aircraft misses one of the routine “ops normal” reports, as well as if the flight plan has not been closed by 12:30 UTC.

**RMK/SARFAXxETAyHR** (where FAXx is the Air Traffic Service Unit, and y is the number of hours after the ETA at the destination that Search and Rescue should start if it has not been cancelled by the pilot with the nominated ATSU). This can only be specified when there is no ATSU in operation at the destination ( **unmanned airfield** ). SAR will start when the aircraft:

- ! is 5 minutes overdue at reporting points in **controlled** airspace, or
- ! “y” hours after the ETA at the destination, calculated from the actual take-off time plus the estimated time en-route from the Flight Plan.

An aircraft flying fairly low level (out of contact with any ATSU) from one uncontrolled airfield to another might want to specify RMK/SARFACTETA1HR. SAR will only be provided if it is not cancelled 1 hour after the ETA (actual take-off time as advised when activating the flight plan plus stipulated en-route time in item 16 of the Flight Plan plus 1 hour).

**RMK/SARNIL** can only be specified when the destination is an **unmanned airfield** . The aircraft will still be subject to Search and Rescue if it:

- ! is 5 minutes overdue at reporting points in **controlled** airspace.

A pilot flying from Cape Town to Swellendam specifies RMK/SARNIL. He will still be subject to SAR if he fails to report while still in controlled airspace.

When choosing options 3, 4 and 5 above (when requesting SAR at a specified time) the Flight Plan must include an after-hours mobile number to assist the Aeronautical Rescue Co- ordination Centre with the search, if necessary. For obvious reasons this should be a phone other than the pilot’s, i.e. a family member, close friend or the operator (for commercial flights).

SAR is cancelled by closing the flight plan – see paragraph 6.5. Flight Plans are automatically closed and SAR cancelled when the destination is a manned airfield.

## 12.2 Search and Rescue Phases

AIP GEN 3.6, AIC 40.2

When an aircraft is overdue, Air Traffic Control will start looking for it by any means at their disposal. This would initially involve an attempt to make contact with the missing aircraft, possibly via a relay through another aircraft in the vicinity. Depending on circumstances, these preliminary procedures could also involve a telephone call to the destination airport or to a police station close to a remote destination airfield, requesting the police to drive out and check if the aircraft has perhaps landed safely.

After that, and depending on circumstances, the following phases of Search and Rescue can be declared by the station co-ordinating the search:

### **12.2.1 Uncertainty Phase (INCERFA)**

INCERFA - if uncertainty exists as to the safety of an aircraft and its occupants. This phase will be declared when:

- ! the aircraft is not in radio contact within 5 minutes after the ETA at the next reporting point or a pre-arranged or scheduled reporting time and preliminary measures fail to re-establish contact within 15 minutes, or
- ! it is known or believed to be subject to irregular operation, is not maintaining the correct track or flight level, is not in normal communications, is experiencing navigational difficulties or hazardous weather conditions, or
- ! the aircraft experiences technical difficulties, but not to the extent that the flight cannot be completed, or
- ! an aircraft operating on a flight plan and stipulating alerting action at a specified time has not contacted an ATSU by the pre-arranged time, and preliminary checks fail to reveal the whereabouts of the aircraft, or
- ! an aircraft is about to make or has made a landing other than a forced landing, where the position is in doubt, or on an unprepared surface or on an aerodrome that is operationally unsuitable, or
- ! information is received that an aircraft is missing.

### **12.2.2 Alert Phase (ALERFA)**

ALERFA - if apprehension exists as to the safety of an aircraft and its occupants. This phase will be declared when:

- ! the actions taken during an INCERFA have been unsuccessful, or
- ! information is received which indicates that the aircraft is experiencing difficulties but not to the extent that a forced landing is likely, or
- ! the aircraft has made a PAN-call, or
- ! an aircraft is not operating under normal circumstances or is lost, and there is reason to believe that the aircraft's safety is in jeopardy.

### **12.2.3 Distress Phase (DETRESFA)**

DETRESFA - when there is reasonable certainty that an aircraft and its occupants are threatened by grave and imminent danger or require immediate assistance. A Detresfa is declared when:

- ! the actions taken during an ALERFA have been unsuccessful, or
- ! information is received which indicates that the aircraft is experiencing difficulties to the extent that a forced landing is likely, or
- ! information is received that the aircraft has made a forced landing, has ditched or crashed, or

- ! the fuel as stated in the flight plan is exhausted, or
- ! the aircraft has made a MAYDAY-call, or
- ! a downed aircraft has been sighted or located through an ELT-transmission, or
- ! when an aircraft which has been given approach or landing clearance fails to land within 5 minutes of the estimated landing time and no contact can be made, or
- ! an aircraft after take-off fails to report within five minutes after being expected to do so, and no contact can be made.

## 12.3 Procedures to be Followed

AIP GEN 3.6, AIC 40.2

### 12.3.1 The Five Golden Rules

When experiencing difficulties of any kind pilots should remember and adhere to the following 'five golden rules':

- ! **CONFESS** your predicament to ATC. It may be embarrassing to admit a mistake and it might be tedious to fill in an incident report afterwards, but it is far preferable to other possible outcomes of the situation.
- ! **COMMUNICATE** with ATC, passing on as much pertinent information as possible in the first message. It wastes everybody's time if you have 20 minutes' worth of fuel left and ATC, blissfully unaware of this fact, starts to direct you to an airfield 30 minutes flying time away.
- ! **CLIMB** for improved radar coverage, direction finding and map reading.
- ! **COMPLY** with instructions and advice given, but don't be afraid to state if you are unable to comply. ATC might ask you to climb for better radar coverage, but they might be unaware that a climb would put a VFR pilot into cloud.
- ! **CONSERVE** fuel and other resources. Slow down, use the power setting for maximum endurance and lean the mixture.

Also remember "ANC" – Aviate, Navigate, Communicate – in that order.

### 12.3.2 Declaring an Emergency

Do not be afraid to declare an emergency. You might have to fill in a form or two after landing, but you will have to fill in a lot more forms after you have crashed the aeroplane – provided you are still capable of filling in forms at all.

Quite a number of serious accidents with loss of life have occurred because the pilot was reluctant to declare an emergency. Pilots seem to think that their licences will be withdrawn or they will be prosecuted or other dreadful things will happen when they admit to a mistake. Usually, though, the worst consequence is an inquiry into the problem, which is infinitely preferable to possible loss of life, damage to aircraft or even the amount of paperwork after an accident.

When you have that niggling feeling that things might not be quite right, they probably are not right. **When in doubt, there is no doubt** – be safe rather than sorry.

12.3.3 Action After a Forced Landing

After a pilot has successfully executed a forced landing, but cannot make contact with anybody by radio or phone, he must use his discretion as to what action to take.

In a populated area and if the aircraft was not operating on a SAR flight plan, it might be better to leave the aircraft, if possible, and go looking for help.

If SAR was requested or the forced landing took place in a sparsely populated area it will normally be better to stay with the aircraft. Removable parts of the aircraft can be placed in open, conspicuous positions, reflecting the rays of the sun. If possible, transmit on the COM radio or an Emergency Locator Transmitter (see paragraph 12.4) at quarter past and quarter to every hour, i.e. at 0715, 0745, 0815, 0845 etc., but leave the ELT on continuously if you see or hear search aircraft in the vicinity. Use flares or other signalling devices only when search aircraft can be seen or heard. And conserve the strength of survivors, water, food and other resources.

12.3.4 Ground/Air Emergency Signalling Code

AIP GEN 3.6

For every flight of more than 20 nm from base a set of signal strips must be carried, obtainable from pilot supply shops. These are four strips of white material, each at least 60 cm wide and 2.5 metres long. In case of a forced landing, when a search aircraft flies overhead but radio communications are not possible, the strips can be used to lay out the following signals:

|                              |    |
|------------------------------|----|
| Require assistance           | V  |
| Require medical assistance   | X  |
| Proceeding in this direction | \$ |

|              |   |
|--------------|---|
| No, negative | N |
| Yes, affirm  | Y |

## 12.4 Emergency Locator Transmitter (ELT)

An Emergency Locator Transmitter is a battery-powered box installed in aircraft, which starts transmitting automatically in case of a high deceleration (a crash), but can also be manually activated by a switch mounted either on the box or remotely somewhere in the cockpit.

All ELTs transmit a high-pitched modulated siren-type signal on 121.5 MHz and 243 MHz when activated. This can be picked up by any VHF radio tuned to 121.5 MHz and in range of the transmission. Most Search and Rescue ground stations do not monitor this frequency anymore, but airliners and other high-flying aircraft usually do, and can alert the relevant authorities when a distress signal is heard. The signal can then be used as a homing device by rescue parties.

The more modern 406 MHz ELT, in case of activation, transmits a digital signal every 50 seconds, which is received by several geosynchronous satellites and low-earth polar satellite repeaters. This signal gives the identity of the ELT (hard-coded into the particular “box”), the country of registration and a GPS position (if the ELT is fitted with a GPS receiver).

From this signal the **COSPAS-SARSAT** (COSPAS is the Russian acronym for “Space System for the Search of Vessels in Distress” and SARSAT stands for **Search And Rescue Satellite-Aided Tracking**) organisation, formed by the USA, Canada, France and Russia, can establish the identity of the aircraft from a register of all 406 MHz ELTs kept by the registering country ( if the ELT has been registered by the aircraft owner). Rescue parties can then be dispatched and home into the siren-type signal transmitted on 121.5 MHz described above.

Pilots of aircraft not equipped with a fixed 406 MHz ELT could consider purchasing a Personal Locator Beacon from marine or outdoor shops. These smaller units operate exactly as the built-in 406 MHz ELT described above and need to be registered in the same manner.

However, the success rate of ELTs actually activating in case of a crash is rather low (the crash itself often damages the box, or the battery, or the antenna ! ), and it is more useful for aircraft owners to rather invest in a more useful (and cheaper!) Satellite Tracking Device:

## 12.5 Satellite Tracking Devices

Satellite Tracking Devices are clever little boxes that “phone home” at intervals set by the operator, usually every two minutes. The “phoning” takes place via one of the commercial satellite networks, and “home” is a website hosted by the manufacturer

of the unit. Anybody with a username and password can log into that website and follow the real-time progress of the aircraft overlaid on a map.

Several makes of these devices add further value with panic buttons, which, when activated by the pilot, trigger a text message to predetermined phone numbers. Some more expensive ones have other useful tools such as satellite text message or phone facilities.

A satellite tracking device should be installed in every aircraft, be it a Boeing 777 or a homebuilt single-seater. If anything ever happens to an aircraft equipped with a satellite tracking system, its last transmitting position is instantly known, and a search for the aircraft or survivors can be focused on a very small area. Even if a device like this cannot save the lives of crew or passengers of an unfortunate aircraft, it can decrease the often enormous costs for the search and, more importantly, shorten the period of uncertainty for the loved ones of the victims.

There are also several (free) cellphone apps that similarly track position. These rely on cellphone coverage and are therefore only useful in fairly populated areas.

## 12.6 Emergency Frequencies

Satellite processing from 121.5 MHz and 243 MHz locator beacons has been discontinued, but 121.5 MHz is still used as an aeronautical international distress frequency for voice communications, and 243 MHz is still used by NATO.

The most commonly used HF emergency communications frequencies are 2182 kHz and 8364 KHz.

Any aircraft engaged in searches should monitor as many as possible of the abovementioned frequencies.

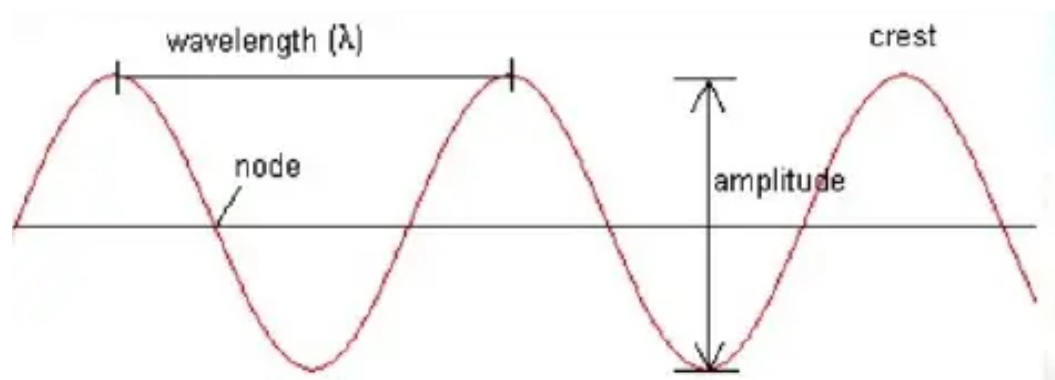
# 13

# Radio Technical

Candidates for the Restricted Certificate only need to know parts of this chapter – please refer to the Restricted ROC syllabus in the Introduction to this book.

## 13.1 Basic Radio Principles

Electromagnetic radio waves are created by a source of alternating voltage connected to a wire antenna. This will set up an oscillating electric field in the antenna, which in turn induces a magnetic field. Through this electromagnetic oscillation a wave is propagated through space and can be received by an appropriate receiving antenna.



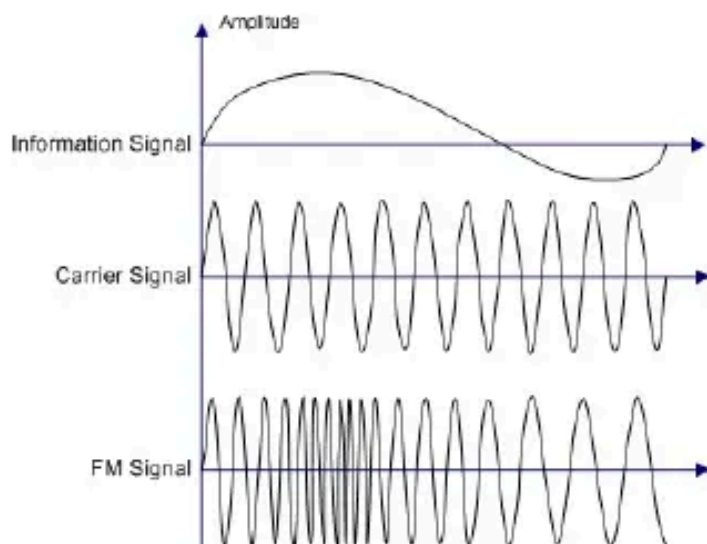
Radio waves travel at the speed of light of 300 million metres per second (m/s). The frequency of a radio wave is the number of waves emitted per second (cycles per second, measured in

Hertz). The frequency of the wave determines its wavelength, which in turn determines the optimal length of the antenna: short waves (high frequencies) need shorter antennae than longer wavelengths. The relationship between wavelength (λ), speed (c) and frequency (f) is:

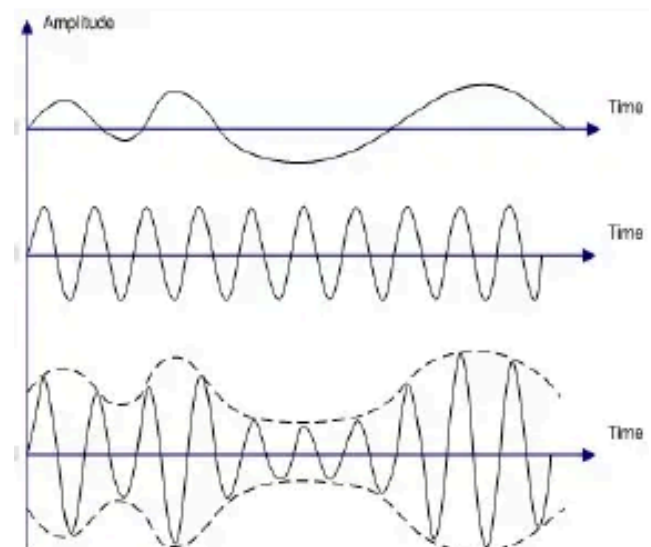
$$\text{wavelength } (\lambda) = \frac{\text{speed } (c)}{\text{frequency } (f)} \quad \text{frequency } (f) = \frac{\text{speed } (c)}{\text{wavelength } (\lambda)}$$

A plain radio wave (carrier wave) conveys no information; it can be compared to the blank pages of a book. To imprint information in the form of speech onto it, the carrier wave must be modulated. This can take the form of **amplitude modulation (AM)** or **frequency modulation (FM)**. In both cases an audio signal is superimposed on the carrier wave, thus modulating either its amplitude or its frequency. Commercial radio and TV stations in the VHF band use mostly frequency modulation, but all aviation

radiotelephony is amplitude modulated, which prevents stronger stations from blocking out weaker ones. Once the modulated wave is propagated from the transmitter through the air and received by a suitable antenna, it needs to be demodulated, that means the superimposed speech (the modulation) is extracted again (demodulation) to be understood by the human ear.



Frequency Modulation (FM)

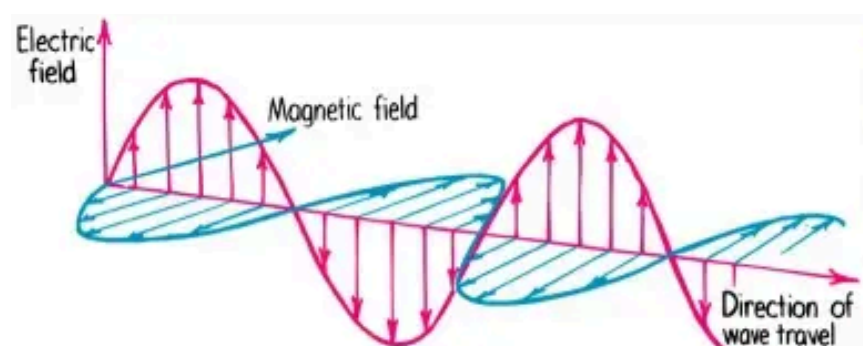


Amplitude Modulation (AM)

As can be seen from the AM drawing above, the modulated wave consists of two mirror images (double sideband), one above the base line (upper sideband), and one below (lower sideband). Depending on application, either only one sideband or both can be used. Single sideband (SSB) generally requires less electrical power for transmissions and uses less bandwidth, so that more channels are available within a certain frequency band.

In addition to AM and FM there is pulse modulation, which is achieved by interrupting (starting and stopping) the carrier wave in a certain pattern. This may be used in simple radar, or to transmit a Morse signal for the identification of navigational beacons.

The electromagnetic wave consists of an oscillating electric field ( $\mathbf{E}$ ) at right angles to the travel of the wave plus a magnetic field ( $\mathbf{H}$ ), oscillating at right angles to both the electric field and the direction of travel. The wave is normally polarised and it is the electric field that determines the wave's polarisation. The polarisation can be either linear (Horizontal or Vertical with reference to the earth's surface), or circular. Circular polarisation consists of two distinct cases, RHCP (right hand circular polarisation) or LHCP (left hand circular polarisation). Circular polarisation can be thought of a helix where RHCP is defined as a electromagnetic wave rotating clockwise when viewed from the source in the direction of travel. Polarisation is of little concern for HF signals



as they are bent and tumbled by the ionosphere as they pass through - but it is used to improve channel reliability for VHF/UHF line-of-sight signals to cut out unwanted noise and interference from the opposite or cross channel. Practically all communication and navigation signals between the aircraft and the ground in aviation use linear polarisation: VHF COM frequencies are vertically polarised and COM antennas are mounted vertically on the aircraft, while the VOR is horizontally polarised and VOR antennas are mounted horizontally, usually on the tail. All satellite based systems (such as GPS) use circular polarisation.

### 13.2 Frequency Bands

Radio frequencies are measured in Hertz, which is number of cycles per second. Since that is a pretty small unit, we mostly deal with larger units like Kilohertz (KHz) = 1000 Hz, Megahertz (MHz) = 1 million Hertz or Gigahertz (GHz) = 1000 million Hertz.

The waves used in aeronautical radio applications vary in length from 1 mm to 100 km and in frequency from 3 kHz to 300 GHz. According to their different behaviour, radio waves are grouped into frequency bands as follows:

| Band           |     | Frequency    | Wave length | Range                                                                                 | Static                                                                                | Attenuation                                                                           |
|----------------|-----|--------------|-------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Very low       | VLF | 3-30 kHz     | 100-10 km   | long                                                                                  | severe                                                                                | slight                                                                                |
| Low            | LF  | 30-300 kHz   | 10-1 km     |  |  |  |
| Medium         | MF  | 300-3000 kHz | 1000-100 m  |                                                                                       |                                                                                       |                                                                                       |
| High           | HF  | 3-30 MHz     | 100-10 m    |                                                                                       |                                                                                       |                                                                                       |
| Very high      | VHF | 30-300 MHz   | 10-1 m      |                                                                                       |                                                                                       |                                                                                       |
| Ultra high     | UHF | 300-3000 MHz | 100-10 cm   |  |  |  |
| Super high     | SHF | 3-30 GHz     | 10-1 cm     |                                                                                       |                                                                                       |                                                                                       |
| Extremely high | EHF | 30-300 GHz   | 10-1 mm     | short                                                                                 | none                                                                                  | severe                                                                                |

The range of frequencies (that is the difference between the lowest and highest frequency carried) that make up a radio signal which occupies a channel is called its **bandwidth** .

Waves have different characteristics, depending on their frequency and wavelength. Frequencies for the various aviation applications have been carefully selected to take advantage of the desirable phenomena and minimise the undesirable ones for the particular application.

**Attenuation** is the loss of energy by **absorption** to the earth's surface or the atmosphere (water vapour, rain etc.), and results in the wave losing its strength and finally becoming undetectable. Lower frequency waves suffer less from attenuation and are therefore utilised for long range applications.

**Diffraction** is the bending of radio waves when encountering obstacles. Diffraction is at its maximum at the lowest end of the frequency spectrum, resulting in ground waves, which follow the curvature of the earth.

**Fading** is caused by receiving two or more waves of the same signal with varying phase differences. This could lead to the two signals re-enforcing each other, but also to the cancelling out of the two signals.

**Interference** results from two transmitters of the same or similar frequency overlapping. Therefore, the COM 1 and COM 2 radio in an aircraft should never be tuned to the same frequency. Interference can also be caused by other electric systems in the aircraft creating electric noise.

**Reflection** means that radio waves can be reflected by objects commensurate with their wave lengths. This could lead to fading or interference.

**Refraction** is the change of direction a wave experiences when travelling from one medium to another one of a different density. This causes, for example, the coastal effect in the ADF application and the long range of HF waves.

**Static** is caused by electromagnetic disturbances in the atmosphere, resulting in 'radio noise'. Static is severe in VLF waves, decreases as the frequency increases, and becomes negligible in the VHF band.

## 13.3 Common Aviation Applications

**NDBs (Non-Directional Beacons)** operate in the low to medium frequencies from 200 to 415 KHz. Waves are propagated by ground waves during the day and ground and sky waves at night. They have a long range of up to 1500nm, but are subject to pronounced static. Installations require large antennae on the ground. They are used for Automatic Direction Finding (ADF), now largely obsolete - see paragraph 15.2.

**HF (High Frequency) Voice Communications** utilise the 2850 to 22000 KHz range. Propagated by sky waves, HF range can be up to 4000 nm, but is subject to noticeable static. HF is used for long-range communications by aircraft overflying the oceans or sparsely populated areas. See paragraph 13.4.2.

**ILS (Instrument Landing System) localiser** operates in the VHF band from 108.1 to 111.9 MHz. Their range is limited to line of sight. See paragraph 15.4.

**VOR (Very High Frequency Omni Range)** beacons use the frequencies from 112.0 to 117.95 MHz in the VHF band. Range is limited to line of sight. See 15.3.

**VHF (Very High Frequency) Voice Communications** utilise the 118.0 to 136.975 MHz range. Propagated by direct waves, VHF range is limited to line of sight

and has very little static. VHF range is often increased by relay stations, and is used for all normal air-to-ground communications, VOR navigational aids and GNSS Ground Based Augmentation Systems (GBAS). See paragraph 13.4.1.

**ILS (Instrument Landing System) glide path** operates in the UHF band from 329.3 to 335.0 MHz. Glide path frequencies are paired with the localiser. See 15.4.

**DME (Distance Measuring Equipment)** works in the UHF band at frequencies between 962 and 1213 MHz. See paragraph 15.5.

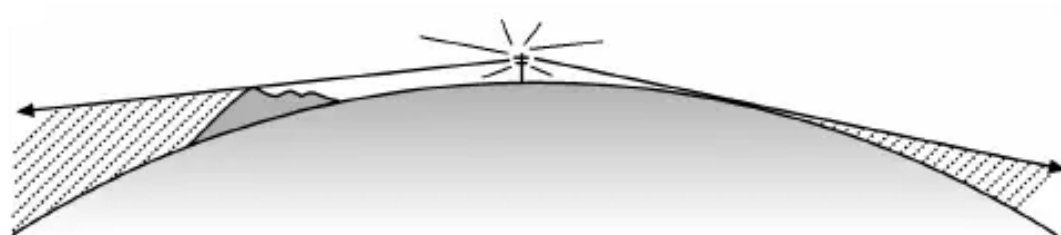
**GNSS (Global Navigation Satellite System)** as well as various **radar applications** (radio altimeters, weather radar, secondary surveillance radar, airfield surface radar) operate in the UHF to EHF bands. See paragraph 15.6.

## 13.4 Range

The range of a radio transmission depends mainly on its frequency, which in turn determines the type of propagation as well as phenomena like attenuation.

### 13.4.1 VHF

VHF transmissions such as VHF voice and VOR transmissions are mainly propagated as direct or line-of-sight waves (and to some small extent as ground reflected waves). This type of transmission is therefore blocked by buildings or mountains and by the curvature of the earth.



The formula for radio range in VHF communications is:

$$\text{Range [nm]} = 1.23 \left( \sqrt{\text{height of aircraft AGL [ft]}} + \sqrt{\text{height of transmitter [ft]}} \right)$$

Example: an aircraft at 9500' ALT, flying over ground with an elevation of 4500' wants to talk to a ground station with a transmitter that is 65' high. At what distance from the ground station can the pilot expect to make contact?

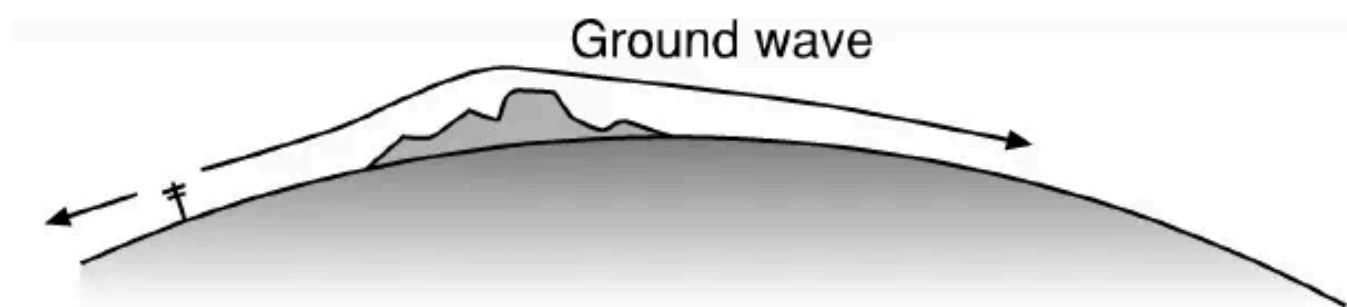
$$\text{Range [nm]} = 1.23 \left( \sqrt{5000} + \sqrt{65} \right) = 1.23 (70 + 8) = 95 \text{ nm}$$

If the aircraft is more than approximately 95 nm from the station, radio communications are probably not possible, and the pilot should either try to contact another aircraft for a relay or wait until he is closer to the station.

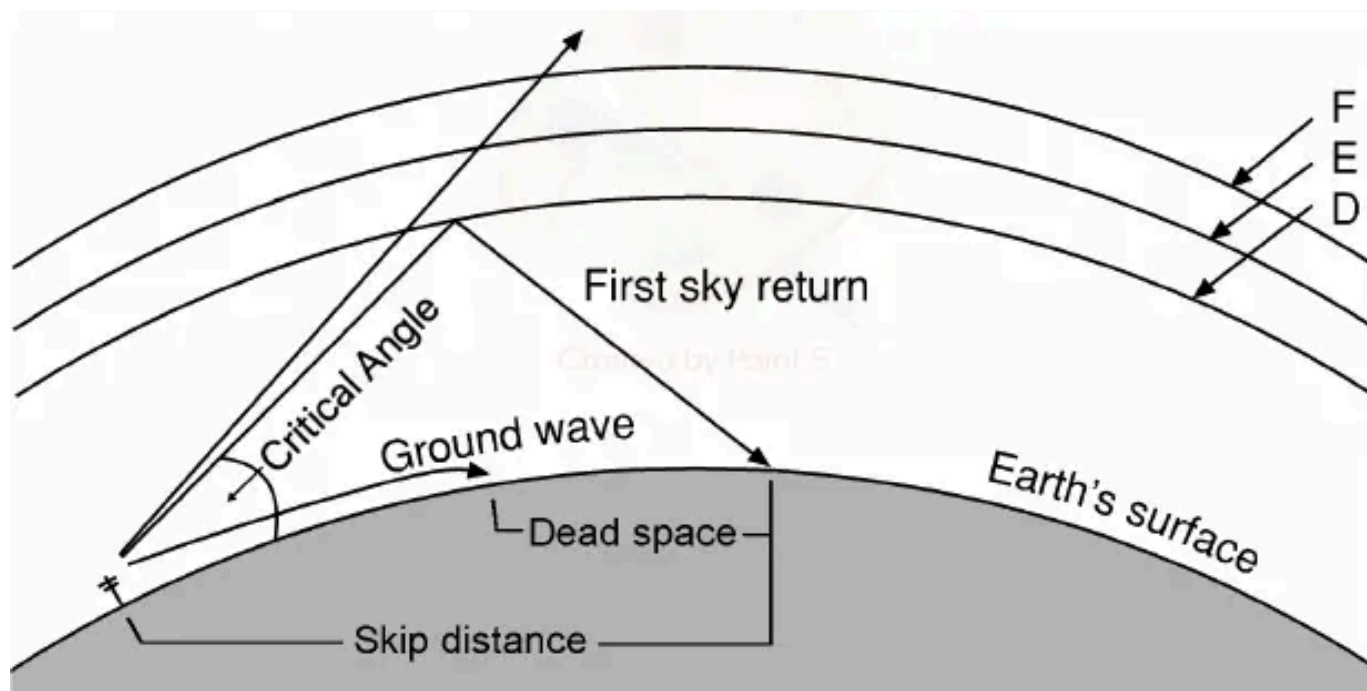
VHF frequencies are spaced either 25 kHz or (in busy airspaces) 8.33 kHz apart. With 8.33 kHz spacing more frequencies can be “crammed” into the total of 19 MHz allocated to VHF voice communications and is used in areas where airports are close together and each airport uses several VHF frequencies for the different controller positions (Ground, Clearance, Tower, Approach, Departure etc. etc.). However, the aircraft radio needs to allow 8.33 kHz spacing without interference with close-by frequencies. See also paragraph 7.3.2 for pronunciation of VHF frequencies.

### 13.4.2 HF

HF transmissions use frequencies in the 2850 to 24890 kHz range, which are propagated by ground and sky waves.



**Ground waves** (surface waves): bend due to diffraction and so follow the curvature of the earth.



**Sky waves:** waves in the LF to HF bands are refracted by ionised layers in the ionosphere above the earth, since the ionised layers are of a different refractive index to the atmosphere. The three ionised layers, D (only present during the day), E (Kennelly-Heaviside) and F (Appleton) vary in height above the surface from 50 to 225 km and move diurnally and seasonally. They also exhibit a lesser electron

density at night. Depending on the degree of refraction (which varies with the electron density in the layers and the frequency of the wave) and the angle at which the radio wave strikes the layers, these waves may either escape into space or return to earth. The angle between the wave that gives the first sky return and the earth is called the **critical angle**, and the distance from the transmitter at which the first sky wave returns to earth is called **skip distance**. The area between the last surface wave and the first sky wave is termed **dead space**, and no reception will be possible in that area. In the HF band the surface wave is fairly short and the skip distance relatively high.

Because of the amount of refraction of the HF waves and the critical angle (which depends on the frequency) and the movement of the ionised layers (which depends on the time of day, the season, sunspot activity etc.) different frequencies will give better results at different times. From these factors the Maximum useable Frequency (MuF) and Lowest useable Frequency (LuF) can be predicted by computer models.

| DAR<br>INFORMATION |       |
|--------------------|-------|
| AFI 3              |       |
| 5517               | 11300 |
| AFI 4              |       |
| 8903               | 13294 |
| INO 1              |       |
| 8879               | 13306 |
| ADNL               |       |
| 8870               |       |
| SELCAL             |       |

| GABORONE<br>RADIO |       |
|-------------------|-------|
| AFI 4             |       |
| 5493              | 13294 |
| ADNL              |       |
| 6586              | 8888  |

HF frequencies can be found on En-route charts, depicted in blocks in the area to which they apply.

**During the day the higher frequency will generally give the better range, and at night the lower frequency.**

" AFI (Africa) and INO (Indian Ocean) are communication regions. ADNL means "additional". SELCAL means the station is Selcal-capable (see paragraph 14.3.4).

HF often uses the Single Sideband (SSB) technique, whereby one of the sidebands of amplitude modulation is suppressed, thus saving power and using bandwidth more efficiently.

## 13.5 Radio Installation in Aircraft

The apparatus that comprises the average small aircraft radio station is:

- ! an **antenna** system with co-axial cable to carry the RF electromagnetic signals to and from the transceiver.
- ! a **transceiver** (transmitter and receiver) with the associated controls for volume, frequency selection etc.
- ! a **microphone** to convert sound waves into audio electronic signals that can be modulated onto the carrier wave by the transmitter.
- ! a **speaker / headphone** to convert audio electronic signals generated by the demodulator in the receiver back into sound waves.

### 13.5.1 Antenna

An antenna in its simplest form consists of a length of conductor (wire rod) called the radiator with a matching conductor called the counterpoise or ground-plane (normally the aircraft itself). It converts radio-frequency (RF) fields impinging on it into alternating currents (AC) in the conductors. The receiving antenna intercepts RF energy and delivers this AC to electronic equipment, and the transmitting antenna is fed with AC from electronic equipment and generates an RF field. A specialized conduit called a feedline (co-axial cable) normally transfers the RF energy between the antenna and the equipment. Usually the same antenna is used for both reception and transmission. Each radio installation of any kind will have its own antenna, of a polarisation and size commensurate with the frequency used.

An antenna's polar diagram is a representation of its E field radiation pattern in both azimuth (360 degrees compass direction when viewed from above) and elevation (180 degrees with respect to the earth's surface). Antennas can be designed to have either an omni-directional or directional radiation pattern, in either azimuth or elevation, or both. For instance the VHF COM system matches vertically polarised dipoles at the ground station with a mostly vertically polarized dipole on the aircraft. In each case the antenna radiation pattern is shaped like a toroid or donut – the azimuth radiation pattern is a circle ( i.e. it is omni-directional and radiates equally in all compass directions) and most of the energy in elevation is directed to the first 30 degrees of the surface. There is a null directly above the ground station.

A VHF radio antenna is essentially a metal rod, sometimes encased in fibreglass to make it more streamlined, of about half a metre length. It is usually mounted on top of the fuselage, with a second antenna sometimes mounted on the belly of the aircraft. Some modern aircraft, especially those built of composite materials have the antenna invisibly embedded in the fuselage.



An HF antenna needs to be much longer, due to the longer wavelength of HF waves. It may be strung from the cockpit roof to the tail, but is nowadays more often embedded in the fuselage. Some older aircraft had trailing HF antennae, which could be reeled in or out and trailed freely behind the aircraft.



A UHF antenna (left) as used for DME or the transponder is short with a “ball” at the tip, and is usually mounted under the fuselage.

A GNSS (GPS) antenna (right) consists internally of a ceramic patch or helix mounted above a ground plane. They are always RHCP (right hand circular polarised), have a radiation pattern that points towards the sky where the GNSS satellites are, and are mounted on top of the fuselage.



### 13.5.2 Transceiver

The transceiver is the actual “box” mounted in the instrument panel of a small aircraft. In larger aircraft the transceiver is often remotely mounted, under the floorboards or in a separate radio stack, with only the control box accessible in the cockpit.

When the transceiver is powered up and the pilot speaks into the microphone while pressing a 'press-to-talk' (PTT) button, the transmitter circuits amplify and broadcast, via the antenna system, the selected output frequency - 126.7 MHz for example - modulated with the audio frequencies from the microphone.

An aircraft antenna continually collects all passing RF energy in the band for which it is designed, which at any time will normally consist of many transmissions. The receiver tunes out all transmissions on all frequencies except one - the selected, or active, frequency. Signals on this frequency are demodulated to isolate the voice information from the carrier, amplified and passed to the speaker system to convert to the sound waves heard in the earphones or speaker.

The control box or front face of the transceiver has knobs or buttons to change the frequency, and often a second frequency window to select a stand-by frequency and a flip-flop switch to flip the active and stand-by frequencies. In modern EFIS-installations the transceiver is often controlled from the display screens.

All transceivers have some form of ON/OFF/TEST or SQUELCH/VOLUME control. The 'squellch' control, if fitted, is an adjustable filtering device, which, for operator comfort, can be set to just filter out the weak atmospheric background noise but still allow strong signals to be switched through. Modern transceivers have an automatic squellch control. In this case, pressing the test facility will override the squellch, allowing the background hash to be heard. Sometimes, when communication is difficult with a far-away station, pressing the “test” facility will make communications possible.

South Africa is using frequencies with 25 KHz spacing at the moment; for example 118.000 - 118.025 - 118.050 - 118.075 – 118.100 etc. Some older radio sets only show 2 decimal places, but they will actually tune to, for example, 118.075 even when the face only shows 118.07. On some sets the tuning knob needs to be pulled out to be able to change the last digit.

In congested airspaces such as in Europe, where the demand for the allocation of more and more frequencies is great, aircraft must have radios fitted which can tune frequencies with 8.33 KHz spacing, i.e. 118.000 - 118.008 - 118.017 - 118.025 – 118.033 – 118.042 – 118.050 etc.

HF radios include a TUNE control button, which must be activated after each frequency change. They also have an SSB/USB/LSB switch to select single sideband, upper sideband or lower sideband.

### 13.5.3 Headphone or Speaker

The loudspeakers or headphones receive electronic signals from the receiver and convert them to audible signals. Normal headsets rely solely on passive noise reduction - creating a physical barrier around the ear to attenuate noise - which usually works quite well for middle to high-frequency sound but doesn't block low-frequency engine noise and background rumble.

Active noise reduction technology uses electronics to determine the amount of low-frequency (50–600 Hz) engine and other noise entering the system and generates out-of-phase noise in the same frequency range; this counters the background noise and leaves a soft 'white' noise in the headphones.

### 13.5.4 Microphone

The microphone, either hand-held or integrated as a boom mike into the headset, converts the sound waves created by speech into weak electronic signals.

Most modern microphones are noise cancelling. This means that in addition to the main opening into which the pilot speaks, there is a little hole at the back of the microphone case, through which sound may enter. In a noisy cockpit environment, the overall engine and cockpit noise will enter through both openings and is cancelled out. To obtain the best results it is therefore imperative that the microphone is placed very close to the mouth.

### 13.5.5 Avionics Master Switch and Audio Control Panel

Most modern aircraft have an Avionics Master Switch, which must be switched ON to supply power to the radios. It should normally be switched ON after starting the engine(s) and OFF before shutting down the engine(s).

Aircraft with more than one communication and/or navigation radio usually also have an Audio Control Panel fitted, which determines on which radio the pilot speaks and which radio(s) he listens to. An audio panel usually has a MIC switch to select whether the pilot wants to transmit on COM 1 or COM 2, as well as selector keys to determine whether he can receive on COM 1, COM 2, or both simultaneously.

Modern “glass cockpits” with an Electronic Flight Instrumentation System (EFIS) often have the Audio Control Panel integrated in the EFIS.

Pilots must familiarise themselves with the operation of the audio panel fitted to the aircraft they are flying. Most suspected “radio failures” are simply caused by “finger trouble”, i.e. the pilot(s) not setting the system up correctly.

## 13.6 Power Supply

Aircraft radios need electric power to operate. In a small aircraft the system is usually rather simple with a battery operated by a master switch, an alternator operated by the alternator switch, and possibly an avionics master switch to control power to all communications and navigation radios.

In a larger aircraft the electric installation can be quite complicated with several batteries, alternators, generators, inverters, electric distribution busses and a whole host of other features.

It is imperative that a pilot knows the electric system of his aircraft and how to deal with failures of any of its components. This forms part of the conversion course (familiarisation or differences training) or type rating course.

### 13.6.1 Batteries

A battery is a storage device for electrical energy. While a battery is being charged, electrical energy is converted into chemical energy, which will then be converted back into electrical energy when current is drawn from the battery.

Batteries consist of a number of cells with two plates (positive and negative) and a fluid (electrolyte) in between. The two types of batteries commonly used in light aircraft are Lead-Acid and Nickel-Cadmium (NiCad). Each cell in a Lead-Acid battery usually stores 2 Volts, in a NiCad battery 1.2 Volts. A 12 V Lead-Acid battery therefore consists of 6 cells, a 24 V battery of 12 cells. The Lead-Acid battery has negative lead and positive lead dioxide plates with a solution of sulphuric acid and distilled water as an electrolyte. NiCad batteries use positive nickel peroxide and negative iron plates with an electrolyte of potassium hydroxide and water.

Batteries, due to their chemical composition, give off toxic and flammable fumes, and are generally vented overboard. Smoking and having open fires around batteries is not recommended.

In addition to supplying current to the starter motor for starting the engine(s), a battery also serves as a backup device. Should the alternator or generator fail in flight, a battery will supply current for anything from 5 minutes to several hours, depending on its state of health and the total electrical load.

In case of alternator failure, the pilot should switch OFF all unnecessary electrical equipment and conserve power by keeping radio transmissions to a minimum (transmitting draws more current than receiving). Remember that an aircraft may not fly in controlled airspace without two-way radio communications, so an alternator failure might mean that the pilot will have to divert to an unmanned airfield.

### 13.6.2 Alternators and Generators

To recharge the battery in flight and to provide continuous power to the electric equipment in the aircraft, one or more generators or alternators are fitted to all but the very antique aeroplanes. A drive belt connected to the crankshaft on the one side and the alternator or generator on the other side is usually situated close behind the propeller of light aircraft and should be checked for proper tension during the pre-flight checks.

Modern small aircraft are fitted with alternators, which start charging the battery even when the engine is idling, while the older generator installations will only charge at fairly high RPM settings. The disadvantage of an alternator is that it needs some charge in the battery to excite it before it will start charging the battery, while a generator will charge even a completely flat battery.

It is important to watch the health of the electrical system in flight. The ammeter should be included in the normal instrument scan to detect any problems at an early stage. Simple ammeters come in two basic types: one showing only the current supplied by the alternator to the aircraft systems (and if it is zero, there might be a problem with the alternator), and the other one showing the charge to or discharge from the battery (if that one is on zero, all is well and the battery is fully charged). Pilots need to know exactly what the ammeter fitted to their aircraft is telling them.

A completely flat battery in an aircraft fitted with an alternator must be charged from an external power source such as a battery cart (or a car of the same voltage) until it has at least some charge in it. Then the aircraft can be started by external means and flown. An aircraft with a generator and a flat battery can be started by prop-swinging or from a battery cart and can then be safely flown. The generator will charge the completely flat battery. Provided the flight is long enough the alternator or generator in either aircraft will have charged the battery enough to facilitate the next start.

In larger aircraft the above can get a lot more complicated. Again, pilots are urged to study the electrical system fitted to the aircraft they fly.

# 14

# IFR Procedures

This chapter gives a few of the terms, definitions and procedures commonly used in Instrument Flying, and the radio phraseology associated with it. **Candidates for the Restricted Radio Licence do not need to know this chapter.**

## 14.1 Definitions

AIP ENR 1.5, AIP ENR 1.8

The **Aircraft Approach Category**: depends on the stalling speed of an aircraft at MAUW in the landing configuration multiplied by 1.3:

A = less than 91 kts

B = between 91 and 120 kts

C = between 121 and 140 kts

D = between 141 and 165 kts

E = above 165 kts.

The aircraft Approach Category influences the DA or MDA as well as maximum speeds on an instrument approach procedure. Do not confuse the Approach Category with Performance Class.

**Clearance Limit:** the point to which a flight may proceed when experiencing radio failure. The clearance limit may be the destination or the point where the aircraft leaves controlled airspace after departure. For an instrument approach the clearance limit is either the Missed Approach Point (when landing) or the end of the missed approach procedure (when executing the missed approach).

**Cloudbreak Procedure:** an instrument procedure where either the runway is not aligned with the final approach track, local QNH is not available, or the procedure does not comply with ICAO requirements for an instrument approach procedure. Minima should not be less than circling minima.

**En-route Safe Altitude:** The altitude that will ensure a separation of at least 1500' above the highest obstacle within 5 nm of the aircraft in flight. Note: an aircraft at night or when flying under IFR has to cruise at least 1000' above the highest obstacle within 5 nm of the aircraft if the obstacle is up to 5000' high - and 2000' above obstacles higher than 5000' MSL.

**Expected Approach Time (EAT):** the time at which an aircraft may expect to be given clearance to leave the holding stack and commence the instrument approach. Usually given by ATC to aircraft in a heavy traffic situation.

**Grid Minimum Off-Route Altitude (Grid MORA):** a Jeppesen term, the minimum altitude in a latitude-longitude grid, which gives the required obstacle clearance.

**Localizer Sensitive Area (LSA):** a defined strip either side of the centreline of an ILS-equipped runway. This must be kept clear of aircraft, cars etc. during ILS operations to avoid distortion of the localiser signal.

**Low Visibility Operations / Procedures (LVO/LVP):** when the RVR decreases to below 600m, or the cloud ceiling is less than 200', ATC will institute Low Visibility Operations, taking special care with taxiing aircraft, keeping the Localizer Sensitive Area (LSA) clear of traffic etc.

**Minimum En-route Altitude (MEA):** the minimum altitude on a route segment that provides reception of the relevant navigation facilities, adequate ATC communication coverage, complies with airspace requirements, and provides the required obstacle clearance for IFR flights.

**Minimum Obstruction Clearance Altitude (MOCA):** an ICAO term, the minimum altitude on a route segment that provides the required obstacle clearance of 1000' for obstacles up to 3000' elevation, 1500' for obstacles between 3000 and 5000' elevation, and 2000' for obstacles above 5000' elevation.

**Minimum Off-Route Altitude (MORA):** a Jeppesen term, the minimum altitude off airways between beacons that gives 1000' clearance for obstacles up to 5000' and 2000' clearance for obstacles higher than 5000' within 10 nm either side.

**Minimum Sector Altitude (MSA):** the lowest altitude giving a minimum terrain clearance of 1500' in South Africa (ICAO recommends minimum 1000') above the highest obstacle within 25 nm of the Aerodrome Reference Point (ARP), radio aid or significant point. Descent below this altitude may only be done when under radar control or when commencing the Initial Approach Segment for an approach based on a radio navigation aid such as a VOR or NDB (AIP ENR 1.7). - See also TAA for RNAV approaches.

**Onward Clearance Time (OCT):** the time at which an aircraft may expect to be cleared to leave the holding pattern at an en-route fix and proceed with the flight.

**Required visual reference:** elements of the approach light system, the threshold, threshold markings or lights, the touchdown zone or touchdown zone markings or touchdown zone lights, or the runway edge lights - must be visible to a pilot before continuing an instrument approach past the missed approach point and land.

**Runway Visual Range (RVR):** the distance at which the pilot of an aircraft on the centre line of a runway can see the runway surface markings or the lights delineating the runway or identify its centre line. It is measured by a transmissometer or by a human observer. RVR is only reported when the visibility decreases below 1500m. The maximum RVR that will be reported is 2000m.

**Segment Minimum Altitude (SMA) or Segment Minimum Safe Altitude (SMSA):** the lowest safe altitude on a segment of a non-precision approach (may be lower than procedure altitudes to achieve a stabilised descent).

**Terminal Arrival Altitude (TAA):** the lowest altitude giving a minimum terrain clearance of 1000' above the highest point within 25 nm of the Aerodrome Reference Point (ARP) or significant point. Descent below this altitude may only be done when cleared for and following an RNAV approach procedure (AIP ENR 1.7).

**Visibility:** for aeronautical purposes is the greater of the distances within at least half the horizon circle that a black object, situated near the ground, can be seen when observed against a bright background, or that lights (1000 candelas) can be seen and identified against an unlit background.

**Visual Approach :** IFR flights arriving at an airfield in good weather can avoid the longer track distance associated with flying an instrument approach procedure by requesting and being cleared for a Visual Approach. A flight can only be cleared for a visual approach when:

- ! within 25 nm of the aerodrome, **and**
- ! the RVR is better than 1500 metres, **and**
- ! the reported cloud ceiling is at or above the initial approach level, **or**  
the pilot has the airfield in sight and it is reasonably assured that the approach can be completed with visual reference to the ground.

The pilot would ask for a visual approach with the words: Request visual approach . Pilots executing a Visual Approach will **not be separated from terrain** , but will still be **separated from other traffic** as long as they remain in controlled airspace (CATS 91.07.5.7, AIP ENR 1.5-2) .

**VMC Climb, Descent or Approach:** An IFR flight in VMC by day may wish to fly below Sector Safe Altitude or below radar vectoring altitude. The pilot can request a VMC Climb or Descent and may be cleared for it, in which case Air Traffic Control **will not provide separation from terrain** . Traffic separation will still be provided in controlled airspace.

Example 1: An IFR flight arrives at a controlled non-radar equipped airfield by day in perfect sunshine, but above a solid cloud layer with the tops below MSA. The pilot can see down to the cloud tops, but must do an instrument approach to get below cloud. He may request a "VMC descent to cloud tops" to arrive over the approach facility above cloud but below the initial approach altitude, saving a long descent in the hold.

Example 2: A pilot flying IFR to an airfield situated just beyond a high mountain range. If it is VMC and the pilot can see the airfield and the mountains, he might want to descend below the published safe altitude to avoid a very steep descent for reasons of passenger comfort or engine handling. He can ask for a “VMC Approach”, but must maintain his own terrain separation.

**Vertical Path Angle (VPA):** the required descent angle on a non-precision approach.

## 14.2 Departure Procedures

AIP ENR 1.5

The following IFR departure procedures apply at busy airports with separate Clearance Delivery, Ground and Tower frequencies. At smaller airfields some or all of those might be combined, but the procedures essentially remain the same. When departing from uncontrolled airfields, IFR flights follow the same procedures as given in paragraph 8.10.2 for VFR flights.

- ! Listen to the **ATIS** to obtain the runway in use, QNH, temperature (for take-off performance calculations), and any other pertinent information. Write it down, you might want to refer to it later.
- ! Call **Clearance Delivery** , or, if there is no separate clearance delivery frequency, call Ground to obtain your after departure clearance. Write it down and make sure you understand exactly what the controller wants you to do. The departure clearance must be read back exactly as given, and the controller must acknowledge that the read back is correct. Study the plate for the SID (see paragraph 14.2.1), if applicable, and set up the frequencies and avionics as far as possible at this stage.
- ! Call the **Surface Movement Control** (“Ground”) to obtain the start clearance. Even if parked in an uncontrolled hangar area, all IFR flights should ask for start clearance to avoid lengthy delays after engines have been started. A pilot should ask for start approximately 15 minutes before the allocated slot time, depending on how long it will take to accomplish the start and after-start checklist and the taxi to the runway. After starting the engines and going through the after start checks, set up frequencies and avionics, if not done previously. Get out the airport diagram for the departure airport and/or bring it up on the MFD to help you find your way and not get lost on the taxiways.
- ! After engine start call **Ground** again to ask for taxi. Make sure you understand exactly which way the controller wants you to taxi, and where you might have to hold and wait for further instructions. If unsure, stop and ask the ground controller for clarification. To cross a runway you need permission, such as

“ABC, cross runway 23”. Never taxi over red stop bars - a line of red lights across a taxiway! If you encounter red stop bars, stop before them, and talk to the ground controller. If he wants you to cross, he must switch them off. Be sure you know taxiway signage (yellow-and-black), runway signage (red-and-white) and runway holding point markings. Make sure you stop at the correct holding point as cleared – see paragraph 7.11.2.

- ! When ready at the holding point and after having been transferred by Ground to the aerodrome controller, call **Tower** and report “ready”. You might still have to hold, or you might be cleared to “line up and wait”, or “behind the 737 on final, line up behind” , but eventually you will be “cleared for take-off”.
- ! After you have taken off make sure you follow the after departure clearance exactly as given, contacting **Approach** or **Departure** at the point stipulated in the after departure clearance and on the given frequency with something like “Approach, good morning, ABC passing 6000 feet” . The altitude should be stated so that the approach controller can verify that he is looking at the correct aircraft on his radar screen and that the indicated altitudes correspond.
- ! After the initial call **Approach** will give you further instructions with regard to headings, routings, altitudes or flight levels to climb to and will eventually transfer you to the next controller.

### 14.2.1 Standard Instrument Departures (SID)

Standard Instrument Departures (SIDs) have been devised to ensure an orderly and efficient traffic flow for departing IFR flights and to save time issuing lengthy clearances over the radio channels. SIDs are predetermined routes from the airport to a navigational aid or other significant point. They are published in the AIP and commercial instrument guides, and have their own published communication failure procedures. Points on a SID can be defined by conventional navigational aids (VOR), or by GNSS-points.

The ATIS should mention when SIDs are in operation. Secondary Surveillance Radar must be operational when SIDs are in use.

The clearance for a SID will be given to the pilot by Clearance Delivery (if operational) or by the Ground controller before start-up, e.g. “ABC is cleared Johannesburg to Bloemfontein. After departure runway 03L to comply with the RAGUL 3A departure. Departure frequency 124.5, squawk 3456” .

The SID plate will give the tracks or radials to be flown after take-off, the altitude at which the pilot must contact the departure frequency, the flight level to which the aircraft may climb, any restrictions with respect to speeds, climb gradients or minimum altitudes at specific points and any other pertinent information. The pilot needs to study the SID plate carefully, preferably before starting the engine(s).

If a pilot cannot comply with any of those stipulations he must notify ATC, who will then issue a non-standard clearance. The standard climb gradient aircraft must be able to achieve is 3.3%, unless the SID plate states a steeper gradient. To work out the climb gradient in feet per minute multiply the anticipated groundspeed by the gradient in percent (climb at 100 kts, gradient 3.3%, needs a 330 fpm climb minimum). Sometimes climb gradients might be too steep for small piston aircraft, in which case the controller must be notified that you “cannot comply”. Note that any given climb gradient assumes all engines operating. Should a pilot suffer an engine failure while on a SID, he should inform the controller and request radar vectors.

## 14.3 IFR En-Route

AIP ENR 1.8

IFR en route flying is really simple, with much less talking than when flying VFR. As long as the route has been properly planned, taking into account:

- ! SID termination point at the departure airfield,
- ! MEA and MOCA (see definitions) restrictions en route,
- ! avoidance of prohibited or restricted airspace, and
- ! STAR entry point at the destination airfield,

there should be few surprises.

If the IFR flight is in contact with ATC and in a radar environment right through the flight, the controller will usually pre-empt the pilot and hand him off to the next controller at the correct point. Should a pilot approach a reporting point where he knows that he should be speaking to another controller without being pre-empted by ATC, he just needs to announce his position with “Johannesburg, ABC approaching XOTOP” and he will be handed over to the next controller. The first call to the next controller will be something like “Central, ZS-ABC, XOTOP, FL260”. Note that there is usually no need for the full AIREP or estimates, since the controller has the aircraft on radar and can drag the mouse on his radar screen to any point to find out the aircraft’s ETA (at present groundspeed).

In a non-radar environment or if the aircraft is still below radar coverage, the pilot should give the full AIREP as explained in paragraph 8.7. This is the case in large parts of Africa north of our borders. In HF communications it is imperative to establish contact every time and to state the nature of the call on first contact. To give a routine position report say something like “Brazzaville, ZS-ABC, position”, wait for a reply, then go ahead with the AIREP.

Any change of level or track in controlled airspace, for example to avoid thunderstorms, needs permission from the controller. Ask for it with “Johannesburg, ZS-ABC, request right of track for weather”. On an HF frequency, due to the usually less clear communications, establish contact first with “Brazzaville, ZS-ABC, request”, wait for a reply, then state your request.

14.3.1 Reduced Vertical Separation Minima (RVSM)

**Between flight levels 290 and 410** most airspace in the world has been declared Reduced Vertical Separation Minima (RVSM) airspace. For flights in RVSM airspace aircraft and pilots have to be “RVSM compliant”, which means:

- ! aircraft need to be approved with two independent, calibrated altimetry systems,
- ! aircraft need to be fitted with an autopilot with altitude hold capabilities,
- ! pilots have to be trained in the use of these systems,
- ! certain pre-flight and in-flight checks have to be carried out on each flight, and
- ! pilots and controllers need to be trained in contingency procedures, i.e . what to do if any of these systems fail.

Furthermore, there are special procedures for certain airspaces, e.g. the very busy North Atlantic routes. These procedures exceed the scope of this book, and this paragraph just serves to remind pilots of special RVSM phraseology.

|                            |                                                                     |
|----------------------------|---------------------------------------------------------------------|
| Negative RVSM              | My aircraft is not RVSM approved                                    |
| Affirm RVSM                | My aircraft is RVSM approved                                        |
| Unable RVSM due turbulence | I cannot maintain the height-keeping requirements due to turbulence |
| Unable RVSM due equipment  | I cannot maintain RVSM requirements due to equipment failure        |
| Ready to resume RVSM       | I can now comply with RVSM requirements again.                      |

14.3.2 Strategic Lateral Offset Procedures (SLOP)

AIP ENR 1.8(8)

With the wide use of GNSS for navigation aircraft are now following absolutely precise tracks, which has increased the probability of mid-air collisions. Applying an offset (by default always to the right of track) significantly reduces the collision risk between aircraft on the same and on opposing tracks, as well as lateral spacing from aircraft above and below.

Strategic Lateral Offset Procedures can be programmed into a modern Flight Management System (FMS) to steer the aircraft parallel to but a certain distance away from the centre of the route.

SLOP may only be applied in certain airspaces or on certain routes as published in the AIP and is typically up to 2.0 nm in oceanic airspace (FAJO) and up to 0.5 nm in continental South African Airspace (FACA, FAJA). The crew would randomly select the desired offset in increments of a fraction of a mile up to the permissible maximum for the relevant airspace. ATC clearance for this procedure is not required and it is not even necessary to advise ATC.

When exiting the area or route where SLOP is authorised, the aircraft must immediately return to the nominal track.

### 14.3.3 Airborne Collision Avoidance System (ACAS)

Several Traffic Information Systems (TIS) or Traffic Collision Avoidance Systems (TCAS I) are available even for smaller aircraft. These use the transponder signals generated by other aircraft to show the traffic's position on a GPS screen or on a MFD (Multi Function Display), relative to their own position.

Aircraft powered by turbine engines with a maximum mass of more than 15,000 kg or more than 30 passenger seats must be equipped with the more capable TCAS II, also called Aircraft Collision Avoidance System (ACAS). This will supply Traffic Advisories (TA) as in TCAS I, but will also give Resolution Advisories (RA), for more serious conflicts. The latter command the pilot to climb or descend into a green band on the vertical speed indicator to avoid other traffic. **RAs take priority over controller instructions**, but ATC must be notified with the following phraseology:

|                                   |                                                    |
|-----------------------------------|----------------------------------------------------|
| TCAS RA                           | I'm deviating from my clearance due to a TCAS RA   |
| Clear of conflict, returning to ! | I am resuming the original clearance level         |
| Clear of conflict, ! resumed      | I have returned to the original clearance level    |
| Unable, TCAS RA                   | I cannot comply with your instruction due to an RA |

Pilots flying aircraft equipped with TCAS II need special initial and recurrent training.

### 14.3.4 Selective Calling System (SELCAL)

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On long intercontinental flights it can be very tiresome for pilots to listen out on a frequency with a lot of static interference or radio chatter meant for other traffic. In these instances, and if both aircraft and ground station are suitably equipped, the **Selective Calling System (SELCAL)** may be used.

SELCAL uses the existing HF or VHF radios, which are modified for SELCAL operations. Should ATC want to talk to the aircraft, the ground station will transmit a group of coded tone pulses. The decoder in the aircraft for which the message is intended will accept the signal and activate the cockpit call system by flashing a light or sounding a chime.

## 14.4 Arrival Procedures

AIP ENR 1.5, AIP ENR 1.8

An IFR arrival in VMC into an uncontrolled airport or into a small airport without published arrival routes will be very much like a VFR arrival. However, if the destination airport has published arrival procedures or the weather is bad, the pilot will have to fly the published arrival route and possibly do an instrument approach.

An instrument approach consists typically of 5 segments:

- ! the **Arrival Segment** is the route from the last en-route fix to the first fix associated with the approach (Initial Approach Fix - IAF). During the Arrival Segment the aircraft must maintain the higher of either the Minimum Safe Sector Altitude (MSA) or the Initial Approach Altitude, unless under radar control. If the en-route tracking ends at the IAF there is no Arrival Segment.
- ! the **Initial Approach Segment** commences at the IAF and may consist of a holding pattern or reversal procedure to lose altitude. If it is a racecourse pattern the pilot should use standard timing, i.e. one-minute legs. The Initial Approach Segment ends at the Intermediate Approach Fix, which may be the same as the Initial Approach Fix, but at a lower altitude.
- ! the **Intermediate Approach Segment** blends the Initial Approach into the Final Approach Segment. In this segment the aircraft speed, configuration and track are adjusted for the final approach. If these are racecourse patterns they are usually published with a certain maximum length (typically 5 to 12 nm). The Intermediate Approach Segment ends at the Final Approach Fix (FAF), which sometimes is again the same facility as the IAF, but at a lower altitude.
- ! the **Final Approach Segment** commences at the Final Approach Fix, which could be an NDB, a VOR, a significant point or a certain distance on a specified bearing from a VOR. For an ILS approach the Final Approach Fix is where the final approach altitude intersects the glide slope. The Final Approach Segment ends at the Missed Approach Point (MAP or MAPt).
- ! the **Missed Approach Segment** starts at the published Missed Approach Point for Non-Precision Approaches or where the glide slope intersects the Decision Altitude (DA) for Precision Approaches. The Missed Approach Segment involves a climb and usually a turn back to the facility. It ends with the commencing of another approach, the return to a holding pattern or the resuming of en-route flight to an alternate aerodrome.

An arriving IFR flight should listen to the ATIS of the destination airport as soon as possible to form an idea of weather conditions, runway in use etc. The aircraft will usually receive an inbound clearance on first contact with the area control or information frequency serving the destination airport. If STARs are not in operation,

this will be something like: “ABC is cleared to GRV, FL220, information Hotel on the ATIS, report for descent”.

#### **14.4.1 Standard Terminal Arrival Routes (STAR)**

To ensure smoother traffic flow and easier routing for arriving instrument traffic, Standard Terminal Arrival Routes (STARs) have been devised. These are predetermined routes from the end point of an air route to the destination airport. STARs are published in the AIP and commercial instrument guides, and have their own published communication failure procedures. Their use ensures an orderly and efficient traffic flow, avoids lengthy clearances and thus makes communications easier and saves time. Like with SIDs, the defining points of STARs can be conventional navigational aids (VOR), a DME distance from a VOR or GNSS-points.

The ATIS should mention when STARs are in operation. Secondary Surveillance Radar must be operational when STARs are in use.

The clearance for a STAR will usually be given when an aircraft first makes contact with the control or information frequency serving the destination airport, with words such as: “ABC route direct to AVILO for the AVILO 1A arrival, landing runway 03 RIGHT, information Foxtrot on the ATIS, report for descent”.

The pilot now needs to study the STAR chart carefully. Once he reaches the STAR entry point, he will follow the route as laid down on the STAR chart. He will also request descent according to his planning, which might be before or after the STAR entry point. The area controller and later the approach controller will clear the pilot to successively lower levels, keeping him separated from other traffic and terrain.

STARs often have published speed limits, on the AVILO 1A arrival those are: 250 kts at AVILO, 210 kts at 30 nm from JSV, and 180 kts on the localizer to 10 nm JSV. ATC may waive those speed restrictions with the words “cancel speed restrictions”, or the colloquial “no speeds”.

Communication failure procedures are published on the STAR chart. These apply not only when an aircraft has been cleared for a STAR, but also for an aircraft inbound to an airport where STARs are in operation.

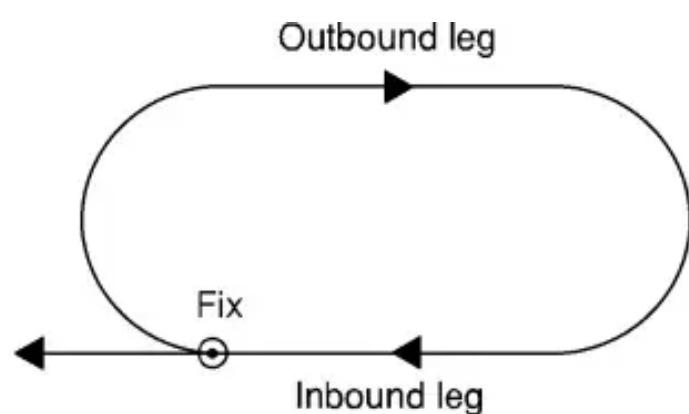
A STAR ends by either the pilot positioning himself on the ILS, or by the controller radar vectoring the pilot to a position and altitude, from which the ILS can be intercepted.

## 14.4.2 Holding Patterns

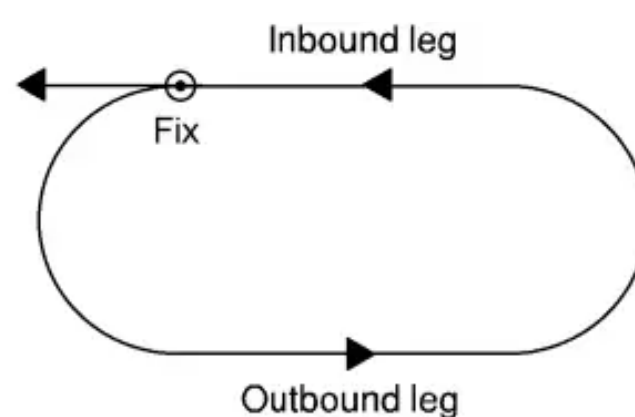
A holding pattern is a race-course pattern established at a certain fix used to delay aircraft for resolving traffic congestion or for spacing aircraft for an instrument approach.

Every holding pattern is defined by the following four items:

- ! **the fix** (NDB, VOR, DME distance on a specific VOR radial, GNSS position).
- ! **the inbound track** (as a track inbound or as a radial)  
If the hold is specified on the, say, 180 radial of a VOR the inbound track will be the reciprocal, 360 in this example.
- ! **the direction of the turns** (left or right)  
A right hand pattern is standard, but due to terrain or airspace restrictions non-standard left hand patterns may be established.
- ! **the timing**  
Standard timing is 4 minutes for holds up to and including FL140 - and 5 minutes for holds above FL140. ATC can, however, ask for any other time to be flown on the outbound track. Since the two 180° turns will take 1 minute each, the standard time on the outbound leg is 1 minute up to FL 140 and 1 & minutes above that. Allowance must be made for the effects of a head- or tailwind wind by shortening or lengthening the time on the outbound leg and by applying drift correction for a crosswind.



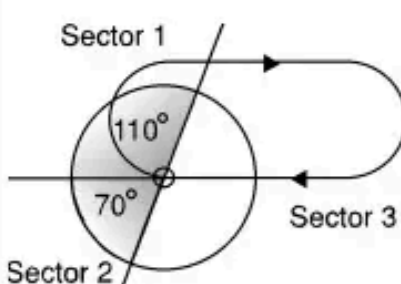
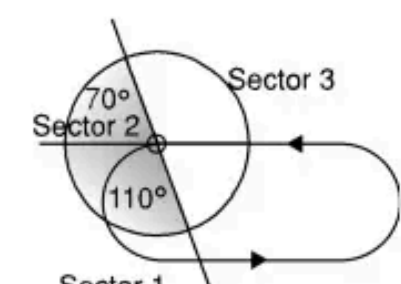
Right hand pattern



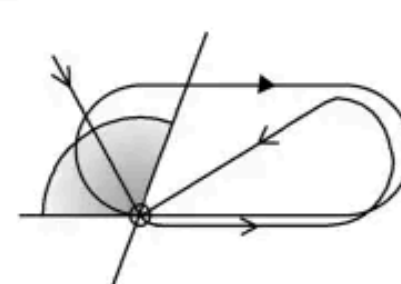
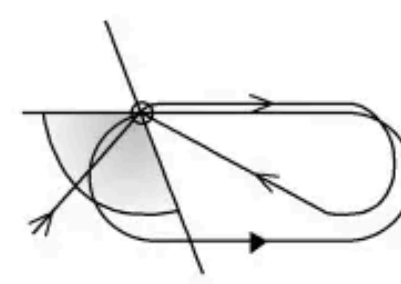
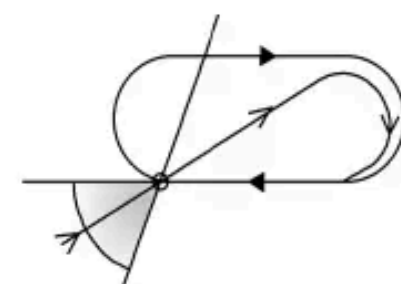
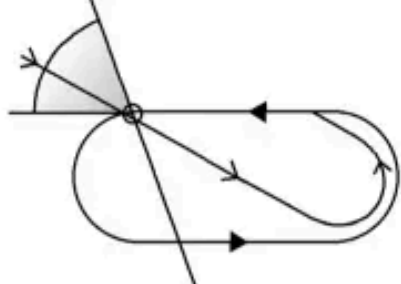
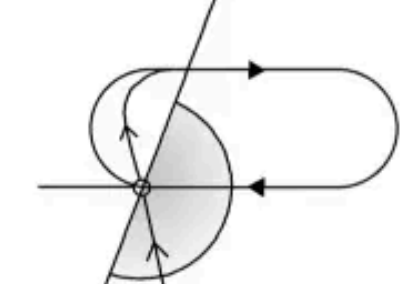
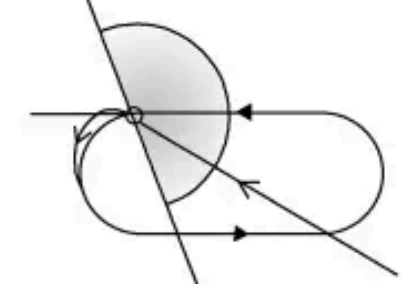
Left hand pattern

Holding patterns are published on approach plates or en-route charts as well as in the AIP. However, a controller could instruct a pilot to hold anywhere with something like the following phraseology: "ABC, hold at WRV, FL210, inbound track 068 degrees, right hand pattern, outbound time 2 minutes, onward clearance time is 1035".

An aircraft arriving at a holding pattern must position itself in the pattern. Depending on the direction of arrival and the direction of the holding pattern, certain entry procedures must be flown.

| Right hand pattern                                                                |                                                                                                                                                | Left hand pattern                                                                   |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | <p>To join a holding pattern a certain entry manoeuvre has to be flown, depending on the direction from which the aircraft reaches the fix</p> |  |

The table below illustrates the three types of entry: parallel, offset, or direct:

| Right hand pattern                                                                  |                                                                                                                                                                                                                                 | Left hand pattern                                                                     |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  | <p><b>Sector 1: Parallel Entry</b><br/>at the fix turn onto the outbound track to fly parallel to the inbound track for 1 minute, then turn into the pattern to return to the fix.</p>                                          |  |
|  | <p><b>Sector 2 : Offset Entry</b><br/>at the fix turn into the pattern on the outbound track <math>\pm 30^\circ</math>. Fly for 1 minute, then turn in the direction of the holding pattern to intercept the inbound track.</p> |  |
|  | <p><b>Sector 3: Direct Entry</b><br/>at the fix turn to follow the holding pattern.</p>                                                                                                                                         |  |

En-route holds are usually aligned with the inbound track, so that the pilot is already established more or less on the inbound holding track and at the fix just needs to turn right or left for a direct sector 3 entry.

Holding patterns are designed to be flown at the following **maximum** speeds:

| Flight Level   | Aircraft / Conditions                                                     | Speed               |
|----------------|---------------------------------------------------------------------------|---------------------|
| up to FL140    | Cat A and B aircraft:                                                     | 170 kts             |
|                | Other aircraft, calm conditions:                                          | 230 kts             |
|                | Other aircraft in turbulence:                                             | 280 kts             |
|                | If the hold is followed by an approach which necessitates a higher speed: | that higher speed   |
| FL150 to FL200 | In turbulent conditions or in an en-route hold:                           | 280 kts or Mach 0.8 |
|                | Otherwise:                                                                | 240 kts             |
| FL210 to FL340 | In turbulent conditions or in an en-route hold:                           | 280 kts or Mach 0.8 |
|                | Otherwise:                                                                | 265 kts             |
| above FL340    | Always:                                                                   | Mach 0.83           |

14.4.3 Instrument Approaches

An instrument approach is a published procedure with regard to specific tracks over ground and given altitudes and descent patterns. It is used to position the aircraft for landing in Instrument Meteorological Conditions.

Positioning for an instrument approach can be done by the pilot, following published approach procedures with the five segments of an instrument approach as explained above. When Secondary Surveillance Radar is available, the controller can also guide the pilot directly to the intermediate approach segment by assuming radar control and then issuing radar vectors, i.e. headings for the pilot to fly. The controller will adjust these headings for any wind drift.

An aircraft will be cleared for an approach with the words: “ABC, cleared for the VOR approach runway 23” . This allows the pilot, once established on the correct track, radial or localizer to descend as specified on the plate. In case of a radio failure during the approach, the pilot may continue to MDA or DA, and execute the full published missed approach procedure if he cannot see the runway.

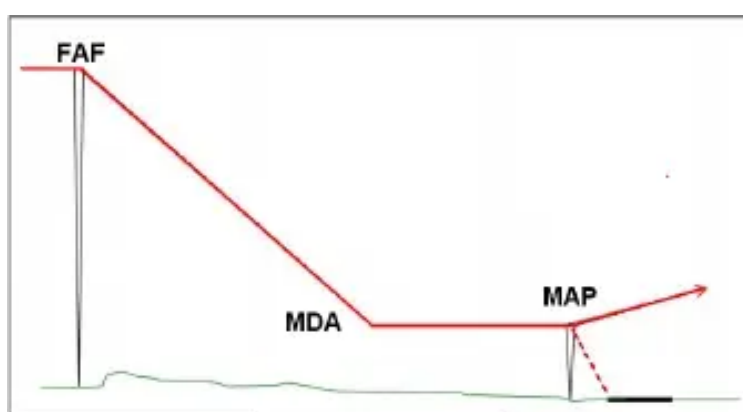
14.4.3.1 Non-Precision Approach Without Vertical Guidance – 2D

A non-precision approach without vertical guidance is an instrument approach giving horizontal (azimuth) guidance only, without vertical guidance, i.e. a 2D-approach.

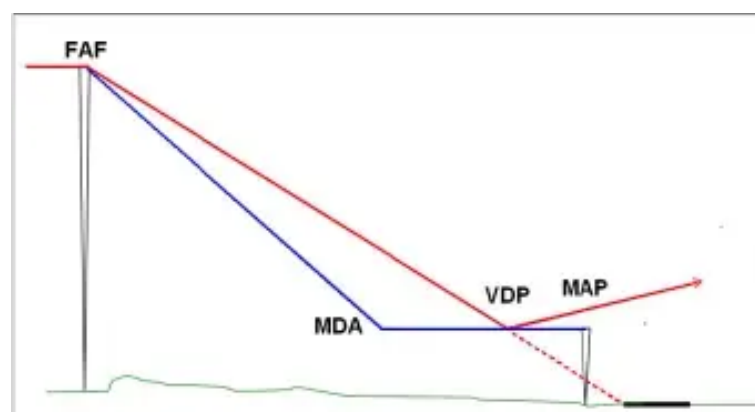
The horizontal guidance is provided by either a ground-based navigational aid ( **NDB** and **VOR** approaches) or by GNSS ( **LP** = Localizer Performance and **LNAV** = Lateral Navigation approaches).

Non-precision approaches without vertical guidance have certain step-down fixes, which need to be overflown at or above the published altitudes using the altimeter. Once the published **Minimum Descent Altitude (MDA)** or **Minimum Descent Height (MDH)** is reached the pilot may fly level until the runway is in sight and continue to land, or until reaching the **Missed Approach Point (MAP or MAPt)**, where the missed approach must be initiated if the runway is not in sight.

Step descents (colloquially called “drive and dive”) necessitate constant power and attitude changes, leading to unstabilised approaches and possibly a final dive towards the runway for landing. It is far safer to fly this type of approach as a Constant Descent Final Approach (CDFA) instead, with a constant descent angle to the **Visual Descent Point (VDP)**. At the VDP the aircraft is in the correct configuration and position for a normal landing or, if the runway is not in sight, executes a missed approach.



Conventional non-precision approach



CDFA non-precision approach

#### 14.4.3.2 Non-Precision Approach With Vertical Guidance (APV) – 3D

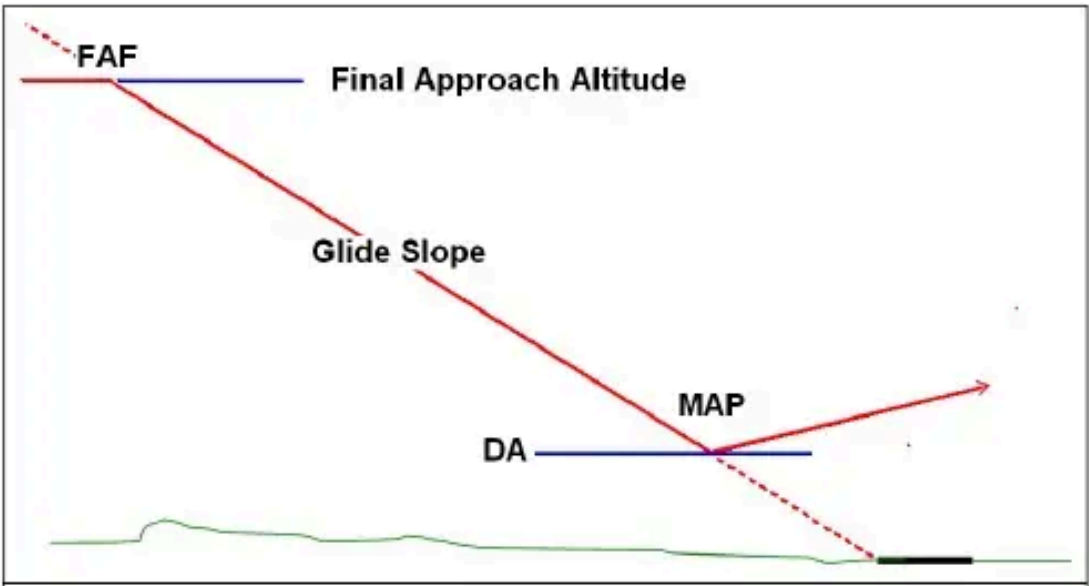
An **LNAV/VNAV** (= lateral **navigation** + vertical **navigation**) GNSS approach, also called a Baro/VNAV GNSS approach, derives altitude information solely from a barometer (the altimeter).

An **LPV** (**L**ocalizer **P**erformance with **V**ertical **G**uidance) approach requires a Space Based Augmentation System (SBAS) such as the Wide Area Augmentation System (WAAS) in the USA, the European Geostationary Navigation Overlay Service (EGNOS) in Europe and North Africa or others in other parts of the world. These systems rely on geostationary satellites to calculate and broadcast small errors in the satellite information and correct these, thereby providing greater accuracy. Unfortunately, SBAS is not (yet) available in Southern Africa.

Although to the pilot the computed vertical guidance looks exactly the same as the glideslope on an ILS, and the missed approach will be initiated at the published **Decision Altitude (DA)** or **Decision Height (DH)** rather than at a defined MAPt, APVs are still classified as non-precision approaches – since the wide area (whole countries or even continents) space based augmentation does not fulfil the accuracy required for a precision approach.

14.4.3.3 Precision Approach - 3D

A precision approach is an instrument approach giving guidance both in azimuth (localizer) and vertical profile (glide slope). The Final Approach Fix (FAF) is the point where the glide slope intersects the final approach altitude and the Missed Approach Point (MAP) is the point where the glide slope intersects the



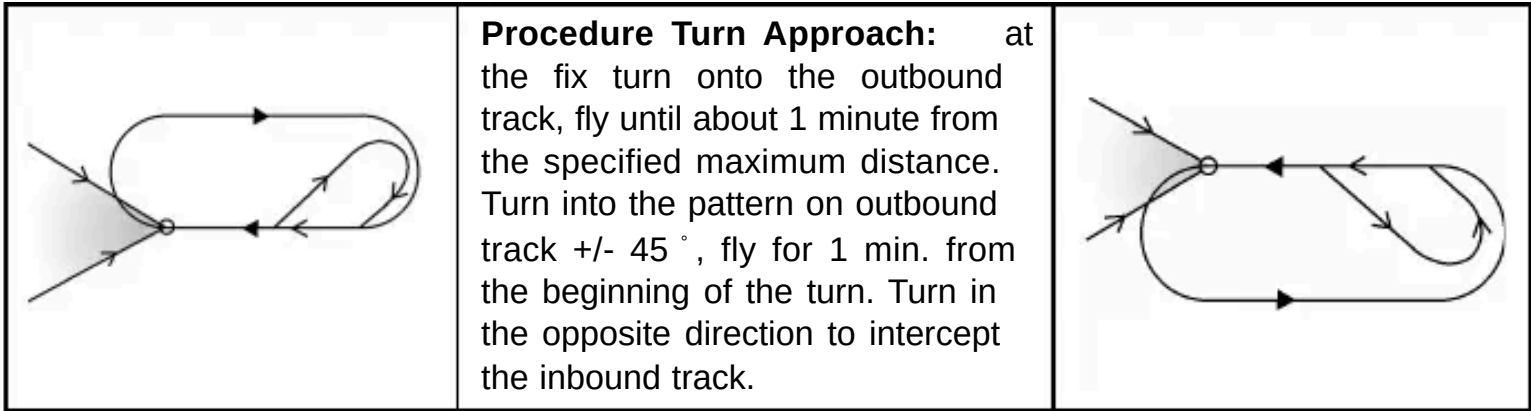
Decision Altitude (DA) or Decision Height (DH) .

The **Instrument Landing System (ILS)** relies solely on a ground-based installation for each runway, which sends out signals for lateral as well as vertical guidance to the touch-down point. See also paragraph 15.4.

A **GNSS Landing System (GLS)** uses ground-based augmentation (GBAS) to GNSS, where a single airport ground station transmits the required GNSS-corrections on a separate VHF frequency to suitably equipped aircraft to enable them to fly a precision approach very similar to the ILS. There are no GLS approaches available in South Africa yet at the time of writing.

14.4.3.4 Procedure Turn Approach

If an aircraft approaches a fix from within 30° of the outbound track, i.e. pretty much opposite to the final approach path, the pilot may request a procedure turn approach with the words “request procedure turn approach” . When cleared, the pilot will follow the procedure below to establish himself on the inbound final approach track, while descending to the final inbound altitude.



#### 14.4.3.5 Circling Approach

A circling approach is an extension of an instrument approach procedure to allow visual manoeuvring prior to landing, where the final segment of the instrument approach procedure is not within the 30° of the runway direction required by a straight-in approach. This could be by design (due to high terrain close to the airport) or it could be because there is only one approach available, but wind conditions require the landing to be made in the opposite direction or any other runway. Provided the cloudbase is high enough and the visibility good enough, a pilot can elect to fly the available approach procedure until he is below cloud and can see the airfield. He will then visually manoeuvre (fly a visual circuit) to land on a runway that is not aligned with the initial approach path of the instrument procedure.

#### 14.4.3.6 Approach Ban

CAR 91.07.26

An approach may be commenced even if the Runway Visual Range (RVR) is less than the minimum required for that particular approach and the chance of being able to land off that approach is small.

However, if ATC advises the pilot before he has reached a point 1000' above threshold elevation that the RVR is less than the published minimum, the pilot must execute a missed approach at the 1000' point.

If the pilot has already passed the 1000' point when ATC advises that the RVR is below minima, the approach may be continued. If at the missed approach point the pilot has the required visual reference, i.e. can see well enough for landing, he may land.

#### 14.4.3.7 Missed Approach

When a pilot on an instrument approach reaches the Missed Approach Point and does not see enough of the runway environment to land (see "required visual reference" in paragraph 14.1), he must execute the missed approach.

Usually, the missed approach will be flown exactly as stipulated on the approach plate. However, ATC may advise a pilot at some point during the approach that the missed approach procedure will be non-standard, with words such as "ABC, in case of a missed approach, maintain runway track to 5000', then turn right to GRV, climbing to 8000'".

After the pilot has initiated the missed approach, and when it is safe to do so, he must advise ATC that he has missed the approach and what his intentions are. An example would be: "Upington, ABC, missed approach, request FL150 to Kimberley".

Alternatively, if the pilot wants to try a second approach, he will say something like:  
“Bloemfontein, ABC, missed approach, request return to BLV for another approach”

14.4.3.8 Speed Limits on an Instrument Approach

Instrument approaches are carefully designed to protect aircraft from terrain. The approach designer (procedures specialist) will have taken into account various factors such as induced altimeter errors due to temperature variances and possible pilot control problems due to mountain turbulence. He will have designed the procedure with turn radii based on the following maximum indicated speeds. To remain protected against terrain pilots must not exceed the following speed limits on an instrument approach.

| Aircraft Approach Category        | A                           | B      | C       | D       | E       |
|-----------------------------------|-----------------------------|--------|---------|---------|---------|
| V <sub>at</sub> (Threshold speed) | < 91                        | 91-120 | 121-140 | 141-165 | 166-210 |
| Approach segment                  | All speeds in kts indicated |        |         |         |         |
| Initial Approach                  | 150                         | 180    | 240     | 250     | 250     |
| Reversal / racetrack              | 110                         | 140    | 240     | 250     | 250     |
| Final Approach                    | 100                         | 130    | 160     | 185     | 230     |
| Circling Approach                 | 100                         | 135    | 180     | 205     | 240     |
| Intermediate Missed Appr.         | 100                         | 130    | 160     | 185     | 230     |
| Final Missed Approach             | 110                         | 150    | 240     | 265     | 275     |

" The **aircraft approach category** (A to E) must not be confused with **aircraft performance class** (A to D), which is based on number and type of engines and aircraft weight, and influences required runway length, climb profiles and cruise altitudes.

14.5 Associated Radio Procedures

14.5.1 A GNSS LNAV Approach in Uncontrolled Airspace

Approach Plate FAPG RNP RWY 30

ABC is approaching Plettenberg Bay from the north-east at FL100 for the RNP (LNAV) approach runway 30, routing towards position GENUV. The aircraft is in uncontrolled airspace in contact with Cape Town East Information on frequency 127.575 MHz. The pilot wants to start his descent at 30 nm before Plettenberg Bay and advises Information accordingly:

A/c: Information, ABC starting the descent now to 4200’ routing GENUV for the RNP runway 30 at Plettenberg Bay.  
Info: ABC, no reported traffic for your descent, George QNH 1020, report at GENUV.  
A/c: George QNH 1020, report at GENUV.

The aircraft descends initially to 7800' (the MEA), at 25 nm from Plettenberg Bay to 6700' (the MSA 25 nm), and when passing the 110 'radial' from Plettenberg Bay to 4200' (the minimum altitude for GENUV). At GENUV the pilot reports:

A/c: Information, ABC at GENUV 4200'.

Info: ABC, no reported traffic, continue TIBA, and don't forget to cancel Search and Rescue

A/c: TIBA, and we'll cancel by telephone, ABC.

On 124.8, the pilot says:

A/c: Traffic in Plettenberg Bay, ABC, a Cessna 210, GENUV 4200' for the RNP approach runway 30.

At PG2T1, the initial approach fix at 2800':

A/c: Plettenberg Bay, ABC, Cessna 210, PG2T1 2800' final approach runway 30.

And at PG2F1, the final approach fix:

A/c: Plettenberg Bay, ABC final runway 30.

The pilot now descends to the MDA of 860', cannot see the runway and initiates the missed approach:

A/c: Plettenberg Bay, ABC, Cessna 210, missed approach, climbing to 3500', diverting to George, changing frequency to 128.2.

The aircraft will now contact George Approach and request routing to George.

## 14.5.2 A procedural ILS Approach

ABC is on an IFR flight inbound to Kruger Mpumalanga International Airport from the west. The pilot is on the descent, has been speaking to Lowveld Information on 119.0, and has been handed over to Kruger Approach on 119.2. The pilot establishes contact with Kruger Approach, who provides both Approach and Aerodrome control.

A/c: Kruger, ABC, good morning

TWR: ABC, Kruger, good morning, route PKV, descend to 8200' on a QNH of 1031, report at PKV, no delay expected for the ILS runway 05. Kruger is overcast at 4000', visibility 5000 m in light rain, wind light and variable.

A/c: Route PKV, descend to 8200' on 1031 for the ILS runway 05, report PKV, ABC.

Overhead PKV at 8200', turning outbound:

A/c: Kruger, ABC, PK 8200'

TWR: ABC is cleared for the ILS approach runway 05, report established inbound

A/c: Cleared for the ILS 05, report established inbound, ABC.

The aircraft now turns on the outbound track of 234 and descends to 5800' as specified on the approach chart, turns right at 12 nm from PKV, and when established inbound on the localizer, says:

A/c: ABC established on the localizer

TWR: ABC cleared to land runway 05, wind 060, 5 kts, caution runway wet

A/c: Cleared to land, ABC.

### 14.5.3 An IFR Flight From Johannesburg to Cape Town

This flight is conducted entirely in controlled airspace; a flight plan has been filed.

On the apron, in parking bay G7, prior to start-up, the pilot contacts the OR Tambo Clearance Delivery on 121.7 to obtain his after departure clearance:

A/c: Clearance Delivery, ZS-ABC

121.7: ABC, Clearance Delivery

A/c: ABC in G7, request Level 240 to Cape Town, endurance 0420

121.7: ABC, Information Juliet with QNH 1017, runway 03L. Your clearance: ABC is cleared Johannesburg to Cape Town, after departure runway 03L to follow the standard Ragul Three Alpha Departure. Departure frequency 123.7, squawk 2273, and your CTOT is 59

A/c: Cleared Johannesburg to Cape Town, after departure runway 03 left to follow the Ragul Three Alpha. Departure frequency 123.7. Squawk 2273, CTOT 59, ABC

121.7: ABC, read back correct.

The aircraft is ready for start-up and contacts the Surface Movement Control on 121.9:

A/c: Ground, ZS-ABC in G7, request start-up, Information Juliet

GND: ABC, Tambo Ground, start-up approved, time 48, Information Juliet is current with QNH 1017, runway 03 left

A/c: Start-up approved, runway 03 left, ABC.

After start-up, when ready to taxi (Golf, Alpha and Bravo are taxiways):

A/c: Tambo Ground, ABC requests taxi

GND: ABC, taxi via Golf, Alpha, hold short of Hotel

A/c: Taxi Golf, Alpha, hold short of Hotel, ABC

After the aircraft has stopped at the intersection of Alpha and Hotel taxiways:

GND: ABC, continue taxi on Alpha to the holding point runway 03 left. Contact Tower on 118.1 when ready for departure

A/c: Taxi via Alpha and Bravo to the hold 03 left. Tower 118.1 for departure, ABC.

When the aircraft is approaching the holding point on 118.1:

A/c: Tambo Tower, good morning, ZS-ABC is ready at the hold 03L

TWR: ABC cleared take-off 03L, wind 050 at 13

A/c: Cleared take-off 03L, ABC.

When passing 6500 feet (the altitude specified in the Standard Departure at which the departure frequency must be contacted):

A/c: Johannesburg Radar, ZS-ABC, passing 6500'

RAD: ABC, radar identified 2 miles north of JSV

A/c: ABC .

A little later:

RAD: ABC, climb FL110, contact Johannesburg on 128.3

A/c: Climb FL110, contact Johannesburg South West 128.3, good day, ABC.

The pilot changes frequency, listens out, then says:

A/c: Johannesburg, ZS-ABC

JS: ABC, climb to FL260, report at the sector boundary

A/c: Climbing to FL260, report at the boundary, ABC.

A while later, when approaching APMIN on the RNAV Route Q10, the boundary between the Johannesburg South West Sector and the Johannesburg Central Sector of the Johannesburg FIR:

JS: ABC, contact Johannesburg Central on 120.3, good day

A/c: Johannesburg Central on 120.3, ABC.

The pilot changes frequency:

A/c: Johannesburg Central, ZS-ABC, level 240

JS: ABC, Johannesburg Central, report EGNOM

A/c: Report EGNOM, ABC .

Shortly before EGNOM (the FIR boundary between Johannesburg and Cape Town):

JS: ABC, contact Cape Town West 125.1, good day

A/c: Cape Town West 125.1, ABC, good day.

Immediately, changing frequency, having listened to the ATIS before hand:

A/c: Cape Town West, good morning, ZS-ABC with Golf, EGNOM, level 240

CT: ABC, under radar control, route direct ERDAS for the ERDAS One Bravo arrival, landing runway 19, information Golf current

A/c: To route ERDAS for the ERDAS One Bravo arrival 19, ABC

When the aircraft wants to descend, probably about 20 nm or so before ERDAS:

A/c: Cape Town, ABC, request descent

CT: ABC, descend to FL150

A/c: Descend to FL150, ABC.

A little later, as the aircraft approaches FL150, probably in the vicinity of Wolseley:

CT: ABC, contact Approach now on 119.7

A/c: Approach 119.7, ABC.

Changing frequency:

A/c: Cape Town Approach, ZS-ABC, good morning, approaching FL150

APP: ABC, descend to FL90, you are number three for the approach runway 19  
A/c: Descend to FL90, ABC.

From ERDAS, the aircraft flies the published arrival tracks. A little later:

APP: ABC, under radar control, vectors for the ILS, descend to 6000 feet, QNH 1027, turn right heading 270, base leg  
A/c: Under radar control, right heading 270, descend to 6000 feet, QNH 1027, ABC.

The aircraft is now radar vectored and flies heading 270. A short while later:

APP: ABC, turn left heading 220, descend to 3000 feet, further descent on the glide slope, cleared for the ILS approach runway 19, report established  
A/c: Left heading 220, descend to 3000', cleared ILS approach 19, report established, ABC.

When the aircraft is established on the ILS:

A/c: ABC established on the localizer  
APP: ABC, radar service terminates, contact Tower now on 118.1  
A/c: Tower 118.1, ABC.

The pilot changes frequency, listens out and says:

A/c: Cape Town Tower, ZS-ABC  
TWR: ABC, cleared to land, the wind is 220 at 10 knots  
A/c: Cleared to land, ABC.

After landing:

TWR: ABC, vacate right taxiway Charlie, contact Ground 121.9 runway vacated  
A/c: Right at Charlie, Ground 121.9, ABC.

After changing frequency when clear of the runway:

A/c: Ground, ZS-ABC  
GND: ZS-ABC, give way to a Baron on Bravo, taxi Hotel to the hangars  
A/c: Hotel to the hangars, we'll give way to the Baron, ZS-ABC.

# 15

# Navigational Aids

**Paragraph 15.1 (VHF Direction Finding)** is part of the syllabus for both the Restricted and the General R/T Certificate.  
The remainder of this chapter has been included for interest only.

## 15.1 VHF Direction Finding (VDF)

Most air traffic control centres have Direction Finding equipment, which works on the normal VHF communications radio. Whenever the pilot presses the push-to-talk button, a readout in the control room or tower will give the magnetic bearing of the aircraft. Since DF works in the VHF band it is limited to line of sight and ranges may be assessed by the already mentioned formula for VHF communications.

A pilot uncertain of his position and within range of a VDF station may ask for a magnetic heading to the station (QDM) with “Please provide me with a VDF heading”. To account for possible drift, this heading should be updated from time to time. Since most of South Africa now has radar coverage, the VDF steer is becoming obsolete.

## 15.2 Automatic Direction Finding (ADF)

Automatic Direction Finding (ADF) using ground-based Non-Directional Beacons (NDB) is old technology, and NDBs are not supported in South Africa anymore. However, they still exist in other parts of the world, IFR pilots still may need to fly NDB approaches and must be thoroughly acquainted with their operation.

## 15.3 VHF Omnidirectional Range (VOR)

VHF Omnidirectional Range (VOR) ground-based beacons work in the VHF (Very High Frequency) band. They are marked on aviation charts with a compass rose

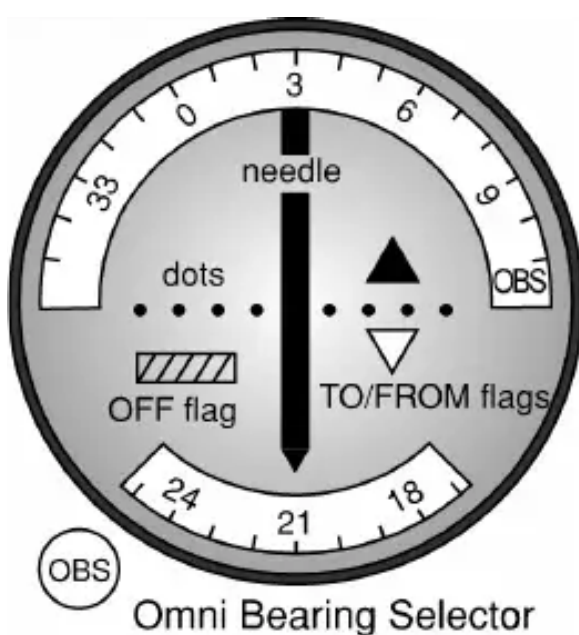
oriented to magnetic north. VORs in South Africa are allocated a three letter identification code, the last letter usually being a V, e.g. BLV for the Bloemfontein VOR, frequency 114.1 MHz, Morse code identification BLV ( - . . . / . - . . / . . . - ).

A VOR station transmits two signals of alternating phase through 360°. At any point around the VOR station the airborne receiver compares the two signals and can establish the aircraft's bearing from the station. The tracks radiating out from the station are called **RADIAL**s.

VOR beacons transmit in the VHF band between 112.0 and 117.95 MHz and are therefore limited to line-of-sight. For range see paragraph 13.4.1.

The aircraft equipment consists of an antenna, a receiver and an indicator.

The antenna consists of two small dipoles, usually mounted horizontally on the vertical fin of the aircraft.



The receiver usually has an on/off switch, sometimes integrated with the volume control, and frequency select knobs – or it may be integrated into an EFIS. To use a VOR station the pilot must enter the appropriate frequency and listen to the identification in Morse (or read the station identifier in the radio window of the EFIS).

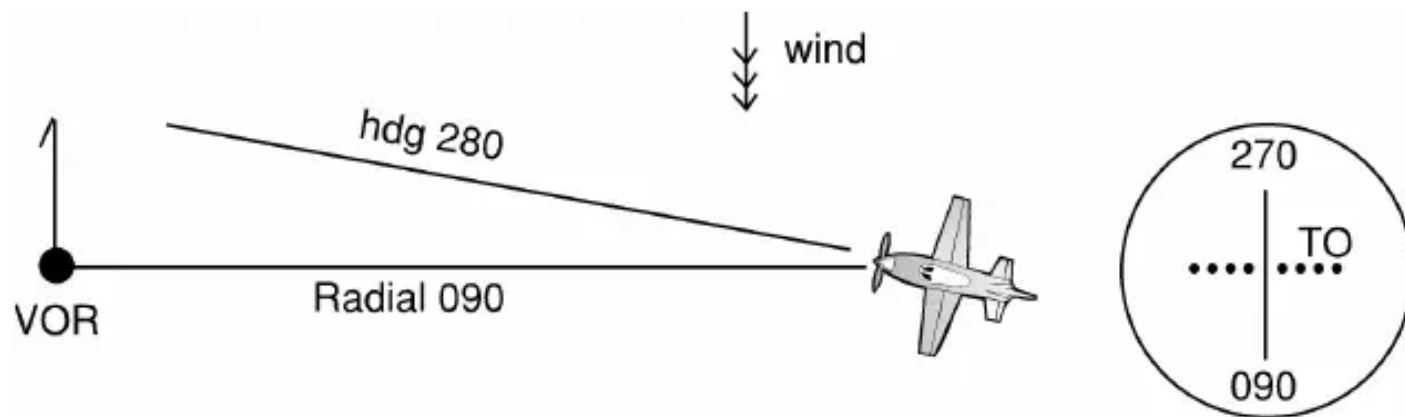
The VOR indicator has an omni-bearing scale, rotatable with the OBS knob through 360°, a TO/FROM indication and a needle that can move left or right from a centre position, usually four dots either way for a total 10° deflection, i.e. each dot is equal to a 2° deflection from the selected radial.

An OFF flag may be incorporated into the TO/FROM indication or may be a separate function on a digital display. Either way it alerts the pilot that the VOR equipment is not receiving any useable signals.

The needle deflection has to be interpreted as a command, i.e. needle left of centre = fly left. In other words, fly towards the needle. The command is given by comparing the number set in the OBS with the radial on which the aircraft is positioned. It is the pilot's responsibility to have the correct course selected on the OBS. A pilot not fully conversant with the operation and principles of the VOR will not benefit from the instrument and might even become seriously disoriented.

To fly to a station: turn the OBS until the needle is centred and the TO flag shows. Now the needle should be kept centred by making small corrections to the aircraft's heading left or right as the needle commands. In no wind the heading showing on the DI and on the top of the OBS will be the same. In a wind other than a direct tail- or

headwind, the heading flown (on the DI) will vary by the drift correction angle from the OBS course.



This aircraft tracks the 090 radial from the VOR, maintaining a heading of 280 to correct for wind.

Two simple rules to 'un-confuse' the VOR:

- ! always have the aircraft's present heading set in the top half of the OBS
- ! read the RADIAL up top if FROM is showing and at the bottom of the OBS if TO is showing. A mnemonic may help here: "FROM top - TO bottom".

## 15.4 Instrument Landing System (ILS)

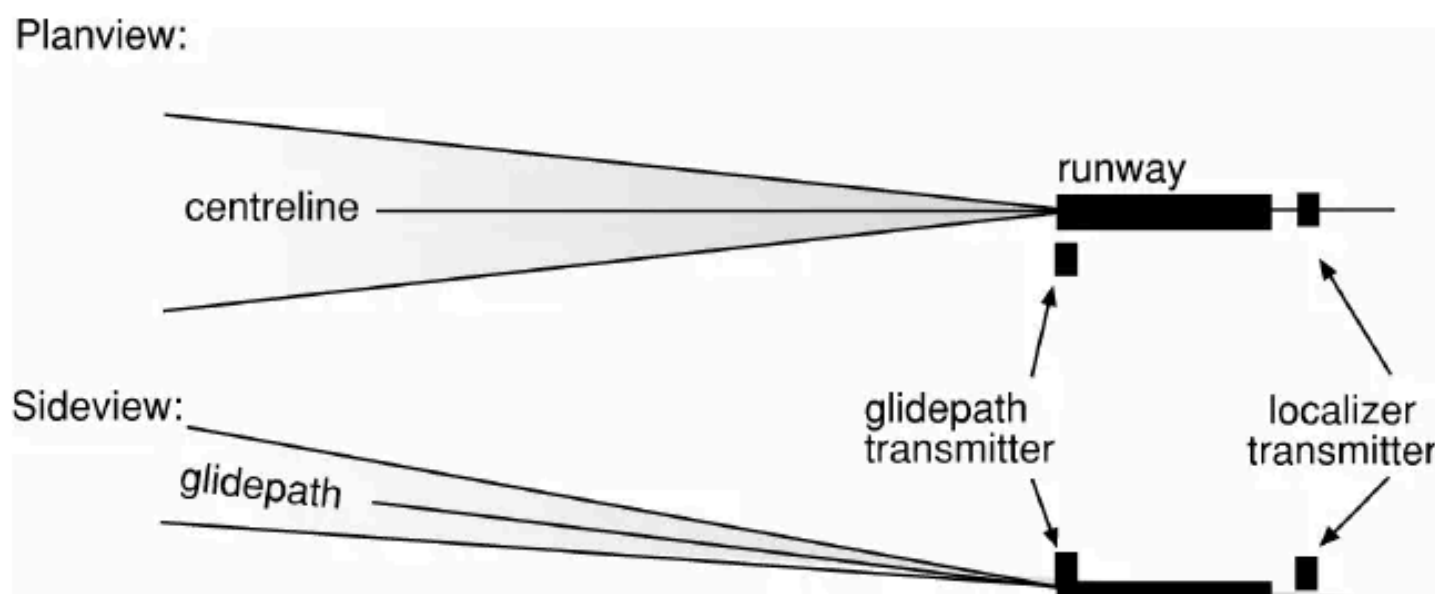
An Instrument Landing System (ILS) provides the pilot with visual instructions to fly along a predetermined path and with vertical guidance down to the threshold of a runway; this is called a precision approach.

ILS facilities fall into one of three categories according to their accuracy:

- CAT I      accurate to 200 feet above the runway threshold
- CAT II     accurate to 50 feet above the runway threshold
- CAT III    accurate to the surface of the runway

An ILS transmitter consists of two parts: the localizer (situated beyond the far end of the runway on the extended centreline), supplying approach guidance in the horizontal plane along the extended centreline of the runway - and the glide path (situated close to the threshold), which supplies guidance in the vertical plane along a usually about 3 ° slope towards the beginning of the runway.

The localizer uses one of 40 frequencies in the VHF band from 108.10 to 111.95 MHz, e.g. 109.10 or 110.35. The glide path operates in the UHF band on frequencies between 330.95 and 334.70 MHz. Localizer and glide path frequencies are always paired, thus eliminating the need to tune two frequencies. The selection of the appropriate localizer frequency for a particular runway will automatically also select the correct glide path frequency.



The ILS system uses the same indicator as the VOR, providing guidance in azimuth (localizer guidance) via the same vertically mounted needle that the VOR uses. An additional, horizontally mounted needle gives glide path guidance.

Like the VOR the ILS is a command instrument: if the pilot sees his glide path needle deflected upward, he is below the glide path and must fly up to regain the glide path.

## 15.5 Distance Measuring Equipment (DME)

DME works on the principle of secondary radar with the airborne equipment being the interrogator and the ground station being the responder. The aircraft's transmitter sends out a continuous series of paired pulses with the time interval between the pairs varied at random. When a DME ground station receives these pulses it will reply on a carrier frequency 63 MHz removed from the interrogation frequency. Once the airborne receiver picks up a response from a ground station it locks on and tracking commences.

DME operates in the UHF band in the frequency range of 962 to 1213 MHz. The range of DME is dependent on line-of-sight as given in paragraph 13.4.1.

The distances given by DME are slant ranges, not ground distances. Far from a station this 'error' is negligible, but closer to the ground station the inaccuracy is noticeable. An aircraft directly overflying a DME ground station at, say, 12000 feet will show a distance to the station of  $12000 / 6080 = 2$  nm (1 nm = 6080 feet), while the distance shown on the GPS to that waypoint will be zero.

When a controller asks an aircraft to report its distance from a facility the pilot should reply with the distance and the word "DME", for example: "ABC is 10 DME from BLV". - It is permissible to use GPS-distances instead of the DME, but the phraseology differs: "ABC is 10 nm GPS BLV" (AIP ENR 1.3-24 ).

The aircraft's DME equipment consists of a transceiver, an indicator and an antenna. All civilian DMEs in South Africa are co-located with VOR or ILS stations, and the DME equipment is channelled by the selection on the navigation radio in the aircraft.

The military version of a VOR with a co-located DME is called TACAN ( **TAC** tical **Air Navigation**). If a civilian VOR is frequency-paired with a military DME we call it VORTAC. System operation is identical.

DME indicators come in all shapes and sizes, displaying slant distance to the station. They often also compute groundspeed and time to the station at present speed.

## 15.6 Global Navigation Satellite System

### 15.6.1 GNSS Basics

The Global Navigation Satellite System (GNSS) depends on signals broadcast from satellites in medium earth orbits. Several satellite networks are in operation: the Global Positioning System (GPS) operated by the United States of America, the Russian GLONASS system, the European Union's Galileo and the Chinese BeiDou. India, France, Japan and others are in the process of developing regional and/or augmentation systems. Most GNSS-receivers can utilise these various systems indiscriminately.

Each of the major GNSS systems consists of between 20 and 50 satellites orbiting approximately 20,000 km above the surface of the earth. These satellites continually broadcast a time signal as well as their orbital details on frequencies between 1.2 and 1.6 GHz. The GNSS receiver in the aircraft, whether panel-mounted or hand-held, receives the broadcasts from the satellites and computes its position in space by spherical trigonometry. Just as position lines from an NDB or VOR fix the aircraft's position where they intersect, so 'position spheres' from the satellites are used to obtain a GNSS fix. Three satellites are needed for a two dimensional position, and at least four satellites for an altitude fix. The accuracy of the fix depends on the GNSS system, the number of satellites received and on their position relative to the aircraft; typically one can count on an accuracy of several metres.

The various GNSS receivers have a multitude of features. All units let you navigate from one point to another, giving magnetic track, distance, groundspeed and time to go. Routes can usually be pre-planned and stored, with the GNSS automatically switching to the next leg when a waypoint is passed. Most units have a database with airports, NDBs, VORs, significant points, intersections etc. stored in memory. Many units alert the pilot when he is about to enter some "special use airspace", for example a TMA or a CTR. Most GNSS receivers have moving map displays, where the pilot can see the exact position of the aircraft in relation to airfields and navigational beacons, roads, rivers and towns.

No doubt there will still be tremendous development, and GNSS will become the navigation system of the future, superseding ground-based navigational aids as we have known them.

A word of caution, though, about the use of single, non-IFR approved GPS units in small aircraft: a GPS can lose sight of sufficient satellites, it can run out of batteries or fail for many other reasons. Therefore, the VFR pilot must always back up his GPS navigation with navigational beacons or pure pilotage and map reading. Secondly, with GNSS navigation, everybody now flies very exact tracks between two points far more accurately than ever before - with an increased risk of mid-air collisions.

With a backup map and all eyes outside the cockpit looking for other traffic, VFR cross-country flying with GNSS navigation is pure pleasure.

## 15.6.2 GNSS-based IFR Flight

IFR navigation is now done almost exclusively with GNSS. With that a new concept was needed to let aircraft fly certain routes or approach procedures depending on the capabilities of the aircraft's systems (and the pilots' training). Thus, **Performance Based Navigation (PBN)** was born. Although the word is new, the intent is actually not: you always needed a working ILS and be instrument-rated to fly an ILS. Equally, you now need certain aircraft and pilot capabilities to fly in different airspaces, routes or GNSS-based instrument approach procedures.

There are two types of Performance Based Navigation, namely RNAV and RNP:

**Area Navigation (RNAV)** - simply means that an aircraft can navigate without going from ground-based beacon to beacon. "Old" RNAV was based on Loran, INS etc. In small aircraft it is now based exclusively on GNSS. There is no automatic monitoring and the equipment does not alert the pilot in case of degradation of the system.

**Required Navigation Performance (RNP)** – is a PBN system that includes onboard performance monitoring and alerting capability.

Performance monitoring and alerting may be done internally within the receiver: **Receiver Autonomous Integrity Monitoring (RAIM)** – a function whereby the receiver monitors its own position and alerts the crew when either:

- the system knows it is outside of the required accuracy, or
- it simply cannot verify its accuracy (possibly due to insufficient satellites).

Simple GNSS is already fairly accurate, but can be further enhanced by **Space Based Augmentation Systems (SBAS)**. One of those is the American Wide Area Augmentation System (WAAS), which uses a network of ground-based reference stations in North America and Hawaii to measure small variations in the GPS satellites' signals in the western hemisphere. Measurements from the reference stations are routed to master stations, which queue the received Deviation Correction

and send the correction messages to geostationary WAAS satellites every 5 seconds or so. Those satellites broadcast the correction messages back to Earth, where WAAS-enabled GPS receivers use the corrections to improve accuracy and monitor integrity. – Other SBAS are the European Geostationary Navigation Overlay Service (EGNOS) in Europe and North Africa, the Japanese MSAS, the Indian GAGAN and various others. Unfortunately, at the time of writing, SBAS is not available in Southern Africa.

Various airspaces, routes or instrument approach procedures now specify the accuracy required of on-board navigation systems to safely fly (in) them using GNSS-based navigation. In South Africa at the moment the following general rules apply:

- **RNAV 5** : en-route and initial STAR outside 30 nm and above MSA  
CDI sensitivity 5 nm
- **RNAV 1** : terminal airspace within 30 nm of the ARP until 2 nm prior to the FAF in a radar environment  
CDI sensitivity 1 nm
- **RNP 1** : terminal airspace within 30 nm of the ARP until 2 nm prior to the FAF with or without radar coverage  
CDI sensitivity 1 nm and alert generated if inaccuracy > 1 nm
- **RNP APCH** : instrument approach procedures from 2 nm prior to the FAF  
CDI sensitivity 1 nm in initial, intermediate and missed approach segments and 0.3 nm in the final approach segment – alert generated if inaccuracy > 0.3 nm
- **RNP AR APCH** (AR = special authorization required): from 2 nm < FAF  
CDI sensitivity 0.1 to 1 nm in initial, intermediate and missed approach segments and 0.1 to 0.3 nm in the final approach segment

" The number after RNAV or RNP denotes the accuracy in nm that must proven to be maintained 95% of the time for the aircraft to be certified.

To fly in these airspaces or the relevant procedure the aircraft must be individually approved and the pilot(s) appropriately trained.

### 15.6.3 GNSS-based Instrument Approach Procedures

Paragraph 14.4.3 of this book already gave a brief overview of Approach Procedures. The following serves as a summary of the currently available GNSS-based approaches. No doubt there will be further development in the future.

Conventional instrument approach procedures are either precision approaches with a Decision Altitude (or Height), at which point the missed approach must be initiated if the runway is not in sight – or non-precision approaches, where the aircraft levels off at the Minimum Descent Altitude (or Height) and flies level up to the Missed Approach Point, where the missed approach is initiated. This distinction has become somewhat blurred with the advent of GNSS procedures. It might be better to rather refer to 2D and to 3D approaches in future.

| Type of approach  | Precision Approach  | Non-precision Approaches |                        |
|-------------------|---------------------|--------------------------|------------------------|
| Vertical Guidance | Yes (3D)            | Yes (3D)                 | No (2D)                |
| Conventional      | ILS<br>MLS*<br>PAR* | -                        | NDB*<br>VOR<br>VOR/DME |
| GNSS-based        | GLS*                | LPV*<br>LNAV/VNAV        | LP*<br>LNAV            |
| Type of minima    | DA                  | DA                       | MDA                    |

\* not available in South Africa at present

**GLS** (GNSS Landing System) approaches require a Ground Based Augmentation System (GBAS), where a ground station constantly compares its own (known) position with the information supplied by GNSS-satellites and broadcasts corrections on a separate VHF frequency to aircraft flying the approach. From that the aircraft systems generate a localizer and glideslope much like the ILS does. Contrary to the conventional ILS only one ground station is needed for all runways of an airport or even other close-by airports. GLS is not available in South Africa yet at the time of writing.

**LPV** (Localizer Performance with vertical guidance) approaches require SBAS and are therefore not available in South Africa. The system generates a vertical path, which presents itself and is flown much like the glideslope of a conventional ILS. The horizontal guidance tapers down at the threshold to 100 m either side of the runway centreline.

**LP** (Localizer Performance) approaches are the same as LPV without the vertical guidance. They are used where vertical guidance cannot be constructed, possibly due to terrain. They require SBAS, are not available in South Africa, and the horizontal guidance tapers down to 100 m either side of the runway centreline.

**LNAV/VNAV** , also called LNAV with BARO (barometric) VNAV approaches rely on the altimeter to construct the vertical guidance. Again, the information is presented to the pilot much like the glideslope of a conventional ILS. The horizontal guidance does not taper towards the runway, but remains at 0.3 nm (approximately 555 metres) either side of the runway centreline – right up to the runway threshold.

**LNAV** approaches give horizontal guidance only – pilots must descend on their altimeter and remain at or above the altitudes stipulated on the approach plate. The horizontal guidance remains at 0.3 nm (approximately 555 metres) either side of the runway centreline – right up to the runway threshold.

# Appendix I

## Airport List

Correct as per the published information at the time of printing (November 2025).

Airfields in South Africa need to be **licenced** (in accordance with CAR Part 139 with regard to equipment, procedures and systems) , **approved** , or at least **registered** (a quick and easy process). Commercial traffic may only use licenced or approved airfields, while private flights may also use registered ones. No aircraft is allowed to take off from or land at an airfield that is not licenced, approved or at least registered.

This Appendix lists licenced or approved airfields, and those with an ICAO location indicator ( AIP GEN 2.4) , navigational aid ( AIP GEN 2.5) , or allocated frequency ( AIP ENR 2.2).

**Licenced airfields with published Instrument Approach Procedures and/or ATC** (and some others) are listed in the South African Aeronautical Information Publication AIP in the third volume Aerodromes (AP), published on the SACAA website (caa.co.za > Industry > Aeronautical Information > Aeronautical Information Publication (AIP) > Aerodromes (59 aerodromes at the time of writing).

Instrument Approach charts for these airports can be found at: caa.co.za > Industry > Aeronautical Information > Aeronautical Charts.

Smaller **Licenced airfields without Instrument Approach Procedures or ATC** can be found on the SACAA website (caa.co.za > Industry > Aeronautical Information > Aerodromes and Helistops Directory). 43 airfields at the time of writing.

**Approved Airfields are** listed on the CAA website (caa.co.za > Industry > Airports > Approved Aerodromes). 109 airfields as of November 2025.

**Registered airfields** are on the CAA website (caa.co.za > Industry > Airports > Registered Aerodromes (27 airfields) and ! > New List of Registered Aerodromes (313 airfields).

Before flying to any unfamiliar airfield the pilot must look up and make himself acquainted with all available information. This includes, but is not limited to radio frequencies, operating hours and restrictions, special local procedures, permissions required, circuit direction and altitudes, night operations, payment of fees etc. etc.

The following abbreviations are used in this Appendix:

|             |                                                                                           |
|-------------|-------------------------------------------------------------------------------------------|
| <b>Intl</b> | = <b>Licenced International Aerodrome (point of entry/exit for international flights)</b> |
| lic         | = other licenced aerodrome                                                                |
| app.        | = Approved aerodrome                                                                      |
| mil         | = Military airfield – unlicensed - not to be used without permission of the Air Force     |
| reg.        | = registered airfields, prior permission usually required                                 |
| Apron       | = Apron Office, callsign Alpha Oscar                                                      |
| Clear       | = Clearance Delivery (IFR only, but also VFR flights in Cape Town and OR Tambo)           |
| GND         | = Surface Movement Control (Ground)                                                       |
| TWR         | = Aerodrome Control (Tower)                                                               |
| APP         | = Approach Control (Approach)                                                             |
| AFIS        | = Aerodrome Flight Information Service (Radio)                                            |
| ATIS        | = Automatic Terminal Information Service                                                  |
| o/h         | = office hours (weekdays, usually from about 7:00 to 17:00)                               |
| a/h         | = after hours (usually at night, weekends, public holidays)                               |

| Place                         | Typ | ICAO | COM                    | NAV       |
|-------------------------------|-----|------|------------------------|-----------|
| <b>A</b>                      |     |      |                        |           |
| Aberdeen                      |     | FAAE |                        |           |
| Aggeneys                      | lic | FAAG | 122.7 unmanned         | 116.7 AGV |
| Alexander Bay                 | lic | FAAB | 118.7 unmanned         | 112.1 ABV |
| Aliwal North                  | lic | FAAN |                        |           |
| Alkatpan Test Range           | lic | FACO |                        |           |
| Alldays                       | reg | FAAL |                        |           |
| Amsterdam                     | reg | FAAM | 121.2 Makhado App.     |           |
| Andrew's Field                |     | FAAF |                        |           |
| Arathusa Safari Lodge         | app | FAAR |                        |           |
| Arnot                         |     |      | 122.2 unmanned         |           |
| Ashkam                        |     | FAAK |                        |           |
| Aviator's Paradise            |     | FAAP |                        |           |
| <b>B</b>                      |     |      |                        |           |
| Ballito Microlight Airfield   | app |      |                        |           |
| Barberton                     | reg | FABR |                        |           |
| Barkley East                  |     | FABF |                        |           |
| Barrange                      | app |      |                        |           |
| Bartelsfontein                | reg |      |                        |           |
| Bat Hawk City                 | reg |      | 130.350 unmanned       |           |
| Beacon Airfield               | reg |      |                        |           |
| Beaufort West – Karoo Gateway | lic | FABW | 125.6 AFIS Karoo Radio |           |
| Belfast                       |     | FABH |                        |           |
| Belladonna                    | reg |      |                        |           |
| Bergplaats                    | reg |      |                        |           |
| Bergtop                       | app |      |                        |           |
| Bethlehem                     | lic | FABM |                        |           |
| Bhanoyi Aero Park             | reg |      |                        |           |
| Bhisho                        | lic | FABE |                        |           |
| Big Water                     | lic | FAGA |                        |           |
| Black Rock                    |     | FABP |                        |           |

| Place                                                   | Typ          | ICAO | COM                                                                                                                         | NAV                                       |
|---------------------------------------------------------|--------------|------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Bloemfontein - Bram Fisher International Airport</b> | lic<br>Intl  | FABL | 124.3 APP<br>120.8 TWR<br>122.65 Apron<br>126.45 ATIS                                                                       | 114.1 BLV                                 |
| Bloemfontein New Tempe                                  | lic          | FATP | 131.3 unmanned                                                                                                              |                                           |
| Bloubank                                                | app          |      |                                                                                                                             |                                           |
| Bona Bona Game Lodge                                    | app          |      |                                                                                                                             |                                           |
| Boschkop (Kitty Hawk Aero Estate)                       | app          | FAKT | 120.650 unmanned                                                                                                            |                                           |
| Boscobel                                                | reg          |      |                                                                                                                             |                                           |
| Bothaville                                              |              | FABO |                                                                                                                             |                                           |
| Braam Fisher International Airport – see Bloemfontein   |              |      |                                                                                                                             |                                           |
| Brakpan-Benoni                                          | lic.         | FABB | 122.7 unmanned                                                                                                              |                                           |
| Brits                                                   | lic          | FABS | 124.2 unmanned                                                                                                              |                                           |
| Buffelshoek                                             |              | FABG |                                                                                                                             |                                           |
| Bultfontein                                             | reg          | FABU |                                                                                                                             |                                           |
| Bundu Fly Inn                                           | app          |      | 124.8 unmanned                                                                                                              |                                           |
| Burgersdorp                                             |              | FABD |                                                                                                                             |                                           |
| Bushmans Kloof                                          | app          | FABK | 124.8 unmanned                                                                                                              |                                           |
| Butterworth                                             |              | FABV |                                                                                                                             |                                           |
| <b>C</b>                                                |              |      |                                                                                                                             |                                           |
| Caledon                                                 |              | FACG |                                                                                                                             | 114.9 CDV                                 |
| Calvinia                                                | lic          | FACV |                                                                                                                             |                                           |
| Campbell                                                | app          |      |                                                                                                                             |                                           |
| <b>Cape Town International Airport</b>                  | lic.<br>Intl | FACT | 119.7 APP<br>124.350 Director<br>118.1 TWR<br>121.9 GND<br>122.1 Clearance<br>122.650 Apron<br>127.0 ATIS<br>127.6 ATIS SLV | 115.7 CTV<br>110.3 ILS 01<br>109.1 ILS 19 |
| Cape Winelands                                          | lic          | FAWN | 131.1 unmanned                                                                                                              |                                           |
| Carletonville                                           |              | FACR |                                                                                                                             |                                           |
| Carnarvon                                               | lic          | FACN |                                                                                                                             |                                           |
| Cathcart                                                |              | FACX |                                                                                                                             |                                           |
| Cedarville                                              |              | FACZ |                                                                                                                             |                                           |
| Ceres                                                   |              | FACE |                                                                                                                             | 114.2 CSV                                 |
| Chief Dawid Stuurman – see Port Elizabeth               |              |      |                                                                                                                             |                                           |
| Chitwa Chitwa                                           | app          | FACC |                                                                                                                             |                                           |
| Clanwilliam                                             |              | FACW |                                                                                                                             |                                           |
| Comre                                                   | app          |      |                                                                                                                             |                                           |
| Contermanskloof                                         | reg.         |      |                                                                                                                             |                                           |
| Cradock                                                 |              | FACD |                                                                                                                             |                                           |
| Cradock Farms Wilgebosch                                | reg          |      |                                                                                                                             |                                           |
| <b>D</b>                                                |              |      |                                                                                                                             |                                           |
| Darlington Dam                                          |              | FADP |                                                                                                                             |                                           |
| De Aar                                                  | lic          | FADA |                                                                                                                             |                                           |
| De Doorns                                               |              | FADS |                                                                                                                             |                                           |
| De Hoop                                                 | app          |      |                                                                                                                             |                                           |
| De Panne                                                | reg          |      |                                                                                                                             |                                           |
| Deel Vier                                               | app          |      |                                                                                                                             |                                           |
| Delareyville                                            | lic          | FADL |                                                                                                                             |                                           |
| Delmas                                                  |              |      | 125.4 unmanned                                                                                                              |                                           |
| Delta 200                                               |              | FADX |                                                                                                                             |                                           |

| Place                                            | Typ      | ICAO | COM                                                                              | NAV                                       |
|--------------------------------------------------|----------|------|----------------------------------------------------------------------------------|-------------------------------------------|
| Dendron                                          |          | FADO |                                                                                  |                                           |
| Diemerskraal                                     | reg      |      | 124.2 unmanned                                                                   |                                           |
| Dinaka                                           | app      |      |                                                                                  |                                           |
| Docks-Waterfront                                 | Heli     | FADW |                                                                                  |                                           |
| Donkerhoek                                       | reg      |      |                                                                                  |                                           |
| Douglas Cape                                     | reg      | FADC |                                                                                  |                                           |
| Drakensberg-Gardens                              |          | FADZ |                                                                                  |                                           |
| Driefontein / Fochville                          |          | FADF |                                                                                  |                                           |
| Dukuduku                                         |          | FADK |                                                                                  |                                           |
| Dullstroom                                       |          | FADU |                                                                                  |                                           |
| Dundi Lodge                                      | app      |      |                                                                                  |                                           |
| <b>Durban – King Shaka International Airport</b> | lic Intl | FALE | 125.750 APP<br>118.450 TWR<br>121.650 GND<br>122.650 Apron Control<br>127.0 ATIS | 115.6 TGV<br>109.7 ILS 24<br>111.3 ILS 06 |
| Durban Virginia                                  | lic      | FAVG | 120.6 TWR                                                                        |                                           |
| Dwaalboom                                        | lic      | FADB |                                                                                  |                                           |
| <b>E</b>                                         |          |      |                                                                                  |                                           |
| Eagle Flight Academy                             | reg      |      |                                                                                  |                                           |
| Eagle Nest                                       | reg      |      |                                                                                  |                                           |
| Eagle's Creek Aviation Estate                    | app      |      | 125.8 unmanned                                                                   |                                           |
| East London – King Phalo Airport                 | lic.     | FAEL | 120.1 APP<br>118.3 TWR<br>122.650 Alpha Oscar<br>126.650 ATIS                    | 114.5 ELV<br>110.9 ILS 11<br>109.9 ILS 29 |
| Ekland Ranch                                     | app      | FAER |                                                                                  |                                           |
| Elgin                                            |          | FAEG |                                                                                  |                                           |
| Elephant Plains                                  | app      |      |                                                                                  |                                           |
| Elliot                                           | reg      | FAET |                                                                                  |                                           |
| Ellisras                                         |          |      |                                                                                  | 116.3 ERV                                 |
| El Mirador                                       | reg      | FALQ | 124.8 unmanned<br>123.4 Gliding                                                  |                                           |
| Empangeni                                        |          | FAEM | 118.9 Richards Bay                                                               |                                           |
| Entabeni                                         |          | FAEN |                                                                                  |                                           |
| Ermelo                                           | lic      | FAEO |                                                                                  |                                           |
| Eshowe                                           |          | FAES |                                                                                  |                                           |
| Estcourt                                         |          |      | 123.4 unmanned                                                                   |                                           |
| Eva's Field                                      | reg      |      | 124.2 unmanned                                                                   |                                           |
| <b>F</b>                                         |          |      |                                                                                  |                                           |
| Famazengwenya                                    | app      |      |                                                                                  |                                           |
| Far North                                        | app      |      |                                                                                  |                                           |
| Fatcats                                          | reg      |      |                                                                                  |                                           |
| Ficksburg SentraOes                              | lic      | FAFB |                                                                                  |                                           |
| Firolaz Airfield                                 | reg      |      | 125.4 unmanned                                                                   |                                           |
| Fort D Acre                                      | reg      |      | 124.4 unmanned                                                                   |                                           |
| Freeway                                          | lic      | FAFW |                                                                                  |                                           |
| <b>G</b>                                         |          |      |                                                                                  |                                           |
| Garden Route Aero – see Plettenberg Bay          |          |      |                                                                                  |                                           |
| Gariep Dam                                       |          |      | 124.8 unmanned<br>123.4 Gliding                                                  |                                           |
| Garonga Safari Camp                              | app      |      |                                                                                  |                                           |
| Gelukshoop                                       | reg.     |      |                                                                                  |                                           |

| Place                                  | Typ  | ICAO | COM                                                                               | NAV                                       |
|----------------------------------------|------|------|-----------------------------------------------------------------------------------|-------------------------------------------|
| George Airport                         | lic. | FAGG | 128.2 APP<br>118.9 TWR<br>122.650 Alpha Oscar<br>126.225 ATIS<br>130.350 for fuel | 116.6 GRV<br>110.1 ILS 29<br>109.5 ILS 11 |
| George Dick Montshioa – see Mahikeng   |      |      |                                                                                   |                                           |
| Gerniso                                | reg. |      |                                                                                   |                                           |
| Glen Lyon                              | app  |      |                                                                                   |                                           |
| Glendower                              | app  |      |                                                                                   |                                           |
| Graaff-Reinet                          | lic  | FAGR |                                                                                   |                                           |
| Grahamstown                            | reg  | FAGT | 123.5 unmanned                                                                    |                                           |
| Grand Central                          | lic  | FAGC | 122.8 TWR<br>130.350 Apron parking                                                |                                           |
| Grasmere                               |      |      |                                                                                   | 115.5 GAV                                 |
| Grasslands Airfield                    | reg  |      | 125.8 unmanned                                                                    |                                           |
| Gravelotte                             |      | FAGV |                                                                                   |                                           |
| Greater Kuduland Tshipise              | app  |      |                                                                                   |                                           |
| Greefswald                             |      |      |                                                                                   | 117.9 GWV                                 |
| Greytown                               | lic  | FAGY | 123.5 unmanned                                                                    | 113.5 GYV                                 |
| Grobblersdal – Kobus de Villiers       | lic  | FAGL | 122.4 unmanned                                                                    |                                           |
| Groot Sleutelfontein                   | app  |      |                                                                                   |                                           |
| Grootfontein                           | reg  | FAGF |                                                                                   |                                           |
| Grootvlei                              |      | FAGI |                                                                                   |                                           |
| <b>H</b>                               |      |      |                                                                                   |                                           |
| Halfweg                                | lic  | FAHI |                                                                                   |                                           |
| Harding                                |      | FAHJ |                                                                                   |                                           |
| Hattingspruit Airstrip                 | app  |      |                                                                                   |                                           |
| Heidelberg                             | lic  | FAHG | 125.9 unmanned                                                                    | 116.7 HGV                                 |
| Heilbron                               |      | FAHO |                                                                                   |                                           |
| Henneman                               |      | FAHN |                                                                                   |                                           |
| Henry's Flats                          |      | FAHF |                                                                                   |                                           |
| Hippo Pools                            |      | FAHD |                                                                                   |                                           |
| Hluhluwe                               |      | FAHL |                                                                                   |                                           |
| HMS Bastard Memorial                   |      | FAHU |                                                                                   |                                           |
| Hoedspruit –<br>Air Force Base         | mil  | FAHS | 126.4 APP<br>119.6 TWR<br>133.450 Hoedspruit<br>Ops<br>114.0 ATIS                 | 114.0 HSV<br>109.5 ILS 18                 |
| Hoedspruit Civil                       | app  | FAHT | 126.4 Hoedspruit APP                                                              |                                           |
| Hofmeyer                               |      |      |                                                                                   | 113.4 HMV                                 |
| Hoopstad                               |      | FAHP |                                                                                   |                                           |
| Howick                                 | lic  | FAHC | 123.4 unmanned                                                                    |                                           |
| Hunters Moon Game Ranch                | app  |      |                                                                                   |                                           |
| <b>I</b>                               |      |      |                                                                                   |                                           |
| Idutywa                                |      | FAID |                                                                                   |                                           |
| Ingwavuma                              |      | FAIV |                                                                                   |                                           |
| Ingwelala                              | reg  | FAIW |                                                                                   |                                           |
| Inmalkaar                              | app  |      |                                                                                   |                                           |
| Inyati Game Lodge                      | app  |      |                                                                                   |                                           |
| Isithebe                               |      | FAIS |                                                                                   |                                           |
| Itala                                  |      | FAIA |                                                                                   |                                           |
| <b>J</b>                               |      |      |                                                                                   |                                           |
| Jack Taylor Airfield – see Krugersdorp |      |      |                                                                                   |                                           |

| Place                                                      | Typ         | ICAO | COM                                                                                                                                                                      | NAV                                                                           |
|------------------------------------------------------------|-------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Jackalberry                                                |             | FAJC |                                                                                                                                                                          |                                                                               |
| Jan Kempdorp                                               |             | FAJK |                                                                                                                                                                          |                                                                               |
| Jansenville                                                |             | FAJV |                                                                                                                                                                          |                                                                               |
| Jejane Private Game Reserve                                | app         |      |                                                                                                                                                                          |                                                                               |
| <b>Johannesburg –<br/>O.R. Tambo International Airport</b> | lic<br>Intl | FAOR | 123.7 Radar W oh<br>124.5 Radar S & E<br>Radar W ah<br>121.4 Director<br>118.1 TWR West<br>118.6 TWR East<br>121.9 GND<br>121.7 Clearance<br>122.650 Apron<br>126.2 ATIS | 115.2 JSV<br>110.3 ILS 03L<br>109.1 ILS 03R<br>109.9 ILS 21L<br>110.9 ILS 21R |
| Johannesburg Rand                                          | lic         | FAGM | 118.7 TWR<br>123.650 GND<br>130.5 Fuel                                                                                                                                   | 117.7 RAV                                                                     |
| Joubertina                                                 |             | FAJP |                                                                                                                                                                          |                                                                               |
| <b>K</b>                                                   |             |      |                                                                                                                                                                          |                                                                               |
| Kagga Kamma                                                |             | FAKA |                                                                                                                                                                          |                                                                               |
| Kalahari Dunes Private Game Res.                           | app         |      |                                                                                                                                                                          |                                                                               |
| Kalahari Oryx                                              | app         |      |                                                                                                                                                                          |                                                                               |
| Kapama Privete Game Reserve                                | app         |      |                                                                                                                                                                          |                                                                               |
| Karan's Camp                                               | app         |      |                                                                                                                                                                          |                                                                               |
| Karoo Gateway – see Beaufort West                          |             |      |                                                                                                                                                                          |                                                                               |
| Karreekloof                                                | app         |      |                                                                                                                                                                          |                                                                               |
| Kateka                                                     | app         |      |                                                                                                                                                                          |                                                                               |
| Kei Mouth                                                  |             | FAKE |                                                                                                                                                                          |                                                                               |
| Kenton-on-Sea                                              | reg         | FAKX |                                                                                                                                                                          |                                                                               |
| Kersefontein                                               | reg         | FAKF | 122.5 Langebaanweg                                                                                                                                                       |                                                                               |
| Khaki                                                      | app         |      |                                                                                                                                                                          |                                                                               |
| Kiboko                                                     | reg         |      | 124.2 unmanned                                                                                                                                                           |                                                                               |
| Kimberley Airport                                          | lic.        | FAKM | 119.4 APP<br>118.2 TWR<br>122.650 Kimberley<br>Apron                                                                                                                     | 113.2 KYV                                                                     |
| King Phalo – see East London                               |             |      |                                                                                                                                                                          |                                                                               |
| King Shaka International – see Durban                      |             |      |                                                                                                                                                                          |                                                                               |
| Kitty Hawk Aero Estate – see Boschkop                      |             |      |                                                                                                                                                                          |                                                                               |
| Klaserie                                                   | reg         | FAKC |                                                                                                                                                                          |                                                                               |
| Klein Geelbek                                              | reg         |      |                                                                                                                                                                          |                                                                               |
| Klein River                                                | reg.        |      |                                                                                                                                                                          |                                                                               |
| Kleinzee                                                   | reg.        | FAKZ | 123.5 unmanned                                                                                                                                                           |                                                                               |
| Klerksdorp P.C.Pelser                                      | lic         | FAKD | 123.5 unmanned                                                                                                                                                           |                                                                               |
| Knoppieskraal                                              | reg         |      |                                                                                                                                                                          |                                                                               |
| Knysna Airfield                                            | reg         |      |                                                                                                                                                                          |                                                                               |
| Koffyfontein                                               |             | FAKV |                                                                                                                                                                          |                                                                               |
| Koodoolake                                                 | app         |      |                                                                                                                                                                          |                                                                               |
| Kosi Bay                                                   |             | FAKB |                                                                                                                                                                          |                                                                               |
| Kosmos Aerodrome                                           | reg         |      |                                                                                                                                                                          |                                                                               |
| Koster                                                     |             | FAKO |                                                                                                                                                                          |                                                                               |
| Kriel                                                      |             | FAKL | 122.2 unmanned                                                                                                                                                           |                                                                               |
| Kroonstad                                                  | lic         | FAKS | 122.4 unmanned                                                                                                                                                           |                                                                               |
|                                                            |             |      |                                                                                                                                                                          |                                                                               |

| Place                                                      | Typ      | ICAO | COM                                                                | NAV                        |
|------------------------------------------------------------|----------|------|--------------------------------------------------------------------|----------------------------|
| <b>Kruger Mpumalanga International Airport - Nelspruit</b> | lic Intl | FAKN | 119.2 APP/TWR<br>122.650 Apron                                     | 112.5 PKV<br>109.1 ILS 05  |
| Krugersdorp (Jack Taylor)                                  | lic      | FAKR | 122.0 unmanned                                                     |                            |
| Kuruman                                                    |          | FAKU |                                                                    |                            |
| Kwa Siza Bantu                                             | reg      |      | 124.2 unmanned                                                     |                            |
| Kwandwe                                                    | lic      | FAKG |                                                                    |                            |
| <b>L</b>                                                   |          |      |                                                                    |                            |
| Ladysmith                                                  | lic      | FALY |                                                                    | 116.5 LYV                  |
| Lamberts Bay                                               | reg      | FALB |                                                                    |                            |
| Lander Strip                                               | reg      |      | 124.4 Pietermaritzburg GFA                                         |                            |
| Langebaanweg Air Force Base                                | mil      | FALW | 133.5 FIS<br>122.5 APP<br>120.8 TWR<br>128.8 TWR<br>121.750 GND    | 117.0 LWV<br>108.9 ILS 20L |
| <b>Lanseria International Airport</b>                      | lic Intl | FALA | 124.0 TWR<br>121.650 GND<br>122.850 Lanseria Apron<br>127.650 ATIS | 117.4 LIV<br>110.7 ILS 07  |
| Lalalala Wilderness                                        | app      |      |                                                                    |                            |
| Leppan                                                     | reg      |      |                                                                    |                            |
| Lichtenburg                                                | lic      | FALI |                                                                    |                            |
| Lime Acres Finsch Mine                                     | app      | FALC | 123.5 unmanned                                                     |                            |
| Lochinvar Farm                                             | reg      |      |                                                                    |                            |
| Loeriesfontein                                             |          | FALF |                                                                    |                            |
| Londolozi                                                  |          | FALD |                                                                    |                            |
| Lumarie                                                    | app      |      |                                                                    |                            |
| Louis Trichardt civil                                      | lic      | FALO | 121.2 Makhado APP                                                  |                            |
| Lydenburg                                                  | reg      | FALL |                                                                    |                            |
| <b>M</b>                                                   |          |      |                                                                    |                            |
| Mabalingwe                                                 |          | FAMA |                                                                    |                            |
| Mabula Runway                                              | app      |      |                                                                    |                            |
| Maclear                                                    |          | FAMF |                                                                    |                            |
| Madikwe East                                               | app      | FAMK |                                                                    |                            |
| Madikwe West                                               | app      |      |                                                                    |                            |
| Madimbo                                                    |          | FAMP |                                                                    |                            |
| Magwa                                                      | reg      |      |                                                                    |                            |
| Mahikeng - George Dick Montshioa                           | lic      | FAMM | 119.3 Mafikeng TWR                                                 | 112.9 MMV                  |
| Majuba Power Station                                       | app      | FAMJ | 122.2 unmanned                                                     |                            |
| Makhado Air Force Base                                     | mil      | FALM | 121.2 APP<br>119.8 TWR<br>121.750 GND<br>133.450 Ops<br>115.0 ATIS | 115.0 LTV<br>111.1 ILS 10  |
| Mala Mala                                                  | lic      | FAMD | 122.250 unmanned                                                   |                            |
| Malelane                                                   | lic      | FAMN | 130.350 unmanned                                                   |                            |
| Mankwe Kalahari Lodge                                      | app      |      |                                                                    |                            |
| Manyeleti Aerodrome                                        | app      |      |                                                                    |                            |
| Marble Hall - FAMI                                         | reg      | FAMI |                                                                    |                            |
| Margate                                                    | lic      | FAMG | 122.7 unmanned                                                     |                            |
| Maritz Farm Strip                                          | reg      |      |                                                                    |                            |

| Place                                  | Typ  | ICAO | COM                                | NAV                       |
|----------------------------------------|------|------|------------------------------------|---------------------------|
| Matatiele                              | reg  |      |                                    |                           |
| Matla-Mamba                            | app  |      |                                    |                           |
| Mazeppa Bay                            | reg  | FAMY |                                    |                           |
| Meadow Lane                            | reg. |      |                                    |                           |
| Microland Flight Park                  | app  | FABA |                                    |                           |
| Middelburg - Eastern Cape              | lic  | FAMC |                                    |                           |
| Middelburg - Mpumalanga                | lic  | FAMB | 127.950 unmanned                   |                           |
| Middeldrift Farm                       | reg  |      |                                    |                           |
| Millvale                               | lic  | FAMS |                                    |                           |
| Minesota Ranch                         | app  |      |                                    |                           |
| Mkambati                               | app  |      |                                    |                           |
| Mkuze                                  | lic  | FAMU |                                    |                           |
| Montague                               |      | FAMQ |                                    |                           |
| Morningstar Flying Club                | reg  |      | 126.8 CT SRA North                 |                           |
| Mosita Wildlife                        | app  |      |                                    |                           |
| Mosselbay                              | lic  | FAMO | 124.2 unmanned                     |                           |
| Msauli                                 |      | FAMZ |                                    |                           |
| Mthatha                                | lic  | FAUT | 121.3 TWR                          |                           |
| Munster                                | reg  |      |                                    |                           |
| Murray Ranches                         | app  |      |                                    |                           |
| Musina                                 |      | FAMH |                                    |                           |
| <b>N</b>                               |      |      |                                    |                           |
| Nampo Park                             | app  |      |                                    |                           |
| Nandoni Dam                            | reg  |      |                                    |                           |
| Naranja Airfield                       | reg  |      |                                    |                           |
| Nduno                                  |      | FANO |                                    |                           |
| Nelspruit                              | lic  | FANS | 125.2 unmanned<br>119.2 Kruger APP |                           |
| Newcastle                              | lic  | FANC |                                    |                           |
| New Tempe – see Bloemfontein New Tempe |      |      |                                    |                           |
| Ngala                                  | app  | FANG | 122.25 unmanned                    |                           |
| Ngodwana / Sappi                       |      | FAND |                                    |                           |
| Nieuwoudtville                         | -    |      |                                    | 116.1 NVV                 |
| Nirvana Aero Club                      | reg  |      | 130.350 unmanned                   |                           |
| Nongoma                                |      | FANA |                                    |                           |
| Ntsiri                                 | reg  | FANI | 124.8 and 119.0 LASS               |                           |
| Nyara Game Farm                        | app  |      |                                    |                           |
| Nyathi                                 | app  | FANT |                                    |                           |
| Nylstroom                              |      | FANY |                                    |                           |
| Nyoka Ridge Airfield                   | reg  |      |                                    |                           |
| Nyumbu                                 | app  |      |                                    |                           |
| <b>O</b>                               |      |      |                                    |                           |
| Olifantshoek                           |      | FAOF |                                    |                           |
| Onrus-Witgat                           | app  |      |                                    |                           |
| Orient                                 | lic  | FAOI | 123.4 unmanned                     |                           |
| OR Tambo – See Johannesburg OR Tambo   |      |      |                                    |                           |
| Othawa                                 |      | FAOL |                                    |                           |
| Oudtshoorn                             | lic  | FAOH | 131.1 AFIS                         |                           |
| Overberg Air Force Base                | mil  | FAOB | 119.8 TWR/APP                      | 115.4 OBV<br>110.9 ILS 35 |
| <b>P</b>                               |      |      |                                    |                           |
| PF Farmers (Pty) Ltd                   | reg  |      |                                    |                           |
| Paarl                                  |      | FAPU |                                    |                           |
| Pafuri                                 | app  |      |                                    |                           |

| Place                                                              | Typ      | ICAO | COM                                                                      | NAV                                       |
|--------------------------------------------------------------------|----------|------|--------------------------------------------------------------------------|-------------------------------------------|
| Paradise Beach                                                     | app      | FAPX |                                                                          |                                           |
| Parys                                                              | reg      | FAPY | 123.5 unmanned                                                           |                                           |
| Paulpietersburg Mondi                                              | reg      | FAPC |                                                                          |                                           |
| Pearly Beach                                                       | app      |      |                                                                          |                                           |
| Petit                                                              | reg      | FARA | 133.2 unmanned                                                           |                                           |
| Petrusville                                                        |          | FAPV |                                                                          | 112.7 PVV                                 |
| Phala Phala Wildlife Airstrip                                      | app      |      |                                                                          |                                           |
| Phalaborwa - Kruger Park Gateway                                   | lic      | FAPH |                                                                          | 115.3 PHV                                 |
| Phinda                                                             | app      | FADQ | 122.5 unmanned                                                           |                                           |
| Piesanghoek                                                        |          | FAQB |                                                                          |                                           |
| Piet Retief                                                        | lic      | FAPF |                                                                          |                                           |
| Pietermaritzburg                                                   | lic      | FAPM | 122.0 TWR                                                                | 117.9 PMV                                 |
| Pietersburg civil                                                  | lic      | FAPI | 119.75 Polokwane                                                         |                                           |
| Pietersrus                                                         |          | FAPW |                                                                          |                                           |
| <b>Pilanesberg International Airport</b>                           | lic Intl | FAPN | 118.4 TWR                                                                | 112.6 PNV                                 |
| Pilgrim's Rest                                                     |          | FAPO |                                                                          |                                           |
| Plettenberg Bay                                                    | lic      | FAPG |                                                                          |                                           |
| Pofadder                                                           |          | FAPD |                                                                          |                                           |
| <b>Polokwane International Airport</b>                             | lic Intl | FAPP | 119.750 TWR/APP                                                          | 114.5 PPV                                 |
| Polson                                                             | reg      |      |                                                                          |                                           |
| Pomfret                                                            |          | FAQF |                                                                          |                                           |
| Pongola                                                            | reg      | FAPL |                                                                          |                                           |
| Port Alfred                                                        | lic      | FAPA | 122.0 AFIS<br>118.4 GND                                                  |                                           |
| <b>Port Elizabeth - Chief Dawid Stuurman International Airport</b> | lic Intl | FAPE | 120.4 APP<br>118.1 TWR<br>122.650 Apron<br>126.8 ATIS<br>122.1 Clearance | 112.9 PEV<br>109.5 ILS 26<br>111.1 ILS 08 |
| Port St. Johns                                                     | app      | FAPJ |                                                                          | 115.9 PJV                                 |
| Postmasburg                                                        | reg      | FAPT |                                                                          |                                           |
| Potchefstroom                                                      | lic      | FAPS | 123.0 unmanned                                                           |                                           |
| Potgietersrus                                                      | reg      | FAQR |                                                                          |                                           |
| Pretoria Wonderboom                                                | lic      | FAWB | 118.350 TWR<br>120.6 GND                                                 | 112.2 DPV                                 |
| Prieska                                                            | lic      | FAPK |                                                                          |                                           |
| Progress                                                           | lic      | FAPZ | 122.7 unmanned                                                           |                                           |
| Pullenhope (Hendrina)                                              |          |      | 122.2 unmanned                                                           |                                           |
| Pumba Ranch Airfield                                               | reg      |      | 119.750 Polokwane TWR                                                    |                                           |
| Punda Maria                                                        |          | FAPQ |                                                                          |                                           |
| Pyramid Airfield                                                   | reg      |      |                                                                          |                                           |
| <b>Q</b>                                                           |          |      |                                                                          |                                           |
| Queenstown                                                         | lic      | FAQT |                                                                          |                                           |
| <b>R</b>                                                           |          |      |                                                                          |                                           |
| Rand Airport – see Johannesburg Rand                               |          |      |                                                                          |                                           |
| Ratelfontein                                                       | app      |      |                                                                          |                                           |
| Red Sands                                                          | app      |      |                                                                          |                                           |
| Reivilo                                                            | lic      | FARI |                                                                          |                                           |
| Reitz                                                              |          | FARZ |                                                                          |                                           |
| Richards Bay                                                       | lic      | FARB | 118.9 TWR                                                                | 113.0 RBV                                 |
| Richmond                                                           |          | FARM |                                                                          |                                           |

| Place                         | Typ  | ICAO | COM                                   | NAV       |
|-------------------------------|------|------|---------------------------------------|-----------|
| Rietfontein                   |      | FARF |                                       |           |
| Robertson                     | app  | FARS | 124.8 unmanned                        |           |
| Rock Fig Lodge                | app  | FATD |                                       |           |
| Royal Karoo Safaris           | app  |      |                                       |           |
| Royal Malewane                | app  |      |                                       |           |
| Royal Thabane                 | app  |      |                                       |           |
| Rustenburg                    | lic  | FARG |                                       |           |
| Rusticana                     | reg  |      |                                       |           |
| <b>S</b>                      |      |      |                                       |           |
| Sabie Lodge                   | app  |      |                                       |           |
| SabiSabi Aerodrome            | app  | FASE | 124.8 unmanned                        |           |
| Saldanha-Vredenburg           | lic  | FASD | 122.7 unmanned<br>122.5 Langebaan APP |           |
| Samara Karoo reserve          | app  |      |                                       |           |
| Sampour Ranch                 | reg  |      |                                       |           |
| Sandhurst                     | app  | FASA |                                       |           |
| Sandymount Park               | app  |      |                                       |           |
| Satara Airstrip               | app  |      |                                       |           |
| Schweizer Reneke              |      | FASG |                                       |           |
| Seaview Aerobatics            | app  | FAEW | 130.35 unmanned                       |           |
| Sebarakoma Airfield           | reg  |      |                                       |           |
| Secunda                       | lic  | FASC | 122.0 unmanned<br>121.0 ATIS          |           |
| Senekal                       |      | FASN |                                       |           |
| Serapa                        | app  |      |                                       |           |
| Shamwari Private Game Reserve | app  |      |                                       |           |
| Shelanti Gane Reserve         | app  |      |                                       |           |
| Shikwaru Lodge                | reg  |      |                                       |           |
| Singita / Sabie Sand          | reg  | FASP |                                       |           |
| Sirkel N                      | reg  |      |                                       |           |
| Sishen                        | lic  | FASS | 123.5 unmanned                        | 113.6 SSV |
| Siteka                        |      | FASM |                                       |           |
| Skoorsteenbergh Runway        | app  |      |                                       |           |
| Skukuza                       | lic  | FASZ | 125.25                                |           |
| Somerset East                 | reg  | FAST |                                       |           |
| Somersveld                    | mil. | FALS |                                       | 113.0 SVV |
| Southern Cross Game Reserve   | app  |      |                                       |           |
| Sparta Alma                   | app  |      |                                       |           |
| Springbok                     | lic  | FASB |                                       |           |
| Springbok Oog                 | reg  |      |                                       |           |
| Springs                       | lic  | FASI | 122.4 unmanned                        |           |
| Square Kilometre Array Karoo  | app  |      |                                       |           |
| St. Francis Field             | lic  | FACF | 124.8 unmanned<br>126.8 ATIS          |           |
| Standerton                    |      |      |                                       | 116.0 STV |
| Star                          |      | FAZQ |                                       |           |
| Steelport                     |      | FASO |                                       |           |
| Stellenbosch                  | lic  | FASH | 119.3 unmanned                        |           |
| Sterkfontein                  | reg  |      |                                       |           |
| Sterkloop                     | reg  |      |                                       |           |
| Steytlerville                 | reg  | FALR |                                       |           |
| Stilbaai                      | app  | FACY | 124.8 unmanned<br>119.8 Overberg AFB  |           |
| STS Meletse                   | app  |      |                                       |           |

| Place                                     | Typ         | ICAO | COM                           | NAV       |
|-------------------------------------------|-------------|------|-------------------------------|-----------|
| Stutterheim                               | reg         | FATM | 125.2 East London<br>SRA      |           |
| Sutherland                                |             | FASL |                               | 113.3 SLV |
| Swartberg                                 |             | FAZG |                               |           |
| Swartkop Air Force Base                   | mil         | FASK | 123.2 TWR<br>121.750 GND      |           |
| Swellendam - Hendrik<br>Swellengrebel     | app         | FASX |                               |           |
| Syferfontein                              |             | FASY | 122.35 unmanned               |           |
| <b>T</b>                                  |             |      |                               |           |
| Tankwa Tented Camp                        | reg         |      |                               |           |
| Tedderfield Air Park                      | lic         | FATA |                               |           |
| Tembani                                   | app         |      |                               |           |
| Thaba N'chu                               |             | FATN |                               |           |
| Thaba Tholo                               | lic         | FATG |                               |           |
| Thabazimbi                                | app         | FATI |                               |           |
| Thohoyandu                                | reg         | FATH |                               |           |
| Thorngrove                                | reg         |      |                               |           |
| Thorny Bush Game Lodge                    | app         | FATB | 126.4 Hoedspruit              |           |
| Thuru Lodge                               | app         |      |                               |           |
| Tommy's Field                             | lic         | FATF |                               |           |
| Tshikondeni                               | app         |      |                               |           |
| Tsitsikama Fly Inn                        |             | FATK |                               |           |
| Tswalu Game Reserve                       | lic         | FATW |                               |           |
| Tutuka Power Station                      | lic         | FATT | 122.2 unmanned                |           |
| Twee Rivieren                             | reg         | FATR |                               |           |
| Tzaneen                                   | lic         | FATZ |                               |           |
| <b>U</b>                                  |             |      |                               |           |
| Ugie                                      | reg         | FAUG |                               |           |
| Uitenhage                                 | lic         | FAUH | 124.2 unmanned                |           |
| Ulco                                      |             | FAUC |                               |           |
| Ulundi - Prince Mangosuthu<br>Buthelezi   | lic         | FAUL |                               |           |
| Ulusaba                                   | lic         | FAUS | 128.35 unmanned               |           |
| Umkomaas                                  | reg         |      | Durban GFA 124.4              |           |
| Umtshezi Municipality                     |             | FAEC | 123.4 unmanned                |           |
| <b>Upington<br/>International Airport</b> | lic<br>Intl | FAUP | 121.3 TWR/APP<br>122.65 Apron | 116.5 UPV |
| Utopia Airfield                           | reg         |      | 123.4 Gliders                 |           |
| Utrecht                                   | reg         | FAUR |                               |           |
| Uys Huis                                  | reg         |      |                               |           |
| <b>V</b>                                  |             |      |                               |           |
| Vaalputs                                  |             | FAVA |                               |           |
| Vanderbijlpark                            |             | FAVP | 122.3 unmanned                |           |
| Vastrap                                   | mil         | FAVS |                               |           |
| Venetia Mine                              | lic         | FAVM |                               |           |
| Verborgfontein                            |             | FAVF |                               |           |
| Vereeniging                               |             | FAVV | 122.1 unmanned                |           |
| Victoria West                             | reg         | FAVW |                               | 113.7 VWV |
| Virginia – see Durban Virginia            |             |      |                               |           |
| Volksrust                                 | reg         | FAVU |                               |           |
| Vredendal                                 | lic         | FAVR |                               |           |
| Vryburg                                   |             | FAVB |                               |           |
| Vryheid                                   | app         | FAVY |                               |           |

| Place                                | Typ | ICAO | COM                             | NAV                       |
|--------------------------------------|-----|------|---------------------------------|---------------------------|
| <b>W</b>                             |     |      |                                 |                           |
| Wagtail Aviation                     | reg |      |                                 |                           |
| Warden                               |     |      |                                 | 112.8 WRV                 |
| Warmbaths                            | lic | FAWA |                                 |                           |
| Waterberg                            | reg |      |                                 |                           |
| Waterkloof<br>Air Force Base         | mil | FAWK | 124.1 TWR<br>121.850 GND        | 116.9 WKV<br>111.3 ILS 01 |
| Watts Landing                        | reg | FAWG | 125.2 unmanned                  |                           |
| Wavecrest                            |     | FAWR |                                 |                           |
| Welgevonden Main Gate                | app | FAWE |                                 |                           |
| Welgevonden Western                  | app |      |                                 |                           |
| Welkom                               | lic | FAWM | 118.0 unmanned                  | 116.4 WMV                 |
| Wenlock                              | app |      |                                 |                           |
| Wesselsbron                          |     | FAWS |                                 |                           |
| White River (Mercy Air)              | app | FAWV |                                 |                           |
| Wilgebosch Farm                      | app |      |                                 |                           |
| Williston                            | lic | FAWL |                                 |                           |
| Willowmore                           |     | FAWO |                                 |                           |
| Winterveld Mine                      | reg | FAWT |                                 |                           |
| Wintervogel Flight Park              | reg |      | 124.2 GFA D69                   |                           |
| Witbank                              | lic | FAWI | 123.5 unmanned                  | 113.3 WIV                 |
| Wolmaranstad                         |     | FAWD |                                 |                           |
| Wonderboom – see Pretoria Wonderboom |     |      |                                 |                           |
| Woodlands Farm                       | reg |      |                                 |                           |
| Worcester                            | lic | FAWC | 124.8 unmanned<br>123.4 Gliding |                           |
| <b>Y</b>                             |     |      |                                 |                           |
| Ysterplaat Air Force Base            | mil | FAYP | 125.6 TWR<br>121.850 GND        |                           |
| <b>Z</b>                             |     |      |                                 |                           |
| Zandspruit                           | reg |      | 126.4 Hoedspruit APP            |                           |
| Zebula                               | reg |      |                                 |                           |
| Zeerust                              |     | FAZR |                                 |                           |
| Zilkaats                             | reg |      |                                 |                           |
| Zonderwater                          |     |      |                                 | 114.3 CZV                 |
| Zuikerkop Lodge                      | reg |      |                                 |                           |
| Zululand Anthracite<br>Colliery      |     | FAZC |                                 |                           |
| Zuney                                |     | FAZU |                                 |                           |

# Appendix II

# Frequencies

It is not always easy to determine who you should be talking to and on what frequency. The following hopefully helps to give some clarity.

The list on the following pages is split into airfields for take-off and landing and into en-route, and can be used essentially like a flowchart. Read the first heading and decide if it applies. If yes, read the paragraph. If not, continue to the next point.

## AT AIRFIELDS

**1. Is there a frequency listed for that airfield in Appendix I of this book and/or in the AIP ENR 2.2-54?**

**NO – go to point 2. YES – read this:**

**In CTRs and ATZs Class C or D (controlled airspace)**

when leaving: talk to Ground (GND) if a frequency for GND is listed, if not, talk to Aerodrome Control on the Tower (TWR) frequency. When arriving, talk to the TWR frequency. Maintain two-way radio contact and follow instructions. If the frequency is not manned (after hours for example), broadcast on the same frequency. These frequencies are also listed in the AIP AGA.

**In ATZs Class G (uncontrolled airspace)**

talk to Aerodrome Flight Information Service (AFIS) on the AFIS frequency given for the particular airfield. Maintain two-way radio contact, but remember that AFIS gives advice only, and does not control air traffic. If the frequency is not manned (after hours for example), broadcast on the same frequency. These frequencies are also listed in the AIP AGA.

**At unmanned airfields which have been allocated a dedicated frequency**

broadcast your intentions within 5 nm of the airfield on the allocated TIBA frequency as given in Appendix I as “unmanned”. These frequencies are also (sometimes) listed on the CAA website under the airfield information as explained in the intro tot Appendix I of this book.

**2. Is the airfield situated within a General Flying Training Area (GFA)?**

**NO – go to point 3. YES – read this:**

**Check under EN ROUTE on the next pages if the GFA has a dedicated frequency.** You can also check AIP ENR 5.1 under Danger Areas.

If the GFA has a dedicated frequency broadcast your intentions within 5 nm of the airfield on the frequency for the General Flying Area (GFA) as listed.

If the GFA does not have a dedicated frequency broadcast your intentions within 5 nm of the airfield on **124.4 MHz** – the default frequency for GFAs.

**3. Is the airfield is within a Special Rules Area (SRA)?**

**NO – go to point 4. YES - read this:**

Broadcast your intentions on the frequency for the specific Special Rules Area (SRA) as listed on the next pages (EN ROUTE) or in AIP ENR 2.2.

**4. Broadcast on **124.8 MHz** - the default uncontrolled airfield frequency.**

## EN ROUTE

### In TMAs (Class C airspace)

Talk to Approach Control on the Approach (APP) frequency given in Appendix I for the particular airport or centre. Maintain two-way radio contact and follow instructions. If the frequency is not manned (after hours for example), broadcast on the same frequency.

### In CTAs, Air Service Routes, RNAV Routes (Class A or C airspace)

Talk to Area Control (Control Service) on the Area frequency as given below for the particular centre. Maintain two-way radio contact and follow instructions.

#### Frequencies for Area Control in CTAs and other Class A airspace:

| FIR  | Sector / CTA            | FL          | Call sign           | Freq.    |
|------|-------------------------|-------------|---------------------|----------|
| FAJA | North                   | FL110-FL460 | Jbg Area North      | 126.7    |
|      | North-East              | FL110-FL460 | Jbg Area North-East | 134.4    |
|      | South-West              | FL110-FL460 | Jbg Area South-West | 128.3*   |
|      | South East              | FL110-FL460 | Jbg Area South-East | 132.150* |
|      | Namibia part of FAJA    | FL145-FL195 | Windhoek Area       | 124.7    |
|      | West (west of E0195439) | FL200-FL460 | Jbg Area West       | 118.5    |
|      | West (east of E0195439) | FL200-FL460 | Jbg Area West       | 118.550  |
|      | Johannesburg Central    | FL200-FL460 | Jbg Area Central    | 120.3    |
|      | Bloemfontein CTA        | FL145-FL195 |                     |          |
|      | Johannesburg East       | FL200-FL460 | Jbg Area East       | 129.1    |
|      | Durban CTA A            | FL145-FL200 |                     |          |
|      | Durban CTA B            | FL105-FL145 |                     |          |
| FACA | Cape Town West          | FL200-FL460 | CT Area West        | 125.1    |
|      | Cape Town CTA A         | FL145-FL200 |                     |          |
|      | Cape Town CTA B West    |             |                     |          |
|      | Cape Town East          | FL200-FL460 | CT Area East        | 124.7    |
|      | Cape Town CTA B East    | FL145-FL200 |                     |          |

\* after hours of operation use 126.7

### In the Lowveld Military Airspace Sector (LASS)

Between 1500' AGL and FL195 talk to "Lowveld Information" on **119.0 MHz**  
or on the secondary frequency **129.4 MHz**

Maintain two-way radio contact and follow instructions. When the frequency is not manned at night or on weekends, broadcast on the same frequency.

In sector A1 at 7500' ALT or above, in sector A2 at 1500' AGL or above and in sector B at 7100' and above talk to "Matsapha Approach" on **119.0 MHz**  
and after hours to "Johannesburg North-East" on **134.4 MHz**

Below the altitudes given for the three sectors broadcast on **125.0 MHz**

IFR traffic wanting to enter Johannesburg controlled airspace must talk to Johannesburg North-East as per the table above (on 128.9 MHz above FL105 by day and at all levels at night or on 127.4 MHz below FL110 by day).

### **In General Flying Training Areas (GFAs) with a dedicated frequency**

Listen out and broadcast your position, altitude and intentions on the frequencies listed below:

|                                                                                                            |                    |
|------------------------------------------------------------------------------------------------------------|--------------------|
| Brits / Grand Central Flying Training Area (FAD129)                                                        | <b>124.8 MHz</b>   |
| Johannesburg Flying Training Area (FAD182)                                                                 | <b>122.350 MHz</b> |
| Magaliesberg General Flying Training Area (FAD70)                                                          | <b>124.8 MHz</b>   |
| Nylstroom Flying Training Area (FAD60)                                                                     | <b>131.1 MHz</b>   |
| Oudtshoorn Flying Training Area (FAD194 and FAD195)                                                        | <b>122.350 MHz</b> |
| Port Elizabeth and Uitenhage Civil General Flying Area (FAD193)                                            | <b>124.2 MHz</b>   |
| Stellenbosch General Flying Area (FAD69)                                                                   | <b>124.2 MHz</b>   |
| Syferfontein Aerobatic Area (FAD184) – call sign RABBIT                                                    | <b>122.350 MHz</b> |
| Active Wed, Sat, Sun. Liaise with JHB Radar 124.5 MHz or 123.7, also broadcast JHB Special Rules 125.6 MHz |                    |

But in the:

|                                |                                |                  |
|--------------------------------|--------------------------------|------------------|
| Kimberley GFA (FAD108) > 5000' | talk to Kimberley Tower on     | <b>118.2 MHz</b> |
| Phalaborwa GFA (FAD125)        | talk to Lowveld Information on | <b>119.0 MHz</b> |

### **In all other General Flying Training Areas**

Listen out and broadcast your position, altitude and intentions on **124.4 MHz**

### **In Special Rules Areas (SRAs):**

Listen out and broadcast position, altitude and intentions on the frequencies:

|                    |                                |                    |
|--------------------|--------------------------------|--------------------|
| Cape Town SRA      | east                           | <b>124.8 MHz</b>   |
|                    | west                           | <b>125.8 MHz</b>   |
|                    | north                          | <b>126.8 MHz</b>   |
| Durban SRA         |                                | <b>124.2 MHz</b>   |
| East London SRA    |                                | <b>125.2 MHz</b>   |
| George SRA         | west                           | <b>124.2 MHz</b>   |
|                    | east                           | <b>124.8 MHz</b>   |
| Hoedspruit SRA     | talk to Lowveld Information on | <b>119.0 MHz</b>   |
| Johannesburg SRA   | east                           | <b>125.4 MHz</b>   |
|                    | south                          | <b>125.6 MHz</b>   |
|                    | west                           | <b>125.8 MHz</b>   |
| Kruger SRA         |                                | <b>130.350 MHz</b> |
| Malelane SRA       |                                | <b>129.350 MHz</b> |
| Port Elizabeth SRA |                                | <b>130.350 MHz</b> |
| Sabi-Sands SRA     |                                | <b>128.350 MHz</b> |

**In all other uncontrolled airspace above 1500’ AGL**

Maintain two-way radio contact with Information on the frequencies given in the table below. The person you will be speaking to will provide you with Flight Information Service (FIS), warn you about other traffic or maybe adverse weather, but will not control the flight.

**Frequencies for Flight Information Service (uncontrolled airspace):**

| FIR  | Sector                  | Alt. / FL      | Frequency    |             |
|------|-------------------------|----------------|--------------|-------------|
|      |                         |                | Office hours | After hours |
| FAJA | North and North-East    | 1500’ to FL105 | 124.7        | 126.7       |
|      | North                   | FL110 to FL195 | 126.7        |             |
|      | North-East              | FL110 to FL195 | 134.4        |             |
|      | South and South-West    | 1500’ to FL105 | 119.5        | 126.7       |
|      | South-West              | FL110 to FL195 | 128.3        | 126.7       |
|      | South East              | FL110 to FL195 | 132.150      | 126.7       |
|      | Namibia part of FAJA    | 1500’ to FL145 | FYWF 123.8   | FYWF 124.7  |
|      | West (west of E0195439) | 1500’ upwards  | 118.5        |             |
|      | West (east of E0195439) | 1500’ upwards  | 118.550      |             |
|      | Johannesburg Central    | 1500’ upwards  | 120.3        |             |
|      | Johannesburg East       | 1500’ upwards  | 129.1        |             |
| FACA | Cape Town Info West     | 1500’ upwards  | 131.125      | unmanned    |
|      | Cape Town Info East     | 1500’ upwards  | 127.575      | unmanned    |

Use the call-sign “Information” for the stations above, either on its own or prefixed by the name of the FIR sector, e.g. “Central Information”

**In all other uncontrolled airspace at or below 1500’ AGL**

Listen out and broadcast your position, altitude and intentions on **124.8 MHz**

There are two other useful general frequencies:

**Glider pilots**

Communicate with each other on **123.4 MHz**

**GENERAL CHAT FREQUENCY – “Numbers”**

If you need to speak to another pilot about the braai arrangements for the evening or anything else that is not related to position reporting, ask him to “go numbers” and then chat to him on: **123.450 MHz**

- “ Be careful what you say, though: other pilots, cruising on autopilot with little to do, might also tune in and listen to your conversation.

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